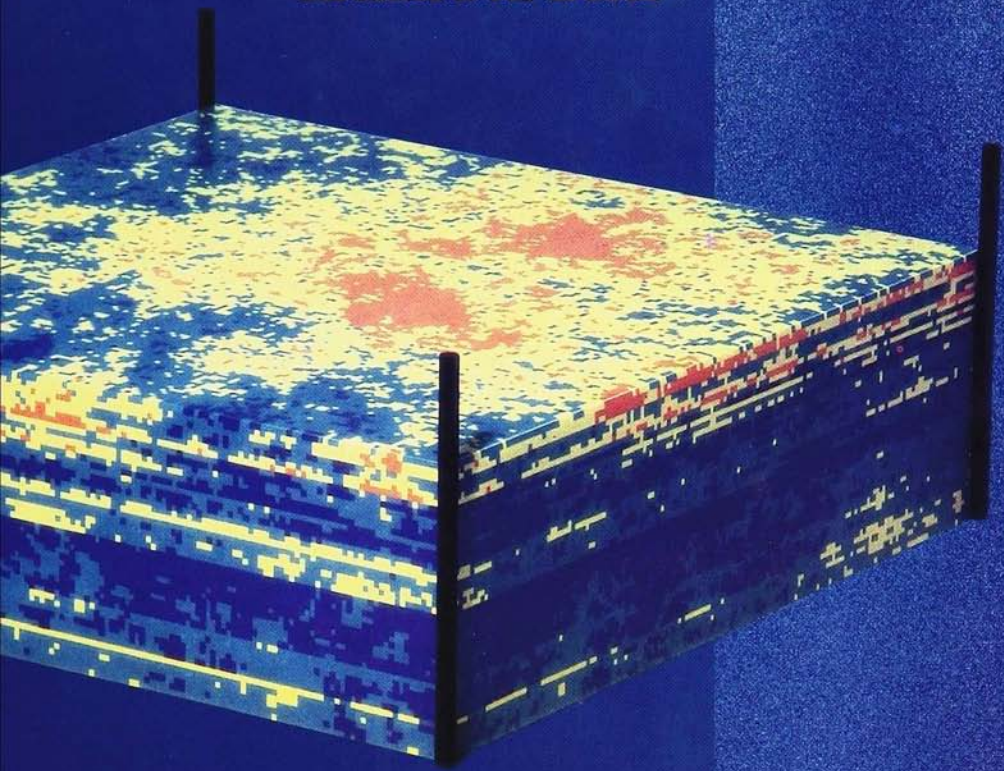
FRACTALS IN RESERVOIR ENGINEERING

H. H. Hardy
Richard A. Beier



World Scientific

FRACTALS IN RESERVOIR ENGINEERING

This page is intensionally left blank

FRACTALS IN RESERVOIR ENGINEERING

H. H. Hardy
Richard A. Beier
Conoco Inc.

World Scientific
Singapore • New Jersey • London • Hong Kong

Published by

World Scientific Publishing Co. Pte. Ltd.

P O Box 128, Farrer Road, Singapore 9128

USA office: Suite 1B, 1060 Main Street, River Edge, NJ 07661

UK office: 73 Lynton Mead, Totteridge, London N20 8DH

Library of Congress Cataloging-in-Publication Data

Hardy, H. H.

Fractals in reservoir engineering / H.H. Hardy, Richard A. Beier.

p. cm.

Includes bibliographical references and index.

ISBN 9810220693

1. Petroleum--Geology--Mathematical models. 2. Fractals.

I. Beier, Richard A. II. Title.

TN870.53.H37 1994

622'.3382--dc20 94-24457

CIP

Copyright © 1994 by World Scientific Publishing Co. Pte. Ltd.

All rights reserved. This book, or parts thereof, may not be reproduced in any form or by any means, electronic or mechanical, including photocopying, recording or any information storage and retrieval system now known or to be invented, without written permission from the Publisher.

For photocopying of material in this volume, please pay a copying fee through the Copyright Clearance Center, Inc., 27 Congress Street, Salem, MA 01970, USA.

Printed in Singapore.

To Joan, Tabitha, Chip, and Kara

HHH

To Carol, Matthew, and John

RAB

This page is intensionally left blank

Foreword

Mandelbrot's book, *The Fractal Geometry of Nature*, has captured the imagination of laymen and scientists alike—it has changed how we "see" Nature. The literature on fractals is huge and growing at a stunning rate. Many interesting results are found in the literature. However, much is confused. There is a lack of guidance of where to start for one who wishes to apply fractals to their respective field. The potential user may be fascinated with the many published visual images but still uncertain how to extend these concepts to an application. Now here is a useful book, which is clear and concise on the application of fractals. It can be used as a textbook as well as a book for self-study.

This book describes the use of fractals in reservoir engineering. Everything in this book has been tested by the authors, and everything is explained carefully. Open questions are spelled out at the end of each chapter.

We all depend on the flow and production of oil (and water), and dramatic progress in reservoir engineering and reservoir simulation has been made possible with the development of modern computers. Information on reservoir properties (such as porosity, permeability, water saturation, and so on) is available from core samples, well logs, and seismic surveys, but this information is insufficient for a good reservoir simulation. The authors discuss in detail to what extent porosity and permeability are fractal and how fractal models of reservoir properties can be used to estimate flow properties between wells. Fractional Brownian motion and fractional Gaussian noise are the main tools. These tools are applied in one dimension to well logs, in two dimensions to correlated heterogeneity seen in core samples and outcrops, and in three dimensions to porosity and permeability in reservoirs.

Upscaling of permeability is a non-trivial exercise, and a real space renormalization procedure, FRACTAM, is discussed. This method is also combined with standard methods in the chapter on reservoir simulations.

This book is a bridge between reservoir engineering and the physics of transport in disordered systems. The engineer should, with this background, gain access to the physics literature. The physicist will find many interesting questions and problems to explore.

Jens Feder
Department of Physics
University of Oslo
Oslo, Norway
June 1944

Preface

Benoit Mandelbrot introduces fractals in his classic book, *The Fractal Geometry of Nature*, with the observation that, "Clouds are not spheres, mountains are not cones, coastlines are not circles, and bark is not smooth, nor does lightning travel in a straight line." To this we might add, "... and hydrocarbon reservoirs are not cylindrical tanks."

There has always been an effort in the oil industry to improve the description of oil and gas reservoirs. This effort is based on the hope that better reservoir descriptions will result in better predictions and, therefore, a more economical production of oil and gas. This book describes the application fractal mathematics as a tool to improve the description of hydrocarbon reservoirs.

The description of hydrocarbon reservoirs encompasses the characterization of rocks from the pore scale (10^{-8} m) to the basin scale (10^{5} m). A composite review from the literature of fractal applications to hydrocarbon production would span this entire range of length scales. We have chosen to focus on the length scales between a few feet (10^{0} m) and the distance between wells (10^{3} m). Ultimately, progress at any length scale may have important consequences. Still, in order to limit the scope of this book, we have chosen to focus on the interwell scale.

We thank our many colleagues within Conoco Inc. for their support and collaboration. Paul Meakin, while working for DuPont, introduced H. H. Hardy to fractals. Jeff Jurinak and Don Whitfill encouraged our interests and supported us in our applications of fractals. Over twenty engineers and geologists in Conoco have applied these methods to more than a dozen Conoco reservoirs. We thank them for their efforts and their patience while learning together what is found in this book. We especially thank Bruce Reid and Yogi Soni for allowing us to use some of their unpublished results as illustrations.

Our applications of fractals have been influenced by discussions with many people outside of Conoco. In particular, we thank Tom Hewett (Stanford University), Jens

Feder and Torstein Jossang (University of Oslo), and Amnon Aharony (Tel Aviv University).

George Overall drafted the figures. Lisa Stover and Cindy Larmer typed the manuscript. All three had a great deal of patience with us throughout the many revisions of the manuscript. We also wish to thank Conoco management for allowing us to publish this work. We hope that the information contained here will help guide and encourage others to continue their applications of fractals to their respective fields.

H. H. Hardy
Richard A. Beier

Contents

Foreword vii

Preface ix

CHAPTER 1. INTRODUCTION 1
- **1.1 Reservoir Engineering** 3
- **1.2 Definition of a Fractal** 4
- **1.3 Statistical Fractals** 11
- **1.4 When is an Object Fractal** 16

APPENDIX A1 18
- **A1.1 Perimeter of the Triadic Koch Island** 18
- **A1.2 Area of the Triadic Koch Island** 19

CHAPTER 2. STATISTICS 23
- **2.1 Ensembles** 24
- **2.2 Univariant Measures** 28
- **2.3 Multivariant Measures** 32
  - **2.3.1** *Covariance* 36
  - **2.3.2** *Variogram* 36
  - **2.3.3** *Spectral Density* 37
  - **2.3.4** *R/S Analysis* 40
  - **2.3.5** *Relationships Between Multivariant Measures* 40
- **2.4 Fractal Models** 41
- **Unresolved Issues** 47

APPENDIX A2 48
- **A2.1 Equations for Statistical Measures** 48
  - **A2.1.1** *Basic Symbolism* 49
  - **A2.1.2** *Univariant Measures* 50
  - **A2.1.3** *Multivariant Measures* 51
- **A2.2 The Fourier Transform in 1D** 54
- **A2.3 Amplitude, Phase, and Spectral Density** 57

A2.4 Spectral Generators for fBm and fGn 58
A2.5 Some Useful Properties of Fourier Transforms 60
A2.6 Linear Scaling 65
A2.7 Stationarity and Ergodicity of fGn and fBm 65
A2.8 Variogram and Covariance of fGn 67
A2.9 Mandelbrot fGn Versus Spectral Generated fGn 77

CHAPTER 3. FGN IN RESERVOIR PROPERTY DISTRIBUTIONS 79
3.1 Applying Statistical Measures to Data 80
3.2 Well Logs 86
3.3 Core and Outcrop Photos 94
3.4 Outcrop Studies 96
3.5 Two Dimensional Data Sets 102
Unresolved Issues 117

APPENDIX A3 118
A3.1 A More Complete Description of Equation 3.2 118
A3.2 Smoothing Filters 123
A3.3 Spectral Density Roll-Off and Filter Size 124
A3.4 The Normal Score Transform 128
A3.5 The Discrete Inverse 2D Fourier Transform 131
A3.6 2D Covariance From the Spectral Density 133

CHAPTER 4. GENERATION OF POROSITY DISTRIBUTIONS 137
4.1 Steps for Generating Distributions 138
4.2 Identify Zones of Interest 139
4.3 Analyze Well Logs 139
4.4 Generate fGn Cross Sections 141
4.4.1 *Basic Assumptions* 141
4.4.2 *Relaxing Some Assumptions* 144
4.4.3 *Cross Section With Many Units* 146
4.4.4 *Cross Section With Thin Coal Seams* 147
4.5 Checking Results 149
4.6 fBm Cross Sections 151
4.7 3D fGn Distributions 152
4.8 Large Well Spacings 154
4.9 Integration of Large and Small Scale Data 156
4.9.1 *One Dimensional Example* 156
4.9.2 *Two Dimensional Example* 161
Unresolved Issues 162

APPENDIX A4 163
A4.1 Filling "Holes" in Log or Core Traces 163
A4.2 fGn Cross Section Generation 171

A4.3 fBm Cross Section Generation 178
A4.4 The Generation of 3D Distributions 180
A4.5 Combining Small and Large Scale Data in 2D 194

CHAPTER 5. PERMEABILITY DISTRIBUTIONS 197
5.1 Generation of Permeability Distributions 198
5.2 Scaling or Averaging Reservoir Properties 202
5.3 Layers and Random Media 203
5.4 The Renormalization Algorithm--FRACTAM 204
5.5 Horizontal and Vertical Permeability Study 212
5.5.1 *The Effect of Layering* 214
5.5.2 *Porosity-Permeability Relationships* 215
5.5.3 *Permeability Cutoffs* 215
5.5.4 *2D and 3D Distributions* 219
5.5.5 *Grid Block and Porosity-Permeability Crossplots* 220
5.5.6 *Grid Block Shapes and Effective Permeabilities* 223
5.5.7 *Permeability as a Function of Distance* 225
Unresolved Issues 229

APPENDIX A5 229
A5.1 Equations for Solving for k_{eff} 230
A5.2 FRACTAM Equations 232
A5.3 FRACTAM 234

CHAPTER 6. RESERVOIR SIMULATIONS 237
6.1 Fluid Permeability With Two Flowing Phases 238
6.2 Scaling Up Reservoir Properties 240
6.3 Hybrid Finite Difference/Streamtube Method 241
6.3.1 *Limitations* 245
6.3.2 *Reservoir Simulations on Cross Sections* 245
6.3.3 *Fractional Flow and Mobility Calculations* 248
6.3.4 *Generation of Streamtubes* 249
6.3.5 *Streamtube Displacement Calculation* 250
6.3.6 *Comparison With Field Production Data* 251
6.3.7 *Sensitivity Studies* 255
6.4 Pseudorelative Permeabilities 258
6.4.1 *Miscible Displacement* 259
6.4.2 *Comparison With Field Production Data* 262
6.4.3 *Sensitivity Studies* 263
6.4.4 *Numerical Dispersion* 267
6.5 3D Simulations 269
6.5.1 *Fractal Distributions* 271
6.5.2 *Simulations Based on Two Well Logs* 272
6.5.3 *Simulations Based on Four Well Logs* 275

6.6 Limitations in Scaling Up 276
6.7 Ranking Realizations 279
6.7.1 *Methodology* 280
6.7.2 *Waterflood Example* 283
Unresolved Issues 287

APPENDIX A6 287
A6.1 Mappings Between 2D Cross Sections and Areal Models 287
A6.2 Streamtube Method 289
A6.2.1 *Mobility Correlations* 289
A6.2.2 *Generation of Streamtubes* 291
A6.2.3 *Streamtube Displacement Calculation* 292
A6.3 Kyte and Berry Pseudorelative Permeabilities 293
A6.4 Flux-Weighted Pseudorelative Permeabilities 295

CHAPTER 7. OTHER APPLICATIONS 297
7.1 Pressure Transient Tests 297
7.1.1 *Conventional Pressure Transient Model* 298
7.1.2 *Conventional Model Applied to Field Test* 301
7.1.3 *Fractal Permeable Network* 303
7.1.4 *Fractal Model Applied to Field Test* 304
7.2 Faults and Fractures 306
7.2.1 *Scaling Relationships* 307
7.2.2 *Simple Fracture Propagation Model* 309
7.3 Music From Well Logs 315
7.3.1 *Porosity to Frequency Map* 315
7.3.2 *Frequency Corresponding to the C Major Scale* 319
7.3.3 *Rhythm and Phrasing* 322
7.3.4 *Musical Score* 322
Unresolved Issues 323
Pressure Transient Tests 323
Faults and Fractures 324

APPENDIX A7 326
A7.1 Fractal Permeable Network 326
A7.2 Fractal Reservoir Solution 327
A7.3 Type Curves 332
A7.4 Match to Pressure Buildup Field Example 335

REFERENCES 339

NOMENCLATURE 347

INDEX 355

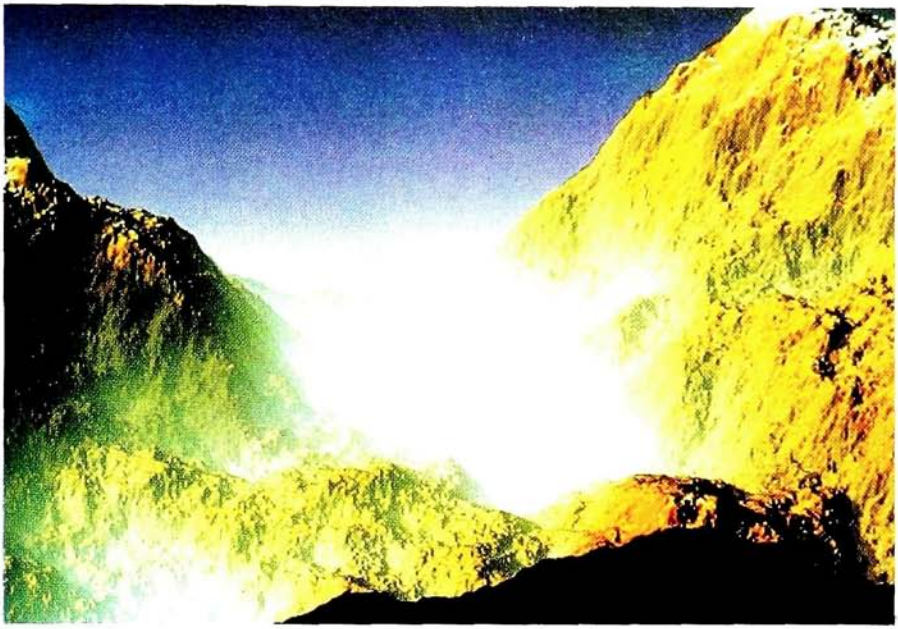


Plate 1. A "mountain" scene generated using 2D fractional Brownian motion.[11]

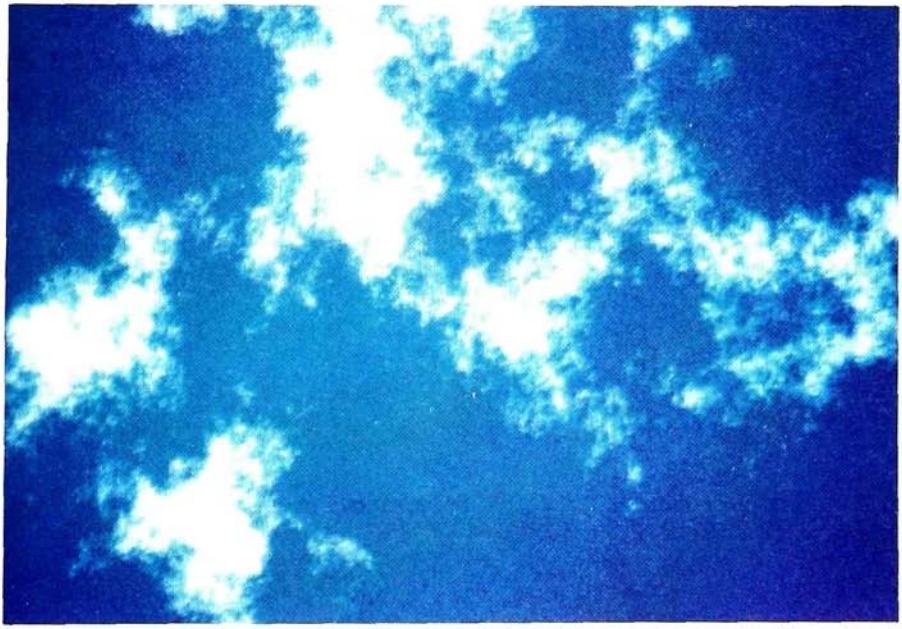


Plate 2. Clouds generated using 2D fractional Brownian motion.[11]

Copyrighted Material

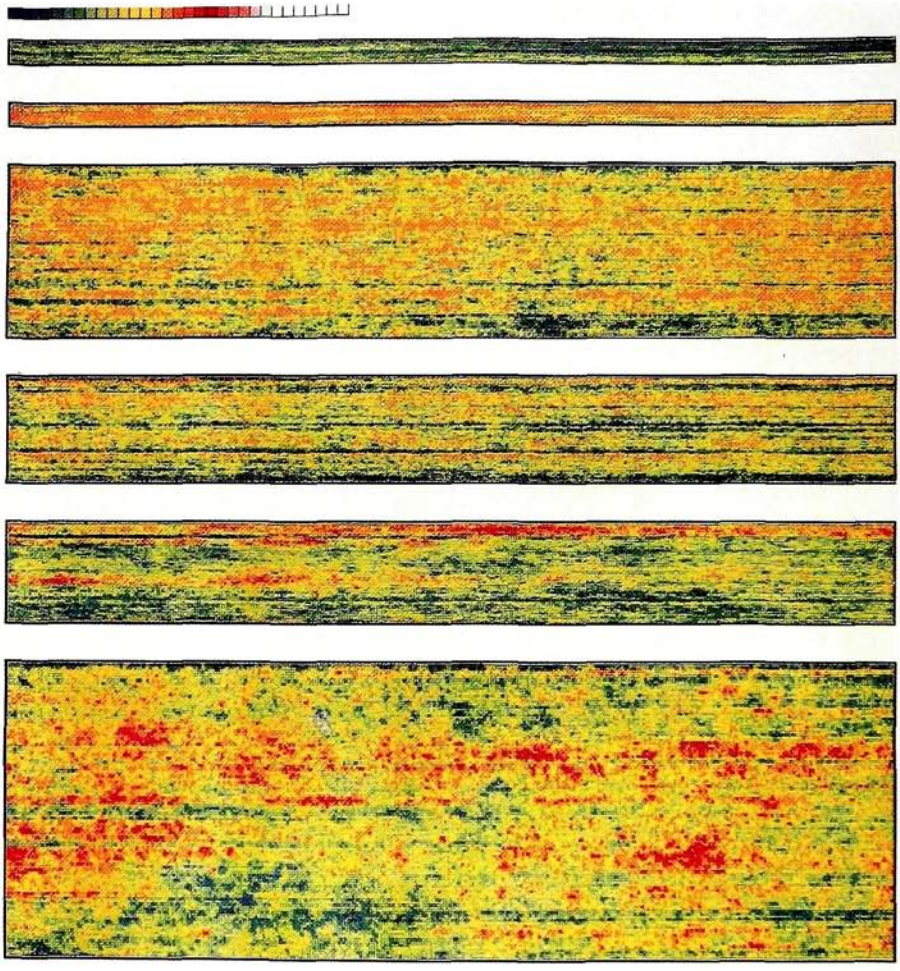


Plate 3. fGn cross sections generated for formations ranging from intertidal mud flats to shallow marine shelf deposits.

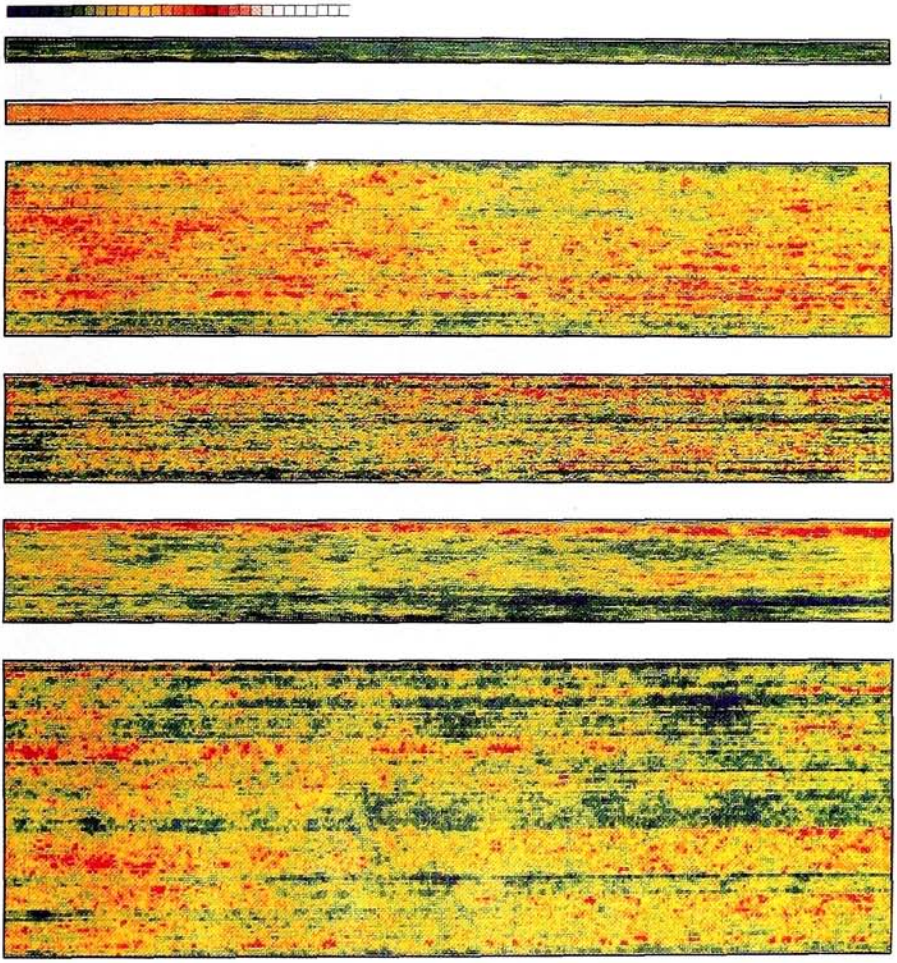


Plate 4. fGn cross sections generated at almost a right angle to Plate 3.

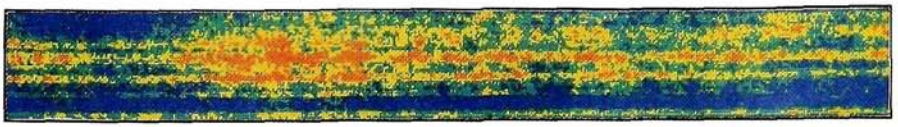


Plate 5a. Realization #1 of vertical cross section.

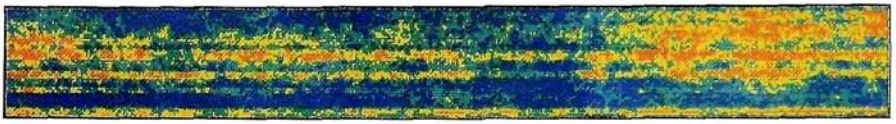


Plate 5b. Realization #2 of vertical cross section.

Plate 5c. Smoothed vertical cross section (Realization #1).

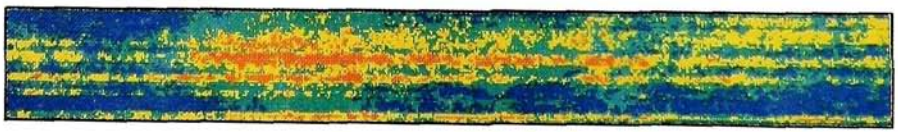


Plate 5d. Realization #2 with low frequency information from Realization #1.

CHAPTER 1

INTRODUCTION

This book is written for reservoir engineers, geologists, geophysicists, and for researchers developing fractal applications. Its purpose is threefold—first, to explain what fractals are and how they are being applied to reservoir engineering problems so that this knowledge can be applied to producing oil and gas more economically; second, to provide for researchers a map of the edge of our understanding so that they may expand the understanding of applying fractals without having to rediscover what is already known; and third, to show the utility of fractal models in one engineering field to encourage others to apply fractals in their own engineering fields.

Most of the applications of fractals by the authors have been centered on the identification of fractal distributions and the application of fractals to reservoir engineering. Thus, most of this book will concentrate on the reservoir engineer's point of view. The reservoir engineer's job is to engineer the production of hydrocarbons. He must deal with much uncertainty because of the sparse sampling of the reservoir that wellbores provide. He uses mathematical models of the earth and the flow of fluids through it, in order to decide between alternative production strategies.

Historically, the mathematical models used by the reservoir engineer have been based on the principle of effective rock properties for fluid flow calculations. Statistical analyses of rock properties challenge the assumptions of homogeneity or complete randomness often used in justifying the use of effective rock properties. Fractals provide an alternative mathematical model of the distribution of rock properties based on scaling laws. Fractals generate distributions that have both

heterogeneity and order. Not all rock property distributions are fractal, but when they are, fractal methods can provide strikingly good models.

This book will show the evidence for fractal distributions and present some of the alternative mathematical models that incorporate these findings. The work presented here has been performed primarily within Conoco, but some effort to incorporate fractal descriptions into reservoir engineering has been made by almost every major oil company.

Ultimately, the description of oil and gas production from reservoirs is an economical question. The models used to address problems need not always be the most complex or even the best available but must be the most cost effective. In the same way that reservoir flow simulations on computers have not completely displaced analytic models, such as material balance and pressure decline analyses, fractals will not completely displace homogeneous models. Still, in oil reservoirs where risks and costs are high, we believe fractal models will take their place along with existing models as a powerful tool for the reservoir engineer. Fractal property distributions are not found in all reservoirs, so other geostatistical methods will also be required when fractal distributions are not representative. We concentrate here, however, on fractals applications. Even so, much of what is presented (e.g., scaling methods, effective properties in reservoir flow simulation, etc.) is also applicable where fractal distributions are not.

This book is organized into seven chapters. Chapters 1 and 2 provide background for the rest of the book. Chapters 3 through 6 discuss the application of fractals to reservoir description, which is the main focus of this book. Chapter 3 describes the data sets tested in developing a reservoir description model based on fractals. Chapter 4 illustrates the application of this model to generating porosity distributions. Chapter 5 describes permeability distributions, and Chapter 6 reviews the results of reservoir flow simulations based on these distributions. Chapter 7 points out some additional applications of fractals.

Each chapter is divided into three parts. The main body of each is a verbal description of the science and engineering concepts. This second part of each chapter is a list of outstanding issues. The third part is an appendix presenting the mathematical formalism. We recommend reading the first part of each chapter, then returning to the outstanding issues or the appendix of each chapter as time and interest permit.

Since this book is designed for readers from various backgrounds, this introduction is designed to provide a simple description of both reservoir engineering and fractals. The first section describes a few basics of reservoir engineering. The next three sections describe a few basics of fractal mathematics. If you are a reservoir engineer or have had a basic reservoir engineering course, you can skip the first section entitled, "Reservoir Engineering." If you are familiar with fractals through other reading, you may wish to skip one or more of the sections on fractals. We'll all get back together at the beginning of the next chapter.

1.1 Reservoir Engineering

The purpose of reservoir engineering is to engineer the production of oil and gas from a reservoir in the most economical way. Once oil or gas is found, the reservoir engineer's job is to manage the production of hydrocarbons from the reservoir. This involves determining the number and location of wells to optimally produce a given reservoir, deciding when and if the injection of fluids (e.g., water or carbon dioxide) is appropriate to enhance recovery, and predicting the rate of production for the design of surface facilities and negotiation of sales contracts.[1]

The reservoir engineer has many tools. Logging tools,[2] subsurface and surface fluid samples,[3] pressure transient tests,[4] coring,[5] and seismic measurements[6] can all be used to help infer the best production strategy. The reservoir engineer must assimilate these data into his reservoir production interpretation. He has a number of numerical modeling tools at his disposal. These range from simple material balance calculations to numerical flow simulations that can involve many thousands of nodes (or grid blocks) and complicated fluid-fluid and fluid-rock interactions.

We obviously cannot address all facets of reservoir engineering in this book. We will not even address all of the reservoir engineering topics to which fractals have been applied. We will concentrate instead on those reservoir engineering areas where the authors have applied fractal mathematics. These include logging measurements, core analyses, pressure test data, and seismic data. Below is a brief explanation of each of these tools.

In general, any tool placed in a well to take a series of measurements at different distances from the surface is called a logging tool. Many different logging tools[2] have been developed to infer lithology, porosity, fluid content, and other

attributes of the subsurface rocks and fluids. In this book we deal especially with tools that estimate porosity of the rock.

While a well is being drilled, cores or cylinders of rock (approximated 4 inches in diameter) can be cut. An array of laboratory tests and geological descriptions can be performed on the cores. Often the reservoir engineer studies oil displacement processes in laboratory experiments using these cores. A big challenge is scaling-up such small scale experiments to the reservoir scale.

Pressure transient tests[4] are performed by making an abrupt change in the production or injection rate of a well. The resulting pressure change with time is measured by a pressure gauge in the well near the depth of reservoir under study. Analysis of the measured transient pressure curve can yield estimates of rock properties such as permeability.

The geophysicist generates seismic (acoustic) waves at the surface[6] and records at the surface reflected waves from the earth strata below. The recorded reflected waves are analyzed to yield information about the geometry and attributes of the subsurface strata.

Because of the interdisciplinary nature of the production of oil and gas, there will be overlap between reservoir engineering and the other disciplines involved. For example, the statistical description of reservoir properties between wells could equally well be considered a geological function, as a reservoir engineering function, and be included in a book entitled, "Fractals in Geology." But because of our backgrounds in reservoir engineering, this book will address each topic from the perspective of the reservoir engineer. Recent books on fractals covering related earth sciences have been written by Turcotte[7] and Korvin.[8]

1.2 Definition of a Fractal

The term "fractal" was coined by Mandelbrot[9] in the 1970's and is rather loosely defined. In its broadest sense, it is a branch of mathematics that deals with objects possessing some scaling property—that is, objects that have some property that is similar to itself after a change in scale. In a more restrictive sense, it is a set of objects that have a fractal (or noninteger) dimension. Both of these "definitions" can be made clearer with a few examples. If a more formal mathematical approach is desired, the reader is directed to other texts.[9-12]

The first example is shown in Figure 1.1.[12] Figure 1.1 looks like a fern leaf, but it is actually a fractal. A number of properties of fractal objects can be illustrated using Figure 1.1. First, the "fern" has a pattern that is repeated at many scales. Second, it is not smooth and looks like a "natural" object. Third, there are limits to the figure's similarity to a real fern plant.

The "leaf" in Figure 1.1 is a repeated pattern. Each branch of the leaf is an exact replication of the overall leaf. This is repeated on finer and finer scales until the details are lost in the printing process. The similarity of an object at different scales is perhaps the most obvious property of a fractal.

The property that the "leaf" is not smooth can be expressed more precisely. In mathematical terms, it is nowhere differentiable. Due to this property, the fractal pattern "looks like" a natural rather than a man-made object. Other natural objects that have scaling properties, or fractal properties, include such diverse objects as trees, human lungs, broccoli, cauliflower, clouds, mountains, and fracture surfaces.[10]

When fractals are used to describe real objects, there are both upper and lower limits to the description. Figure 1.1 looks like a fern leaf, but continued generation of the "leaf" on larger and larger scales produces only a larger and larger fern leaf. A fern "plant" will never be produced. Similarly, there is a smallest scale to which a fractal description applies. As the overall fern leaf pattern is repeated on smaller and smaller scales, eventually the scale of a single fern plant cell will be reached. A single cell does not look like the overall shape shown in Figure 1.1. Finding the upper and lower limits is an important part of using fractals to model nature.

Mathematical definitions of a fractal concentrate on the property that fractal objects have a fractional dimension, that is, a dimension that is not an integer. We are familiar with the concept of 1, 2, and 3 dimensions. A fractal may have a dimension of 1.2 or 2.7. Again, examples are helpful.

Figure 1.2a is a square with a square hole in its center. The hole has sides that are exactly 1/3 the length of the entire square. If this pattern is repeated in each of the four corners and sides of the large square, Figure 1.2b results. Repeating this again produces Figure 1.2c. This process can be repeated on successively smaller scales. The repeated generation process produces a mathematical object, called a "Sierpinski carpet." As with the fern leaf, the resolution of the printing process sets a lower limit of what can be seen. If the figure is enlarged, the enlarged figure looks

Figure 1.1. A fractal which looks like a fern leaf.[12]

exactly like the original, only exposing more detail. The Sierpinski carpet, like the fern leaf in Figure 1.1, is a fractal.

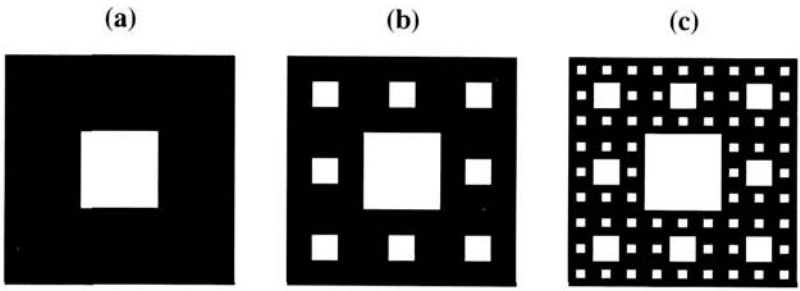


Figure 1.2. A Sierpinski carpet.[9]

By the mathematical definition, the Sierpinski carpet has a dimension of 1.893! The dimension is calculated by noting that the area, A, of a Euclidian (nonfractal) object is related to its "size" or characteristic length, L, by

$$A = c\, L^{\alpha} \tag{1.1}$$

where α and c are constants. For planar objects, α is the integer 2. For example, the equation for calculating the area of a square is

$$A = L^2 \tag{1.2}$$

In this case, Equation 1.2 is Equation 1.1 with c=1, α=2, and L the length of one side of the square (see Figure 1.3). For an equilateral triangle, Equation 1.1 applies with c= $\sqrt{3}/4$ and α=2. A half circle has c=$\pi/2$ and α=2. Each of these objects has a dimension of 2 (the value of α). For 3 dimensional objects, α is 3 and c varies with the shape of the object (e.g., c is 1, 4/3 π, and 1/6 for a cube, sphere, and pyramid, respectively).

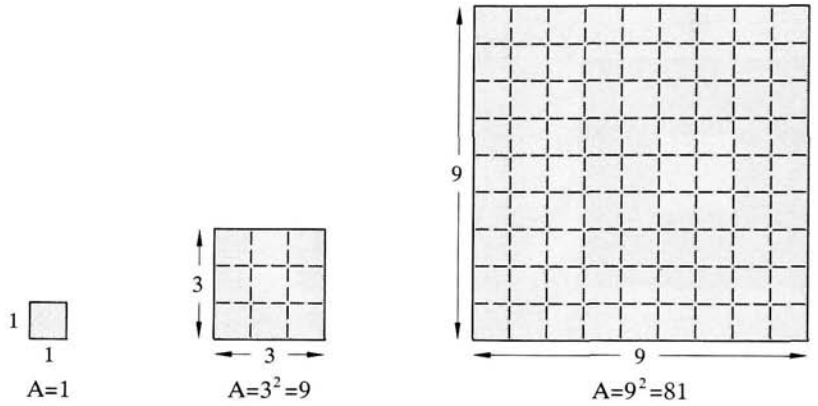


Figure 1.3. The scaling of the area of a square as it becomes larger.

The area of the Sierpinski carpet can be calculated at any scale from the length of one of its sides with c=1.00 and α=1.893 in Equation 1.1 (see Figure 1.4.). Figure 1.4 shows that if the size, L, of a Sierpinski carpet is increased by a factor of 3, the area, A, that can be measured at a given resolution increases by a factor of 8 (i.e., about $3^{1.893}$). The noninteger value of "α" is the essence of the mathematical definition of a fractal in terms of dimensionality.

Another way of finding the dimension of an object is to change the scale of the measuring unit, ℓ. Then the number of measuring units, N, needed to cover the object is related to ℓ by

$$N = b(1/\ell)^{d_f} \qquad (1.3)$$

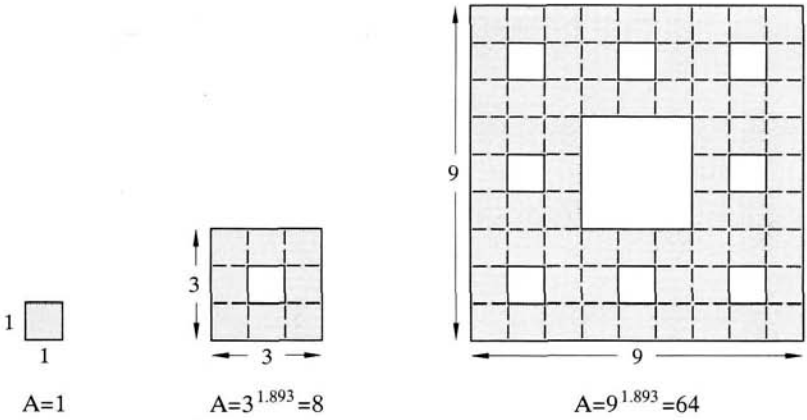


Figure 1.4. The scaling of a Sierpinski carpet as it becomes larger.

where b is a constant and d_f is the dimension of the object. In Figure 1.3, the number of squares needed to cover a unit square is 1. The required number of squares with sides of length 1/3 is $3^2=9$, the number with sides of length 1/9 is $9^2=81$, etc. For the Sierpinski carpet (Figure 1.2), the number of unit squares needed to cover it is 1. Then, the number of unit squares of size 1/3 is $3^{1.893}=8$, then number of length 1/9 is $9^{1.893}=64$, etc.

This covering of an object with different size unit measures is the essence of the Hausdorff-Besicovitch dimension referred to in other texts on fractals (e.g., Reference No. 9). This covering procedure is sometimes called "box counting." The covering description, Equation 1.3, is equivalent to the scaling description, Equation 1.1. You may note that for the Sierpinski carpet, the scaling Equation 1.1 is correct only for certain integer multiples of the length scale. For statistical fractals, described in the next chapter, the scaling equation does not have this limitation.

For objects with "curved" edges, the scaling relationships, Equation 1.1 and Equation 1.3, are correct only if the unit measures are much smaller than the overall object dimensions. For a circle, Figure 1.5 shows a plot of the radius verses the number of units to cover it. Superimposed upon this is a plot of Equation 1.3 for $c=\pi$ and $a=2$. These two plots converge only as the unit size becomes small compared to the circle size.

Equations 1.1 and 1.3 illustrate that fractals can be described by a "scaling law." This statement is equivalent to saying that some property of the fractal can be

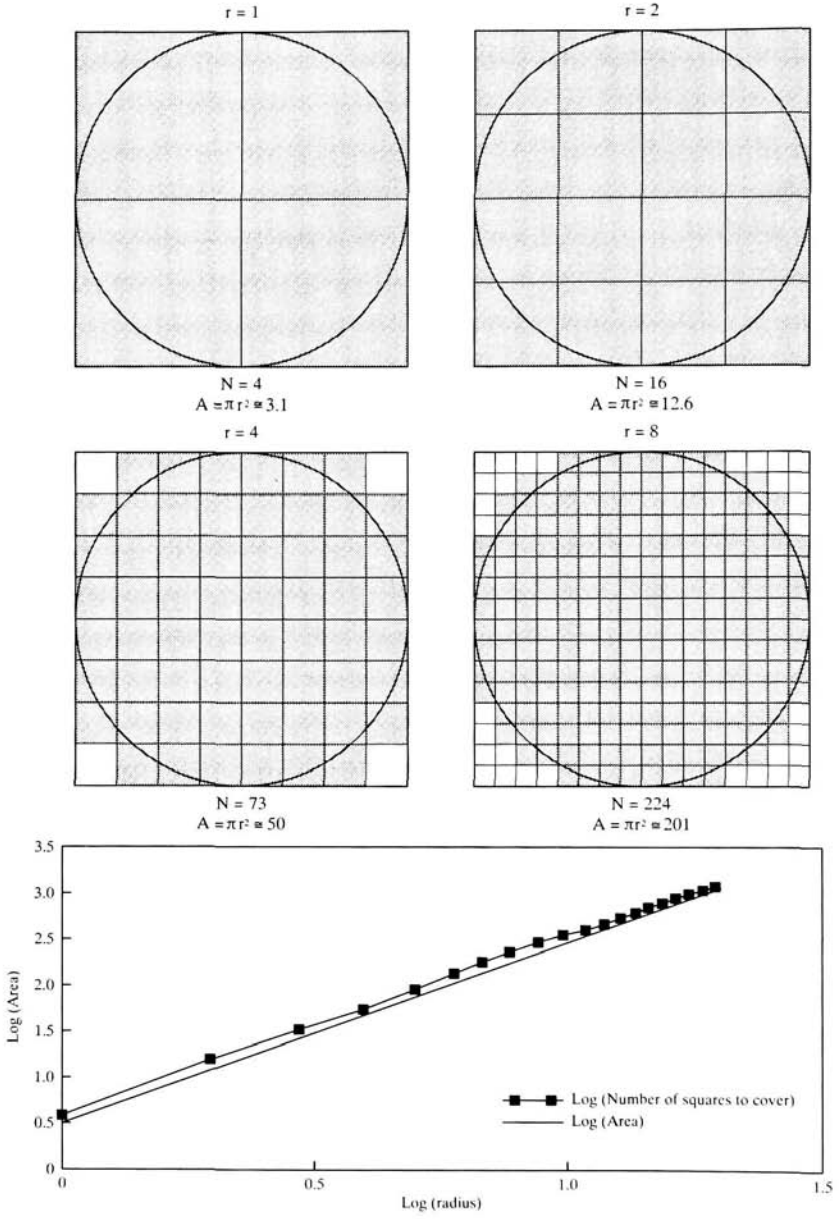


Figure 1.5. Number of squares necessary to cover a circle compared to the area of the circle in square units.

described by a power law equation. The scaling property is so commonly associated with fractals objects that it is sometimes used as an alternative definition of a fractal.

1.3 Statistical Fractals

Not all fractals are "exact" fractals. Fractals can also be defined in terms of their statistical properties. In this case, the average value of an attribute may scale as a power law. For example, the mean, standard deviation, covariance, or spectral density may scale as a power law.

Figure 1.6a shows an example of an object whose average "mass" is fractal. This object looks like the result of injecting a very mobile fluid into a very viscous one in porous media as shown in Figure 1.6b. Figure 1.6a was produced by a computer process called diffusion limited aggregation, or DLA.[10]

DLA is produced by placing a "particle" or seed at the center of a grid. Particles are released one at a time on the edge of the grid and allowed to walk randomly on the grid until they become adjacent to the seed. They then "stick" to the seed particle and become part of the seed. Another particle is released, and the process is repeated many, many times.

The "seed" grown by DLA is not a complete fractal. That would require the irregular pattern seen in Figure 1.6a to be repeated to infinitely small scales. Instead, Figure 1.6a is generated with a finite number of random walkers having a finite size. Sometimes DLA and other incomplete fractal patterns are referred to as "prefractals." For engineering purposes, this is an unnecessary distinction, because as noted earlier, for any real object, there is always a limit to the scale at which it can be considered fractal, so in a strict sense, all fractal models applied to real objects are "prefractals." We will not be so careful and will simply identify a real object as "fractal" with this understanding.

The object in Figure 1.6a has a "mass," m, that scales with its diameter, d, as

$$m = c\, d^{\alpha} \tag{1.4}$$

For the object in Figure 1.6a, α is not an integer, and therefore, the object can be considered a "fractal." The object is statistically fractal, since an infinite number of different figures "like" Figure 1.6a can be generated using DLA. The average mass

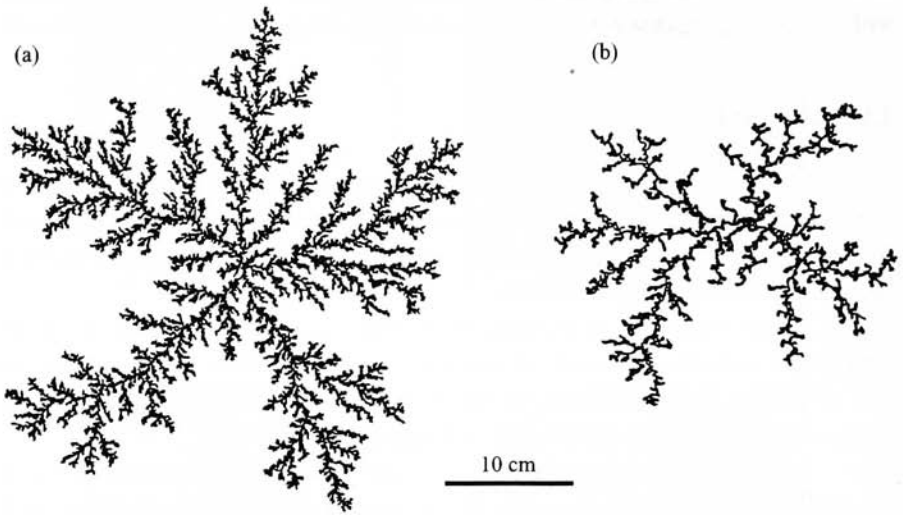


Figure 1.6. (a) A diffusion limited aggregation (DLA) pattern[10] and (b) pattern produced from injecting air into liquid epoxy in a two-dimensional porous medium consisting of 1.6 mm glass spheres.[10]

of the figures scales as Equation 1.4. Note also that successive enlargement of Figure 1.6a does not yield a pattern that is an exact match to its larger piece. It is only "similar." Here the exact similarity is in the scaling law, Equation 1.4.

Now we turn to a different example. For the statistical fractal in Figure 1.7, the standard deviation scales as a power law. This figure looks a bit like a plot of the Dow Jones average of the stock market or the "skyline" projection of the Rocky Mountains. It is called fractional Brownian motion. The standard deviation, σ, of the fBm value varies with the measurement of length, L, as

$$\sigma = c\, L^{\alpha} \tag{1.5}$$

As with DLA, an infinite number of figures can be generated. Each has the statistical property shown in Equation 1.5.

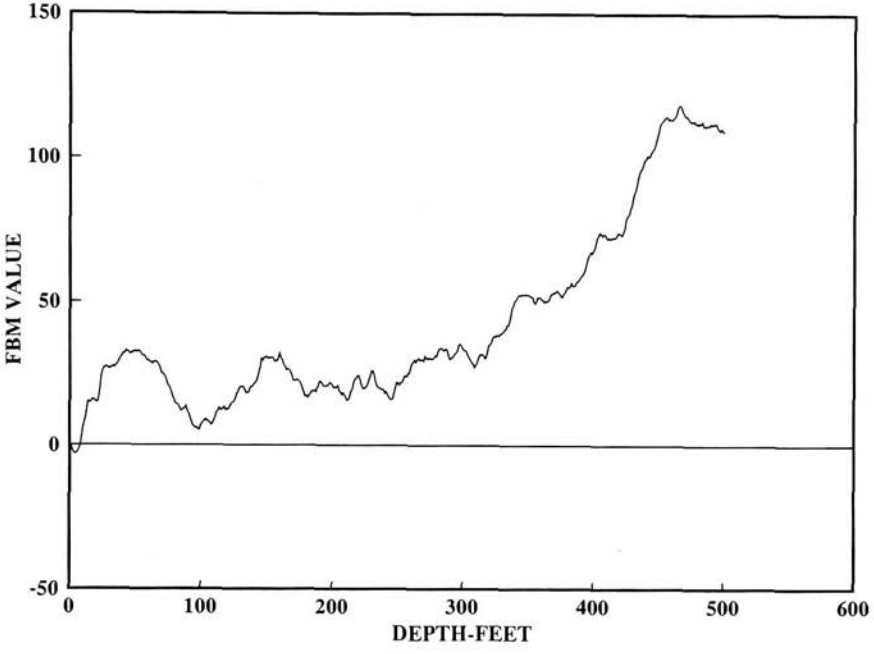


Figure 1.7. Fractional Brownian motion.

Some very beautiful pictures can be generated by extending fractional Brownian motion (fBm) into multiple dimensions. Plate 1 shows a "mountain" scene generated by Voss.[11] Plate 2 shows "clouds."[11] Both of these were generated by two dimensional fractional Brownian motion models.

Fractional Gaussian noise (fGn) is another statistical fractal. It is one that is defined in terms of its covariance. The mathematical definition of the covariance is found in Appendix A2.1. Simply stated, the covariance describes how similar a function is at two points that are separated by a distance, s. The closer to the same value, or the more correlated, the larger the covariance. The more different two values are, the smaller the covariance. Fractional Gaussian noise can be described in terms of its covariance, Cov, as follows:

$$(\mathrm{Cov}(s) - b) = cs^{\alpha} \tag{1.6}$$

where b, c, and α are constants and $-1 < \alpha < 0$.

Figure 1.8 shows a plot of one dimensional fractional Gaussian noise. fGn is more varied than fractional Brownian motion and can be shown to be a derivative (or finite difference) of a fBm sequence. fGn and fBm are both fractal in the sense of having a scaling property, but some care is required in order to apply the "box counting" tests described in Section 1.2 to fGn and fBm.[10]

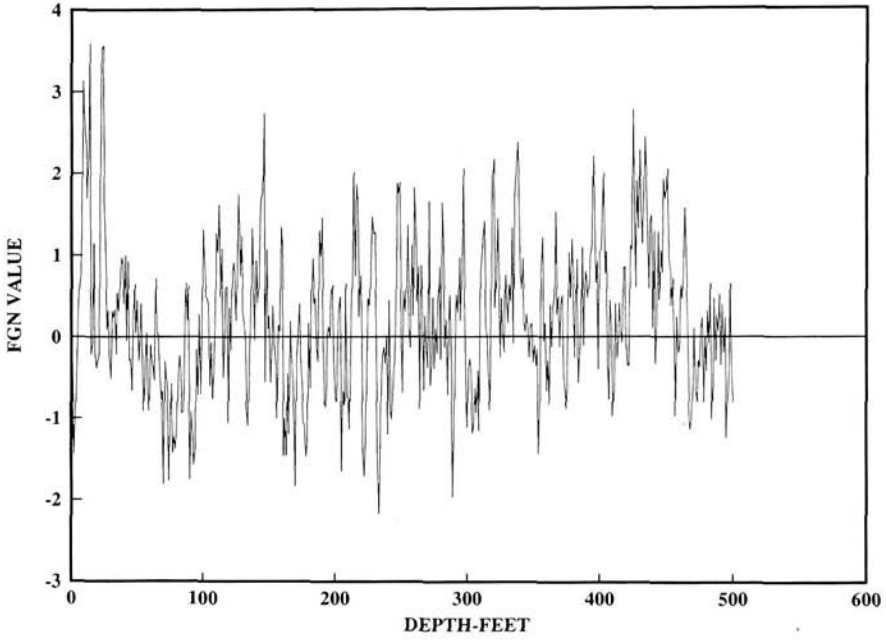


Figure 1.8. Fractional Gaussian noise.

Extending fractional Gaussian noise into multiple dimensions also produces some interesting pictures. Figure 1.9 was generated[13] using fractional Gaussian noise in two dimensions. For comparison, Figure 1.10 is a photo of a cylindrical piece of core that has been cut open with the face of one side of the cut polished smooth. This is called "slabbed" core. These figures are surprisingly similar. In Chapter 3, we will explore the similarity of fractional Gaussian noise and "slabbed" cores in detail.

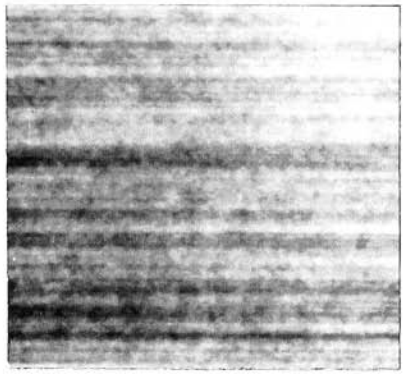


Figure 1.9. A slabbed core photo generated using 2D fractional Gaussian noise.[13]

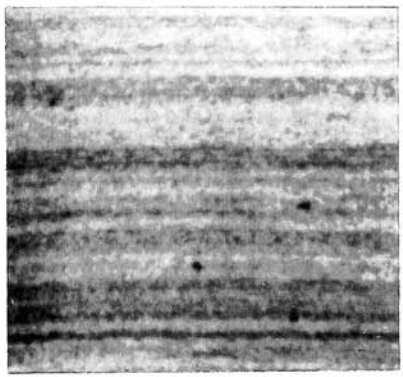


Figure 1.10. A true slabbed core photo of a West Texas Sandstone (about 2 in by 2 in).[13]

1.4 When is an Object Fractal

Figures 1.7 through 1.10, as well as Plates 1 and 2, suggest a similarity between fractal models and natural things. One of the limits of such an association has already been pointed out—that is, that real objects in nature are only fractal over some finite range of scales. There is another caveat when objects are labeled as "fractal." Strictly speaking, a real physical object should not be considered fractal. Only some measured property of the object should carry this designation. This caveat is also true of Euclidean models of physical objects too, but we are so familiar with Euclidean models that we often don't think about this.

For example, consider the question, "Is a coin circular?" Yes, if you are considering its perimeter, looking at it from above. No, if you look at it straight on the edge so that all you see is its edge. From the edge, it is either a line segment or a rectangle, depending on your preference. On the face of the coin is a picture. This picture is not circular either, so what does it mean to say, "a coin is circular?"

We mean that some property of the coin is circular (over some scale and to some degree of accuracy). Note that, strictly speaking, the coin itself is neither "circular" nor "not circular." It is a property of the coin, namely it's perimeter, that is circular. In the same sense, real objects are, or are not, fractal. By saying an object is fractal, we mean that some property of the object is fractal.

An example should help clarify how one property can be fractal, while another property of the same object is not fractal. One of the fractal objects described by Mandelbrot[9] is a triadic Koch island (Figure 1.11). Mandelbrot points out that the perimeter of this object scales like Equation 1.3 and, therefore, is fractal. The area scales with the size of the unit used to measure it, Δ, as (see Appendix A1.2 for details)

$$\text{Area} = b(1 - c\,\Delta^{\alpha}) \tag{1.7}$$

where α is greater than zero.

The area asymptotically approaches b, as finer and finer detail (i.e., $\Delta \rightarrow 0$) is used to measure the area. We can intuitively see that the area is bounded, by simply drawing a circle around the outside of the triadic Koch island. No matter how much resolution is added to the island, no part of the island crosses the circle. Therefore, its area never exceeds the area of the circle. Thus, the area of the triadic

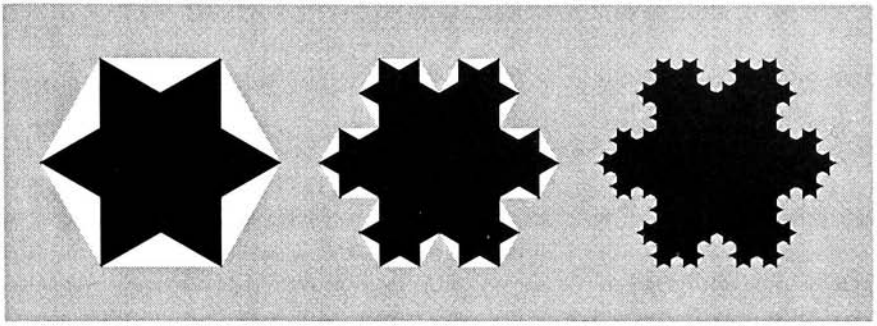


Figure 1.11. A Koch island.[9]

Koch island, which is bounded, is not a fractional power of its size. For the triadic Koch island, then, the perimeter is fractal, but the area is not.

Now, if the triadic Koch island is used as a model of a sand grain, should the sand grain itself be considered fractal? If the sand grain is thought of as the boundary between rock and pore space (the boundary between white and black in Figure 1.11), it is fractal, but if the sand grain is thought of as the rock itself (the shaded portion in Figure 1.11), it is not. This confusion is avoided by realizing that it is the measured property, its area or perimeter, for example, and not the sand grain itself that should be considered fractal or not fractal.

This detail can cause some rather subtle problems. For example, one paper may claim pores of rocks are fractal. The author means that he has measured pore surface areas and found them to be fractal. The statement "pores are fractal" can be easily misinterpreted to mean that the pore size distribution is fractal, that is, the number of pores is a power law distribution in size. This may or may not be the case, but the paper on pore surface areas does not address the question of pore size distribution. Because of this confusion, care should be used when reading papers to note what property of an object is fractal. This should be done in addition to noting over what scale the property is fractal.

APPENDIX A1

A1.1 Perimeter of the Triadic Koch Island

The perimeter of the triadic Koch island, Figure 1.11, can be shown to be fractal by showing that an equation of the form of Equation 1.3 calculates the perimeter for different ruler lengths. If the unit of measure (i.e., the ruler length) is chosen equal to the length of one edge of one of the points in Figure 1.11a, the number of units required to cover the perimeter of the figure is

$$(6 \text{ points})(2/\text{point}) = 12 \tag{A1.1}$$

Decreasing the unit size to 1/3 allows measuring more detail. From Figure 1.11b, we get the number of units required to cover the perimeter is

$$(6 \text{ points})(8/\text{point}) = 48 \tag{A1.2}$$

For unit size 1/9 and Figure 1.11c, the number of units is now

$$(6 \text{ points})(32/\text{point}) = 192 \tag{A1.3}$$

In general, the measuring unit, ℓ, at step n is

$$\ell = (1/3)^{n-1} \tag{A1.4}$$

The number of unit measures, N_n, needed to cover a Koch island with units of size ℓ is

$$N_n = (6)(2)4^{(n-1)} \tag{A1.5}$$

Rearranging to get an equation of the form, Equation 1.3, one finds

$$N_n = b[1/(1/3)^{(n-1)}]^{d_f} = (6)(2)(4)^{(n-1)} \quad \textbf{(A1.6)}$$

d_f and b can be solved for by taking the logarithm of both sizes of Equation A1.6. We get b=12 and d_f=log(4)/log(3) in an equation of the form of Equation 1.3.

A1.2 Area of the Triadic Koch Island

The area of the triadic Koch island is calculated using unit triangles of different size. If the unit of measure (i.e., triangle) is chosen to be the size of one of the points in Figure 1.11a, then the area of Figure 1.11a is covered with 12 unit measures--six points and the center is covered by six triangles. Thus, the area is

$$A_0 = 12.0 \quad \textbf{(A1.7)}$$

The area of Figure 1.11b can be measured using units 1/5 the area of those used in Figure 1.11a. Five times as many units will be required to cover the area of Figure 1.11a, plus an additional 2 triangles for each of the six points. This gives a total number for the area of Figure 1.11b as

$$A_1 = A_0 + (2)(6)(1/5) \quad \textbf{(A1.8)}$$

For Figure 1.11c and triangles 1/5 the size of those needed in Figure 1.11b, we have (2)(2)(6) "new" points that can be seen at this resolution, so

$$A_2 = A_1 + (2)(2)(6)(1/5)^2 \quad \textbf{(A1.9)}$$

In general,

$$A_{n+1} = A_n + (2^n)(6)(1/5)^n \quad \textbf{(A1.10)}$$

or

$$A_n = 12 + 6\sum_{j=1}^{n} (2/5)^j \tag{A1.11}$$

To make further progress, consider the geometric series,

$$s_n = 1 + a + a^2 + \ldots + a^{n-1} \tag{A1.12}$$

then

$$as_n = a + a^2 + a^3 + \ldots + a^n \tag{A1.13}$$

Subtracting,

$$s_n - as_n = 1 - a^n \tag{A1.14}$$

so

$$s_n = \frac{1 - a^n}{1 - a} \text{ for all } a \neq 1. \tag{A1.15}$$

In our case, letting a = 2/5 in Equation A1.12 allows Equation A1.11 to be written as follows:

$$A_n = 12 + 6(s_{n+1} - 1) \tag{A1.16}$$

Substituting Equation A1.15 into Equation A1.16 and rearranging gives

$$A_n = 12 + 6\left(\frac{a}{1 - a}\right)(1 - a^n) \tag{A1.17}$$

Setting a=2/5, we get

$$A_n = 16(1 - \frac{1}{4}(2/5)^n) \tag{A1.18}$$

This is the same as Equation 1.7, if b=16, c=1/4, and

$$(2/5)^n = (1/5^n)^\alpha = \Delta^\alpha \tag{A1.19}$$

By taking the log of both sides of Equation A1.19, we find Equation A1.19 can be satisfied for all n if α = (1- log(2)/log(5)). Thus, the area of the triadic Koch island can be expressed as Equation 1.7.

This page is intensionally left blank

CHAPTER 2

STATISTICS

A large portion of this book deals with statistical fractals. Statistical fractals are objects that have some statistical property that is scaling, or is fractal. To explain the properties of statistical fractals that are important to reservoir description, we first describe some basic statistical concepts and properties. We then apply these to fractal models of fGn and fBm.

Statistical measures are used to describe a set of objects. These measures can be used to infer information about other objects that are similar to those measured. Statistical inference, like all mathematical inferences, is not magic. A set of assumptions is applied to a data set. Then predictions are made based on these assumptions and the data input. If the assumptions do not apply to the problem or if the data input is the "wrong" data, the algorithm will not give good predictions. It is therefore critically important to examine the assumptions and the data input on a case-by-case basis. Human judgements must be made as to the applicability of the assumptions and the use of the data. This critical approach will be a recurring theme throughout this book.

We will begin our study of statistical measures by covering some basic statistical assumptions in the next section entitled "Ensembles." Then some basic statistical measures are described in the section entitled "Univariant Measures." The section entitled "Multivariant Measures" introduces more complicated and less familiar statistical measures that capture ordering. Finally, the last section, "Fractal Models," explains fGn and fBm in terms of these statistical measures.

2.1 Ensembles

The purpose of using statistics is to make predictions about what is not known from what is known. The foundation on which prediction is built is the concept of an ensemble. An ensemble is a collection of similarly prepared sets. For example, an ensemble may be the set of numbers produced by rolling a pair of dice 100 times (or rolling 100 "similar" pairs of dice one time). Since similarity is assumed, the probability of the outcome of a new roll can be calculated from the ensemble of outcomes of previous rolls.

We are immediately faced with a problem in applying this concept to reservoir properties. It is not clear how the distribution of reservoir properties within the earth can be treated as an ensemble. By definition, an ensemble should be made up of "similarly prepared sets." Should this be a reservoir? A geologic layer? A depositional unit? A single sand lens? These are not academic questions, because the definition of the ensemble determines the set to be used to make predictions.

The problem of identifying an ensemble is not a new problem to geologists or engineers. Whenever a geologist defines a depositional unit, he is defining an ensemble (layers of earth deposited in a similar depositional environment). Whenever a reservoir engineer calculates the average porosity of a reservoir, he assumes an ensemble (a volume of the earth with similar properties). Even so, care must be taken.

As an example, consider a reservoir that has porosity gradually increasing from east to west. Let us consider the problem of predicting the average porosity of the reservoir. If porosities measured at the wells are taken as an ensemble, a simple arithmetic average is one estimate of the average reservoir porosity. The result is not the best estimate of the average reservoir porosity, however. The problem with the arithmetic average is that the set of average porosity values measured at well locations is not a good ensemble due to the east/west trend. There may be for example five wells on the high porosity side of the field and only one well on the low porosity side. An arithmetic average would unduely weight the high porosity portion of the field. A better estimate can be made by first mapping the overall porosity trends. Next, multiply porosity by the reservoir volume associated with each well, sum the result, and finally divide by the total reservoir volume. This volume weighting is common reservoir engineering practice.

In reservoir engineering, we never have enough data. Wells and well logs often examine no more than a billionth of the reservoir volume. With this small sampling, any statistical approach is suspect. In order to make progress, we try to leverage our data as much as possible. One way to do this is to assume that ensembles are both stationary and ergodic.

To understand the concepts of stationarity and ergodicity, we first need to examine two types of averages--time average and ensemble average. To illustrate these concepts, consider the four traces shown in Figure 2.1a. These might be a plot of some measurement versus time that resulted from running the same experiment four times (e.g., rolling dice), or the traces could be porosity measurements versus depth from four well wells in the same area. Each trace has 30 time (or depth) values. We will denote each value as $g_j(k)$, where the subscript corresponds to time and the number in parentheses corresponds to the trace number. Thus, j runs from 1 to 30 and k runs from 1 to 4 for the example in Figure 2.1a.

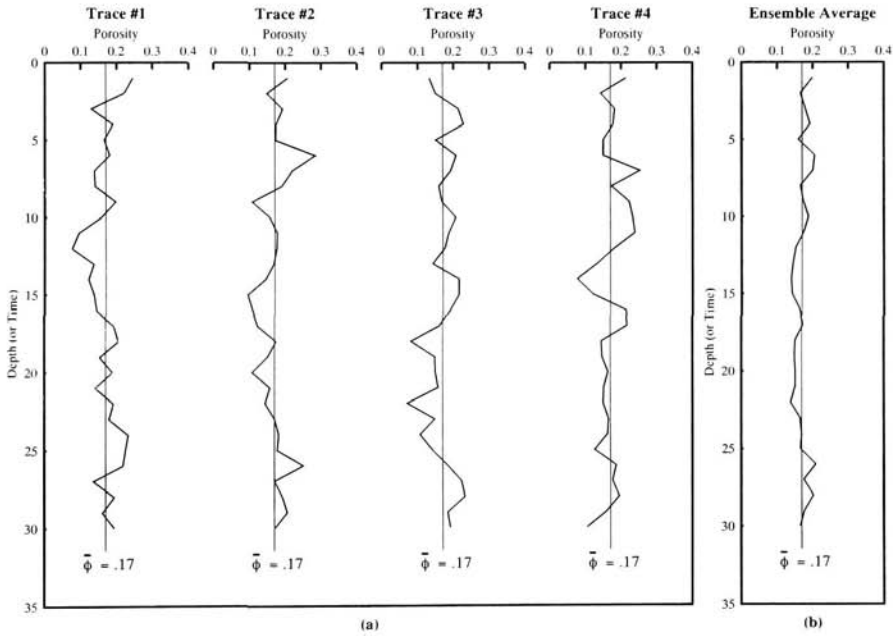


Figure 2.1. Ensemble and time averages for a stationary set of data, (a) four traces of data and (b) ensemble average of the four traces for each time (depth) interval.

The time average, $\{g(k)\}$, of the k^{th} trace is

$$\{g(k)\} = \frac{1}{30} \sum_{j=1}^{30} g_j(k) \tag{2.1}$$

The ensemble average, $\langle g_j \rangle$, for the j^{th} time step is

$$\langle g_j \rangle = \frac{1}{4} \sum_{k=1}^{4} g_j(k) \tag{2.2}$$

Notice that in general, the time average depends on which trace is averaged. In general, the ensemble average is a function of which time step (or depth increment) is averaged.

A stationary ensemble is one whose ensemble average properties are independent of time. In reservoir applications, depth or distance takes the place of time. Thus, in our applications, a stationary ensemble is one whose ensemble average properties are independent of depth or spatial location.

An ergodic ensemble is one whose ensemble and time averages are the same. Note that in order for the ensemble and time averages to have the same value, the ensemble average must be independent of time and the time average independent of the trace. As a result, an ergodic ensemble must be stationary, but a stationary ensemble need not be ergodic. As already mentioned, for our applications, spatial location takes the place of time, so that an ergodic ensemble is one whose ensemble and spacial averages are the same.

Stationarity and ergodicity are assumptions. We need to examine closely what these assumptions mean and when they are likely to be good or bad assumptions. To do this, we turn to our previous example. Consider Figure 2.1a as plots of four porosity logs. We might interpret these porosity distributions as the result of the same depositional process and, therefore, consider these traces to make up an ensemble of four similarly prepared sets. With this assumption, we can predict what another well drilled in the same area might encounter.

In order to see if stationarity and ergodicity are good assumptions for this ensemble, let us associate depth with time and calculate the time and ensemble averages of these porosities. Figure 2.1b shows a plot of the ensemble average of the

four logs versus depth. Below each log is listed the time (or depth) average of each log. The ensemble average seems to show no obvious trend but varies closely around 0.17. The time average is the same for each log and is close to the ensemble average of 0.17. To prove stationarity for this data set, we would have to repeat these tests for all statistical measures. This is not practical. From our single test of averages, however, stationarity and ergodicity seem to be good assumptions. If we assume that a new well drilled in this area would be part of the same ensemble, we would predict the most likely average porosity of the new well to be about 0.17.

Now consider Figure 2.2a. Again, we might interpret these porosity logs as the result of the same sequence of depositional processes. Because of this assumed similarity, we could consider these traces to make up an ensemble of four similarly prepared sets. With this assumption, we could use statistics to predict what another well drilled in the same area might encounter.

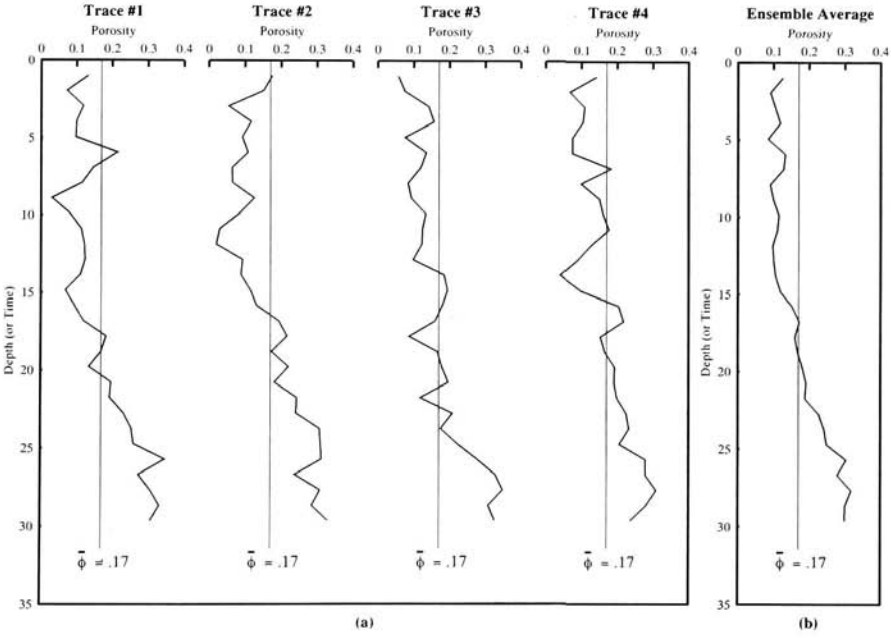


Figure 2.2. Ensemble and time averages for a nonstationary set of data, (a) four traces of data and (b) ensemble average of the four traces for each time (depth) interval.

Figure 2.2b shows a plot of the ensemble average of the four logs versus depth. Below each log is listed the time (or depth) average of each log. For this example, the ensemble average shows a clear trend of increasing porosity with increasing depth. The ensemble is nonstationary, and therefore, it is also nonergodic. The lack of stationarity and ergodicity does not mean that we cannot make statistical predictions. It does mean that we should not use statistical tools that rely on these assumptions.

If we were to assume a stationary distribution for these logs, we would predict an average porosity of 0.17 over the entire logged interval. A better prediction would be that the porosity at the top of the interval is less than the porosity at the bottom of the interval for a new well.

In this example, the lack of ergodicity and stationarity is not such a bad thing. In fact, we could make a more detailed prediction without stationarity and ergodicity than with stationarity and ergodicity! Nevertheless, many mathematical tools depend on stationarity and ergodicity. Because of this dependance, we will make these assumptions unless otherwise noted.

The inferences from Figures 2.1 and 2.2 are a bit trivial. The point of the discussion is not to show the power of statistical prediction--this comes later. The point is to clarify the meaning of ensemble, stationarity, and ergodicity so that we may be aware of the assumptions being made in geostatistical models.

We will now turn to some specific statistical measures. All of the statistical measures described in the next two sections are defined as ensemble averages. The equations for each section are found in Appendix A2.1. For the following two sections, we will assume ergodic ensembles. This assumption will allow us to describe the measures as time averages and keep our examples simple. In particular, with the ergodic assumption, we only need to show one trace instead of a set of traces for each example. We will return to our discussion of nonergodic ensembles in Section 2.4.

2.2 Univariant Measures

Univariant statistical measures describe data without respect to the ordering of the data. They might be called "bucket" statistics, that is, statistics that describe data that are dumped into a bucket and then drawn out in any order. The most

commonly used statistical measures fall into this category. Examples are the mean, the standard deviation, the histogram, and the Dystra-Parson's coefficient.[14]

Consider the mean, or average, of a set of values. The mean is calculated by adding together all of the values in a given sample set (or bucket) and dividing by the number of values. If the first value is placed last or the last value placed in the middle, the mean is unchanged. The mean is a property of the "bucket" of values and is independent of the original ordering of the values.

The same is true of the standard deviation of a sample set. The standard deviation describes how the set of values varies about the mean. The square of the standard deviation is the variance. Again, the ordering of the values is neglected.

The omission of order also applies to a reservoir engineering measure of variance known as the Dykstra-Parsons coefficient.[14] The Dykstra-Parsons coefficient is a function of the standard deviation and the mean. Since the Dykstra-Parsons coefficient is made up of values which are independent of the ordering of the data, it too is independent of the ordering.

The histogram is a plot of the number of values that fall into a set of intervals which spans the entire range of the data. Consider the following data set:

{1,4,12,3,10,9,2,17}

To construct a histogram, we choose the intervals 1 through 5, 6 through 10, 11 through 15, and 16 through 20. We plot the number of values in each interval versus the interval values (Figure 2.3). Choosing a different set of intervals can produce a different histogram. In fact, if the intervals are made small enough, the histogram becomes a complete list of the original data. The ordering of the data set is immaterial to the histogram. The histogram, too, is a "bucket" statistic.

Once a histogram is obtained, we can curve fit the histogram with a chosen equation. Several equations are very commonly used to fit histogram shapes. Some of these are the Gaussian (or normal) distribution, the log-normal, and the p-normal. Equations for each of these are found in Appendix A2.1. Since the histogram is used to determine these equations for distributions, all of these distributions are also "bucket" statistics. The distributions are independent of the ordering of the input data set.

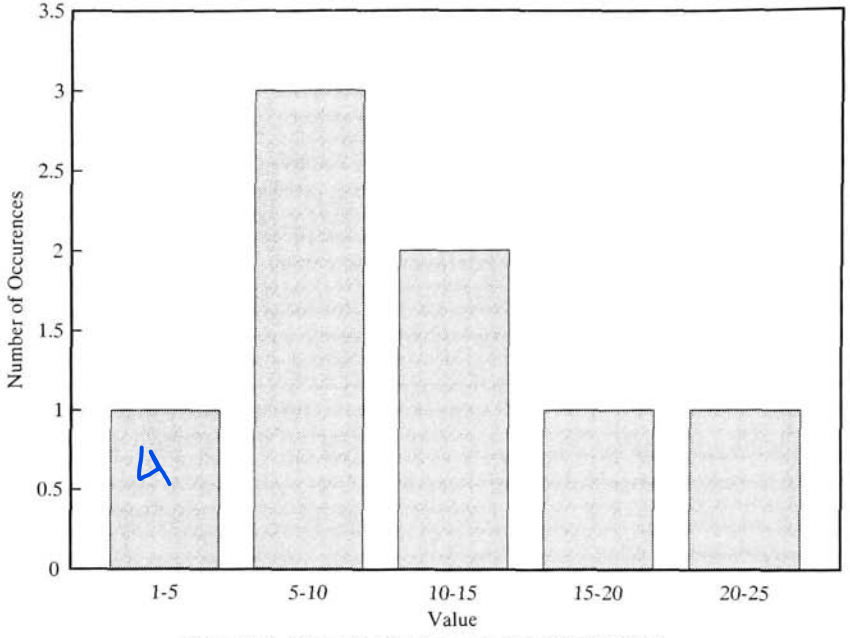


Figure 2.3. Example histogram of {1,4,12,3,10,9,2,17}.

Statistical analysis of reservoir property distributions, e.g., porosity, are usually limited to univariant measures like those just described. Typically, zones or regions of "like" values (usually based on geologic units) are identified. The data from each zone are dumped in a "bucket," and one or more of the univariant measures described above are calculated.

Unfortunately, univariant statistics do not provide sufficient information about porosity and permeability for fluid flow simulations. In particular, the same univariant distributions can produce markedly different responses to fluid flow. Consider Figures 2.4a through 2.4c. For these figures, white corresponds to regions of high permeability and black to regions of low permeability. All three figures contain the same 256 permeability values. Because the underlying data sets are the same, all univariant statistical measures will be the identical. Even so, the horizontal flow of fluids through Figures 2.4a, 2.4b, and 2.4c will be markedly different. Clearly, univariant statistical measures are not sufficient to describe all of the heterogeneity that is important for predicting fluid flow in heterogeneous media. We need statistical measures that include the ordering of the data. The next section describes multivariant measures which measure properties of the ordering of data.

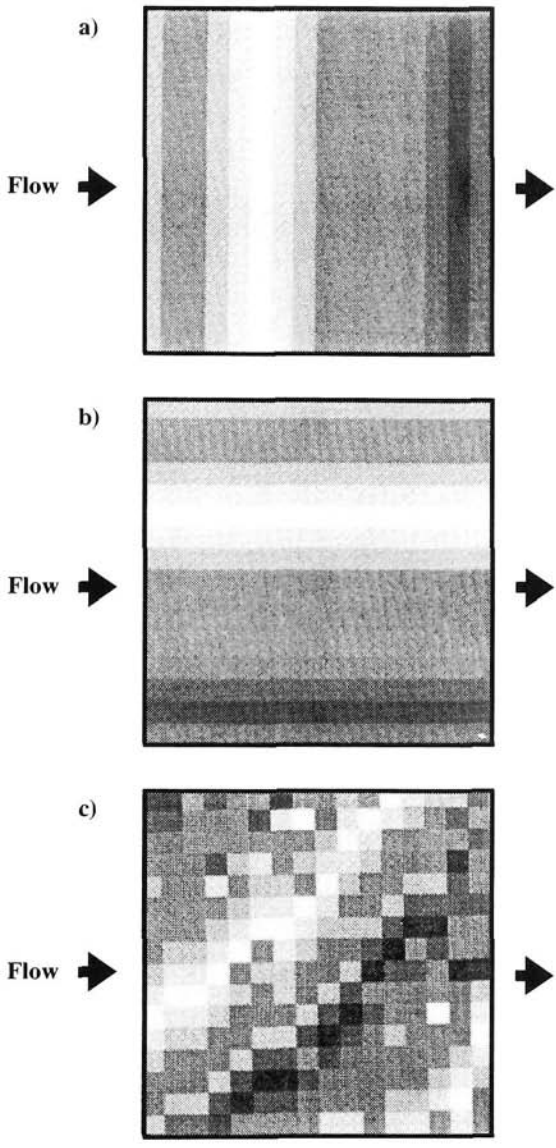


Figure 2.4. Various spatial distributions of the same set of permeability values. White is high permeability; black is low.

2.3 Multivariant Measures

The terminology of multivariant measures is more diverse than that of univariant measures. In particular, several science and engineering disciplines have each developed their own multivariant tools. These tools sometimes describe the same properties but have different names and different formulas for their calculations. This is unfortunate, because it adds to the confusion of trying to learn the basic ideas.

The multivariant measures that we will discuss are the following:

Covariance (or auto covariance)
Variogram (or semivariogram)
Spectral Analysis (through Fourier transforms)
R/S Analysis

The covariance is used primarily in time sequence analysis[15] and statistical physics.[16] The variogram is used primarily in geostatistics.[17] Spectral analysis is used primarily in electrical engineering, seismic processing,[18] and image analysis.[19] The R/S analysis was developed in hydrology and is most commonly used in fractal analyses.[9]

Each statistical measure is defined mathematically in Appendix A2.1. What we will do here is discuss how each of these measures displays a property of the ordering of a data set. We will use three example data sets that all are made up of the exact same set of numbers but just ordered differently. There will be no difference in any univariant measure of these data so that we can concentrate on the multivariant measures.

The example data sets are shown in Figures 2.5a, 2.6a, and 2.7a. The data set in Figure 2.5a is completely random. The data set in Figure 2.6a is completely ordered from smallest to largest value. The data set in Figure 2.7a has some order but not complete order. In each figure, plots of each statistical measure are shown--(b) histogram, (c) covariance, (d) variogram, (e) spectral density, (f) R/S analysis. All three data sets have the same histogram, Figures 2.5b though 2.7b. We will examine the multivariant statistical measures to see what each tells us about our data sets.

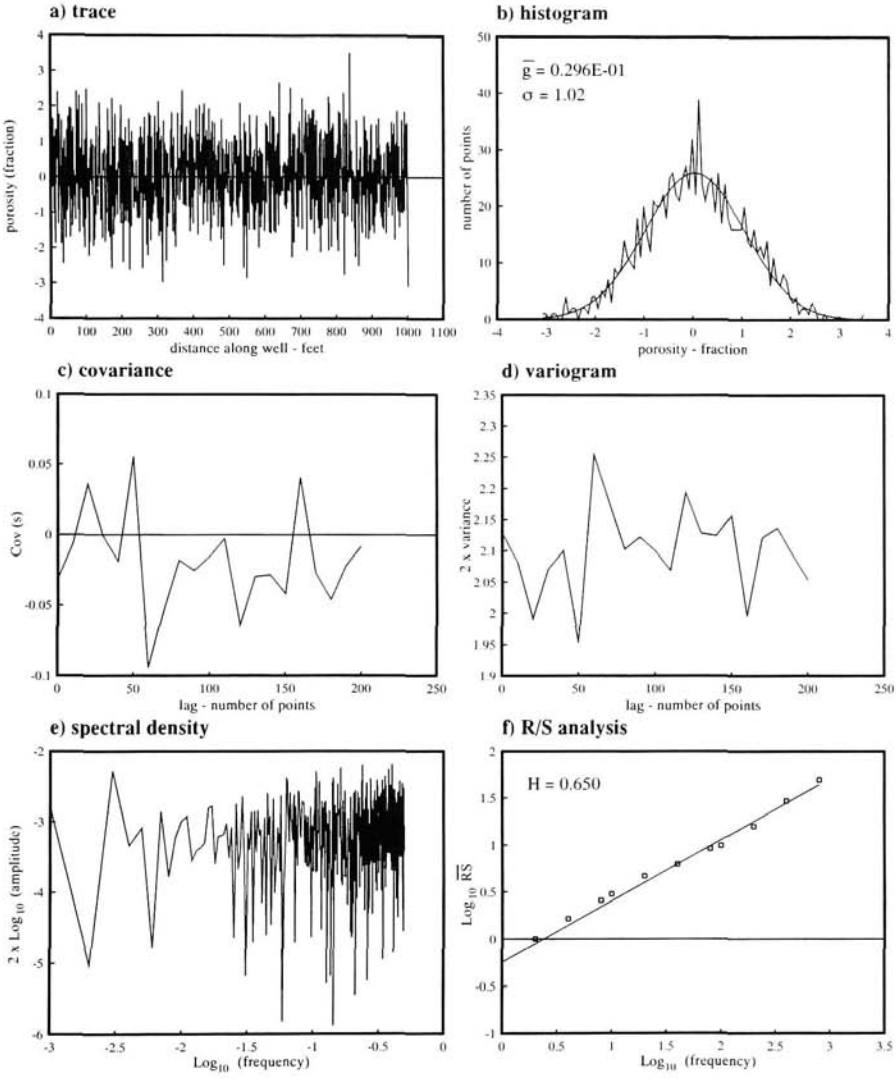


Figure 2.5. A random distribution of values and the following measures of the distribution: (a) distribution itself, (b) histogram, (c) covariance, (d) variogram, (e) spectral density, and (f) R/S analysis.

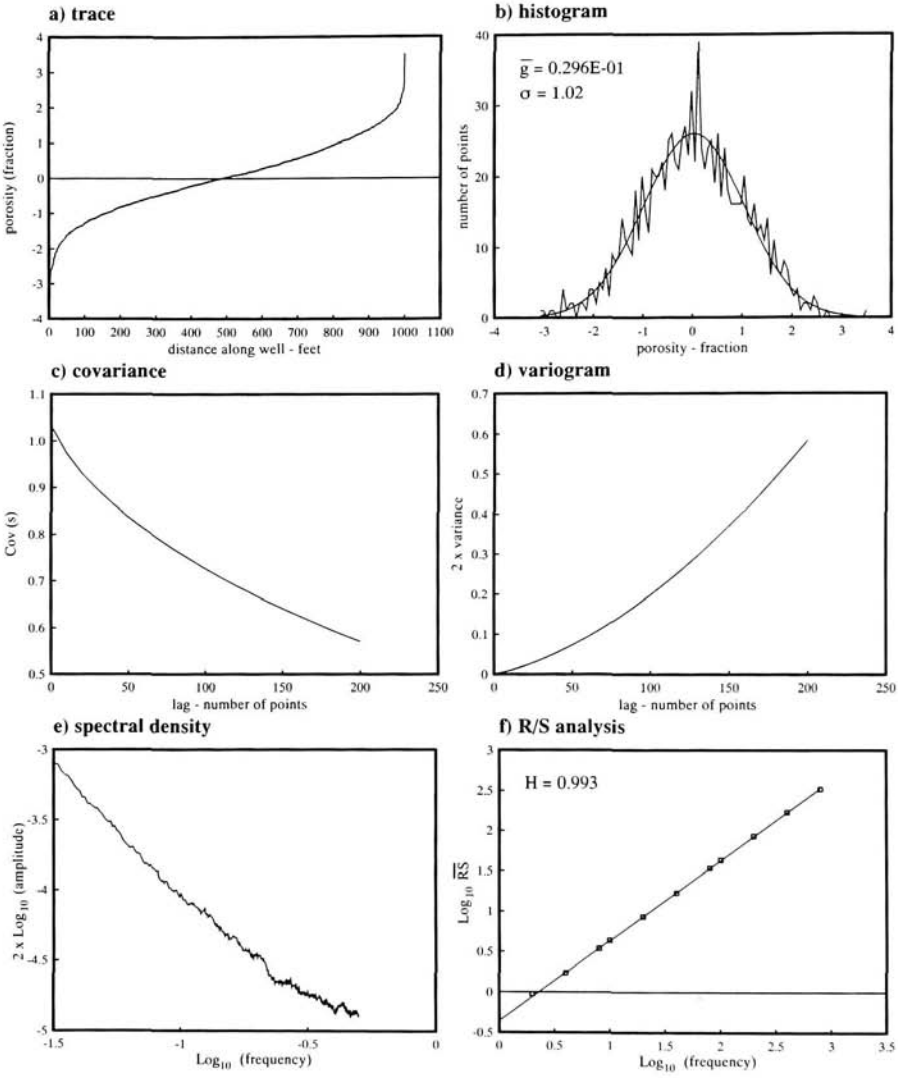


Figure 2.6. A completely ordered distribution of values and the following measures of the distribution: (a) distribution itself, (b) histogram, (c) covariance, (d) variogram, (e) spectral density, and (f) R/S analysis.

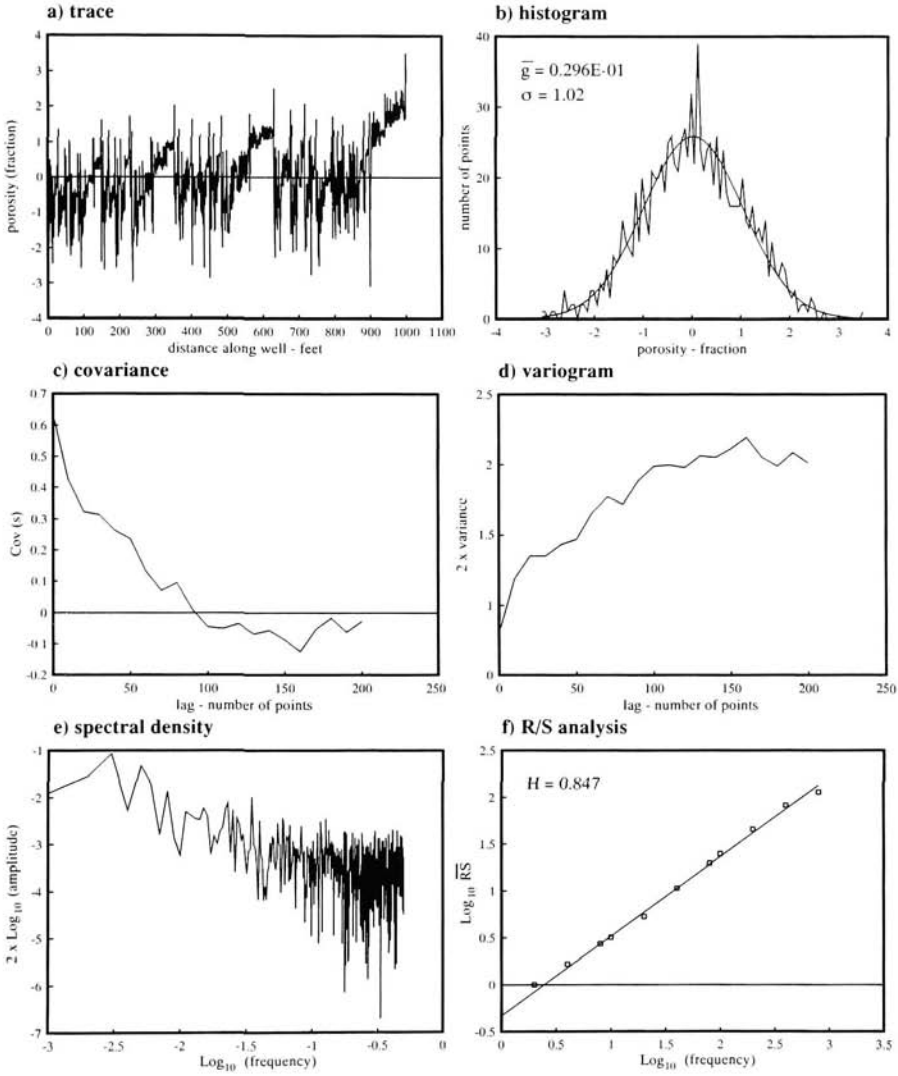


Figure 2.7. A semi-ordered distribution of values and the following measures of the distribution: (a) distribution itself, (b) histogram, (c) covariance, (d) variogram, (e) spectral density, and (f) R/S analysis.

2.3.1 *Covariance*

The covariance describes the amount of correlation among sample values. It is a function of the distance between samples (or the lag). The covariance of zero lag (not plotted) is equal to the variance of the entire sample. As the lag increases, the covariance decreases. It approaches zero for large lags, if samples separated by large distances are uncorrelated. It can become negative if the samples have negative correlation, that is, if a large positive value is likely to be followed by a small positive or large negative value for a given lag. Samples are spatially random, if they have a covariance of zero for all lags greater than zero.

The covariance of each data set shown in Figures 2.5a through 2.7a is shown in Figures 2.5c through 2.7c. For all three figures, the covariance of lag zero is one, the variance of the entire sample. The covariance for every lag greater than zero is approximately zero for Figure 2.5c, corresponding to a random ordering of the data. There is no correlation for any lag except zero. The covariance of the second data set, Figure 2.6, decreases very slowly from one as the lag increases. This trend indicates the values are well correlated. The covariance of the third data set, Figure 2.7, is somewhere between the extremes of 2.5 and 2.6. In Figure 2.7c, there is correlation up to some length, but eventually the sample is approaching no correlation for large lags. That is, as one data point gets far away from another, it has no memory of the value of the point left behind.

One practical note needs to be made about the covariance. It is a statistical measure and requires many points for accurate answers. In any finite data set, as the lag gets longer, fewer pairs of points are available to estimate the covariance at each lag. A good "rule of thumb" is to only consider covariance calculations meaningful up to lags of at most 1/3 of the original data length. In our examples, we have 1,000 points but have only plotted the covariance up to lag 200 (1/5 of the data length).

2.3.2 *Variogram*

The variogram is a measure of how fast the sample sequence varies. It is essentially the covariance curve turned upside down. The variogram is always zero for a lag of zero. There is no variation between the value of a point and itself. As the lag increases, the variogram increases. In practice, sampling error and short-scale variability may cause sample values to be dissimilar even at extremely small lags.

This is referred to in geostatistics as a nugget effect. If the samples become uncorrelated for large lags, the variogram will approach the total sample variance. This asymptote is called the sill.

The variogram of each data set is shown in Figures 2.5d through 2.7d. In the first example, the variogram is approximately constant for all lags. The value of the sill is the variance of the data set. When a variogram has all values (except for zero lag) equal to the sill, it is called a pure nugget. The name originated in the early applications of geostatistics in the mining industry.[17,20,21] In the second example, the variogram increases slowly with increasing lag. The sill is never reached, because the data never become totally uncorrelated. In the third example, the variogram rises, slowly approaching the sill for large lags. Thus, the sample is correlated for small lags, but becomes uncorrelated for large lags (>100). Like the covariance, the variogram should not be considered meaningful for lags more than 1/3 the total sample length.

2.3.3 *Spectral Density*

The spectral density is a statistical measure based on the Fourier transform of the original data. In order to discuss the spectral density, therefore, it is necessary to first explain the Fourier transform.

The Fourier transform turns a sequence of data into a set of sine and cosine curves that, when added together, reproduce the original sequence. Figure 2.8 illustrates how a sequence can be represented as the sum of sine and cosine curves. The plot at the top of the figure is the original sequence. The five sine curves on the left column add up to give the original sequence. The five curves on the right column show the results of progressively adding the sine curves together. Each of the five sine curves has a different height (or amplitude), period (or frequency), and starting angle (or phase).

The mathematical definition of the Fourier transform is found in the Appendix A2.2. Since all our data sets are in digital form, they are discrete and finite. The mathematical definition of a finite, discrete Fourier transform is also found in Appendix A2.2. The Fourier transform generates two data sets--one real and one imaginary. These two sets can, in turn, be used to compute the amplitude and the phase of the sine or cosine curves that reproduce the original sequence (see

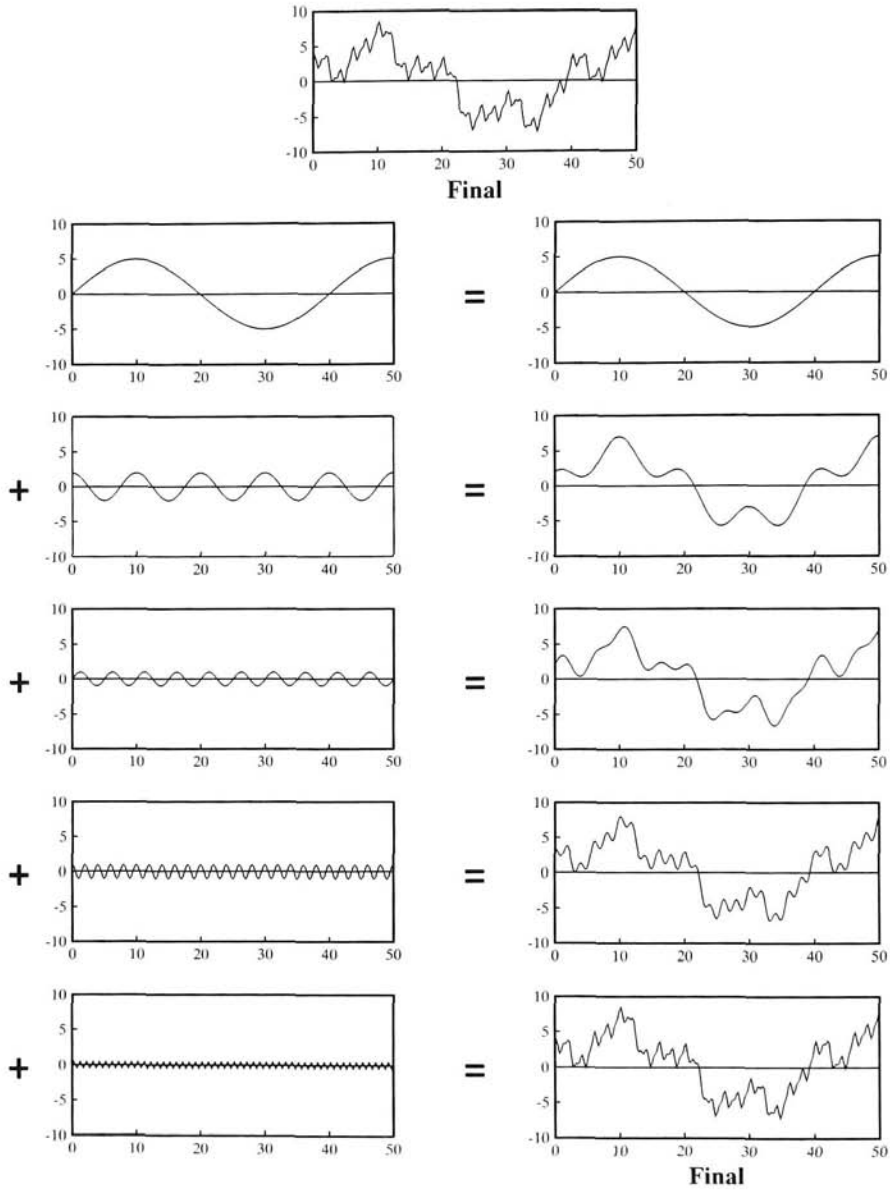


Figure 2.8. The Fourier sine and cosine curves into which an arbitrary distribution can be divided.

Appendix A2.3 for the details). The amplitude and phase relationships vary with the frequency of the associated sine or cosine curves.

All of the information in a discrete sequence is retained in the amplitude and phase. This is proven by the fact that an inverse Fourier transform exists that can transform the amplitude and phase results back into exactly the original sequence. For our application, we do not wish to preserve the entire original sequence but only a statistical measure of the original sequence, so the amplitude alone is often analyzed. The square of the amplitude is called the spectral density. Because the amplitude is a function of frequency, so is the spectral density.

Analysis of the spectral density can provide insights into the ordering of the original sequences. For example, a rapidly varying sequence will have large amplitudes of high frequency (rapidly varying) sine and cosine components. A slowly varying sequence will have larger amplitudes of low frequencies. If there were regular, periodic occurrences of a given frequency in the sequence, the amplitude of the corresponding frequency would be larger than the amplitude of other frequencies. Completely random sequences have all frequencies of equal likelihood, and therefore, the amplitude is independent of frequency.

The spectral density applied to our three examples is shown in Figures 2.5e through 2.7e. We choose to plot the logarithm of the spectral density versus the logarithm of frequency. The spectral density plot of the first example has a completely horizontal trend. This says that all frequencies have about the same amplitude. This trend is characteristic of uncorrelated ordering of data and can be used as the definition of a random data set. (Uncorrelated, or purely random data, is sometimes referred to as "white noise," because white light is composed of light that has about the same amplitude for all frequencies.) The spectral density of the second example decreases continuously for higher frequencies. The largest amplitude is at the lowest frequency, and the lowest amplitude is at the highest frequency. Notice that the original data is "smooth," having little high frequency (rapidly varying) information. The spectral density of the third example decreases but not as rapidly as in the second example. There is some ordering (low frequency information), but it is not as dominant as it is in the second example.

2.3.4 *R/S Analysis*

The rescaled range (R/S) analysis is another measure of how a sequence varies as lag increases. The range is a measure of the cumulative fluctuation of a sequence. The range is rescaled by dividing by the standard deviation. In general, the longer the interval studied (the lag), the greater the rescaled range. In R/S analysis, the log of the rescaled range is plotted versus the log of the lag. If a straight line results for the region of long lag, the slope is denoted as H. The H value is known as the Hurst exponent,[9] or the intermittency exponent.[22] A straight line of slope H=.5 corresponds to uncorrelated or white noise.[10] The R/S analysis is largely insensitive to both the global histogram of the sequence and high frequency fluctuations.

In Figure 2.5f, the slope of H is found to be 0.650. This is near but not exactly the theoretical value of 0.5 for a completely uncorrelated sequence.[10] There are two reasons for this difference. First, H is a statistical parameter and second, artificially generated sequences have H slightly greater than 0.5. Figure 2.9 shows the range of H values produced from artificially generated uncorrelated sequences of different length. For short sequences, with few data points, there is a wide variation in H values from the R/S analysis. As sequences get longer, the variation in H values decreases but approaches 0.525 instead of the theoretical value of 0.5. This discrepancy has been noted by others.[10]

For the second example, Figure 2.6f, H is found to be 0.995. A peculiar feature of the rescaled range is that it is always a number between 0.0 and 1.0 for Gaussian distributions. An H value of 1.0 (or greater) can occur for many sequences and should be considered a "null" result. An H value greater than or equal to 1.0 is not a definitive analysis. For the third sequence, Figure 2.7f, H=0.847. As we will find in the next section, H is greater than 0.5 and less than 1.0 for the fractal sequence called fractional Gaussian noise (fGn).

2.3.5 *Relationships Between Multivariant Measures*

The four measures just described in Figures 2.5 through 2.7 are related. Mathematical proofs for infinite sequences show that the variogram of lag s is the covariance of lag zero minus the covariance of lags.[20] The inverse Fourier transform of the spectral density is the covariance for an ergodic ensemble.[16] In general, the

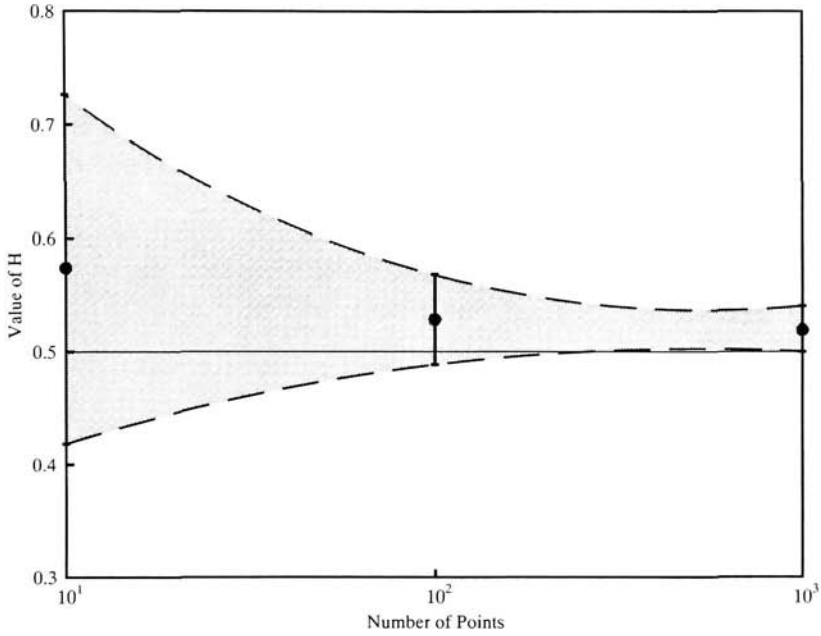


Figure 2.9. Variation of H from R/S analysis as a function of sample size. Results based on a computer-generated Gaussian random sequence.Shaded area represents one standard deviation from mean value.

rescaled range is not directly related to the other tests; however, for fGn, H is related to the slope of the spectral density plot.[9,22]

For finite data sequences, these analyses emphasize different length scales. We have already noted that the covariance and variogram are limited to analyzing short lags (high frequencies). The Fourier transform is a good statistical measure for medium and short lags (medium and high frequencies). The R/S analysis seems to work best for long lags (low frequencies). Thus, a look at the covariance (or variogram), spectral density, and R/S analysis together gives us a more complete evaluation of the statistical properties of a data set than any one analysis alone.

2.4 Fractal Models

In this section, we will apply the statistical tools described in the last section to fractional Gaussian noise (fGn) and fractional Brownian motion (fBm) traces. fGn and fBm are singled out for special treatment, because these are sequences with

fractal character that have been used to describe reservoir property distributions. As can be seen in Figure 2.10, fGn and fBm appear to be very complicated sequences. This apparent complexity is deceiving, because they can be described by only three parameters--the mean, the standard deviation, and the slope of the spectral density plot.

The mean and standard deviations of fGn and fBm can take on any values. The histogram of fGn is Gaussian. The histogram of fBm is "box" shaped. fGn and fBm have spectral densities, S, that scale with angular frequency, ω, as

$$S \propto \omega^{\beta} \tag{2.3}$$

For fGn, β is between +1 and -1, exclusive. For fBm, β is between -1 and -3, exclusive. Figure 2.11 captures this pictorially. For fGn, the slope of the spectral density, β, is related to the slope of the R/S analysis, H, as

$$\beta = -(2H - 1) \tag{2.4}$$

For fBm, this relationship is

$$\beta = -(2H + 1) \tag{2.5}$$

The plots of fGn and fBm in Figure 2.10 are generated using Equation 2.3 and the spectral generation procedure described in Appendix A2.4. fGn is "noisier" than fBm. As β decreases, the sequences become more "smooth." These intuitive impressions can be made more precise by using the statistical tools described in Sections 2.2 and 2.3.

Figures 2.12a and 2.13a show fGn and fBm traces, respectively. These figures also include the statistical measures described above. Note the Gaussian shape of the fGn histogram and the "box" shape of the fBm histogram in Figures 2.12b and 2.13b, respectively. The covariance and variogram capture the more correlated structure of fBm compared to fGn (Figures 2.12c and d and 2.13c and d). The slope of the spectral density plot is 0.4 for fGn and 2.6 for fBm (Figures 2.12e and 2.13e). The R/S analysis of fGn yields H=0.7 (Figure 2.12f), which is consistent with a spectral

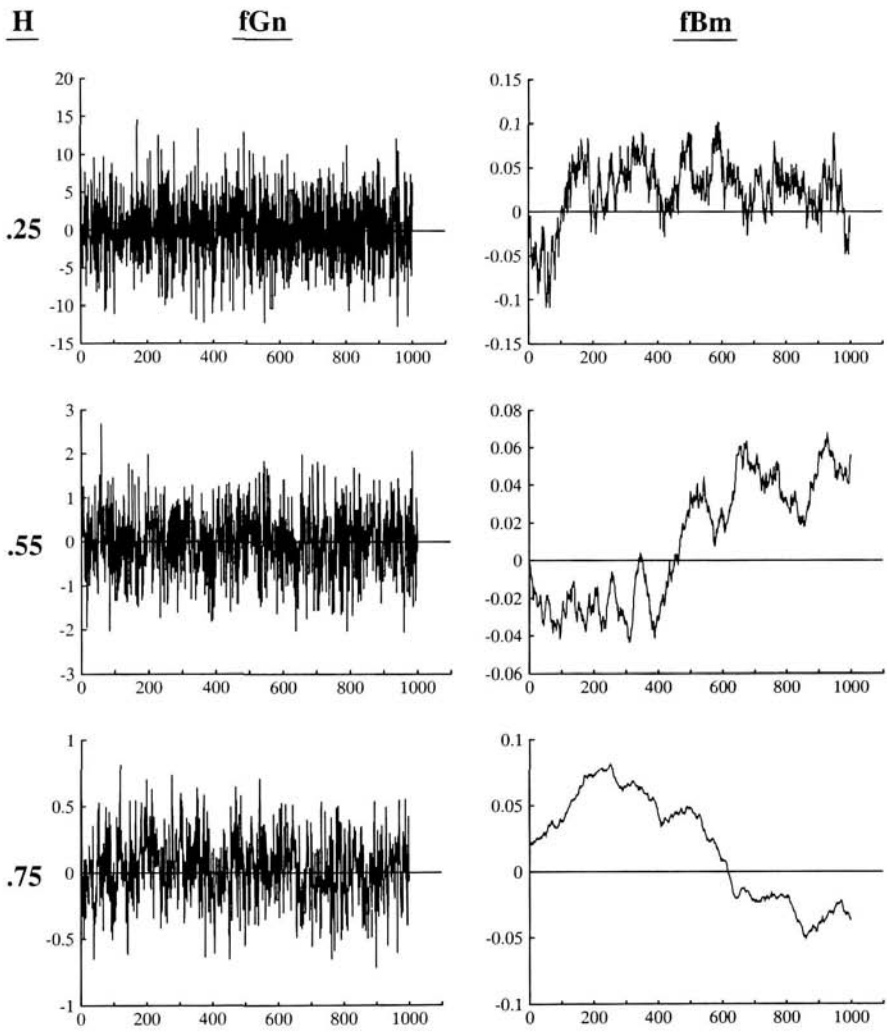


Figure 2.10. Some example fGn and fBm data sets for different H values.

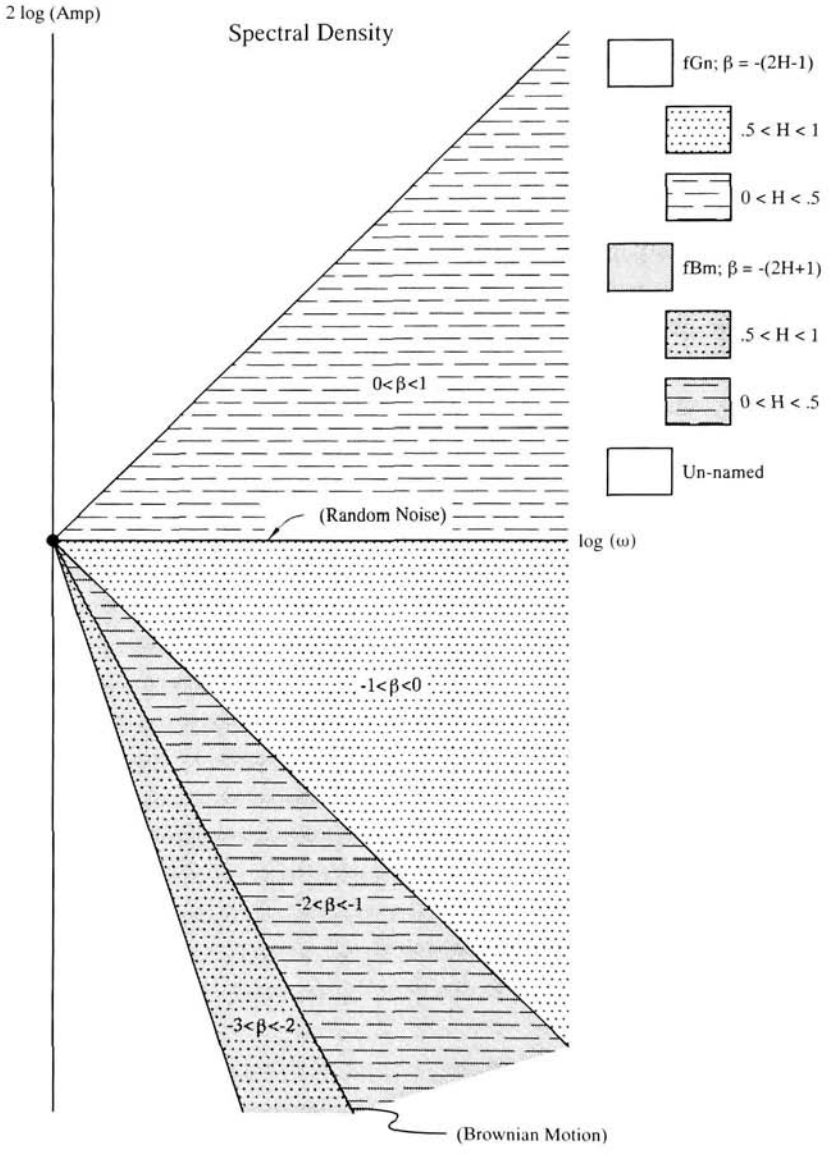


Figure 2.11. The slopes of the spectral densities corresponding to fGn and fBm.

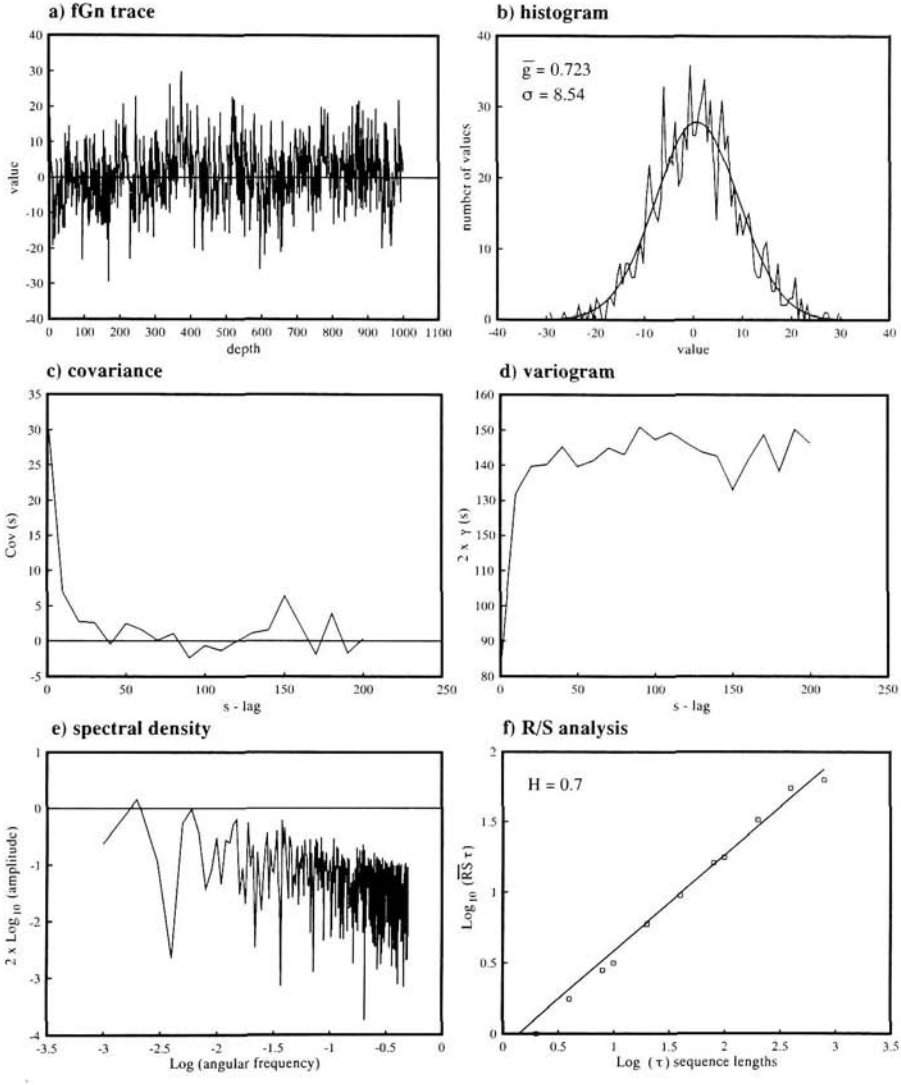


Figure 2.12. An fGn distribution with H=.7 and the following measures of the distribution: (a) distribution itself, (b) histogram, (c) covariance, (d) variogram, (e) spectral density, and (f) R/S analysis.

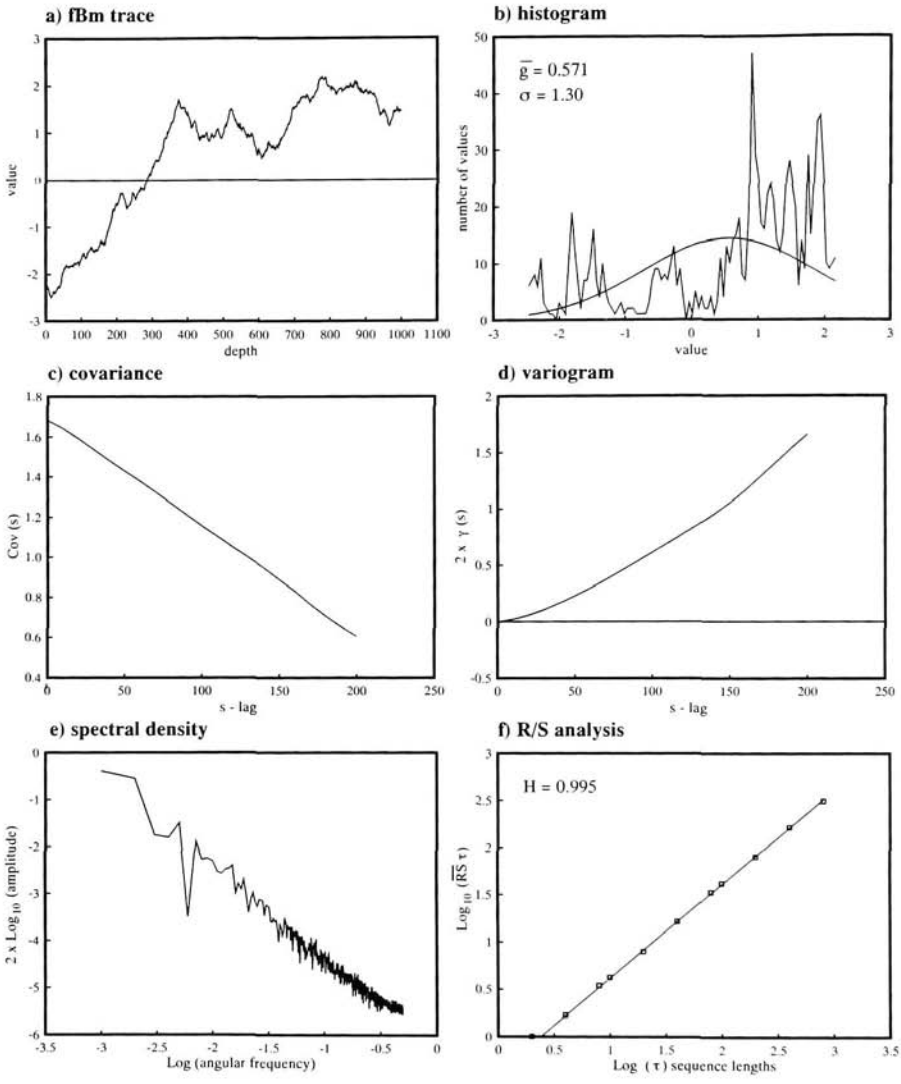


Figure 2.13. An fBm distribution with H=.8 and the following measures of the distribution: (a) distribution itself, (b) histogram, (c) covariance, (d) variogram, (e) spectral density, and (f) R/S analysis.

density slope of -(2H-1). The R/S analysis fails for fBm, since it yields H=1.0 (Figure 2.13f). The functional forms of the variogram and covariance for fGn are derived in Appendix A2.8.

Both fGn and fBm are fractal sequences if the definition of fractal is chosen to be an object with a property that scales as a power law. For fGn and fBm, the spectral density scales as a power law. fBm is also fractal by the box counting technique described in Chapter 1. Care is required if box counting is used. Unlike Figures 1.2 and 1.5, fBm is asymmetric (i.e., its x direction is distance and its y direction is some measured property--stock price, height above sea level, porosity, etc.). As a result, correct box counting "shapes" are required.[10] Fractals that scale differently in the x and y directions are called self-affine fractals, while symmetric fractals are called self-similar.

fBm has some additional peculiar properties. For example, the variance of fBm is unbounded, but fBm is still stationary. Another property is that fBm is nonergodic, even though it is stationary. fGn is much better behaved. It is stationary and ergodic, at least close enough for engineering applications (see Appendix A2.7). Appendices A2.5 and A2.6 describe how fGn and fBm can be generated with any desired mean, standard deviation, and H value.

Unresolved Issues

1. What is the relationship between R/S analysis and the other statistical measures? Is there a general underlying relationship between R/S and spectral methods as there is between spectral methods and the covariance?

2. What is the definition of fBm? Mandelbrot[9] defines it in terms of a divergent integral and then proceeds to present ways to generate approximations to this integral. The problem is that a divergent integral definition presents no mathematical tests to apply to the different approximations, so how do you tell which are "good" approximations and which are "poor" approximations. Voss[11] presented several generators of fGn and fBm. Even a list of the statistical measures applied to these different generators to identify their likenesses and differences would be useful.

3. Is there a simplistic mechanical model that generates fBm or fGn? fBm and fGn seem very common in natural records, but we know of no simple physical model that generates these sequences. Fractional calculus[24] can transform random noise into fBm or fGn, but we know of no simple physical models of fGn or fBm that are described by the fractional calculus. A simple physical model would go a long way in helping find mechanisms that might be producing fGn or fBm in the earth. Right now, fGn and fBm are only descriptive properties.

4. Can the geostatistical process of kriging[20] be reformulated to be based on ensemble rather than time averages? If it could, then it could be used for nonstationary data sets. As an example of its utility, the ensemble data from the first example in this chapter could be used as input. The output would be maps of the average porosity within the formation.

5. Fourier transforms produce both amplitude and phase information. The spectral density analyzes only the amplitude data. It would be useful if a statistical tool similar to the spectral density were available to study the phase data.

APPENDIX A2

Appendix A2 is divided into 8 sections. Section A2.1 defines mathematically the statistical measures described in the text. Sections A2.2 and A2.3 introduce the basic mathematics of the Fourier transform. Sections A2.4, A2.5, and A2.6 show how to generate fGn and fBm sequences with any desired mean, standard deviation, and intermittency exponent by using Fourier transforms. Sections A2.7 and A2.8 describe some basic properties of fGn and fBm. Section A2.9 compares the covariance of Fourier transform generated fGn to the theoretical covariance of fGn.

A2.1 Equations for Statistical Measures

This section of the appendix is divided into three parts. The first part describes some basic symbolism used to define statistical measures. The second part uses the basic symbolism to define a number of univariant measures. The third part uses the same symbolism to define multivariant measures.

A2.1.1 *Basic Symbolism*

The general symbolism for the ensemble and time averages is described in this section.

Let $g_j(k)$ be the jth value of the kth trace of an ensemble. Let N_e be the number of traces and N_p be the number of points in each trace. The ensemble average is

$$\langle g_j \rangle = \frac{1}{N_e} \sum_{k=1}^{N_e} g_j(k) \tag{A2.1}$$

The time average is

$$\{g(k)\} = \frac{1}{N_p} \sum_{j=1}^{N_p} g_j(k) \tag{A2.2}$$

For ergodic ensembles,

$$\langle g_j \rangle = \{g(k)\} = \bar{g} \tag{A2.3}$$

In general, ensemble averages can be applied to any statistical measure - $O[g_j(k)]$. The <.> symbol means

$$\langle O[g_j] \rangle = \frac{1}{N_e} \sum_{k=1}^{N_e} O[g_j(k)] \tag{A2.4}$$

For example, if

$$O[g_j] \equiv (g_j - \bar{g}_e)^2 \tag{A2.5}$$

then <.> becomes

$$<(g_j - \bar{g}_e)^2> = \frac{1}{N_e} \sum_{k=1}^{N_e} (g_j(k) - \bar{g}_e)^2 \qquad \textbf{(A2.6)}$$

where the subscript e denotes ensemble average.

A2.1.2 *Univariant Measures*

Mean:

$$\bar{g}_e = <g_j> \qquad \textbf{(A2.7)}$$

Standard Deviation:

$$\sigma_e = \sqrt{<(g_j - \bar{g}_e)^2>} \qquad \textbf{(A2.8)}$$

Variance or second moment about the mean:

$$Var_e = \sigma_e^2 \qquad \textbf{(A2.9)}$$

Dystra-Parson's Coefficient:

The Dystra-Parsons coefficient[14] is defined in terms of the mean permeability, $\bar{k}$, and the 84.1 percentile of the permeability, k_σ.

$$V_{DP} = \frac{\bar{k} - k_\sigma}{\bar{k}} \qquad \textbf{(A2.10)}$$

The permeability at the 84.1 percentile, k_σ, is the permeability value for which 84.1 percent of the sample permeabilities are more than k_σ.

Gaussian (or Normal) Distribution:

$$f_g(\xi) = \frac{1}{\sqrt{(2\ \pi)}\ \sigma}\ e^{-\frac{(\xi-\bar{\xi})^2}{2\sigma^2}} \qquad \textbf{(A2.11)}$$

where $\bar{\xi}$ is the mean (or average) and σ is the standard deviation of the distribution.

Log-Normal Distribution:

$$f_\ell(\xi) = \frac{1}{\sqrt{2\ \pi}\ \bar{\sigma}\ \xi}\ e^{-\frac{(\ln\xi-\bar{\mu})^2}{2\bar{\sigma}^2}} \qquad \textbf{(A2.12)}$$

where the shape of the distribution is determined by $\bar{\sigma}$ and $\bar{\mu}$.

P-Normal Distribution:[25]

$$f_p(\xi) = \frac{\xi^{(p-1)}}{\sqrt{2\pi}\ \sigma}\ e^{-\frac{(\xi-\bar{\xi})^2}{2\sigma^2}} \qquad \textbf{(A2.13)}$$

A2.1.3 *Multivariant Measures*

Covariance:

Covariance is defined in general for two different data sets, f_j and g_j, as follows:

$$\mathrm{Cov}(f_j,g_j) = \langle(f_j - \bar{f})(g_j - \bar{g})\rangle \qquad \textbf{(A2.14)}$$

In Chapter 2, the covariance is applied to a single data set, g_j. For lag s, the covariance for sequence g_j is

$$\mathrm{Cov}(g_j,g_{j+s}) \tag{A2.15}$$

Other disciplines often use other forms of the covariance. Three examples are the correlation coefficient, moment of inertia, and the correlation function:

Correlation Coefficient:

$$\rho(f_j,g_j) = \frac{\mathrm{Cov}(f_j,g_j)}{\sigma_f\ \sigma_g} \tag{A2.16}$$

Moment of Inertia:

$$\gamma(f_j,g_j) = <(f_j - g_j)^2>/2 \tag{A2.17}$$

Correlation Function:

$$K_c(s) = <g_j g_{j+s}> \tag{A2.18}$$

Variogram:

The variogram or semivariogram is a plot of $\gamma(g_j,g_{j+s})$ versus s. In geostatistics, ergodicity is usually assumed and the variogram is often calculated as follows:

$$\gamma(s) = <\{(g_j(k) - g_{j+s}(k))^2\}>/2.0 \tag{A2.19}$$

The correlogram (or correlation coefficient) is sometimes used as an alternative to the variogram. The correlogram is a plot of $\rho(g_j,g_{j+s})$ versus s.

Spectral Density:

$$S_m = \langle G_m^2 \rangle \tag{A2.20}$$

where the ensemble average <.> is over a set of like frequency values and G_m is the discrete Fourier transform (Appendix A2.2).

R/S Analysis:

The R/S analysis is also called the "rescaled range" analysis. The range is the difference between the maximum and minimum values of a running sum of the data. The range is rescaled by dividing the range by the standard deviation of the data. A very clear description of the R/S analysis in terms of rainfall is given by Feder.[10] For each starting location, j, and for a given analysis length, τ, the range of a set of data, ξ_j, is

$$R_j(\tau) = \max_{j \le \lambda \le j+\tau} \Xi_j(\lambda,\tau) - \min_{j \le \lambda \le j+\tau} \Xi_j(\lambda,\tau) \tag{A2.21}$$

where

$$\Xi_j(\lambda,\tau) = \sum_{k=j}^{j+\lambda} (\xi_k - \bar{\xi}_j(\tau)) \tag{A2.22}$$

and

$$\bar{\xi}_j(\tau) = \frac{1}{\tau + 1} \sum_{m=j}^{j+\tau} \xi_m \tag{A2.23}$$

The standard deviation, $\sigma_j(\tau)$, is

$$\sigma_j^2 = \frac{1}{\tau + 1} \sum_{k=j}^{j+\tau} (\xi_k - \bar{\xi}_k(\tau))^2 \qquad \text{(A2.24)}$$

The rescaled range is

$$RS_j(\tau) = \frac{R_j(\tau)}{\sigma_j(\tau)} \qquad \text{(A2.25)}$$

This function depends on the analysis length, τ, and the starting location, j, over which R/S is calculated. Generally, the average of R/S over a number of starting locations, $\overline{RS}\ (\tau)$, is plotted versus the analysis length or lag, τ. That is,

$$\overline{RS}(\tau) = \frac{1}{N - 1} \sum_{j=1}^{N-1} RS_j(\tau) \qquad \text{(A2.26)}$$

It is very slow to calculate this sum over all starting values and all lags. A subset is usually chosen. In this book, the sequence lengths, τ, are chosen as 1,2,4,6,10,20,40,... up to the sequence length, N. A maximum of twenty starting values are chosen and equally spaced by incrementing j from 1 to N-τ in steps of (N-τ)/20.

A2.2 The Fourier Transform in 1D

All data stored in digital computers are both discrete and finite. We will therefore use the Fourier transform of discrete, finite sets. The Fourier transform is usually defined for infinite, or periodic, continuous functions. We will begin our discussion with the Fourier transform of an infinite, continuous function, g(x).

The Fourier transform is defined as[26]

$$G(\omega) = \int_{-\infty}^{+\infty} g(x) e^{-i\omega x}\, dx \qquad \text{(A2.27)}$$

Here, the angular frequency, ω, is defined in terms of the frequency of a sine or cosine function, f, as

$$\omega = 2\pi f \tag{A2.28}$$

The period, T, and the frequency, f, of a sine or cosine are related as

$$T = \frac{1}{f} \tag{A2.29}$$

The inverse Fourier transform is defined as

$$g(x) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} G(\omega)e^{i\omega x}\, d\omega \tag{A2.30}$$

For a continuous function, g(x), of finite length, L, beginning at x_0 and ending at x_L, we choose

$$g(x) = \begin{bmatrix} g(x) & \text{if} \quad x_0 \le x \le x_L \\ 0 & \text{if} \quad x < x_0 \text{ or } x > x_L \end{bmatrix} \tag{A2.31}$$

with

$$L = x_L - x_0 \tag{A2.32}$$

The Fourier transform, Equation A2.27, becomes

$$G(\omega) = \int_{x_0}^{x_L} g(x)e^{-i\omega x}\, dx \tag{A2.33}$$

Consider a discrete, finite sequence of N equally spaced data values, g_j, given at points x_j. Let x_j be given by

$$x_j = (j - 1)\Delta x + x_o \tag{A2.34}$$

where

$$\Delta x = L/N \tag{A2.35}$$

The value g_j at each point x_j is

$$g_j = g(x_j) \tag{A2.36}$$

The discrete form of Equation A2.33 can now be written as

$$G(\omega) = \sum_{j=1}^{N} g(x_j) e^{-i\omega(j-1)\Delta x} \Delta x \tag{A2.37}$$

Similarly, $G(\omega)$ can be discretized. For

$$\omega_m = (m - 1)\Delta\omega \tag{A2.38}$$

with

$$\Delta\omega = 2\pi/L \tag{A2.39}$$

We assign the discrete value of G_m as

$$G_m = G(\omega_m) \tag{A2.40}$$

The discrete, finite Fourier transform (Equation A2.37) is then

$$G_m = \Delta x \sum_{j=1}^{N} g_j e^{-2\pi i(m-1)(j-1)/N} \quad \textbf{(A2.41)}$$

An exact, discrete inverse Fourier transform exists.[19] It is

$$g_j = \frac{1}{N\Delta x} \sum_{m=1}^{N} G_m e^{2\pi i(j-1)(m-1)/N} \quad \textbf{(A2.42)}$$

From Equations A2.35 and A2.39, this can be written as

$$g_j = \frac{\Delta \omega}{2\pi} \sum_{m=1}^{N} G_m e^{2\pi i(j-1)(m-1)/N} \quad \textbf{(A2.43)}$$

Equation A2.43 is the discrete form of the continuous, inverse Fourier transform, Equation A2.30.

A2.3 Amplitude, Phase, and Spectral Density

Any complex number, G_m, can be written in terms of an amplitude, R_m, and phase, θ_m.

$$G_m = R_m e^{i\theta_m} \quad \textbf{(A2.44)}$$

The amplitude, R_m, is

$$R_m = |G_m| = \sqrt{[\text{Real}(G_m)]^2 + [\text{Imaginary}(G_m)]^2} \quad \textbf{(A2.45)}$$

The phase, θ_m, is

$$\theta_m = \text{Arctan}\left[\frac{\text{Real } (G_m)}{\text{Imaginary } (G_m)}\right] \quad \textbf{(A2.46)}$$

The spectral density, S_m, is

$$S_m = R_m^2 \tag{A2.47}$$

A2.4 Spectral Generators for fBm and fGn

The spectral generators used to generate fBm and fGn sequences shown in this book are based on the discrete, finite Fourier transform. They are similar to those described by Voss.[11] The spectral density scales with angular frequency as described in Equation 2.3

$$S = \lambda_{SD} |\omega|^{\beta} \tag{A2.48}$$

except for $\omega=0$, where we set

$$S(\omega=0) = 0.0 \tag{A2.49}$$

Equations 2.4 and 2.5 give the relationships between β and H for fGn and fBm, respectively. The angular frequency, ω, is discretized as

$$\omega_m = (m - 1)\Delta\omega \tag{A2.50}$$

To generate a sequence with an even number of points, N, the discrete amplitude, R_m, is calculated from Equation A2.47, as

$$R_m = \sqrt{S(\omega_m)} \tag{A2.51}$$

for j=2,3,4,...,N/2.

The phase portion of the array, $\theta(\omega)$, in frequency space is assigned from a set of random numbers between $-\pi$ and $+\pi$:

$$\theta_m = \text{Random } [-\pi,\pi] \tag{A2.52}$$

for m=2,3,4,....,N/2.

The amplitude and phase arrays are converted to a complex array in frequency space as follows:

$$G_m = R_m\cos(\theta_m) + iR_m\sin(\theta_m) \tag{A2.53}$$

for m=2,3,4,...,N/2 and $i=\sqrt{-1}$

Next

$$G_1 = 0 + 0i \tag{A2.54}$$

$$G_{N/2+1} = 0 + 0i \tag{A2.55}$$

Finally,

$$G_{N-m+2} = G_m^* \tag{A2.56}$$

for m=2,3,4,...,N/2, where G*= complex conjugate of G. That is, if

$$G = a + ib \tag{A2.57}$$

then

$$G^* = a - ib \tag{A2.58}$$

To generate an odd number of points, M, the same equations are used with N=(M+1) except that $G_{N/2+1}$ is not set equal to zero, and Equation A2.56 becomes

$$G_{N-m+1} = G_m^* \tag{A2.59}$$

for m=1,2,3...,N/2.

With the above complex array, G_m, the discrete inverse Fourier transform described by Equation A2.42 generates fGn or fBm sequences. As described in Appendix A2.5 (Property 4), setting the value of the spectral density to zero at ω = 0 in Equation A2.49 produces a sequence with mean equal to zero. This is accomplished by Equation A2.54. By a simple linear scaling equation (Appendix A2.6), we can adjust the mean and standard deviation of the sequence to any desired values.

This generation of fGn and fBm introduces correlation between the first and last points of the distributions that should not exist. This undesired correlation can be eliminated by using only 1/2 to 1/4 of the sequence generated.[11] This is done in the examples presented in this book except where noted otherwise.

A2.5 Some Useful Properties of Fourier Transforms

We wish to be able to generate fGn sequences with specified means, standard deviations, and H values. A Gaussian distribution is defined by specifying the mean and standard deviation. Similarly, specifying the mean, the standard deviation, and H specifies a fGn distribution. This section will demonstrate four properties of the Fourier transform. These properties and the linear scaling equation in Appendix A2.6 allow us to rescale fGn sequences generated with λ_{SD}=1.0, $\Delta\omega$=1.0, and S(ω=0)=0.0 into any fGn distribution with the same H. In addition, Appendix A2.8 uses these properties to establish the equations for the variogram and covariance of fGn. The four properties we will demonstrate here are as follows:

1. Multiplying a sequence of numbers by a constant is equivalent to multiplying the spectral density by a constant.

2. Adding a constant to a sequence of numbers is equivalent to changing the spectral density at $\omega=0$.

3. Changing the separation of points (i.e., multiplying Δx by a constant) is equivalent to multiplying $\Delta\omega$ by a constant.

4. The mean of a sequence generated with $S(\omega=0) = 0$ is zero.

The result of the first property is that we can generate a sequence with the spectral density "wrong" by a constant multiple and "fix" it by multiplying all the generated sequence values by a constant. The second property allows us to generate a sequence with the "wrong" spectral density at $\omega=0$ and "fix" it by adding a constant to all the generated sequence values. The third property allows us to build a Fourier transform with the wrong $\Delta\omega$ and "fix" it by simply adjusting the spacing of the final distribution. The fourth property keeps us from having to repeatedly calculate the average of a sequence generated with $S(\omega=0)=0$.

The proofs of these properties follow.

Property 1. Multiplying a sequence of numbers by a constant is equivalent to multiplying the spectral density by a constant.

Let g_j be a sequence generated by an inverse Fourier transform,

$$g_j = \frac{\Delta\omega}{2\pi} \sum_{m=1}^{N} G_m e^{2\pi i(j-1)(m-1)/N} \qquad \textbf{(A2.60)}$$

Calculate a sequence, $\breve{g}_j$, that is a constant multiple, c, of the original sequence, g_j,

$$\breve{g}_j = cg_j = c\frac{\Delta\omega}{2\pi} \sum_{m=1}^{N} G_m e^{2\pi i(j-1)(m-1)/N} \qquad \textbf{(A2.61)}$$

$$\breve{g}_j = \frac{\Delta\omega}{2\pi} \sum_{m=1}^{N} \breve{G}_j e^{2\pi i(j-1)(m-1)/N} \qquad \textbf{(A2.62)}$$

where

$$\check{G}_m = cG_m \tag{A2.63}$$

The spectral densities are related as follows:

$$\check{S}_m = |\check{G}_m|^2 = c^2|G_m|^2 = c^2S_m \tag{A2.64}$$

Property 2. Adding a constant to a sequence of numbers is equivalent to changing the spectral density at $\omega=0$.

Let g_j be a sequence generated by an inverse Fourier transform,

$$g_j = \frac{\Delta\omega}{2\pi} \sum_{m=1}^{N} G_m e^{2\pi i(j-1)(m-1)/N} \tag{A2.65}$$

Calculate a sequence, $\check{g}_j$, that is the original g_j sequence plus a constant, c,

$$\check{g}_j = g_j + c \tag{A2.66}$$

We divide g_j into two parts. The first element of the Fourier series, G_1, and the remaining elements, G_m,

$$\check{g}_j = \frac{\Delta\omega}{2\pi} G_1 + \frac{\Delta\omega}{2\pi} \sum_{m=2}^{N} G_m e^{2\pi i(j-1)(m-1)/N} + c \tag{A2.67}$$

$$\check{g}_j = \frac{\Delta\omega}{2\pi} \left[G_1 + c\frac{2\pi}{\Delta\omega}\right] + \sum_{m=2}^{N} G_m e^{2\pi i(j-1)(m-1)/N} \tag{A2.68}$$

then

$$\check{g}_j = \frac{\Delta\omega}{2\pi} \sum_{m=1}^{N} \check{G}_m e^{2\pi i(j-1)(m-1)/N} \tag{A2.69}$$

where

$$\check{G}_1 = G_1 + c\frac{2\pi}{\Delta\omega} \tag{A2.70}$$

and

$$\check{G}_m = G_m \quad \text{for} \quad m \neq 1 \tag{A2.71}$$

Thus, the spectral density of $\check{g}_j$, $\check{S}_m$, is related to the spectral density of g_j, S_j, as

$$\check{S}_m = \begin{bmatrix} |G_m + c\frac{2\pi}{\Delta\omega}|^2 & \text{if} \quad m = 1 \\ S_m & \text{if} \quad m \neq 1 \end{bmatrix} \tag{A2.72}$$

Property 3. Changing the separation of points (i.e., multiplying Δx by a constant) is equivalent to multiplying $\Delta\omega$ by a constant.

Let

$$g_j = \frac{\Delta\omega}{2\pi} \sum_{m=1}^{N} G_m e^{2\pi i(j-1)(m-1)/N} \tag{A2.73}$$

Calculate $\check{g}_j$ where the separation of points in $\check{g}_j$ is δx,

$$\delta x = c\,\Delta x \tag{A2.74}$$

Using the relationship in Equation (A2.35) for both sequences gives

$$\check{L} = N\delta x = Nc\Delta x = cL \tag{A2.75}$$

The relationship in Equation A2.39 gives

$$\delta\omega = \frac{2\pi}{\check{L}} = \frac{1}{c}\frac{2\pi}{L} = \frac{1}{c}\Delta\omega \tag{A2.76}$$

and

$$\check{g}_j = \frac{\delta\omega}{2\pi}\sum_{m=1}^{N} G_m e^{2\pi i(j-1)(m-1)/N} \tag{A2.77}$$

Property 4. The mean of a sequence generated with $S(\omega=0) = 0$ is zero.

The mean of the sequence g_j consisting of N points is calculated as

$$\bar{g} = \frac{1}{N}\sum_{j=1}^{N} g_j = \frac{1}{N^2\Delta x}\sum_{j=1}^{N}\sum_{m=1}^{N} G_m e^{2\pi i(j-1)(m-1)/N} \tag{A2.78}$$

If $G_1=0.0$,

$$\bar{g} = \frac{1}{N^2\Delta x}\sum_{m=2}^{N} G_m \sum_{j=1}^{N} e^{-2\pi i(j-1)(m-1)/N} \tag{A2.79}$$

but from Walker,[19]

$$\sum_{j=1}^{N} e^{-2\pi i(j-1)(m-1)/N} = 0.0 \quad \text{if} \quad m - 1 \neq 0 \tag{A2.80}$$

Thus, if $G_1 = 0.0$,

$$\bar{g} = 0.0 \tag{A2.81}$$

A2.6 Linear Scaling

In Appendix A2.4, the fGn and fBm sequences are generated with zero mean and arbitrary standard deviation. A linear scaling allows us to rescale these sequences into another sequence with the desired mean and standard deviation. The value of H remains unchanged during the transformation.

To go from a sequence with mean, μ_H, and standard deviation, σ_H, to another sequence with mean, μ_W, and standard deviation, σ_W, each value in the first array, ξ_j, is changed into the corresponding value in the second array, g_j, as follows:

$$g_j = \left[\frac{\sigma_W}{\sigma_H}\right](\xi_j - \mu_H) + \mu_W \tag{A2.82}$$

We use the properties proved in Appendix A2.5 to show that the value of H remains unchanged. First, we are multiplying each value, ξ_j, by a constant, σ_W/σ_H. Property 1 in Appendix A2.5 shows this operation is equivalent to multiplying the spectral density by a constant. Multiplying the spectral density by a constant does not change the exponent of ω (β in Equation A2.48), so the value of H remains unchanged. In the second operation, we are adding a constant to each value, ξ_j. From Property 2 in Appendix A2.5, the second operation is equivalent to adding a constant to the spectral density at $\omega=0$. As with multiplication, adding a constant to the spectral density at $\omega=0$ does not change H, which is defined for all $\omega>0$.

A2.7 Stationarity and Ergodicity of fGn and fBm

Stationarity and ergodicity are impossible to test completely, because they must be tested for every statistical measure. Here we will demonstrate in one statistical measure, the variogram, that fBm is stationary, but nonergodic. We will also demonstrate that fGn is both stationary and ergodic. Results are also stated for another statistical measure, the mean.

Let $g_j(k)$ be the j^{th} point in the k^{th} sequence of fBm or fGn. Then the ensemble variogram of lag, s, is

$$\gamma_e(g_j,g_{j+s}) = <(g_{j+s}(k) - g_j(k))^2>/2 \qquad \textbf{(A2.83)}$$

Here we have added the subscript "e" to our notation to specify the ensemble average variogram. Writing out Equation A2.83 explicitly gives

$$\gamma_e(g_j,g_{j+s}) = \frac{1}{2N_e}\sum_{k=1}^{N_e}(g_{j+s}(k) - g_j(k))^2 \qquad \textbf{(A2.84)}$$

for

$$s = 1, 2, 3,..., N_l \qquad \textbf{(A2.85)}$$

$$j = 1, 2, 3,..., N_{pt}/4 \qquad \textbf{(A2.86)}$$

Here N_e is the number of sequences in the ensemble. N_l is the maximum lag. N_{pt} is the number of values (or points) in each sequence. Here we have chosen to limit j to $N_{pt}/4.0$. Another alternative limit could have been N_{pt}-s.

The time average variogram of lag, s, is

$$\gamma_t(g(k),g_s(k)) = \{(g_{j+s}(k) - g_j(k))^2\}/2 \qquad \textbf{(A2.87)}$$

Here the subscript "t" is used to distinguish the time average from the ensemble average variogram. Writing Equation A2.87 explicitly gives

$$\gamma_t(g(k),g_s(k)) = \frac{1}{2(N_{pt} - s)}\sum_{j=1}^{N_{pt}-s}(g_{j+s}(k) - g_j(k))^2 \qquad \textbf{(A2.88)}$$

for

$$k = 1, 2, 3, \ldots, N_e \tag{A2.89}$$

$$s = 1, 2, 3, \ldots, N_l \tag{A2.90}$$

The spectral generator in Appendix A2.4 was used to generate 50 sequences of fGn and fBm each containing 200 points. The variograms were then computed for N_e=50, N_{pt}=200, and N_l=50 points.

Figures A2.1 and A2.2 are plots of γ_e and γ_t, respectively, for fBm. γ_e is relatively invariant to j, or time, indicating stationarity for fBm in the variogram. γ_t however is very different from γ_e, indicating fBm has a nonergodic variogram.

Figures A2.3 and A2.4 are plots of γ_e and γ_t, respectively, for fGn. γ_e and γ_t are similar and both invariant to j. This result indicates fGn is stationary and ergodic for the variogram. Similar plots of the mean instead of the variogram show that fGn is also stationary and ergodic in the mean.

The definition of the ensemble will determine whether or not the mean of fBm is stationary or not. If we use the spectral density (Equation A2.48) to generate fBm, we can set the spectral density to be zero at $\omega = 0$. Then, the mean will always be zero. In this case, fBm is stationary in the mean. If the initial value of the fBm sequence is fixed, the mean will vary between realizations and fBm will not be stationary in the mean. In either case, fBm is nonergodic in the mean.

A2.8 Variogram and Covariance of fGn

In this section, we will show that the variogram, $\gamma(g_j,g_{j+s})$, of fGn generated as described in Appendix A2.4 can be written for sequences with a large number of points (N>10), as follows:

$$\gamma(g_j,g_{j+s}) = a - b|s|^{2H-2} \tag{A2.91}$$

We will also show that the covariance, $Cov(g_j,g_{j+s})$ is

$$Cov(g_j,g_{j+s}) = c + b|s|^{2H-2} \text{ for } s \neq 0 \tag{A2.92}$$

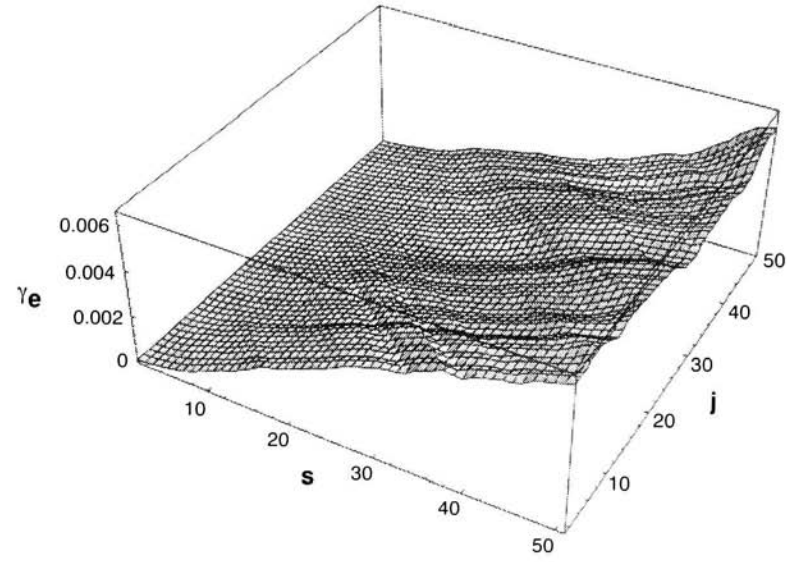


Figure A2.1. Ensemble average of the variance of fBm plotted versus lag, s, and time, j.

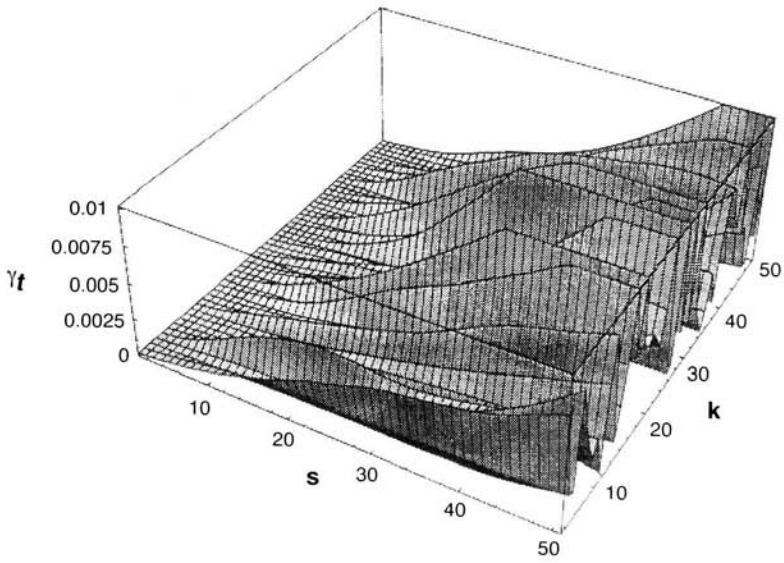


Figure A2.2. Time average of the variance of fBm plotted versus lag, s, and ensemble, k.

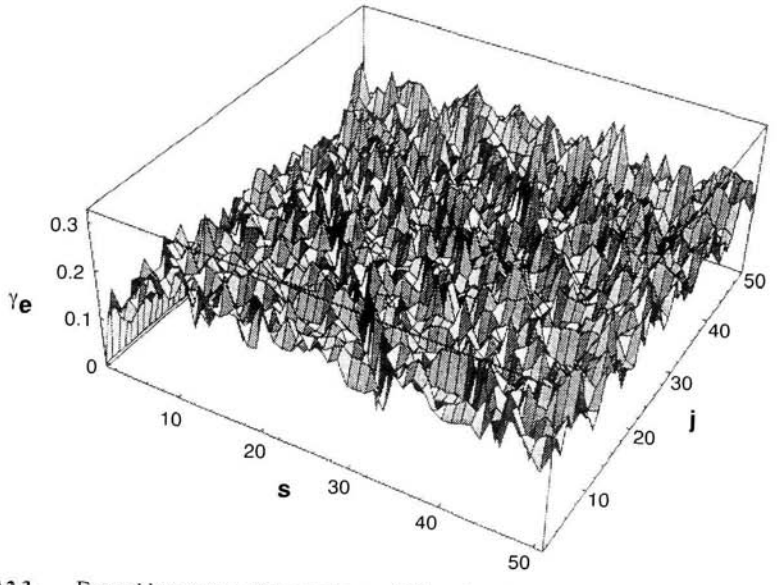


Figure A2.3. Ensemble average of the variance of fGn plotted versus lag, s, and time, j.

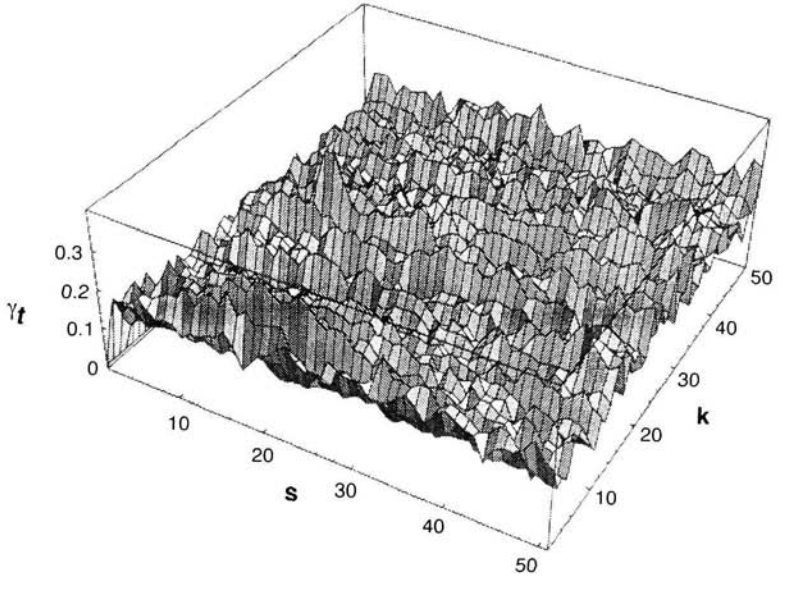


Figure A2.4. Time average of the variance of fGn plotted versus lag, s, and ensemble, k.

The variogram is defined as

$$\gamma(g_j,g_{j+s}) = \langle(g_j - g_{j+s})^2\rangle/2 \tag{A2.93}$$

Expanding,

$$\gamma(g_j,g_{j+s}) = (\langle g_j^2\rangle + \langle g_{j+s}^2\rangle - 2\langle g_j g_{j+s}\rangle)/2 \tag{A2.94}$$

Since fGn generated as described in Appendix A2.4 is stationary in the mean (Appendix A2.7),

$$\gamma(g_j,g_{j+s}) = \{g_j^2\} - \{g_j g_{j+s}\} \tag{A2.95}$$

We will now evaluate

$$\{g_j g_{j+s}\} = \frac{1}{N}\sum_{j=1}^{N} g_j g_{j+s} \tag{A2.96}$$

To do this, we will follow Reif[16] except we will use finite sequences instead of continuous functions. In terms of the Fourier transforms,

$$g_j = \frac{\Delta\omega}{2\pi}\sum_{m=1}^{N} G_m e^{2\pi i(j-1)(m-1)/N} \tag{A2.97}$$

$$g_{j+s} = \frac{\Delta\omega}{2\pi}\sum_{n=1}^{N} G_n e^{2\pi i(j+s-1)(n-1)/N} \tag{A2.98}$$

After substituting into Equation A2.96 an rearranging, we get

$$\{g_j g_{j+s}\} = \left[\frac{\Delta\omega}{2\pi}\right]^2 \frac{1}{N}\sum_{n=1}^{N}\sum_{m=1}^{N} G_n G_m e^{2\pi i(n-1)s/N}\sum_{j=1}^{N} e^{2\pi i(m+n-2)(j-1)/N} \tag{A2.99}$$

But (Reference No. 19, pp. 321-322), if $(m+n-2) \neq 0, \pm N, \pm 2N, \ldots$ then

$$\sum_{j=1}^{N} e^{2\pi i(m+n-2)(j-1)/N} = 0 \qquad \textbf{(A2.100)}$$

and if $(m+n-2) = 0, \pm N, \pm 2N, \ldots$, then

$$\sum_{j=1}^{N} e^{2\pi i(m+n-2)(j-1)/N} = N \qquad \textbf{(A2.101)}$$

With these results,

$$\{g_j g_{j+s}\} = \left[\frac{\Delta\omega}{2\pi}\right]^2 \frac{1}{N} \sum_{n=1}^{N} \sum_{m=1}^{N} G_m G_n e^{2\pi i(n-1)s/N}\, \delta_N(m+n-2) \qquad \textbf{(A2.102)}$$

where

$$\delta_N(m+n-2) = \begin{bmatrix} N \text{ if } (m+n-2) = 0, \pm N, \pm 2N, \ldots \\ 0 \text{ if } (m+n-2) \neq 0, \pm N, \pm 2N, \ldots \end{bmatrix} \qquad \textbf{(A2.103)}$$

To simplify Equation A2.102, we need only $(m+n-2)=0$ and $(m+n-2)=N$. The $(m+n-2)=0$ case produces only one term, $(m=n=1)$. Collecting the nonzero terms where $m=N+2-n$,

$$\{g_j g_{j+s}\} = \left[\frac{\Delta\omega}{2\pi}\right]^2 \sum_{n=1}^{N} G_{N+2-n} G_n e^{2\pi i(n-1)s/N} + \left[\frac{\Delta\omega}{2\pi}\right]^2 G_1 G_1 \qquad \textbf{(A2.104)}$$

But in generating fGn (Appendix A2.4),

$$G_{N+2-n} = G_n^* \qquad \textbf{(A2.105)}$$

and

$$G_1 = G_1^* \tag{A2.106}$$

so that

$$\{g_j g_{j+s}\} = \left[\frac{\Delta\omega}{2\pi}\right]^2 \sum_{n=1}^{N} G_n G_n^* e^{2\pi i(n-1)s/N} \tag{A2.107}$$

and

$$\{g_j g_{j+s}\} = \left[\frac{\Delta\omega}{2\pi}\right]^2 \sum_{n=1}^{N} |G_n|^2 e^{2\pi i(n-1)s/N} \tag{A2.108}$$

To calculate $\{g_j g_{j+s}\}$, we now need to renumber the indices to make the sum more symmetrical. We will demonstrate this for N odd. We will define the new index, m, as follows:

$$m = n - 1 \quad \text{for} \quad n = 1, 2, 3, \ldots, \frac{N+1}{2} \tag{A2.109}$$

$$\tilde{G}_m = G_{n-1} \tag{A2.110}$$

and

$$m = \frac{N+1}{2} - n \quad \text{for} \quad n = \frac{N+3}{2}, \frac{N+5}{2}, \frac{N+7}{2}, \ldots, N \tag{A2.111}$$

$$\tilde{G}_m = G_{\frac{N+1}{2} - n} \tag{A2.112}$$

Using the m index, we have

$$\{g_j g_{j+s}\} = \left[\frac{\Delta\omega}{2\pi}\right]^2 \sum_{m=-\frac{N+1}{2}}^{\frac{N+1}{2}} |\tilde{G}_m|^2 e^{2\pi i m s/N} \qquad \text{(A2.113)}$$

Now we wish to turn the sum into an integral so that we can get an analytic expression. We will do this by reversing the discretization process. To do this, define the following

$$x_s = s\,\Delta x \qquad \text{(A2.114)}$$

$$\omega_m = m\,\Delta\omega \qquad \text{(A2.115)}$$

and

$$\Delta x\,\Delta\omega = 2\pi/N \qquad \text{(A2.116)}$$

Equation A2.112 then becomes

$$\{g_j g_{j+s}\} = \frac{\Delta\omega}{(2\pi)^2} \sum_{m=-\frac{N+1}{2}}^{\frac{N+1}{2}} |\tilde{G}_m|^2 e^{i\omega_m x_s} \Delta\omega \qquad \text{(A2.117)}$$

For fGn,

$$|G(\omega_m)|^2 = |\tilde{G}_m|^2 = \lambda_{SD} |\omega_m|^{\alpha} \quad \text{for} \quad \omega_m \neq 0 \qquad \text{(A2.118)}$$

$$\alpha = -(2H - 1) \qquad \text{(A2.119)}$$

$$0.5 < H < 1.0 \qquad \text{(A2.120)}$$

Notice that we will not make

$$|G(\omega_m)|^2 = |\tilde{G}_m|^2 \quad \text{for} \quad \omega_m = 0 \tag{A2.121}$$

Except for $\omega_m=0$, the right-hand side of Equation A2.117 is

$$h(x_s) = \frac{\lambda_{SD}\Delta\omega}{(2\pi)^2} \int_{-\frac{N+1}{2}}^{+\frac{N+1}{2}} |\omega|^{\alpha} e^{i\omega x_s} d\omega \tag{A2.122}$$

This can be integrated with finite limits, but numerical calculations show that for N>10, the results are almost the same if N→∞, or

$$h(x_s) = \frac{\lambda_{SD}\Delta\omega}{(2\pi)^2} \int_{-\infty}^{+\infty} |\omega|^{\alpha} e^{i\omega x_s} d\omega \tag{A2.123}$$

$h(x_s)$ is evaluated using Lighthill[27] to be

$$h(x_s) = -\frac{\lambda_{SD}\Delta\omega}{2\pi^2} \sin(\alpha\pi/2)\ \Gamma(\alpha + 1)|x_s|^{-\alpha-1} \tag{A2.124}$$

We learned earlier that a difference in the spectral density at $\omega=0$ corresponds to a constant added to the resulting function. Because we did not assign the $\omega=0$ term in the sum in Equation A2.117, h(x) will differ from $\{g_j\ g_{j+s}\}$ by a constant. That is,

$$\{g_j g_{j+s}\} = h(x_s) + c \tag{A2.125}$$

We can evaluate the constant, c, because the discrete sum with $\omega=0.0$ gives a result that has an average of zero (Property 4 in Appendix A2.5). Therefore, the average of $g(x_s)$ plus c must be zero. To evaluate c, we need only find the average of $h(x_s)$, $\{h(x_s)\}$. This is

$$\{h(x_s)\} = \frac{1}{N} \sum_{s=-\frac{N+1}{2}}^{\frac{N+1}{2}} h(x_s) \tag{A2.126}$$

For large N, we get

$$\{h(x_s)\} = \frac{1}{N\,\Delta x} \int_{-(N/2)\Delta x}^{+(N/2)\Delta x} h(x)\,dx \tag{A2.127}$$

Integrating Equation A2.124 and using Equation A2.116, we get

$$\{h(x_s)\} = +\frac{\lambda_{SD}(\Delta\omega)}{2\alpha\pi} \sin(\alpha\pi/2)\ \Gamma(\alpha + 1) \left|\frac{N\,\Delta x}{2}\right|^{-\alpha-1} \tag{A2.128}$$

Then

$$c = -\{h(x_s)\} \tag{A2.129}$$

For large N, we now have

$$\{g_j g_{j+s}\} = \kappa |x_s|^{-\alpha-1} + c \tag{A2.130}$$

where

$$\kappa = -\frac{\lambda_{SD}\Delta\omega}{2\pi^2} \sin(\alpha\pi/2)\ \Gamma(\alpha + 1) \tag{A2.131}$$

Returning to Equation A2.94, we can now write the variogram as

$$\gamma(g_j, g_{j+s}) = a - \kappa |x_s|^{-\alpha-1} \tag{A2.132}$$

where

$$a = \{y_j^2\} - c \tag{A2.133}$$

When a sequence is built using the Fourier transform, a $\Delta\omega$ is specified. This then determines Δx by Equation A2.116. Combining Equation A2.114 and A2.116 gives

$$x_s = s\left(\frac{2\pi}{N\,\Delta\omega}\right) \tag{A2.134}$$

Substituting this into Equation A2.132 gives

$$\gamma(g_j,g_{j+s}) = a - b|s|^{2H-2} \tag{A2.135}$$

with

$$b = \kappa\left(\frac{2\pi}{N\,\Delta\omega}\right)^{2H-2} \tag{A2.136}$$

For a stationary distribution,

$$\mathrm{Cov}(g_j,g_{j+s}) = \{(g_j - \bar{g})(g_{j+s} - \bar{g})\} = \{g_j g_{j+s}\} - \bar{g}^2 \tag{A2.137}$$

Combining this with Equation A2.95 gives

$$\gamma(g_j,g_{j+s}) = \sigma_y - \mathrm{Cov}(g_j g_{j+s}) \tag{A2.138}$$

where

$$\sigma_y = \overline{g^2} - \bar{g}^2 \tag{A2.139}$$

Solving for $Cov(g_j\ g_{j+s})$ and substituting Equation A2.135 yields

$$Cov(g_j,g_{j+s}) = c + b|s|^{2H-2} \quad \textbf{(A2.140)}$$

with

$$c = \sigma_y - a \quad \textbf{(A2.141)}$$

A2.9 Mandelbrot fGn Versus Spectral Generated fGn

Mandelbrot[9] defines fGn as sequences with covariances described by

$$Cov(s) = (1/2)(|s + 1|^{2H} - 2|s|^{2H} + |s - 1|^{2H}) \quad \textbf{(A2.142)}$$

In Appendix A2.8, we demonstrated that the covariance of sequences generated using spectral methods is

$$Cov(s) = c + b|s|^{2H-1} \quad \textbf{(A2.143)}$$

Figure A2.5 shows a plot of Equation A2.142 and Equation A2.143 for H = .8, c = 0, and b = 0.5. Although the spectral method does not exactly match Mandelbrot's definition, it is close enough for our engineering applications.

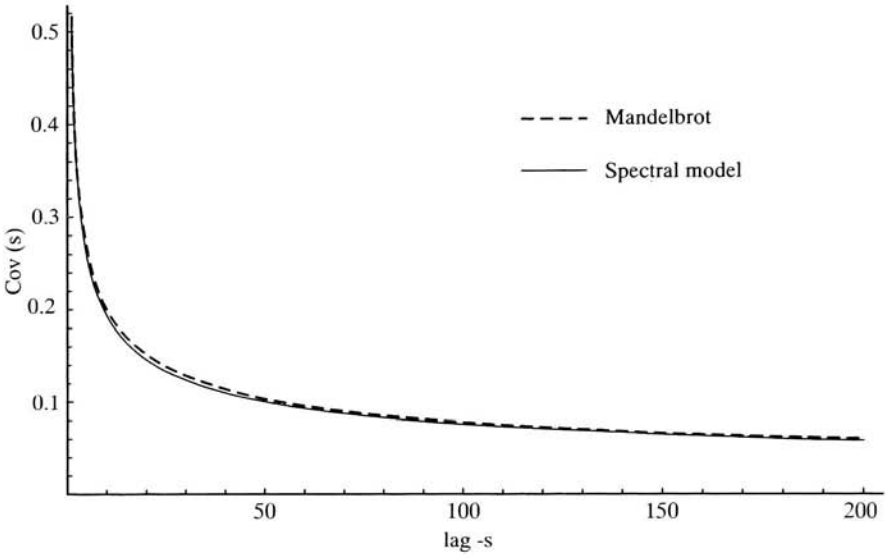


Figure A2.5. Comparison of the covariance of fGn calculated from the spectral (Fourier transform) method and from Mandelbrot's defining equation.

CHAPTER 3

FGN IN RESERVOIR PROPERTY DISTRIBUTIONS

In this chapter, the tools described in Chapter 2 are applied to measurements of reservoir property distributions. We examine well logs from vertical and horizontal wells, core photos, outcrop photos, and outcrop permeability measurements. Often these data sets are very similar to fGn. Each data set has limitations, but all of these together strongly support the careful use of fGn as a model of reservoir property distributions. When fGn is a good model, a 2D spectral density equation produces remarkably good forgeries of core photos. In Chapter 4, we will use these same techniques to generate reservoir property distributions for reservoir simulations from vertical well logs.

A number of reservoir properties and processes that influence reservoir engineering have been found to have fractal properties.[10,28,29] A partial list is found in Table 3.1. We obviously do not have the time or space to study this entire list. We will concentrate instead on well logs, core photos, and minipermeameter outcrop studies. This choice is due more to the authors' familiarity with these than necessarily their relative importance.

In this chapter, we first describe the application of the statistical measures introduced in the last chapter. Next, we will apply these tests to well logs, core photos, and outcrop photos. Then, we will extend the spectral density measure to 2D and apply this 2D spectral measure to core photos. Finally, we will show how forgeries of core photos can be generated using the 2D spectral density. This will prepare us for Chapter 4 where a 2D, fGn spectral model is used to build cross sections of reservoir property distributions for reservoir simulation.

We do not propose a fGn model for all reservoirs. We recommend that the fractal models be applied to reservoir description when well log and other data are fractal. If reservoir measurements do not have a fractal character, other methods should be employed (e.g., variogram,[17] n-point histogram,[30] or Boolean[31] methods.)

3.1 Applying Statistical Measures to Data

We will use the statistical measures of Chapter 2 to test if data sets are fGn or not. On the one hand, this is trivial--simply plot the data and see if it "looks like" the plots of these measures applied to fGn. On the other hand, the application can be confusing--"What is a good fit, how do I fit noisy data, and how does my measuring device distort the actual measured sample?" In this section, we will show one approach of testing for fGn. In most cases, we will also suggest other methods.

We will apply the measures to a porosity well log, Figure 3.1a. As we learned in Chapter 2, the first question to ask is, "Is the well log a stationary ensemble?" This question must be answered by a judgement call. First, any overall trends need to be identified in the data set and removed (e.g., Figure 2.2). Second, the data needs to be divided into like sample distributions. This division will require a combined geological and engineering interpretation.

The next question impacts the first--"Is there enough data?" Figure 2.9 shows that the variation in H depends on the number of points analyzed (i.e., the sequence length). From Figure 2.9, it is clear that if there are less than 50 points, the fluctuations in H values are so large that even a "good" analysis will not yield a meaningful H value. If a log is divided into too many zones, there will not be enough points in any one zone to get a good estimate of H. On the other hand, if geologic units with different H values are mixed together, the H value found for the mixture may not be meaningful. Like most engineering applications, a compromise must be sought based on judgement and perhaps trial and error.

Once we have identified a geological interval that we feel is stationary and representative of a geologic unit, we can continue the analysis by applying the statistical tests of Chapter 2. In Figure 3.1, (a) is a horizontal well porosity log, (b) the histogram of the log, (c) the covariance, (d) the variogram, (e) the spectral density, and (f) the R/S analysis. We will interpret each plot separately.

Table 3.1 Reservoir properties and processes that have been found to be fractal

Fold systems in rocks
Joint and fracture systems
Earthquake seismicity
Geomorphology of rivers and their drainage basins
Geometry of fluid/fluid interfaces
Sediment varves
Sedimentary structures
Pore/rock interfaces
Rainfall and river discharges
Topography
Rock cross-sectional area (and volume)
Pore size-frequency distributions
Perimeters of pore element cross-sections
Net sand in deep-water sand/shale sequences
Clastic sand and shale thickness-frequency distributions
Carbonate sand/carbonate mud thickness-frequency distributions
Sandstone grain size-frequency distributions
Styolite amplitude and residue thickness frequency distributions
Well log traces
Minipermeameter data taken on outcrop faces
Grey scales from slabbed core and outcrop photos
Electrical resistivity in soil
Vegetation cover
Ground water levels
Sand content in soils
Distribution of oil reservoirs

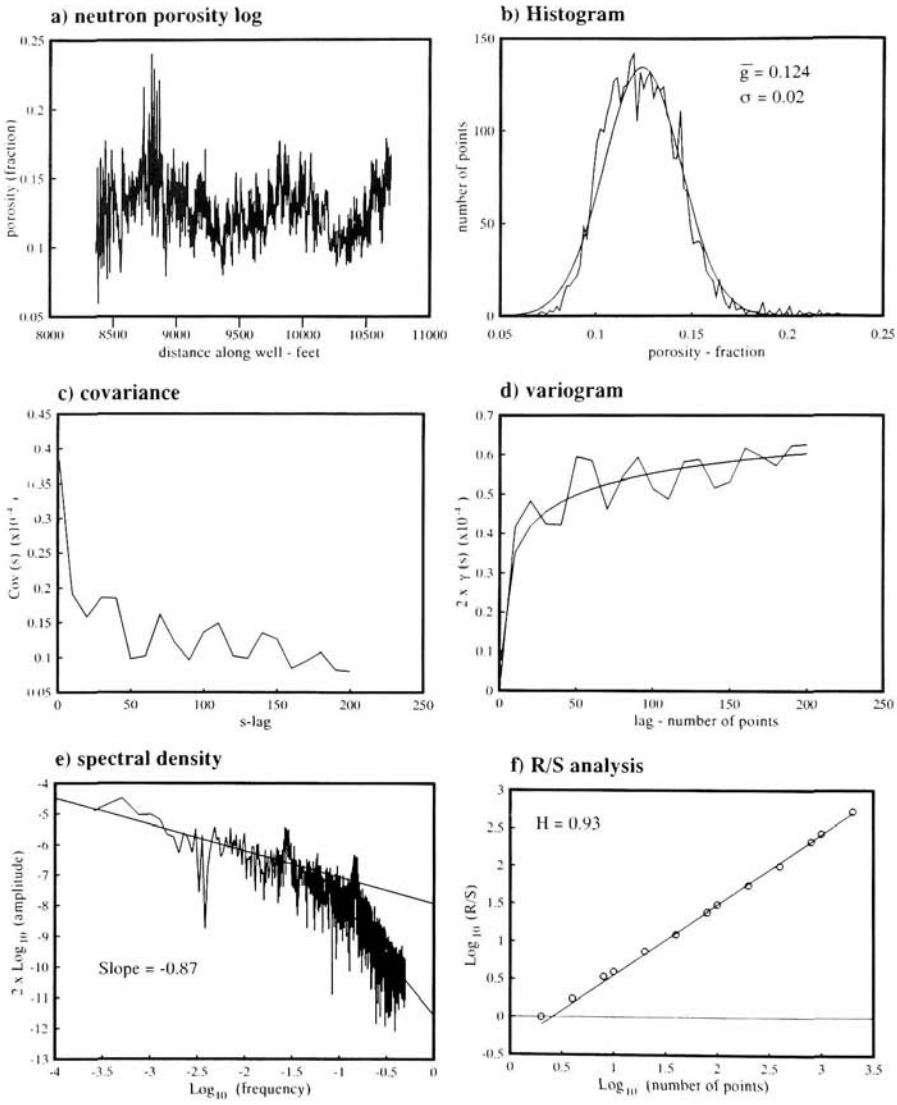


Figure 3.1. Statistical measures of a neutron log from a horizontal well in an aeolion standstone reservoir. (a) log trace, (b) histogram, (c) covariance, (d) variogram, (e) spectral density, and (f) R/S analysis.

The R/S analysis is probably the easiest to interpret. If the data set is fGn, the R/S analysis should produce a straight line of slope H. As mentioned in the last chapter, the R/S analysis describes long lags best. A computer fit is made of a straight line using linear regression on all of the points of lag greater than 20. The computer plots the fitted line and prints the corresponding slope (H value). A visual inspection of the R/S plot indicates if a good fit was obtained. Figure 3.1f shows a reasonably good fit. An example of a poor fit is shown in Figure 3.2. A least squares error analysis could be used to quantify the fit, but such precision is usually not necessary.

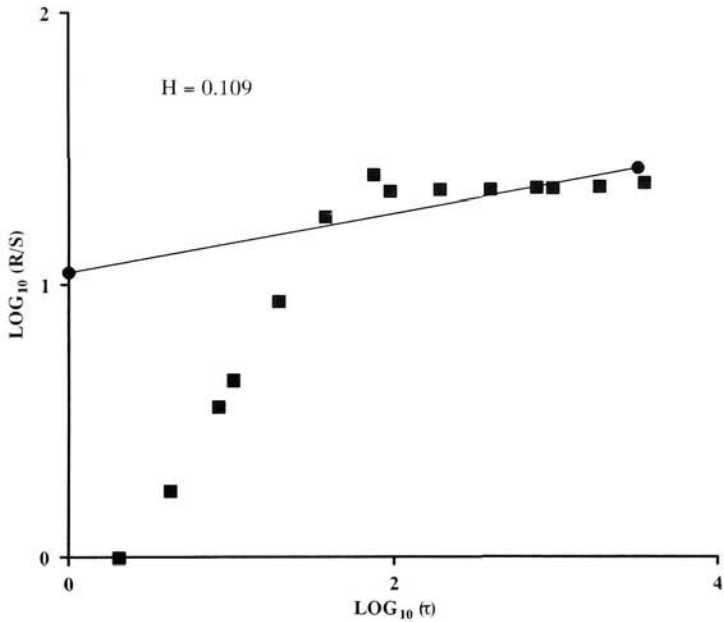


Figure 3.2. A poor straight line fit in the R/S analysis.

The next step is to interpret the spectral density. A line passing through the low and medium frequencies of the spectral density plot should have a slope of -(2H-1). This could be computerized, but the data is so "noisy" that we have used "an eyeball fit" to simply check the H value found by the computer fit in the R/S analysis. Another procedure, which has been suggested by Mandelbrot,[32] is to divide the spectral density plot into bins in the log (frequency) scale. Average all of the spectral density values

in each bin. Plot these averages. Then, fit a straight line to these average points. This would remove much of the scatter of data and make a computer regression much better behaved. We have not tried this procedure, although it seems to be a good suggestion.

An example of the spectral density fit is shown in Figure 3.1e. In this figure, data at the medium and low frequencies match the slope, but the high frequency data does not. The high frequency data "roll-off" from the straight line describing fGn. This roll-off has been identified as a result of the bed resolution of the logging tool.[33,34] In fact, the frequency at which the roll-off starts can be correlated with the size of a running average filter that would produce the roll-off (see Appendices A3.2 and A3.3). This analysis yields a running average filter length that agrees with published "bed resolutions" of different logging tools[35] (Figure 3.3).

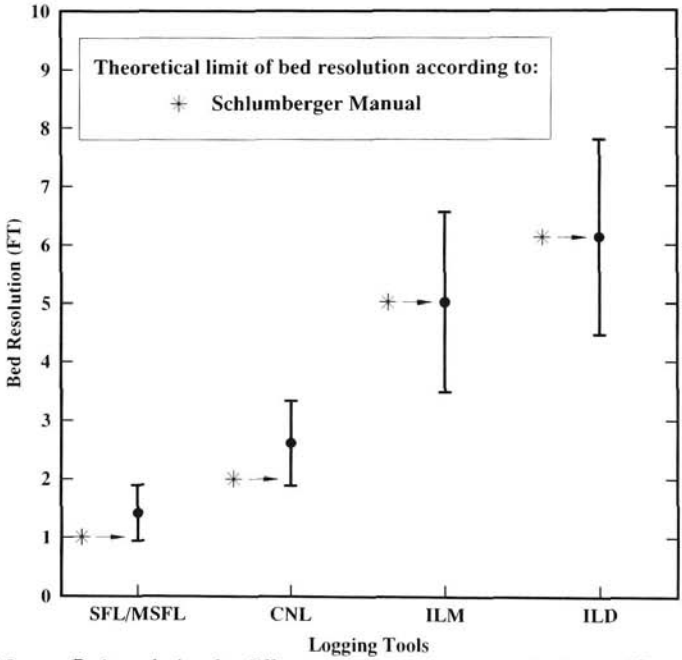


Figure 3.3. Bed resolution for different logging tools compared with the "filter size" estimated from a spectral density analysis.[33]

Next we look at the histogram of the data, Figure 3.1b. A Gaussian curve with the mean and standard deviation of the data is superimposed on the histogram. A simple visual inspection is made to qualitatively compare the histogram of the data to the Gaussian curve. Another possible technique is to generate a cumulative histogram of the input data, and then apply an inverse cumulative Gaussian transform. A plot of the resulting transform versus the measured value produces a straight line if the distribution is Gaussian. Isaaks and Strivastava[17] call the resulting graph a q-q plot and claim it is superior to "an eyeball fit" directly to a Gaussian distribution.

We usually conclude the analysis with a visual inspection of the variogram (Figure 3.1d) and covariance (Figure 3.1c) to see if it has the right shape. If we wish to be a bit more precise, the functional form of the variogram for fGn is (Appendix A2.8)

$$\gamma(g_j,g_{j+s}) = a - b|s|^{2H-2} \qquad (3.1)$$

Similarly, Equation A2.92 is the functional form of the covariance for fGn.

In this case, H from the R/S analysis is used and a and b are fitting parameters. The problem with using the variogram is that it is only good up to lags of 1/3 of the data length, and the smallest lags are those that are most affected by tool resolution. The problem of tool resolution could be avoided if data were chosen from the well logs at intervals greater than the tool's bed resolution, but then the number of data left to be analyzed would be greatly reduced. One way to supplement the reduced data set would be to calculate the ensemble rather than the time averages when analyzing the variogram.

If a poor fit to a Gaussian distribution is found in these analyses, there are transforms that can manipulate the data set to make it Gaussian. If a transform is used, then distributions can be built in this transformed space. Afterwards, an inverse transform can be used to return the original properties. An example of this is the resistivity log. The logarithm of resistivity usually results in a data set that is closer to Gaussian than the resistivity itself. If a distribution of resistivity is desired, we would choose to work with the log (resistivity) and then take an inverse logarithm to return a resistivity value.

We would like any transform used to preserve the functional form of the spectral density and the H value as the logarithm does.[9] If a transform of a data set

is used, it is best to perform the R/S and spectral analysis before and after such a transform. If the transform does not preserve H and the form of the spectral density, the corresponding inverse transform is also not likely to.

A transform does exist that always produces a Gaussian distribution. It is the normal score transform (see Appendix A3.4). The price paid, however, is that the inverse normal score transform may not always preserve H and the form of the spectral density.

3.2 Well Logs

Many studies have been published that describe porosity well logs as fGn.[22,33,34,36,37,38] All of these authors used the R/S analysis, and several used many more of the tools outlined in the first chapter. We will not repeat all of this work here. Instead, we will simply discuss the tests and the findings of these papers as a whole.

The discovery that well logs often have a fGn character was made by Hewett.[22] He analyzed a porosity log from "a late Miocene-early Pleistocene sandstone formation originally deposited in a deep submarine fan." He reported an H value of 0.855.

Crane and Tubman[34] analyzed three horizontal wells and four vertical wells in a carbonate formation. They found all the logs to be fGn. H ranged between 0.88 and 0.89 for the vertical wells, while H ranged between 0.85 and 0.93 for the horizontal wells. Crane and Tubman also analyzed five horizontal wells and seven vertical wells in a sandstone. Again, they found fGn in all cases. H for the vertical wells averaged 0.83. H for the horizontal wells averaged 0.90.

Taggart[37] analyzed a well from the Bowen/Surat Basin, a well from the Hutton Sandstone in the Jackson Field (Eromanga Basin), and two wells from the Windalia Sandstone, Barrow Island (Carnarvon Basin). He found fGn with H values of 0.82, 0.82, and 0.80, respectively.

Aasum et al.[38] analyzed 18 wells from a single unit. They used both R/S and variogram analysis. They found fGn with H varying between 0.70 and 0.94 for logs of both porosity and the logarithm of permeability.

Hardy[33] analyzed 72 porosity logs (and 168 other logs) from 29 wells from a carbonate formation deposited on a shallow-water, mud-dominated carbonate platform. He found all of the porosity logs were fGn with H values shown in

Table 3.2. Hardy[36] also analyzed a single horizontal well from an aeolian sand. This is the analysis shown in Figure 3.1. He found fGn with H=0.93.

Table 3.2 Measured H values for logs

Log Tested	No. of Logs	AVE H	STD H
DEN	26	0.843	0.044
CNL	26	0.834	0.054
SON	22	0.817	0.054
ILD	26	0.844	0.049
ILM	22	0.817	0.062
MSFL	10	0.762	0.748
SFL	10	0.810	0.113
MSFL,SFL	20	0.786	0.099
MSFL,SFL,LLS,RXO,ML1,ML2	25	0.786	0.091
GR	29	0.919	0.053
CAL	26	1.001	0.122

GR: Gamma Ray
CAL: Caliper
DEN: Density
CNL: Compensated Neutron
SON: Sonic
ILD: Deep Induction
ILM: Medium Induction
MSFL: Micro SFL
SFL: Spherically Focused (Resistivity) Log
LLS: Shallow Laterolog (Resistivity)
RXO: Proximity Log
ML1: Microlog (1" x 1" Microinverse)
ML2: Microlog (2" Micronormal)

It is surprising the logs analyzed by so many authors in so many different formations are fGn with H values ranging from 0.7 to 0.95. We do not know the reason for this similarity.

It is also surprising to find the same H values for the porosity and the resistivity logs. In general, we would not expect this to be the case. In the formation Hardy[33] analyzed, however, there is no oil-water contact, so these results may just reflect the fact that the resistivity logs were "tracking" the porosity logs. It should also be noted that the resistivity logs were not Gaussian but log-normal. To fit a fGn model to the resistivity traces, the logarithm of the resistivity values was taken first.

A study was conducted with the logs used by Hardy[33] to see if the same H value calculated for a single formation could be found using an interval larger than the formation of interest. Figure 3.4 shows the results. Only five wells of the 29 wells used in the previous study were logged over the entire well depth (~5,000 feet). Figure 3.5 shows a corresponding histogram of the H values for the 29 porosity logs analyzed in the formation of interest. A comparison of these plots shows that the average H value for the entire interval logged is around 0.95, while that for the formation of interest is clustered around 0.85. Furthermore, the H values for the interval of interest are more clustered about the mean than are the H values for the entire interval logged. The reverse should have been the case, because the more values in a data set, the more accurate should have been the H value for each trace (see Figure 2.9). We conclude, therefore, that the entire logged interval should <u>not</u> be used to estimate H of the formation of interest.

For the logs studied by Hardy,[33] the lower limit of the fractal behavior was somewhere below the bed resolution of the shortest spaced resistivity tool (<1 foot). For all the other published studies, the lower limit was below the resolution of the tools described (<1 to 5 feet).

Not all logs are fGn. Two examples that fail the fGn tests are described in the following two paragraphs.

Figure 3.6 shows a set of logs in a shallow-platform and marginal-bank carbonate formation in Texas (San Andres Formation[39]). Figure 3.7 shows the analysis of one of these logs. These logs are nonstationary, since the bottom interval of each log has higher porosities than the top interval. fGn is stationary, so it is not surprising that these logs failed the tests for fGn. In particular, for the log shown in Figure 3.7, H=1.007 is suspect and the histogram is not Gaussian.

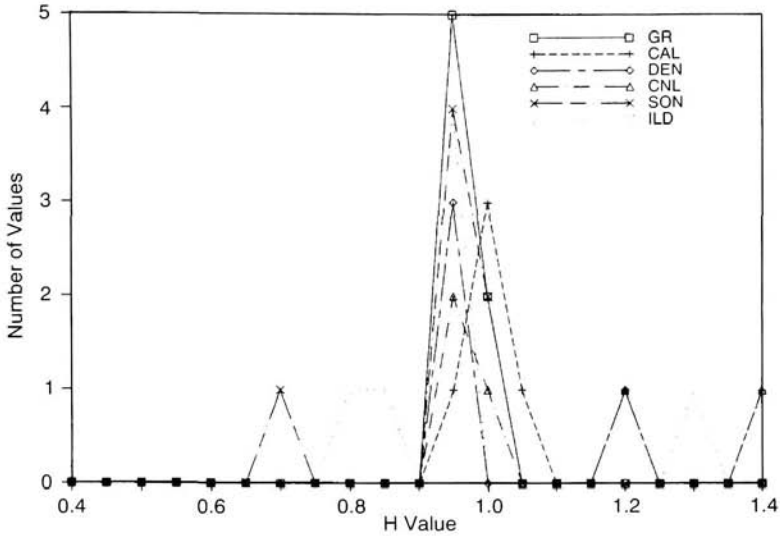


Figure 3.4. Histogram of H values from five wells with 4,000 to 5,000 feet of log from a carbonate formation deposited on a shallow water, mud-dominated carbonate platform.

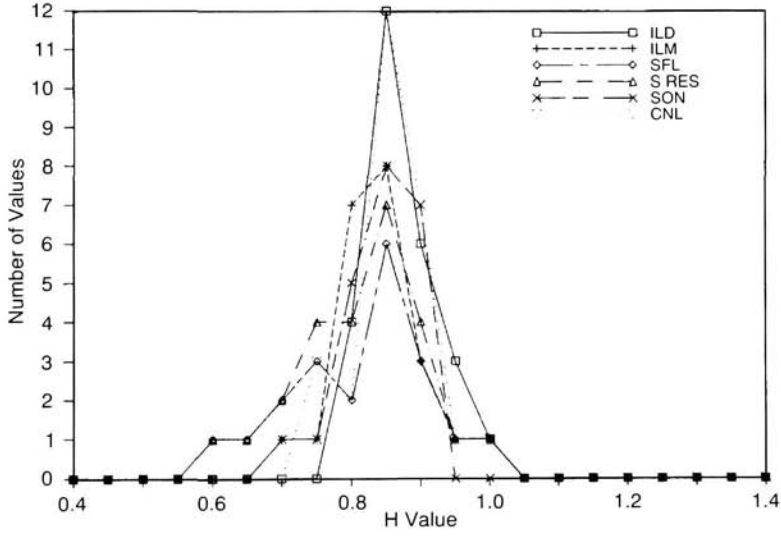


Figure 3.5. Histogram of H values from 29 wells in the formation of interest.

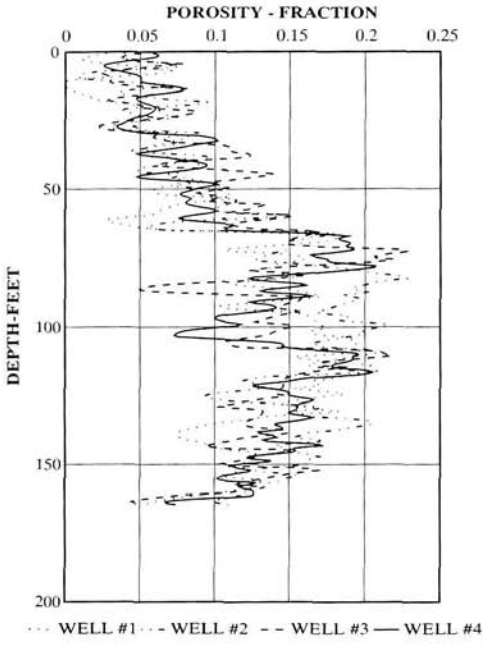


Figure 3.6. A set of logs in a carbonate formation in West Texas.

Figure 3.8 shows the analyses of a single porosity log in a meandering stream formation in the North Sea. It also failed the fGn tests. Stationarity is not an obvious problem (Figure 3.8a), but the histogram is bimodal (Figure 3.8b). Looking at the original log trace again, it can be seen that there are a number of large porosity values and a number of small porosity values but very few in between. One interpretation is that if the log is reading porosity from a channel sand, one log response is found. If it is reading porosity for outside of a channel, a different response is found. The log is thus sampling two different populations. This seems to be a good interpretation, but we have also analyzed logs in other meandering stream deposits and found they have a Gaussian histogram and fit the fGn model.

Other logs are "fairly close" to fGn. An example is shown in Figure 3.9. In this figure, the fGn model does not fit as well as Figure 3.1, but the fit is clearly better than Figure 3.7 or 3.8. Figure 3.9 is one of the logs analyzed by Hardy.[33]

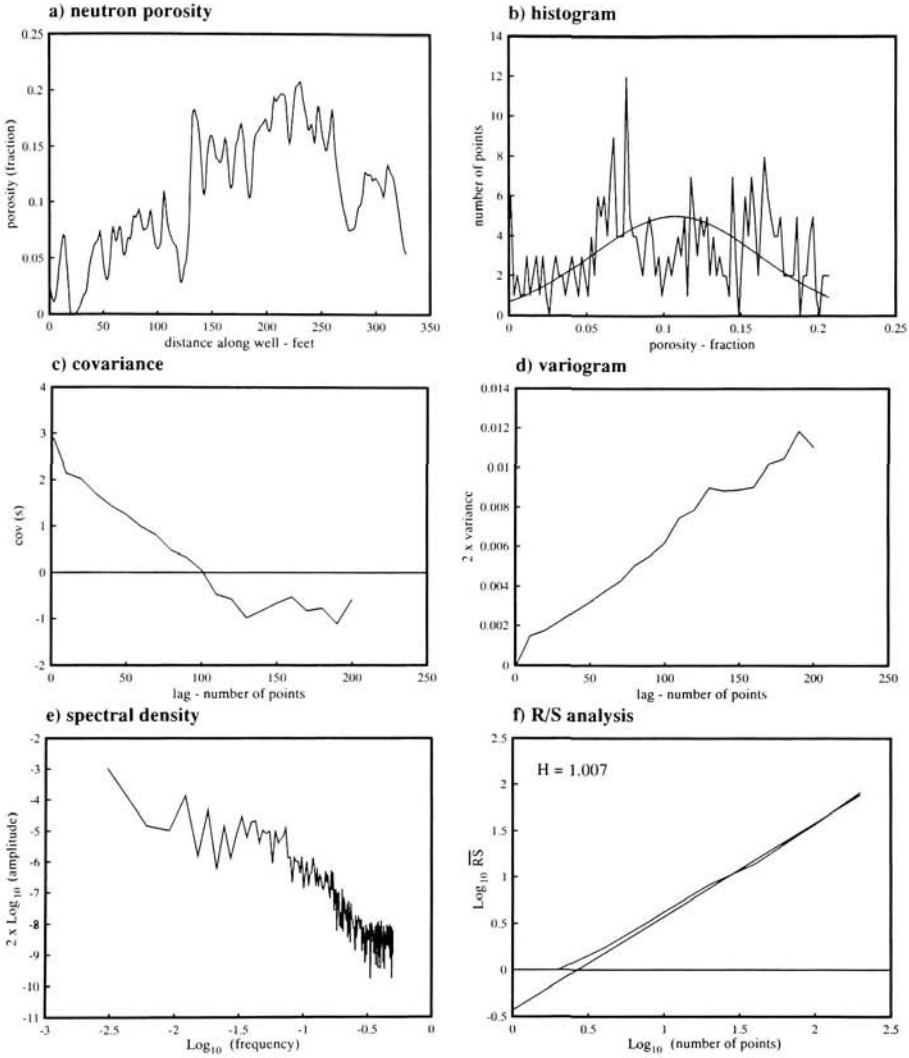


Figure 3.7. Statistical measures of one of the porosity logs in Figure 3.6. (a) log trace, (b) histogram, (c) covariance, (d) variogram, (e) spectral density, and (f) R/S analysis.

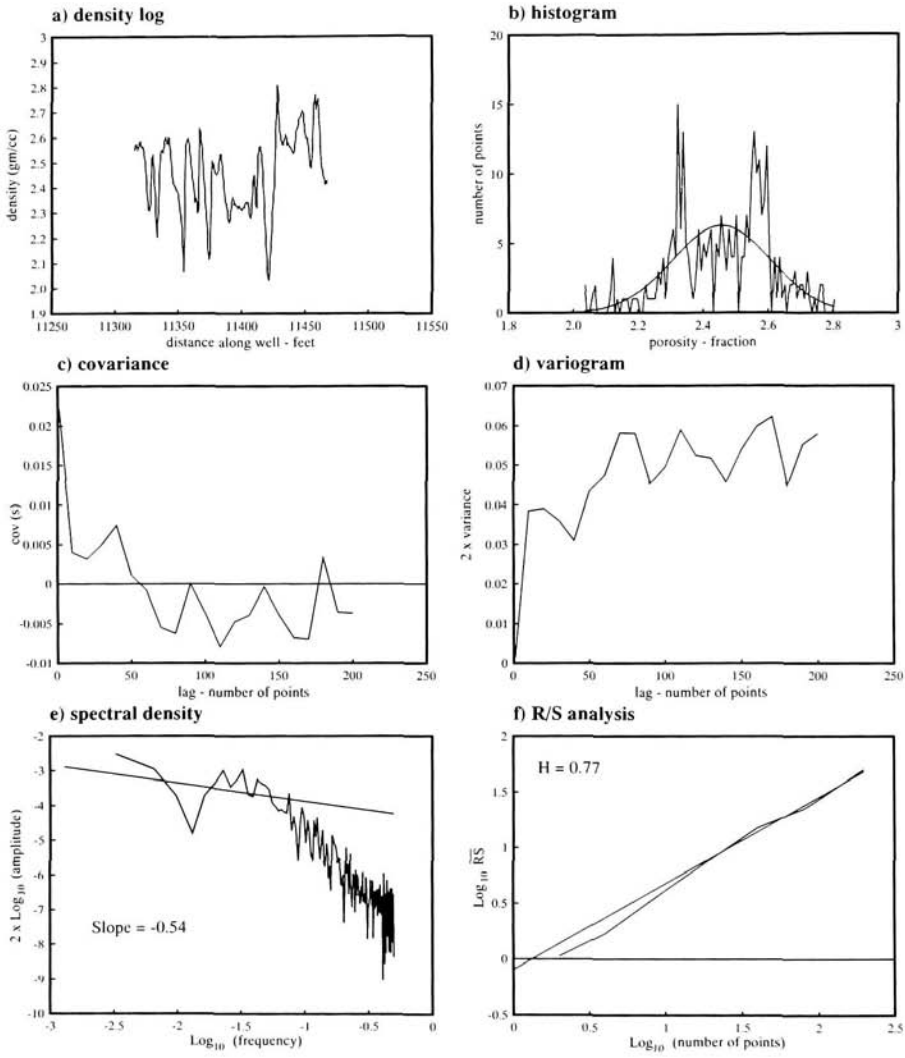


Figure 3.8. Statistical measures of a density log in a meandering stream formation. (a) log itself, (b) histogram, (c) covariance, (d) variogram, (e) spectral density, and (f) R/S analysis.

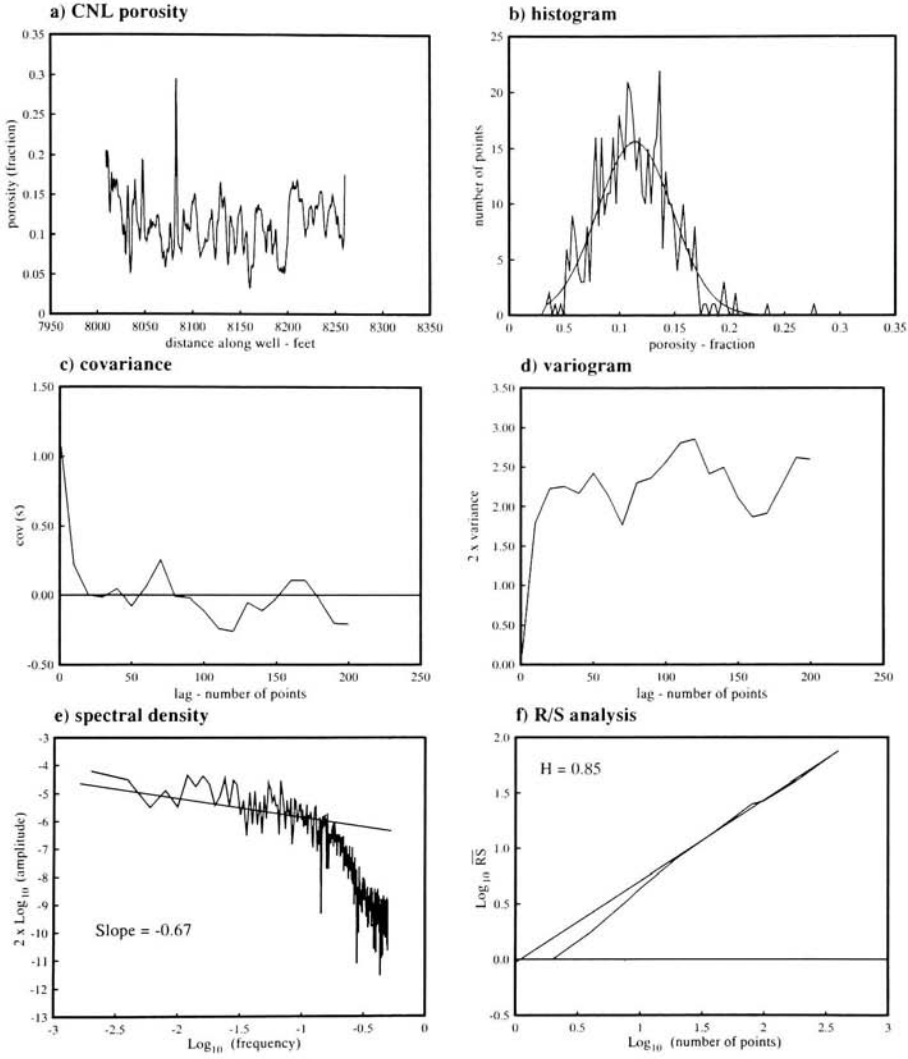


Figure 3.9. Statistical measures of a CNL log from a carbonate formation. (a) log trace, (b) histogram, (c) covariance, (d) variogram, (e) spectral density, and (f) R/S analysis.

Well logs provide the closest thing to in-situ measurement of reservoir property distributions. Well logs are limited, however, in that they provide only one dimensional measures along the wellbores. In addition, vertical wells may not be perpendicular to bedding planes, and horizontal wells may not be parallel to bedding planes. Two dimensional data sets would help this problem. Several are analyzed in the next sections.

3.3 Core and Outcrop Photos

Core and outcrop photos have been analyzed for fGn character.[13,40] These, like many well logs, often indicate fGn. As in the last section, we will briefly review the findings.

Stolum[40] analyzed the number of shales in a marginal marine sand/shale sequence of a long core section. This was done by first digitizing core photos. Next, the light intensities were analyzed from vertical traces along the core. He found fGn with H=0.87.

Hardy[13] analyzed core photos that were 2" x 2" (e.g., Figure 1.10). The bedding planes of the cores were aligned with the horizontal. The photos were digitized as black and white core photos. The light intensities were stored as pixel values between 0 and 256. Analyses of both horizontal and vertical lines were made. Surprisingly, both the horizontal and vertical traces were fGn (Table 3.3). This was the case despite clear horizontal bedding planes in the horizontal direction. The bedding planes were a result of different averages and standard deviations in the horizontal and vertical directions rather than a difference in the type of spatial distribution in the two directions.

As we mentioned before, Crane and Tubman[34] found fGn in both horizontal and vertical logs in the same formation. They also noted that the standard deviations of the vertical logs were greater than the horizontal logs. This is in agreement with Hardy's findings[11] on core photos.

A photo of an outcrop in the Canadian Rockies (Figure 3.10) was also analyzed by Hardy.[36] The results were similar to that found for the slabbed core photo. Table 3.3 lists the findings for both the core photo and the outcrop photo. Both were fGn with H's ranging from 0.7 to 0.97.

Table 3.3 1D statistical measures* of core and outcrop photos

Relation to Bedding Planes	No. of Traces	Ave Pixel Value	Standard Deviation of Pixels	H
Core: Parallel Direction	27	206±5	3.63±.85	.835±.078
Core: Perpendicular	27	206±1	5.94±.45	.762±.040
Core: 45 Degrees	27	206±1	5.61±.71	.819±.082
Outcrop: Parallel Direction	27	169+9	6.93±2.04	0.885±.060
Outcrop: Perpendicular	27	171+3	11.9±1.7	0.967±.020
Outcrop: 45 Degrees	27	172+6	8.83±2.06	0.923±.059

*For each entry, the average value of the 27 traces is given followed by ± the standard deviation of that value.

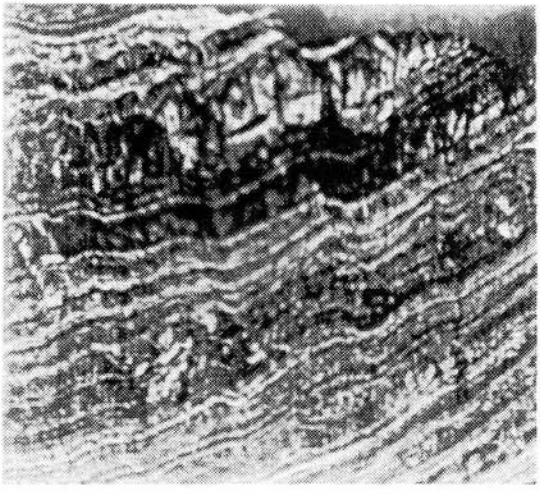


Figure 3.10. Canadian outcrop photo.

Core photos are limited in that they only provide an indirect measure of the reservoir properties needed for reservoir simulation. The relationship of light intensity to porosity or permeability is unknown. Also lighting conditions, type of light, and surface preparation of the core are variables that can affect light intensity. Finally, core photos are not at the reservoir scale.

Outcrop photos have some of the same limitations as core photos as well as others. Problems include lighting and the relationship between light reflectance and reservoir properties. In addition, weathering and even vegetation can distort outcrop photos.

Another type of 2D data is minipermeameter measurements on outcrops. These have their own set of advantages and disadvantages as we will see in the next section.

3.4 Outcrop Studies

Permeability measurements on outcrop surfaces provide additional data sets to test for fractal characteristics. We focus on measurements from an aeolian sandstone outcrop. The analysis of these data has not been previously published. After a discussion of the aeolian data set, we will also mention the results of a study by Goggins et al.[41] on another sandstone outcrop.

Measurements on the first sandstone outcrop were performed by groups from Imperial College and Heriot-Watt University under the direction of Jon Lewis. Measurements sampled a section of sand dune and interdune sediments over 1,280 ft in length. The dune unit has planar crossbeds that are schematically illustrated in Figure 3.11. This dune unit has a maximum preserved thickness of 51 feet and overlies an interdune unit with maximum thickness 10.5 feet. Measurements were made along lateral traces, vertical traces, and individual dune crossbeds. Each lateral trace maintains a constant distance from the dune/interdune boundary.

A fGn character was found for the logarithm of permeability along the lateral trace through the dune. An R/S analysis estimates H to be 0.85 in Figure 3.12. A line with the corresponding slope gives a reasonable match to the spectral density plot. The histogram plot has some departures from a Gaussian shape. A normal score transformation was applied. As can be seen in Figure 3.13, for this data set, the R/S and spectral density plots are insensitive to the transform, and an "exact" Gaussian shape is produced.

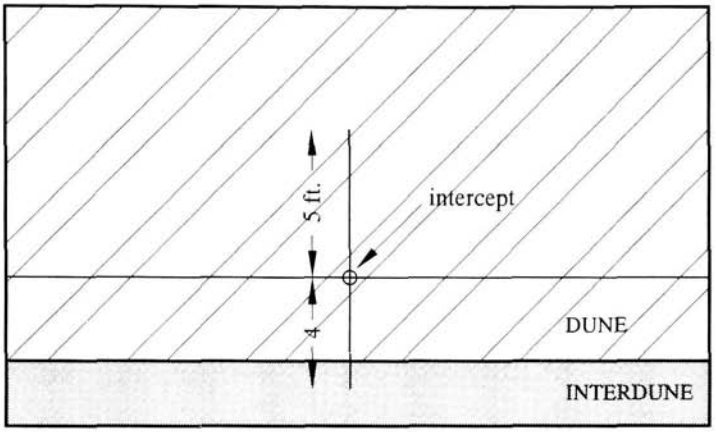


Figure 3.11. A schematic diagram of the bedding pattern.

A vertical trace through the dune also has a fGn character (Figure 3.14). The H value from the R/S analysis is very similar to the horizontal trace. The vertical trace has about 100 points, while the lateral trace has over 700 points. Because of the smaller number of points in the vertical direction, the interpretation in the vertical direction has more uncertainty.

Lateral traces in the interdune unit are not fGn. The spectral density plot in Figure 3.15 has a flat trend except at very low frequencies where a sharp rise occurs. Apparently, these low frequencies dominate the R/S analysis, which has a slope near unity. No vertical traces were made in the interdune unit, because it was so thin.

Outcrop studies were also made by Goggins et al.[41] in another sandstone outcrop. They found the R/S analysis yielded "a behavior atypical of fractal noises." Goggins et al. used the term, "nested fractal behavior." This result may be due to mixing several depositional types into a single analysis.

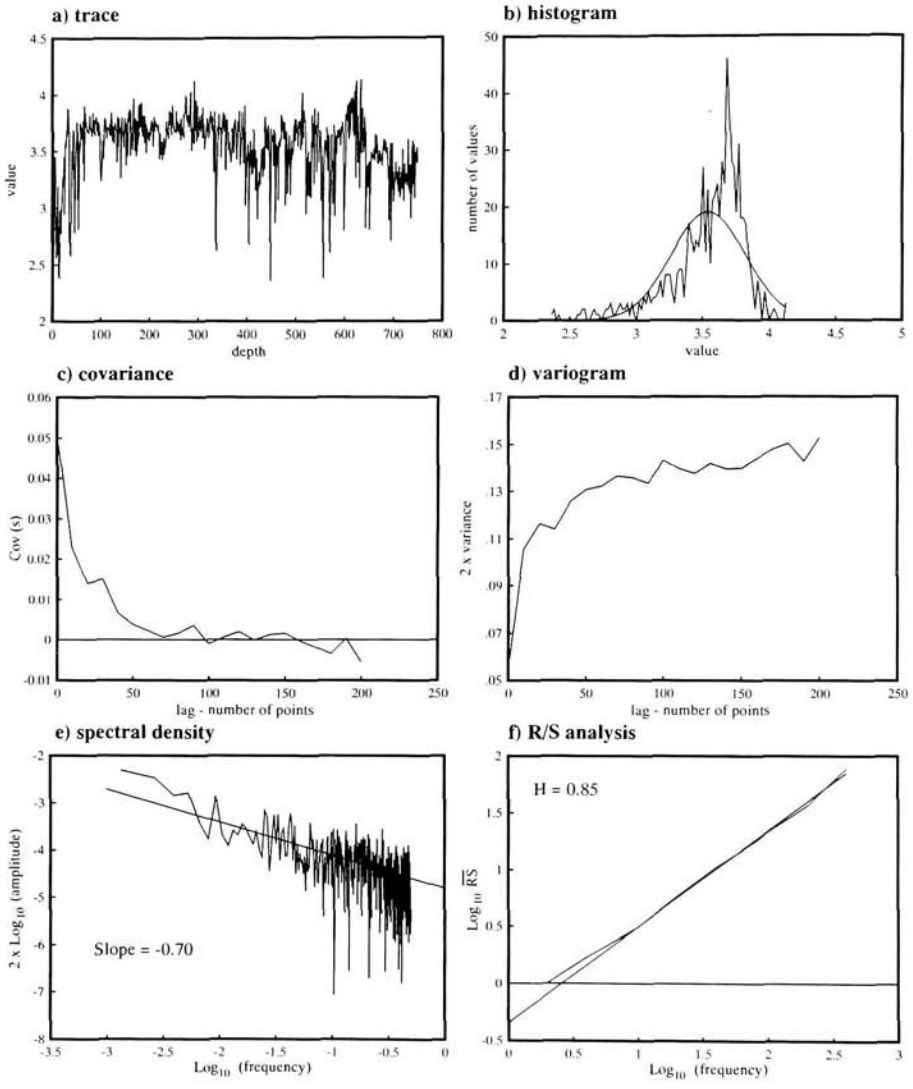


Figure 3.12. Plots for a lateral trace in the dune unit. (a) trace, (b) histogram, (c) covariance, (d) variogram, (e) spectral density, and (f) R/S analysis.

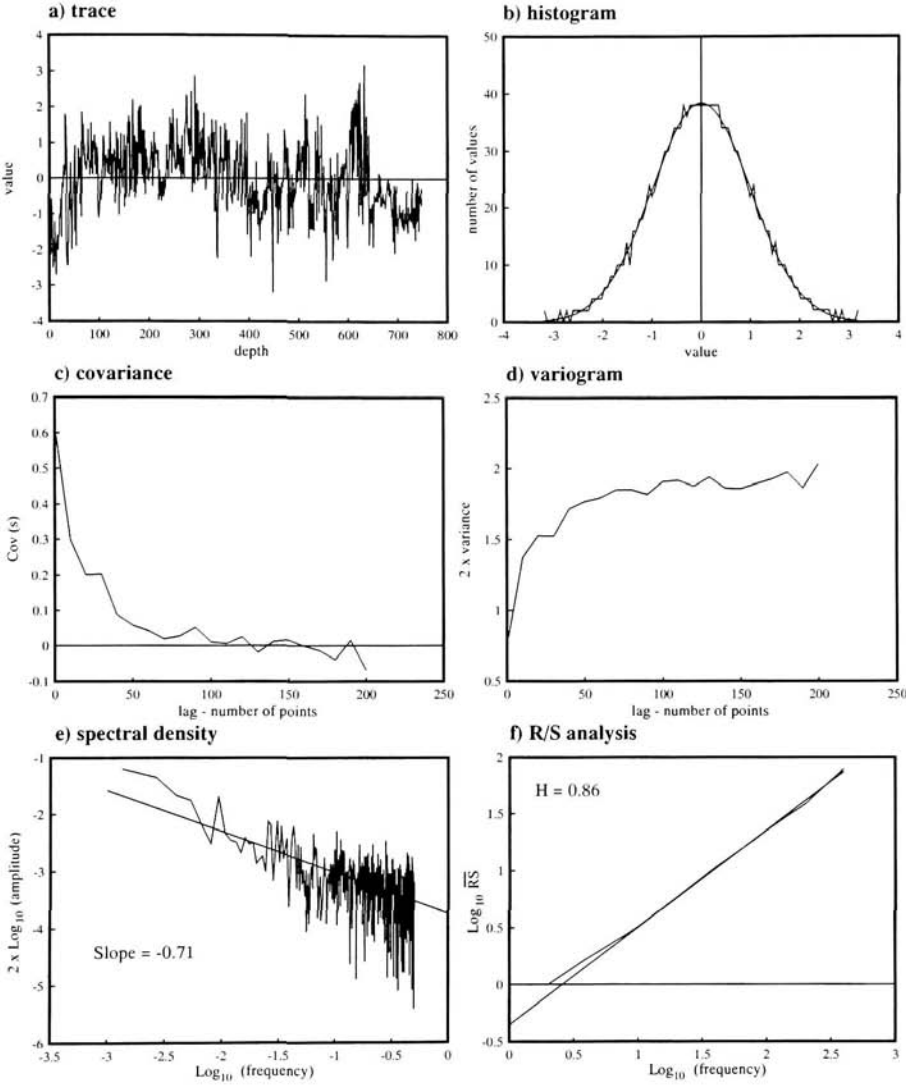


Figure 3.13. Plots for the lateral trace shown in Figure 3.12 after a normal score transform has been applied. (a) trace, (b) histogram, (c) covariance, (d) variogram, (e) spectral density, and (f) R/S analysis.

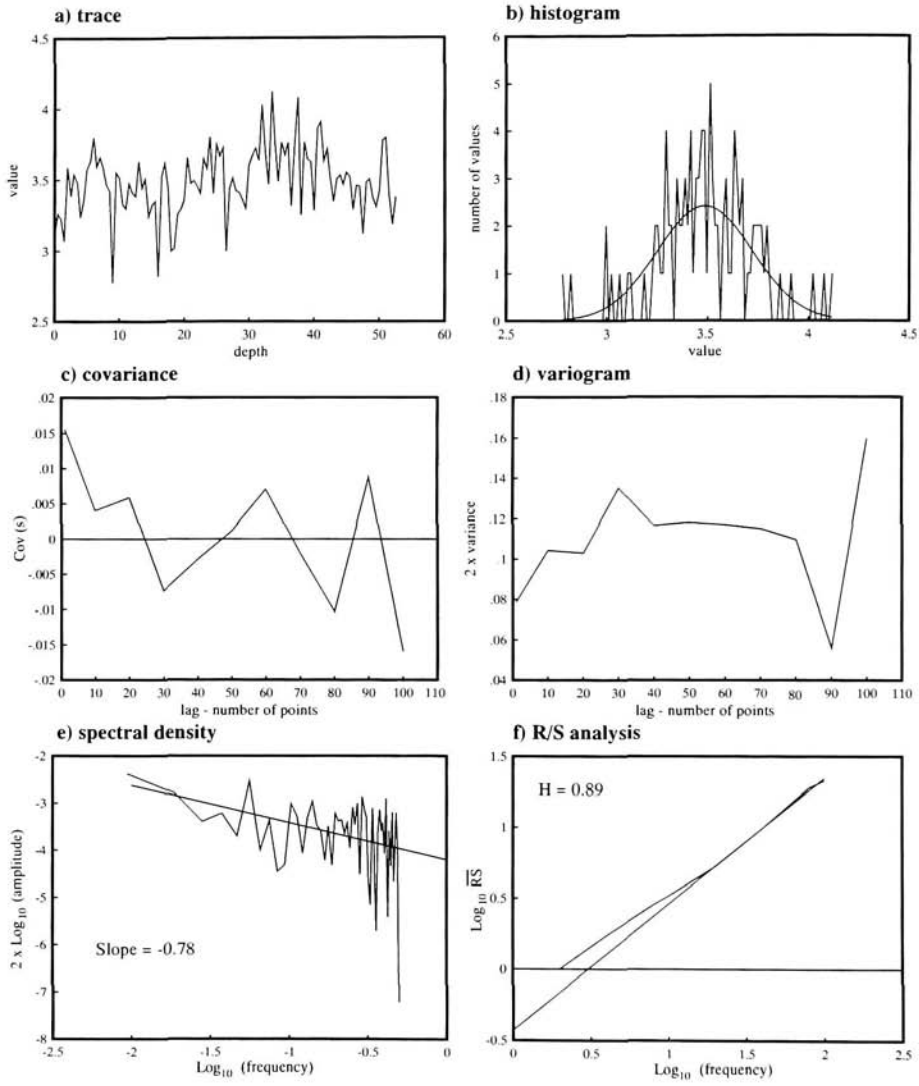


Figure 3.14. Plots for a vertical trace in the dune unit. (a) trace, (b) histogram, (c) covariance, (d) variogram, (e) spectral density, and (f) R/S analysis.

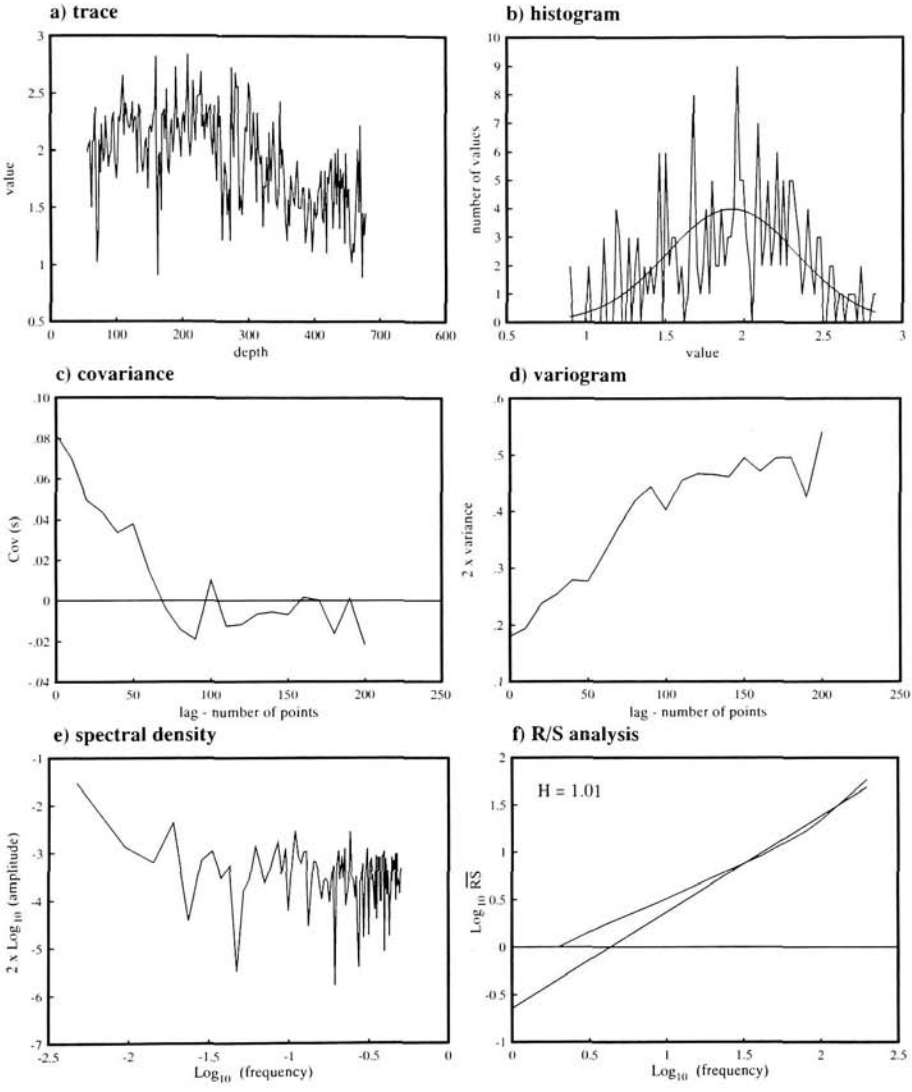


Figure 3.15. Plots for a lateral trace in the interdune unit. (a) trace, (b) histogram, (c) covariance, (d) variogram, (e) spectral density, and (f) R/S analysis.

Outcrop studies provide direct measurements of permeability on the outcrop face but no information about porosity. In minipermeameter studies, surface preparation can be critical to good results. Perhaps the biggest unknown is "how representative of the formation of interest is the outcrop itself?" Does diagenesis affect deeper buried sediment? Have measurements been taken deep enough to avoid weathering effects?

The analyses performed so far have been one dimensional analyses, even when the data sets themselves are two dimensional. There do exist two dimensional statistical measures that more fully capture two dimensional data sets. One of these is applied to core photos in the next section.

3.5 Two Dimensional Data Sets

Core photos can also be analyzed using two dimensional statistical measures. The extension of the covariance, variogram, and spectral density to 2D is straightforward.[17,26] Hardy[13] analyzed the same core photos described in Section 3.3 with 2D Fourier transforms. He found a spectral density model that captured much of the core photos.

That spectral density model is the following:

$$S(\omega_x,\omega_z) = A_{SD}\left[\frac{\delta(\omega_x)}{\omega_z^{\alpha}} + \frac{B_{SD}}{(\omega_x^2 + \omega_z^2)^{\beta}}\right] \qquad (3.2)$$

The details of the derivation of Equation 3.2 can be found in Appendix A3.1 and in Reference No. 13. Equation 3.2 was derived as a fit to the 2D spectral density of Figure 1.10. The spectral density is expressed in terms of the angular frequency on the x and z direction, ω_x and ω_z, respectively.

There are actually only 3 free parameters (α, β, and B_{SD}) in Equation 3.2 for generating core photos. Because of the power law form of Equation 3.2, α and β dictate the fractal behavior of the final distribution. The first term, which contains α, captures the layering seen in Figure 1.10. The second term, containing β, adds fractal noise. Because these are additive, the value of B_{SD} determines the relative amount of layering and noise. The other parameter, A_{SD}, determines the overall standard deviation of the entire sample. The parameter A_{SD} has no effect on the

appearance of the photos, because photos are rescaled when they are generated (see Appendix A3.1).

Figures 3.16 through 3.23 show the surprising variety of core photos that can be simulated using Equation 3.2. Figures 3.16, 3.18, 3.20, and 3.22 are core photos. Figures 3.17, 3.19, 3.21, and 3.23 are "corresponding" computer-generated photos. The "corresponding" photos were generated by varying parameters in Equation 3.2 and applying thresholds to the resulting image. After viewing a number of distributions generated using Equation 3.2, we chose several sets of slabbed core photos that looked "similar" to the computer-generated photos. This comparison is purely subjective. The resulting pairs indicate the flexibility of Equation 3.2. All slabbed cores cannot be simulated with Equation 3.2, but a wide range of core types do seem to be encompassed by Equation 3.2. More details of Equation 3.2 are given in Appendix A3.1.

Two parameter studies indicate the range of distributions that can be generated using Equation 3.2. The first shows the effect of the exponential parameter, β. The second shows the effect of the multiplicative parameter, B_{SD}. For consistency, a 3-point running average filter is used in all pictures (see Appendices A3.2 and A3.3 for a description of 3-point filters). A threshold was chosen for each picture to be 30 percent of the maximum value in the array. All values less than this value are set to black. All values greater than this are set to white.

The effect of β has been studied in Figures 3.24 through 3.30. The parameters are chosen so that the resulting pictures are isotropic, that is, no layering is included. This is done by setting $B_{SD} >> 1.0$. The first term, which adds layering, is then negligible. This leaves only one independent parameter, β. In Figure 3.24, $\beta=0$. This corresponds to white noise. There is no "pattern" to the dots. It looks like pepper sprinkled on a white sheet. β is set to 0.4 in Figure 3.25. At first, it just looks like there are less dots, but on closer examination of the photo, it becomes apparent that the dots are clustered. Some areas have more dots; others have less. This becomes more apparent as β is increased. β is 0.7 in Figure 3.26. β is 0.85 in Figure 3.27. As β is increased further, beyond fGn into fBm, shapes resembling islands and then continents begin to appear (Figure 3.28, β = 0.95, Figure 3.29, $\beta=1.85$). The parameters used in Figure 3.29 are very nearly those used by Voss[11] to produce mountain ranges and cloud formation (Plates 1 and 2). Choosing $\beta=-1$ results in Figure 3.30.

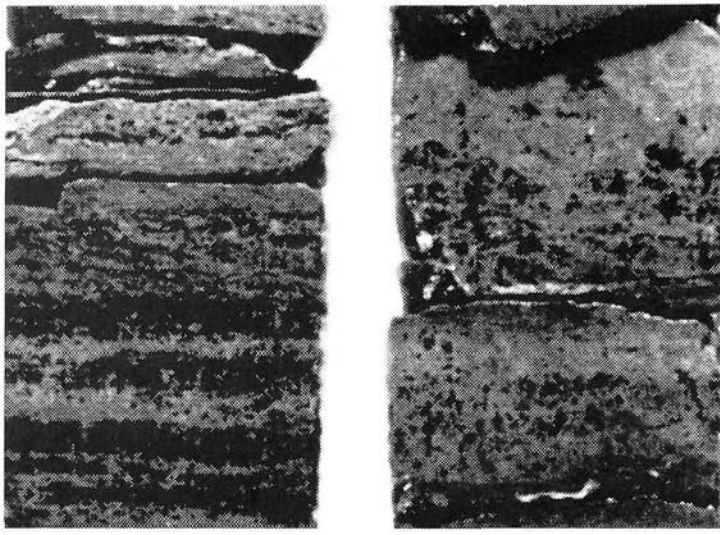


Figure 3.16. Slabbed core photo of a San Andres dolomite with algal laminations and anhydrite cement.[13]

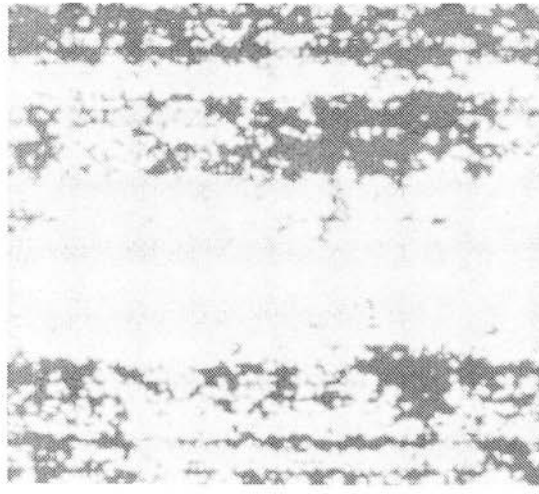


Figure 3.17. Computer-generated[13] core photo: $\alpha=1.9$, $\beta=.95$, and $B_{SD}=1/9$ in Equation 3.2 with a 2D, 5-point filter and threshold of -15.

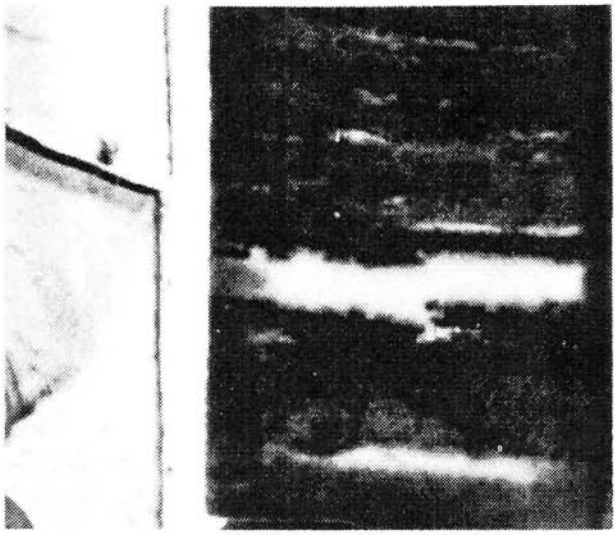


Figure 3.18. Slabbed core photo of a thinly bedded Tensleep Sandstone with anhydrite cement.[13]

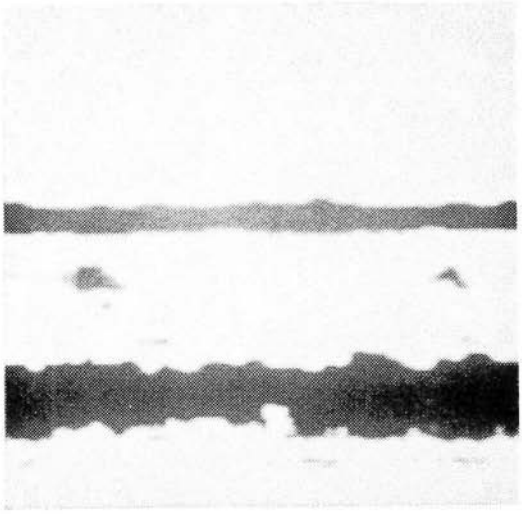


Figure 3.19. Computer-generated[13] core photo: $\alpha=0.9$, $\beta=0.85$, and $B_{SD}=1/4$ in Equation 3.2 with a 2D, 3-point filter and a threshold of -10.

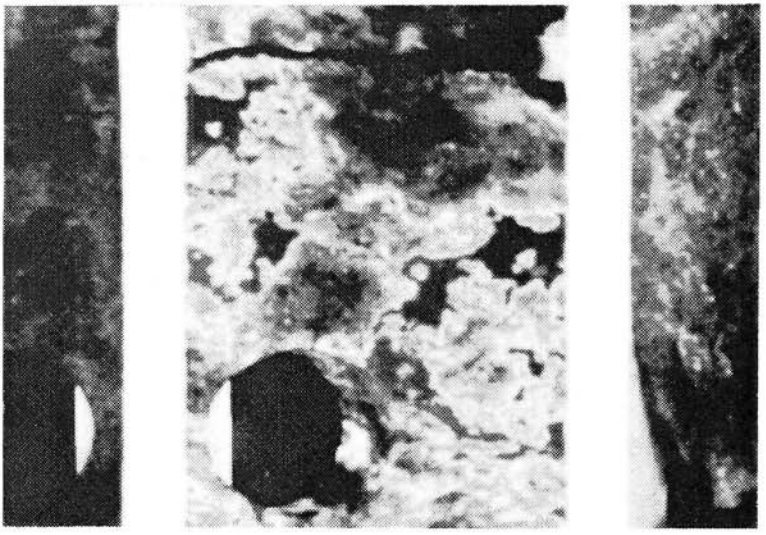


Figure 3.20. Slabbed core photo using UV light of an unevenly oil-stained sandstone.[13]

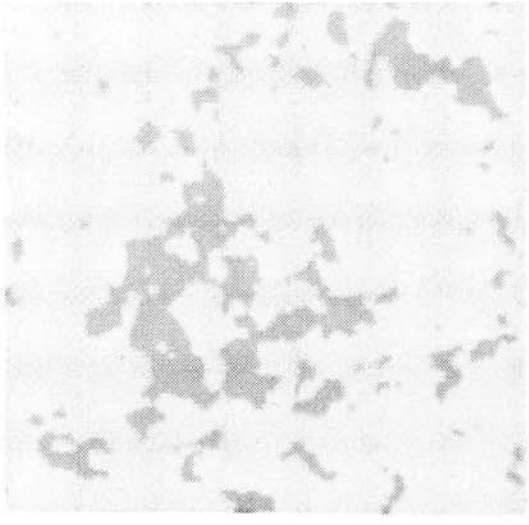


Figure 3.21. Computer-generated[13] core photo: $B_{SD} >> 1$ and $\beta = 0.95$ in Equation 3.2 with a 2D, five-point filter and a threshold of -5.

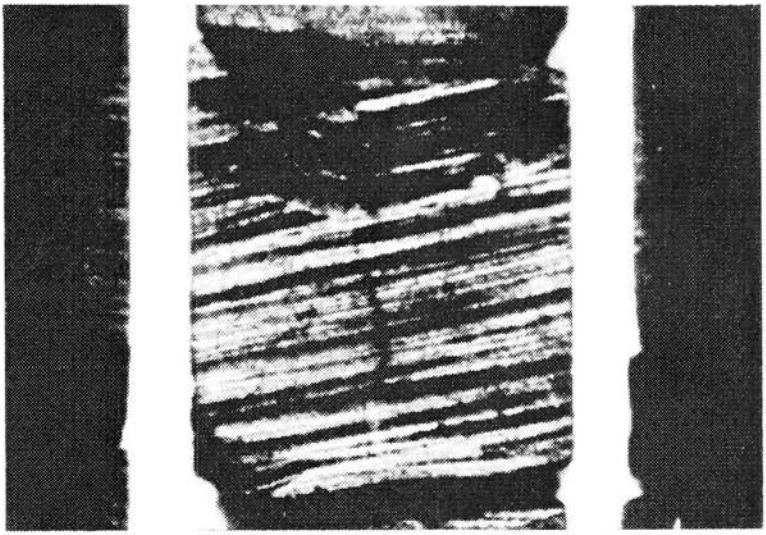


Figure 3.22. Slabbed core photo using UV light of an oil-stained, cross-bedded sandstone.[13]

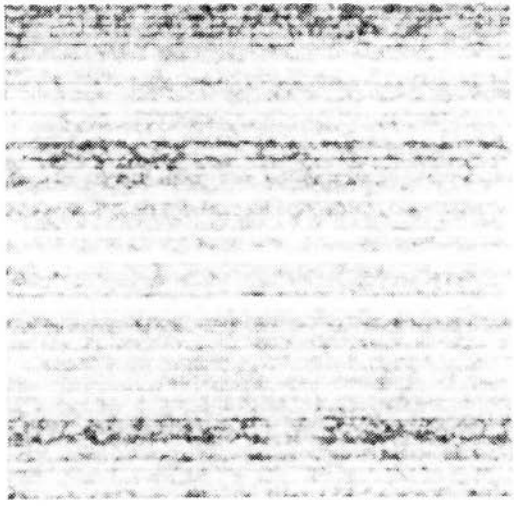


Figure 3.23. Computer-generated[13] core photo: $\alpha=0.8$, $\beta=0.4$, and $B_{SD}=1/9$ in Equation 3.2 with no filter or threshold.

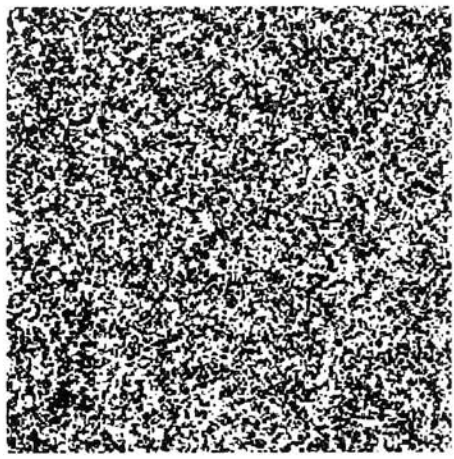


Figure 3.24. Parameter study of Equation 3.2. $B_{SD}>>1$, $\alpha=0.0$, and $\beta=0.0$.

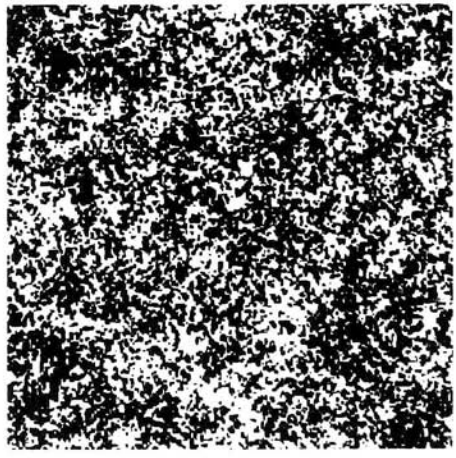


Figure 3.25. Parameter study of Equation 3.2. $B_{SD}>>1$, $\alpha=0.0$, and $\beta=0.4$.

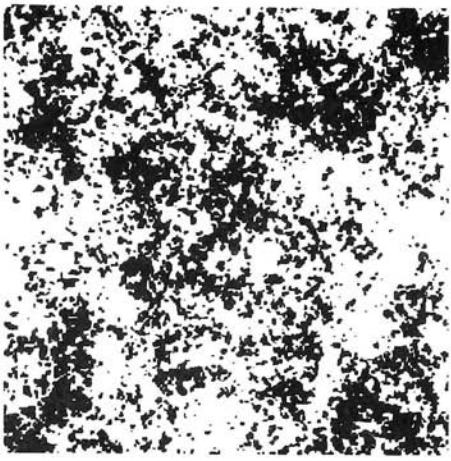


Figure 3.26. Parameter study of Equation 3.2. $B_{SD}>>1$, $\alpha=0.0$, and $\beta=0.7$.

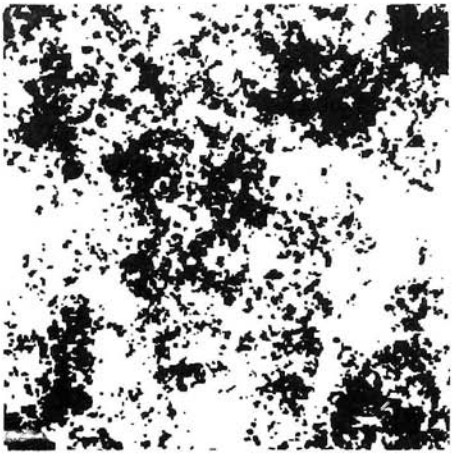


Figure 3.27. Parameter study of Equation 3.2. $B_{SD}>>1$, $\alpha=0.0$, and $\beta=0.85$.

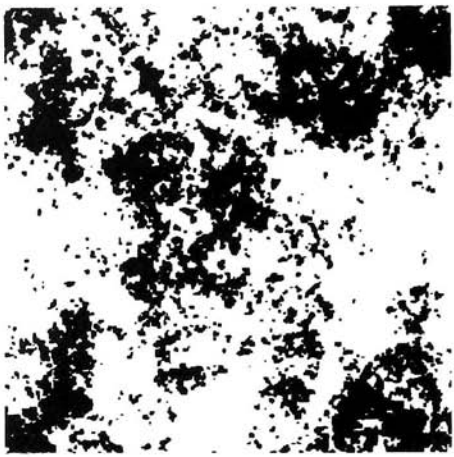


Figure 3.28. Parameter study of Equation 3.2. $B_{SD}>>1$, $\alpha=0.0$, and $\beta=0.95$.

Figure 3.29. Parameter study of Equation 3.2. $B_{SD}>>1$, $\alpha=0.0$, and $\beta=1.85$.

The effect of B_{SD} was studied in Figures 3.31 through 3.33. This was done by setting $\alpha=0.6$, $\beta=0.8$, $A_{SD}=1.0$, and varying B_{SD}. Figures 3.31 through 3.33 show that as B_{SD} decreases, the layering within the photos becomes more dominant. The B_{SD} values are .5, .1, and .01 in Figures 3.31, 3.32, and 3.33, respectively.

All of the layering shown in the artificial photos is horizontal. Often in outcrops, the layering is "bumpy" or tilted (e.g., Figure 3.10). Tilted layering (e.g., crossbedding) can be simulated simply by rotating the distribution resulting from Equation 3.2 so that the layering lies at any chosen angle. "Bumpy" layering requires more work. Several possibilities present themselves. Two of these are described here. Others are described in the section on unresolved issues.

The first possible approach is through modifying the spectral density description. A study of the spectral density of Figure 3.10 indicates an overall similarity to Equation 3.2, but there are additional features (spots) in the spectral density not found in plots of Equation 3.2. Perhaps by extending Equation 3.2 to include these features, the desire of "bumpiness" may be achieved.

Another approach is by pixel compression. Figure 3.34 shows the result of one such deformation. This figure is produced from Figure 1.9 by compressing each pixel in the z direction proportional to its light intensity (i.e., pixel value). The pixels are then restacked one on top of another. The new stack is then sampled at each old pixel location and replotted. Linear interpolation is used where necessary. The picture looks similar to mountain ranges even though it is much too simple a model to capture the effects present in mountain building.

Equation 3.2 consists of two parts. The first part describes the layering, and the second describes "splotches" or background noise. This description can be used to produce photos another way. Interpolation can be used between the edges of the photo to produce the layering (Figure 3.35). Noise generated from Equation 3.2 with $B_{SD}>>1$ can be added until a photo very similar to the original core photo is produced. (Figure 1.9 was produced from Figure 1.10 in this manner.)

There are distributions in core photos that do not look like any distributions produced by this model (e.g., Figures 3.36 and the top portion of Figure 3.37). Other distributions in core photos appear to have different parameters for Equation 3.2 at different depths (e.g., Figure 3.38).

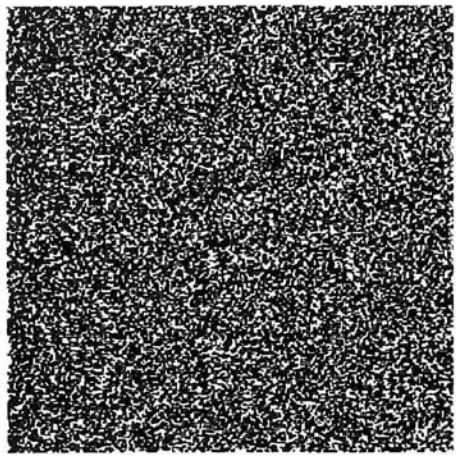


Figure 3.30. Parameter study of Equation 3.2. $B_{SD}>>1$, $\alpha=0.0$, and $\beta=-1.0$.

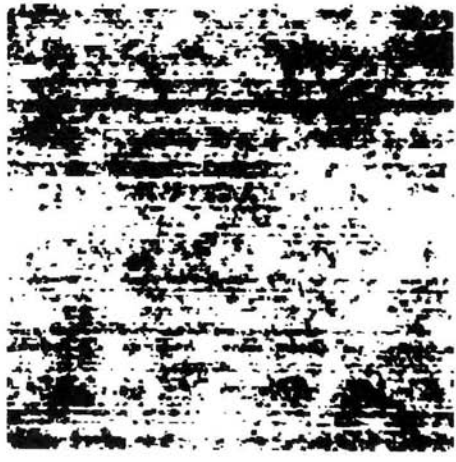


Figure 3.31. Parameter study of Equation 3.2. $B_{SD}=1/2$, $\alpha=0.6$, and $\beta=0.8$.

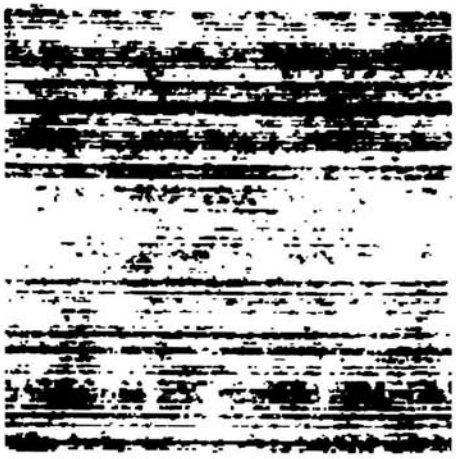


Figure 3.32. Parameter study of Equation 3.2. B_{SD}=0.1, α=0.6, and β=0.8.

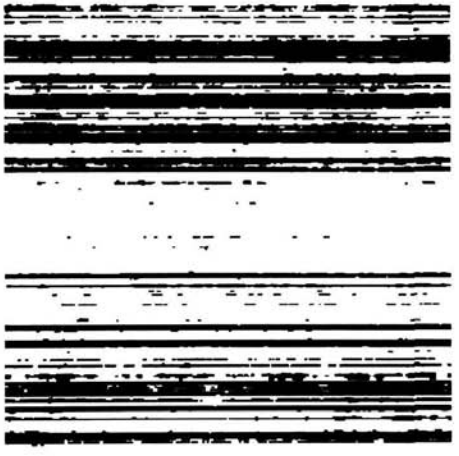


Figure 3.33. Parameter study of Equation 3.2. B_{SD}=0.1, α=0.6, and β=0.8.

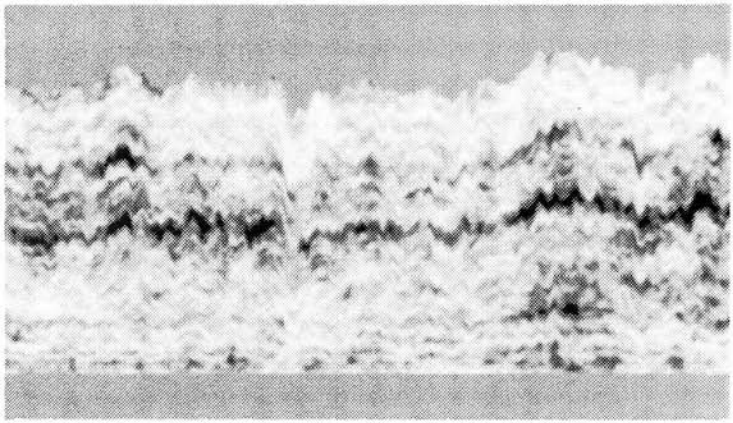


Figure 3.34. Computer-generated "mountains" created by pixel compression of Figure 1.9.

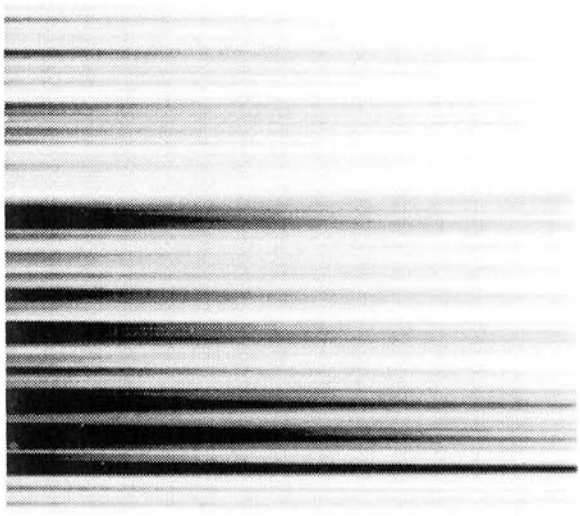


Figure 3.35. Linear interpolation between edges of Figure 1.9.

Figure 3.36. Slabbed core photo showing spacial distribution that cannot be captured by Equation 3.2.

Figure 3.37. Another slabbed core photo showing spacial distribution that cannot be captured by Equation 3.2.

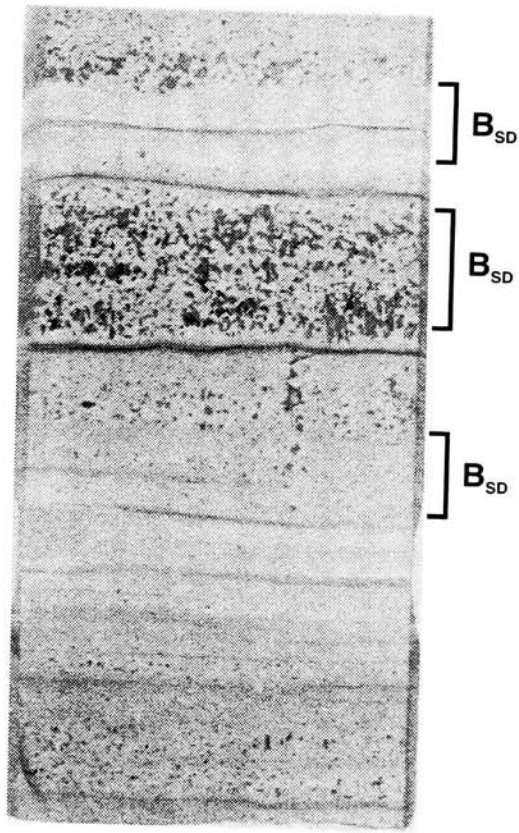


Figure 3.38. Slabbed core photo showing spacial distribution that looks like it is similar to several distributions from Equation 3.2 stacked on top of each other.

In general, the parameters (α, β, B_{SD}) in Equation 3.2 could be made "functions of everything"...e.g., depth, size, position, etc. A much more general model would be produced. But, where would we get the data to define these functions? Usually all we have are vertical wells. Sometimes we may have some horizontal wells, but even then, we never know precisely where they are relative to the bedding planes. Sometimes analogous outcrops are available. Then, we must decide what outcrop information carries over into the subsurface. Without an extensive data set to draw on, Equation 3.2 is probably a good first choice, because the data sets in this chapter suggest a rather wide applicability of Equation 3.2. We will pursue the application of Equation 3.2 to porosity distributions in the next chapter.

Unresolved Issues

1. More work needs to be done on the effect on bivariant measures of using nonlinear transforms on a data set. For example, consider a data set that is not fGn, but after a normal score transform, it is. Now, suppose an artificial fGn distribution is built in the transform space. Then an inverse transform is used to produce a final distribution. Will the final artificial distribution have the same statistical distribution as the original data set? When does this work and when doesn't it?

2. Why do we get fGn so often in reservoir property distributions? When should we expect fGn and when shouldn't we? Is there a mechanism in deposition that can be shown to generate fGn and others that do not?

3. More statistical analyses of outcrops and other data are needed. More outcrops, logs, and cores should be analyzed. The possible data sets include porosity distributions from outcrop samples, minipermeameter from cores on different scales, and CAT and NMR data on cores.

4. What is the best way to add relief to the fractal distributions? Is contouring sufficient? Should we use mid-point displacement?[11] Should we work in spectral space by adding "spots" as described in the text? Can a process be found that mimics mountain building?

APPENDIX A3

A3.1 A More Complete Description of Equation 3.2

Hardy[13] found that the spectral density Equation 3.2 fits the spectral density of core photos. When analyzing the core photos, Hardy only analyzed the spectral density of the core photos. This is common with statistical studies using Fourier transforms. No attempt was made to fit the phase array. The phase array is assumed to be described with random values between 0 and 2π.

2D Fourier transforms and their inverses were taken using IMSL software.[23] The IMSL program uses a fast Fourier transform (FFT) that orders the output arrays in a peculiar way. The arrays are ordered as shown in Figure A3.1. In Figure A3.1, $\omega_x = 2\pi f_x$ and $\omega_y = 2\pi f_y$. Pictures of spectral density arrays will use this peculiar ordering. In addition, the amplitude and phase arrays have particular symmetries, because the original image is real.[42]

To display the spectral densities, the spectral density values were first divided into logarithmically scaled bins. Thresholds were set to $2x10^2$, $1x10^3$, $2x10^3$, and $1x10^4$. The light intensity on the graphics display is assigned according to which bin each amplitude value falls into. For example, if the amplitude is $2x10^{-4}$, the value assigned falls between the second and third threshold and is assigned the value of 3. These bin values are then rescaled for display as a photo. In this manner, a rough approximation of the logarithm of the amplitude is displayed.

Figure A3.2 is a photo of the results of this logarithmic transform of the spectral density of Figure 1.10. A referral to the ordering of the spectral density array (Figure A3.1) will allow the reader to notice a very definite circular pattern to the spectral density. These circular rings suggest a radial functional form.

Another presentation of the logarithm transform is useful in trying to determine the functional form of the spectral density. The camera is rotated 45 degrees, and another photo is taken in the same location as Figure 1.10. Figure A3.3 is a plot of the logarithm of the spectral density versus frequency for this new picture. In addition to the circular rings, note the "finger" of high amplitude data "sticking out" along a 45 degree angle in Figure A3.3. This "finger" is also in Figure A3.2 along the ω_x axis, but it is more difficult to see because it lies on the edge of the photo in that presentation.

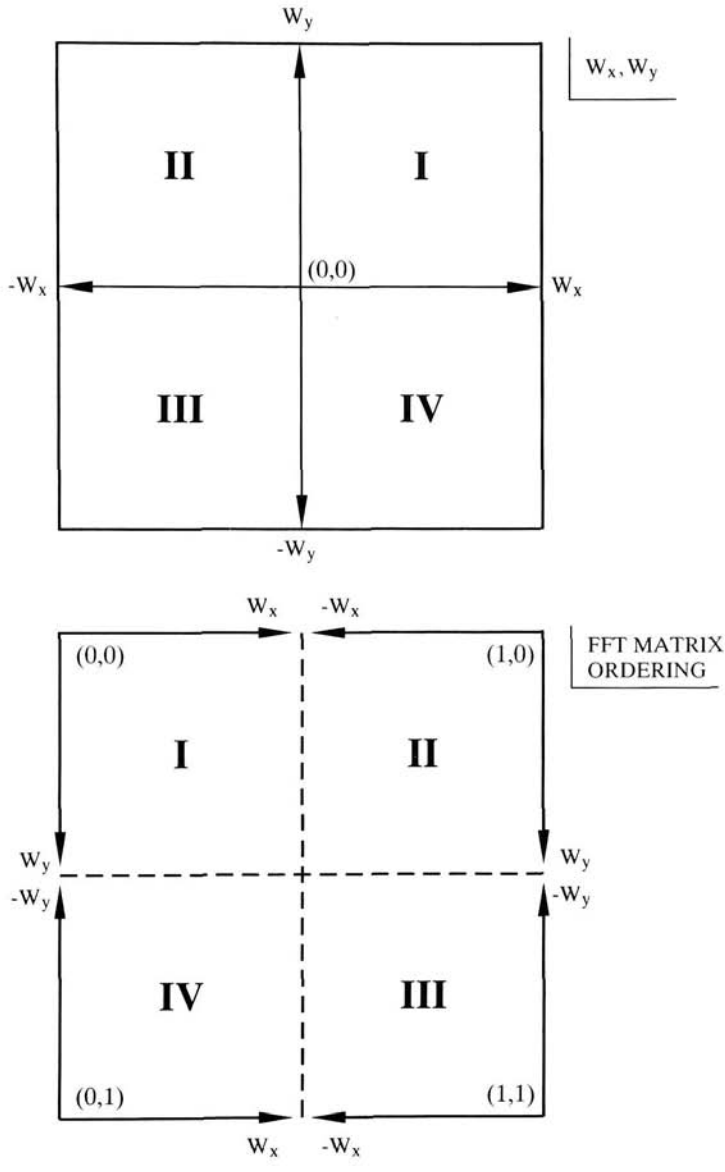


Figure A3.1. For a given input array (top figure), the fast Fourier transform produces an output array (bottom figure) with a different ordering.

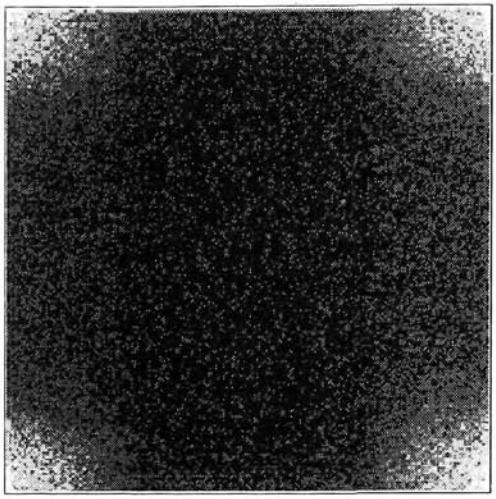


Figure A3.2. Spectral density array of Figure 1.10 displayed using logarithmic bins.[13]

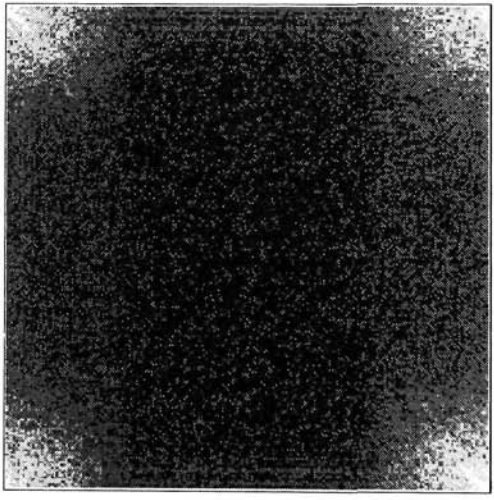


Figure A3.3. Spectral density array of rock shown in Figure 1.10 but rotated 45 degrees.[13]

Although discontinuous fits to data are not appealing, it is apparent from these figures that one line of amplitude data is discontinuous from the rest of the amplitude data. It is this line of data that produces the layering structure. Notice that no discontinuous line exists on the lower left corner of Figure A3.3. This is because there is no layering perpendicular to the bedding planes. The amplitude data are nearly spherically symmetric except for the discontinuous line defining the bedding planes.

Putting together the information presented in these figures, Hardy[13] postulated Equation 3.2. The equation produces photos similar to the slabbed core photo shown in Figure 1.10.

Equation 3.2 is undefined for $\omega_x = \omega_z = 0$. The value of the spectral density at $\omega_x = \omega_z = 0$ defines a constant value that is added to the distribution. We choose this value to be zero by setting

$$S(0,0) = 0.0 \qquad \textbf{(A3.1)}$$

This choice for S(0,0) results in a final distribution with a mean of zero (Property 4 in Appendix A2.5). A distribution of any desired mean and standard deviation can be produced from this distribution by a linear scaling (Appendix A2.6).

The arguments, ω_x and ω_z, of the spectral density equation, Equation 3.2, are defined in terms of the frequencies in Fourier space in the x, f_x, and z, f_z directions as follows:

$$\omega_x = 2\pi f_x \qquad \textbf{(A3.2)}$$

$$\omega_z = 2\pi f_z \qquad \textbf{(A3.3)}$$

The delta function in Equation 3.2 is defined so that for any function, g(x,z),

$$\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} g(x,z)\delta(x)\,dx\,dz = \int_{-\infty}^{+\infty} g(0,z)\,dz \qquad \textbf{(A3.4)}$$

For finite distributions of size N_x by N_z with spacings Δx and Δz, Equation A3.4 becomes

$$\sum_{j=1}^{N_x}\sum_{\ell=1}^{N_z} g(x_j,z_\ell)\delta(x_j)\,\Delta x\,\Delta z = \sum_{\ell=1}^{N_z} g(0,z_\ell)\Delta z \qquad \text{(A3.5)}$$

so that

$$\delta(x) = \begin{bmatrix} \dfrac{1}{\Delta x} & \text{if } x_j = 0 \\ 0 & \text{if } x_j \neq 0 \end{bmatrix} \qquad \text{(A3.6)}$$

Pure fGn in both the x and z directions can be produced by Equation 3.2 if

$$\alpha = 2H - 1 \qquad \text{(A3.7)}$$

and

$$\beta = H \qquad \text{(A3.8)}$$

where

$$0 < H < 1 \qquad \text{(A3.9)}$$

Rescaling these computer-generated distributions is necessary in order to produce gray scale values. First two-dimensional arrays are generated from Equation 3.2 (see Appendix A3.5). Next these array values, $g_{j,\ell}$, are mapped into grey scale values, $h_{j,\ell}$. The grey scale map requires pixel values between 0 and 256. This is done with the following linear transform:

$$h_{j,\ell} = 256\left(\frac{g_{j,\ell} - g_{min}}{g_{max} - g_{min}}\right) \qquad \textbf{(A3.10)}$$

where $g_{j,\ell}$ is the array value, g_{min} is the minimum value in the g array, g_{max} is the maximum value in the g array, and $h_{j,\ell}$ is the pixel value used for display.

A3.2 Smoothing Filters

An m-point smoothing filter is defined in one dimension in the following way. Let $\{f_1,f_2,f_3,...,f_n\}$ be a set of original values and n be an odd integer. An m-point smoothing function produces a set of values $\{g_1,g_2,g_3,...,g_n\}$ such that

$$g_j = \sum_{p=j-(m-1)/2}^{j+(m-1)/2} f_p/m \qquad \textbf{(A3.11)}$$

The question of what to do with the first and last (m-1)/2 points is addressed shortly.

An m-point smoothing filter is defined in two dimensions in the following way. Let

$$f_{j,\ell} \quad \text{for} \quad j = 1,N \quad \text{and} \quad \ell = 1,N \qquad \textbf{(A3.12)}$$

be an original set of N^2 values. An m-point filter for this 2D data set is defined as follows:

$$g_{j,\ell} = \sum_{p=j-(m-1)/2}^{j+(m-1)/2} \sum_{q=\ell-(m-1)/2}^{\ell+(m-1)/2} f_{p,q}/m^2 \qquad \textbf{(A3.13)}$$

again with some question as to what to do with the first and last (m-1)/2 values of j and ℓ. In the examples in this book, periodic boundary conditions are used so that if p (or q) becomes zero or less in Equation A3.11 or A3.13, then p (or q) is replaced by N+p (or N+q). If p (or q) becomes greater than N, then it is replaced by p-N (or q-N).

A3.3 Spectral Density Roll-Off and Filter Size

We will derive the equation needed to relate the roll-off found in the spectral density plots (e.g., Figure 3.1e) to the size of an averaging filter. We will use a one dimensional continuous distribution to explain the analogous effects for a discrete distribution. We will show that the resulting equation applies to 2D discrete distributions by an example.

In one dimension, a 3-point filter can be stated mathematically as a running average of the original distribution, ξ_j. The resulting values, g_j, are from Equation A3.11

$$g_j = (\xi_{j-1} + \xi_j + \xi_{j+1})/3 \tag{A3.14}$$

This can be written in summation notation as

$$g_j = \sum_{p=-\infty}^{p=+\infty} h_{j-p}\xi_p \tag{A3.15}$$

where

$$h_{j-p} = \begin{bmatrix} 0 & \text{if } |j-p| > 1 \\ 1/3 & \text{if } |j-p| \le 1 \end{bmatrix} \tag{A3.16}$$

In integral form this is

$$g(x) = \int_{-\infty}^{+\infty} h(x-\chi)\xi(\chi)\, d\chi \tag{A3.17}$$

where

$$h(x-\chi) = \begin{bmatrix} 0 & \text{if } |x-\chi| > x_0 \\ a & \text{if } |x-\chi| \le x_0 \end{bmatrix} \tag{A3.18}$$

here

$$a = 1/3 \quad \text{and} \quad x_0 = 1 \tag{A3.19}$$

for the 3-point filter described above.

The Fourier transform of the original sequence, $\Xi(f)$, is related to the Fourier transform of the filtered sequence, $G(f)$, by the convolution theorem.[26,43] The convolution theorem states that the Fourier transform of convolved functions, h(x) and $\xi(x)$, is the product of the Fourier transform of each of the functions, $H(f)$ and $\Xi(f)$. That is,

$$G(f) = H(f)\Xi(f) \tag{A3.20}$$

where g(x) is given by Equation A3.17 and

$$G(f) = \int_{-\infty}^{\infty} g(x)e^{-i2\pi fx}dx \tag{A3.21}$$

$$\Xi(f) = \int_{-\infty}^{\infty} \xi(x)e^{-i2\pi fx}dx \tag{A3.22}$$

and

$$H(f) = \int_{-\infty}^{\infty} h(x)e^{-i2\pi fx}dx \tag{A3.23}$$

In this book, the spectral density, $|G(f)|^2$, of the smoothed sequence is evaluated. We have

$$|G(f)|^2 = |H(f)\Xi(f)|^2 = |H(f)|^2|\Xi(f)|^2 \tag{A3.24}$$

In particular, we would like to show how the plot of log $|G(f)|^2$ compares to a plot of log $|\Xi(f)|^2$. From Equation A3.24,

$$\log|G(f)|^2 = \log|H(f)|^2 + \log|\Xi(f)|^2 \qquad \textbf{(A3.25)}$$

For our case,

$$h(x) = \begin{bmatrix} 0 & \text{if } |x| > x_0 \\ a & \text{if } |x| \le x_0 \end{bmatrix} \qquad \textbf{(A3.26)}$$

yields[26]

$$|H(f)|^2 = \left| \int_{-\infty}^{+\infty} h(x) e^{-i2\pi f x} dx \right|^2 = \frac{[a \sin(2\pi x_0 f)]^2}{(\pi f)^2} \qquad \textbf{(A3.27)}$$

Figure A3.4 shows a plot of log $|H(f)|^2$ versus f for x_0 and a=1/3. Note that the plot is a constant up to some value, f', and then decreases. We can find the value of f' by solving for f' where $H(f)$ is less than $\lim\{f\to 0\}$ $H(f)$ by some preset amount, say ϵ. That is, f' is the solution of

$$\log|H(f')|^2 = (1 - \epsilon) \lim_{f \to 0} (\log|H(f)|^2) \qquad \textbf{(A3.28)}$$

For ϵ = 5 percent, f' is the solution of

$$\log\left[\frac{\sin(2\pi x_0 f')}{\pi f'}\right] = 0.95 \log(2x_0) \qquad \textbf{(A3.29)}$$

For x_0 = 1, f'=0.08, and log (f') = -1.1.

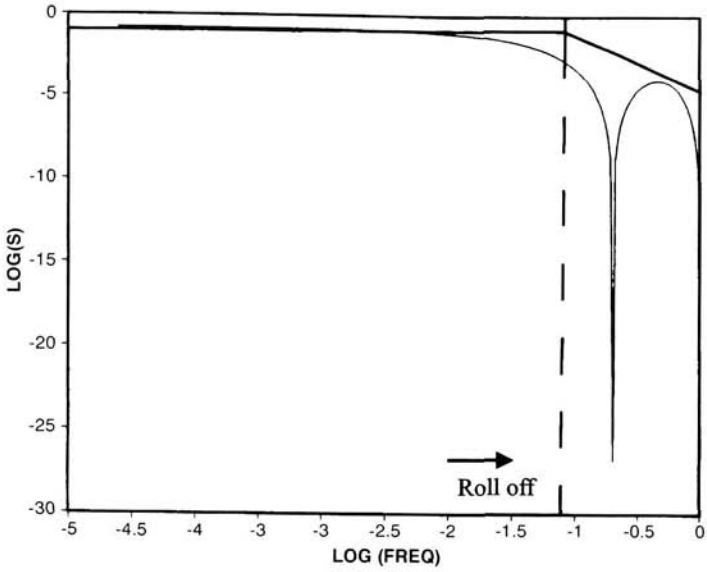


Figure A3.4. Plot showing "roll-off" from white noise in 1D due to a 3-point filter. Curved line shows analytic result. Straight line shows the simple approximation used for bed resolution calculations.[13]

This says that a running average filter adds a constant to within 5 percent to the spectral density of the original sequence from $-\infty$ to f'. Above f', the amount added rapidly decreases.

In 2D, the 3-point filter (Equation A3.13) has been applied to a completely random data set. The data set is generated by setting the spectral density,

$$S(\omega_x,\omega_z) = \begin{bmatrix} 1 & \text{if } \omega_x \neq 0 \text{ or } \omega_z \neq 0 \\ 0 & \text{if } \omega_x = \omega_z = 0 \end{bmatrix} \qquad \textbf{(A3.30)}$$

and using a random phase array. After taking the inverse Fourier transform (described in detail in Appendix A3.5), we apply the 3-point filter to the result. The spectral density of the result is then calculated. A radial cut from the spectral density is plotted in Figure A3.5. Details of this process are found in Reference No. 13. A reference line of slope -2 is shown for frequencies above the roll-off frequency. Note the roll-off departs from the input spectral density around -1.1. This is an idealized

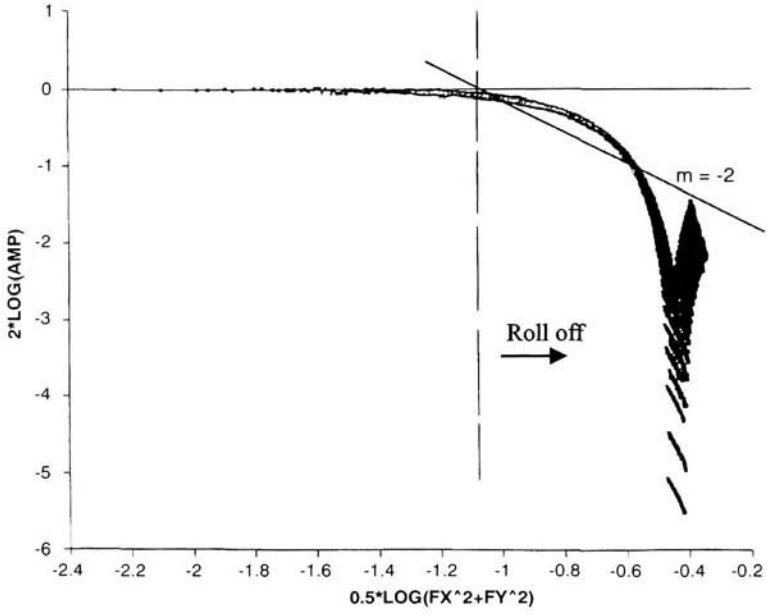


Figure A3.5. Plot showing "roll-off" from white noise in 2D due to a 3-point filter. Plotted points show numerical result. Straight line of slope -2 shows a simple approximation for a 3-point filter.[13]

example of the averaging of the digitizing camera found in the core photos described in Sections 3.3 and 3.5. The 2D smoothing filter has also been applied to the core forgeries in Section 3.3.

A3.4 The Normal Score Transform

The normal score transform produces a Gaussian distribution from any input distribution. The transform is based on the cumulative probability distribution function. The cumulative probability distribution function, $C_d(\xi_j)$, of the data values, ξ_j, is

$$C_d(\xi_j) = \frac{n(\xi_j)}{N} \qquad \textbf{(A3.31)}$$

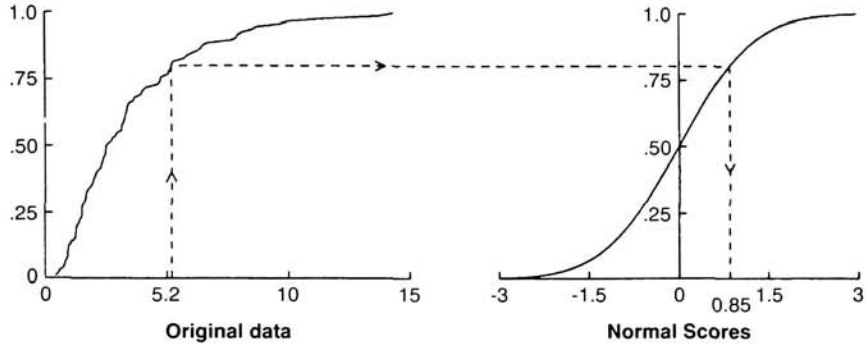


Figure A3.6. A graphical illustration for transforming the values of one distribution into a Gaussian distribution. (a) input data cumulative distribution and (b) Gaussian cumulative distribution.

where

$n(\xi_j)$ = the number of points in the input data set with value less than or equal to ξ_j.
N = the total number of points in the input data set.

An example, $C_d(\xi_j)$, is plotted in Figure A3.6a.

The cumulative probability distribution of a Gaussian distribution, $C_g(\xi)$, is defined as follows:

$$C_g(\xi) = \frac{\int_{-\infty}^{\xi} f_g(\xi')d\xi'}{\int_{-\infty}^{+\infty} f_g(\xi')d\xi'} \tag{A3.32}$$

where

$f_g(\xi')$ = Gaussian histogram equation (Equation A2.11)

A plot of $C_g(\xi)$ is shown in Figure A3.6b.

We need the inverse Gaussian cumulative probability distribution function, C_g^{-1}. That is,

$$C_g^{-1}[C_g(\xi)] = \xi \tag{A3.33}$$

An algorithm for calculating C_g^{-1} is found in Reference 44.

The normal score transformed value, ζ_j, for the input value, ξ_j, is defined as

$$\zeta_j = C_g^{-1}[C(\xi_j)] \tag{A3.34}$$

Figure A3.6 illustrates the procedure graphically.[17] We start on the left-hand side with the cumulative distribution plot of the original data. For any data value, ξ_j, we can read $C(\xi_j)$, the cumulative proportion of values below ξ_j. This corresponds to the vertical dashed line in Figure A3.6a. Next, we find the value ζ_j that is the corresponding value from a Gaussian distribution. To do this, we find the value of $C_g(\zeta_j)$ equal to $C(\xi_j)$. This is shown as the horizontal dashed line connecting Figure A3.6a to Figure A3.6b. Finally, we identify ζ_j as the normal score transform of ξ_j. The vertical line extended down to ζ_j on the abscissa identifies ζ_j in Figure A3.6b. This procedure is applied to each value in the original data set. The new set, ζ_j, is the normal score transform of the original set, ξ_j.

A3.5 The Discrete Inverse 2D Fourier Transform

The core forgeries in Section 3.5 are generated using a discrete inverse Fourier transform in 2D. Equation 3.2 gives the spectral density, $S(\omega_x,\omega_z)$. With the discrete Fourier transform, we write the core image, $g_{j,\ell}$, as

$$g_{j,\ell} = \sum_{m=1}^{N_x} \sum_{p=1}^{N_z} G_{m,p} e^{2\pi i[(j-1)(m-1)/N_x + (\ell-1)(p-1)/N_z]} \qquad \textbf{(A3.35)}$$

Here,

$$G_{m,p} = R_{m,p} e^{i\theta_{m,p}} \qquad \textbf{(A3.36)}$$

where

$$R_{m,p}^2 = S(\omega_x,\omega_z) \qquad \textbf{(A3.37)}$$

and

$$\omega_x = \frac{2\pi(m - 1)}{N_x \Delta x} \qquad \textbf{(A3.38)}$$

$$\omega_z = \frac{2\pi(p - 1)}{N_z \Delta z} \qquad \textbf{(A3.39)}$$

for

$$\frac{N_x}{2} \geq m - 1 \geq 0 \quad \& \quad \frac{N_z}{2} \geq p - 1 \geq 0 \qquad \textbf{(A3.40)}$$

where

Δx = spacing of points in the x direction
Δz = spacing of points in the z direction

except

$$R_{1,1} = 0.0 \tag{A3.41}$$

Because of symmetry requirements between first and fourth quadrants as illustrated in Figure A3.1

$$R_{m,N_z+2-p} = R_{m,p} \tag{A3.42}$$

for

$$\frac{N_x}{2} \geq m > 1 \quad \& \quad \frac{N_z}{2} \geq p > 1 \tag{A3.43}$$

The phase angle, $\theta_{m,p}$, is defined as a random number between $-\pi$ and $+\pi$ for the range of m and p values satisfying

$$\frac{N_x}{2} \geq m - 1 \geq 0 \quad \& \quad \frac{N_z}{2} \geq p - 1 \geq 0 \tag{A3.44}$$

and

$$\frac{N_x}{2} > m - 1 > 0 \quad \& \quad N_y - 1 \geq p - 1 > \frac{N_z}{2} \tag{A3.45}$$

Finally, to generate a real noise array, the 2D inverse Fourier transform requires the following:[42]

$$G_{N_x+2-m,1} = G^*_{m,1} \quad \text{for} \quad \frac{N_x}{2} \geq m > 1 \tag{A3.46}$$

$$G_{1,N_z+2-p} = G^*_{1,p} \quad \text{for} \quad \frac{N_z}{2} \geq p > 1 \tag{A3.47}$$

$$G_{N_x+2-m,N_z+2-p} = G^*_{m,p} \quad \text{for} \quad \frac{N_x}{2} \geq m > 1 \;\&\; N_z \geq p > 1 \tag{A3.48}$$

$$G_{\frac{N_x}{2}+1,N_z+2-p} = G^*_{\frac{N_x}{2}+1,p} \quad \text{for} \quad \frac{N_z}{2} \geq p > 1 \tag{A3.49}$$

Here G^* is the complex conjugate of G. For the most accurate distribution, only one-half to one-fourth of the original distribution should be retained,[11] but for this work, the entire array is used.

The resulting array values from Equation A3.35 are then mapped into gray scale values by using Equation A3.10.

A3.6 2D Covariance From the Spectral Density

In Appendix A2.8, we noted that the inverse Fourier transform of the spectral density produces the covariance for an ergodic data set. We also found that fGn is ergodic. This carries over to 2D distributions. We will perform the Fourier transform of the 2D spectral density to get a 2D covariance. We will then note that this 2D covariance yields a 1D fGn covariance if either the x or the z lag is set to zero. This result verifies that Equation 3.2 generates fGn in both the vertical and horizontal directions and is consistent with the core photo findings in Section 3.3.

We define the 2D inverse Fourier transform as follows:

$$\mathcal{F}[.] = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} [.]e^{i\omega_x x}e^{i\omega_z z}d\omega_x d\omega_z \tag{A3.50}$$

Thus, the 2D covariance function Cov(x,z) is calculated from Equation 3.2 as follows:

$$\text{Cov}(x,z) = \mathcal{F}\left[\frac{A_{SD}\delta(w_x)}{|\omega_z|^{\alpha}} + \frac{A_{SD}B_{SD}}{|\omega_x^2 + \omega_y^2|^{\beta}}\right] \qquad \text{(A3.51)}$$

That is

$$\text{Cov}(x,z) = \mathcal{F}\left[\frac{A_{SD}\delta(w_x)}{|\omega_z|^{\alpha}}\right] + \mathcal{F}\left[\frac{A_{SD}B_{SD}}{|\omega_x^2 + \omega_z^2|^{\beta}}\right] \qquad \text{(A3.52)}$$

The first integral is calculated in Appendix A2.8. The second integral is calculated by first changing to polar coordinates. Next, the integral over angle can be evaluated as a zeroth order Bessel function. The integral over ω_r can then be carried out using Gradshteyn[45] for $1/4 < \beta < 1$ to be

$$\mathcal{F}\left[\frac{1}{(\omega_x^2 + \omega_z^2)^{\beta}}\right] = \frac{\Gamma(1 - \beta)}{2^{(2\beta+2)}\pi^2\Gamma(\beta)} r^{2\beta-2} \qquad \text{(A3.53)}$$

where

$$r = \sqrt{(x^2 + z^2)} \qquad \text{(A3.54)}$$

Adding this to the first term, found like Equation A2.124, we get for Equation A3.52

$$\text{Cov}(x,z) = 2A_{SD}\sin(\alpha\pi/2)\,\Gamma(1 - \alpha)\,|z|^{\alpha-1} + \frac{A_{SD}B_{SD}\Gamma(1 - \beta)}{2^{2\beta+2}\pi^2\Gamma(\beta)} r^{2\beta-2} + c' \qquad \text{(A3.55)}$$

Here we have approximated the finite range of the Fourier transform by an infinite range as was done in Appendix A2.8. We have also added a constant to the infinite integral to take into account the unassigned value of the spectral density at $\omega_x = \omega_z = 0$. This also follows the procedure of Appendix A2.8. This constant could be

evaluated as the constant c was in Appendix 2.8, but we choose to simply leave it unassigned, since we will not need its value to show the result needed in this section.

If $\beta = H$, Equation A3.55 for horizontal traces ($z = 0$) becomes

$$\mathrm{Cov}(x,z = 0) = A_{SD}B_{SD}\frac{\Gamma(1 - H)}{2^{2H+2}\pi^2\Gamma(H)}|x|^{2H-2} + c' \qquad \text{(A3.56)}$$

or

$$(\mathrm{Cov}(x,z = 0) - c') = \mathrm{const}(H)|x|^{2H-2} \qquad \text{(A3.57)}$$

which is fGn in the x direction if $0<H<1$ as shown in Appendix A2.8. Thus horizontal traces are fGn as measured from the 1D analysis.

If $\beta = H$ and $\alpha=2H-1$, then for vertical traces ($x = 0$)

$$\mathrm{Cov}(x = 0,z) = \left[2A_{SD}\sin(\pi/2(2H - 1))\Gamma(2 - 2H) + A_{SD}B_{SD}\frac{\Gamma(1 - H)}{2^{2H+2}\pi^2\Gamma(H)}\right]z^{2H-2} + c' \qquad \text{(A3.58)}$$

or

$$(\mathrm{Cov}(x=0,z) - c') = \mathrm{const}(H)|z|^{2H-2} \qquad \text{(A3.59)}$$

For $0<H<1$ this corresponds to fGn in the vertical direction.

This page is intensionally left blank

CHAPTER 4

GENERATION OF POROSITY DISTRIBUTIONS

The similarity between the vertical and horizontal statistics of many core photos and porosity well logs suggests using fGn to generate porosity distributions between wells. The majority of this chapter will show how this can be done using well logs. The details of applying fractal distributions vary from reservoir to reservoir. In this chapter, we will outline a procedure to follow, but not all possibilities can be explored here.

We will first concentrate on the generation of fGn cross sections[33,46] to illustrate a general procedure. Next, we expand the discussion to fBm cross sections[22,46] and 3D fGn distributions.[47] We will then discuss some special considerations for large well spacing, especially the integration of data sets of different sizes (e.g., seismic and geological maps with well data). All three generation procedures (2D fGn, 2D fBm, and 3D fGn distributions) will be compared in fluid flow simulations in Chapter 6.

The 2D and 3D distributions are generated by using spectral methods that are extensions of the 2D methods already described in Section 2.4 and Appendix A2.4. The methods take advantage of an algorithm known as the fast Fourier transform.[19] This algorithm makes generating distributions very fast computionally. As an example, we find we can generate a distribution of reservoir properties with one million grids in about 3 minutes on an IBM RISC 6000 computer.

4.1 Steps for Generating Distributions

The steps for generating reservoir property distributions from well logs are basically the same whether 2D fGn, 2D fBm, or 3D fGn distributions are generated. The steps are the following:

1. Identify zones of interest.

2. Test well logs in these zones for fractal character.

3. Generate a reservoir property distribution.

4. Check results.

This chapter will first describe each of these steps in detail for fGn cross sections. A number of caveats and applications of fGn distributions are also discussed. This is necessary, because the pictures generated by these methods can be so real "looking" that one must keep in mind what is real and what is simulated. Also, the fGn model does not describe adequately all possible distributions found in the subsurface. We will discuss some variations to the basic approach using some specific examples.

We choose to build fractal distributions of porosity. Porosity well logs are more often fGn than are other logs. We do not know why this is the case. It is especially surprising that Gamma ray logs often do not show the same fGn character as porosity logs. We have chosen to build porosity distributions from logs rather than from outcrop data, core photos, or outcrop photos. Logs provide more direct measurements of subsurface properties than these other data sets. Core measurements can be used by substituting the core measurements for log data; however, core data is usually discontinuous. As a result, missing data must be filled in before these techniques can be used. One way to fill missing data is described in the Appendix A4.1.

Permeability distributions are needed as well as porosity in order to model fluid flow. We will address permeabilities in the next chapter.

4.2 Identify Zones of Interest

Fractal models of porosity distributions are statistical descriptions. They are not substitutes for good geological descriptions. In fact, a good fractal model is dependent on a good geological description.

As discussed in Chapter 2, the application of statistics requires an ensemble. An ensemble must be identified in the earth strata before any statistical techniques can be used to make valid predictions. Good geologic interpretation is essential to this identification. Intervals of like geology need to be identified and "mapped" between wells wherever possible. This is usually done by identifying "markers" on logs. Markers are log responses or core descriptions that are similar in many different wells so that they can be correlated from well to well. These markers can be used to put wells on depth with one another so that the same marker appears at the same depth on all wells. In order to do this, the log from one well is treated as the baseline log, and logs from other wells are stretched or compressed until the markers from all wells are on depth with each other.

Sometimes segments of the geological record in one well may be missing in another well. This can be due to pinching out, faulting, or erosion of sediments. If the missing interval is interpreted as a pinch out, the best approach is either to stretch the logged interval to put marker beds on depth or to treat the pinch out as the result of erosion. If the missing interval is interpreted to be due to faulting or erosion, the missing interval should be accounted for in the log. This can be done by "pasting" the interval from the well that has the missing interval into the log that is missing it. Then the log to be depth adjusted should be stretched and compressed to put the markers on depth. The adjusted logs will be used to build the porosity distribution between wells. After generation of the cross section is complete, the added interval should be removed from the final distribution by removing erosional regions and displacing faulted sections so that the added interval does not appear in the final well location. This can be done in a reservoir flow simulator by adjusting grid-block sizes and locations.

4.3 Analyze Well Logs

Once the geologic units have been identified and placed on depth, the log reading (or core data) through each unit should be analyzed to see if it is fGn or not.

It is best to perform the fractal analysis on both the original logs and the depth-adjusted logs to be used to generate distributions. A fGn distribution needs to be identified before applying the fractal models described here. If this is not done, we may be building a distribution with the wrong statistical properties. As pointed out in Chapter 2, fractal distributions all have upper and lower limits of their applicability to real objects. These limits need to be identified or at least confirmed that they are beyond the scale of interest. The tests needed are described in Chapter 3.

Good matches to fGn are not always found. When not, there are several things that can be tried to obtain a fGn model. Units can be recombined, trends can be removed, a transform can be applied, or the differences in log traces can be analyzed. These processes should be performed with geologic input so that the final unit choice has geological meaning. The individual units that have been identified as fGn should be worked with individually and then restacked to produce the final, complete distribution.

Logs like the traces shown in Figure 2.2 are not stationary and therefore will not be fGn. However, if the overall drift is first removed from the logs, fGn may be left. This is done simply by identifying the trend, subtracting it from the log data, and analyzing the difference. If this is successful, porosity distributions can be generated as described here for the difference, and this difference can be added back to the overall trend before fluid flow simulations are made.

If the logs are not Gaussian, one of the transforms described in Section 3.1 can be applied. The distributions can be generated in the transform space, and an inverse transformed can be used to return to original properties.

The differences in logs can also be tested. The concept of fGn layering and splotches (Equation 3.2) can be modified to include non fGn layers with fGn noise. This approach requires only the difference in the logs to be fGn. To test for this description, put two logs on depth, take differences between log values foot by foot working down the log, and then analyze the differences versus depth as a separate log. If fGn, proceed. If not fGn, the fGn model is not appropriate and should not be used.

If fGn is not appropriate, the model described in this book should not be used to generate reservoir property distributions. Another model is required. Alternative models include layer cake,[46] variogram,[20] and Boolean[31] models, as well as many others. A description of these would take us beyond the scope of this book.

Attempts to use fGn have been made to describe fifteen reservoirs within Conoco. The fGn model was found to be inappropriate in only two cases.

The confirmation of fGn is important, because the distributions are generated purely on a statistical basis. No geologic model input is incorporated except identifying like zones and placing well logs on depth. The fGn spectral model described here will build only fGn. If the input distribution is not fGn, a fGn distribution will be superimposed on the input distribution, resulting in a distribution that is not representative of the data.

4.4 Generate fGn Cross Sections

4.4.1 *Basic Assumptions*

Once we have identified fGn as an appropriate model, we can continue building reservoir property distributions using fGn. We choose to begin with a uniformly spaced grid. The reason for this is twofold. First, the Fourier transforms that we are going to use to generate fGn distributions are defined on uniformly spaced grids. Second, the standard deviation of a set of data generally depends on the grid size of the data. If grids were different shapes and sizes, it is difficult to determine what standard deviation should be used for the final distribution of reservoir properties.

The dependence of the standard deviation on grid size can be seen if we go to the extreme. If all of the porosity log data are described by a single grid, the mean will be the mean of the distribution, but the standard deviation will be zero. For four grids, the mean will be the same, but the standard deviation will be somewhere between zero and the standard deviation of the entire data set. As the size of the grids approach the size of the data sampling volume, the standard deviation of the set of grids will approach the standard deviation of the data. In order to apply the statistical measures of the data to the grids directly, the grids should be the same size as the data samples.

The generation of reservoir property distributions on fGn cross sections is performed in two steps. First, smooth layers are produced that are consistent with any layering found in the correlated well logs. Next, noise (fGn) is added in an amount proportional to the differences between the well log pair that has been placed on depth.

This procedure is based on the assumption that the distribution of properties can be viewed as layers plus noise. This assumption is based on the findings of Chapter 3, as summarized in Equation 3.2. If a formation is perfectly layered, two logs that are on depth with each other will be identical. Their differences will be zero. Therefore, any differences in the logs after they are put on depth are considered a result of noise.

Equation 3.2 has four parameters (A_{SD}, B_{SD}, α, and β). α and β can be expressed in terms of an intermittency exponent H (see Appendix A3.1). Appendix A3.6 showed that if the intermittency exponent were the same for α and β, fGn would result in both the horizontal and vertical directions. For simplicity, we will make that assumption for our modeling of reservoir property distributions. With that assumption, the overall standard deviation of the well logs, the differences between the logs after they are put on depth, and the measured value of H from the logs can be used to calculate A_{SD}, B_{SD}, α, and β.

The mathematics used to generate the cross sections are found in Appendix A4.2. The appendix describes the layering either by using linear interpolation between the wells or by setting a constant value at each depth. Usually, we use the former method, following Hewett.[22] There are advantages and disadvantages of each method. A constant value for each depth produces pure fGn in the horizontal and vertical directions, if the vertical distribution is fGn with the same intermittency exponent, H (see Appendix A3.6) as the horizontal distribution. The linear interpolation mixes a linear trend with fGn noise. The result is not pure fGn. On the other hand, linear interpolation ties the generated distributions at the edges closer to the input well log values. Neither of these two approaches is ideal. Other possible options are suggested in "Unresolved Issues" at the end of this chapter.

Next, we calculate the amount of noise to be added. First, the average value of each well logged interval is calculated and subtracted from each well log value. This calculation removes any overall drift in porosity in the horizontal direction. It is necessary to remove drift, since fGn is stationary. Next, we calculate the standard deviation of the differences between the depth adjusted well pair values. A fGn distribution is generated using Equation 3.2 with $B >> 1$ (splotches only). It is linearly scaled to give zero mean and a standard deviation equal to that calculated for the differences of the well log pairs. The scaled noise array is then added to the smooth interpolation generated between wells.

The last step of the process is to correct the standard deviation and mean of the final distribution to the well logs standard deviation and mean. There are two ways to do this. One method is to force the mean and standard deviation of the distribution at the well locations to equal the mean and standard deviation of the well log data. The other method is to adjust the mean and standard deviation of the entire distribution to the well logs. There are advantages and disadvantages to each.

From a statistical point of view, it makes more sense to make the adjustment based on the generated values at only the well locations. In some cases, however, this method produces large variations in the average porosity of the entire distributions from the well log average--especially in 3D distributions. These large variations seem statistically correct, since as little as 1 billionth of the reservoir has been sampled by the well logs. It is very possible that the mean reservoir properties may vary appreciably from the mean measured at the well logs with such a small sample. From an engineering point of view, however, this can produce a problem.

A significant difference in the average porosity of a distribution from that of the well data can produce a large difference in oil in place. Oil in place is usually calculated based on well log and core porosities. Estimates of oil in place must be jointly agreed upon among partners with a working interest in the field. Estimates also include other data in addition to well logs, such as pressure transient tests, historical pressure measurements at different times in the life of the field, and seismic interpretations. Thus, the engineer often needs to match the mutually agreed upon estimate. If this is the case, then the mean and standard deviation of the entire final distribution should be made to coincide with the mean and standard deviation found from well logs and other data sources.

Adjusting the mean and standard deviation of the final distribution does not change the fGn character of a distribution. The reasons for this are shown mathematically in Appendix A2.5. Conceptually, the fGn character is capturing the ordering of the data (e.g., how likely are large data values to be adjacent to large data values?). Multiplying all the values in a distribution by a constant or adding a constant to all the values does not change this ordering property.

Figure 4.1 shows a fGn cross section for the Grayburg and San Andres formations in southeastern New Mexico. The sediments were deposited along a prograding shoreline and represent near shore marine, intertidal, and supratidal deposits. The two wells used at the edges of the cross sections are 1,060 feet apart. The thickness of the cross section is 300 feet. The Grayburg and San Andres

portions are generated separately and then stacked together. The upper layer (Grayburg) is Gaussian, but the lower layer (San Andres) is closer to a log-normal porosity. The differences in the logs of both the upper and lower formations are Gaussian, however, so a transformation is not needed for the added noise. Linear interpolation is made between the two wells before fractal noise is added. In this case, we match the distribution only at the edges to the mean porosity of the well logs.

Figures 4.2 and 4.3 show two additional realizations. These images are generated using the same input data as Figure 4.1, except a different random number seed is used to create the realizations. Note that the "splotches" are in different locations in the different realizations. This points out that these realizations are only statistical models of the subsurface. They do not describe what is actually there at each point in space. They only generate realizations consistent with the known data. As a result, it is necessary to generate a number of realizations before drawing meaningful conclusions. Each realization becomes a member of the ensemble from which statistical inferences can be made.

4.4.2 *Relaxing Some Assumptions*

A number of assumptions are built into the fGn model described above. If additional geological information is available, some of these assumptions can be relaxed easily within the spectral density framework described here. These include changing the amount of noise added and making the added noise anisotropic. These changes should be made based on geologic interpretations.

The amount of noise to be added to the layers is calculated from the difference between the well logs after they are put on depth. If this produces a distribution that is too layered or too noisy (i.e., too broken up), there are two ways to adjust the results. One is to stretch and compress the logs so that layers are more or less correlated between the logs, depending on whether more or less layering is desired. Another way is to simply introduce an adjustable parameter that multiplies the noise cross section before it is added to the layered distribution.

The noise added to the distributions is isotropic if Equation 3.2 is used for the spectral density model. (Note that even though the noise is isotropic, the final distribution may not be, because the layering component is added to the noise.) Geological models may suggest isotropic noise is not the best representation. For

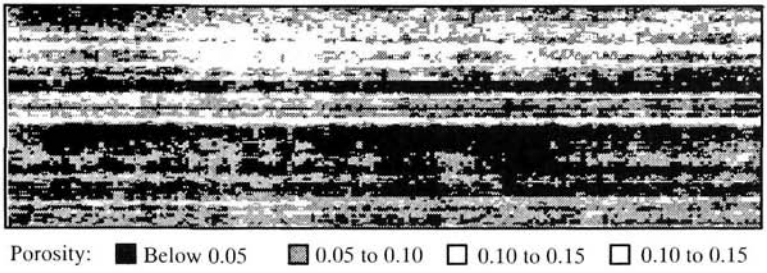


Figure 4.1. fGn cross section generated for the Grayburg and San Andres formations in southeastern New Mexico.

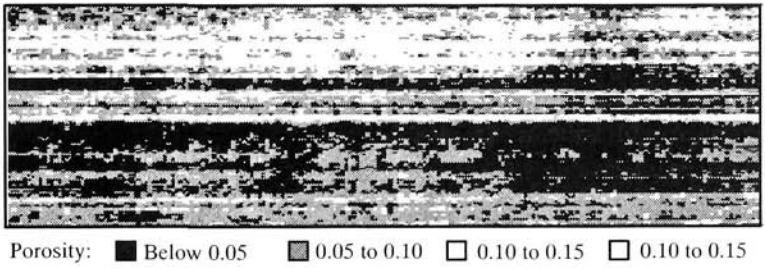


Figure 4.2. Realization #2 of the fGn cross section in Figure 4.1. Only the seed number is different.

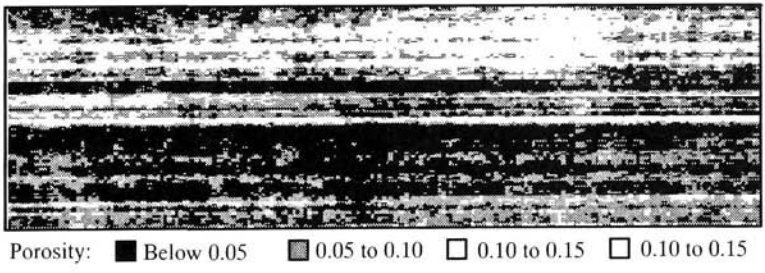


Figure 4.3. Realization #3 of the fGn cross section in Figure 4.1.

example, a stream channel is more likely to have larger variations in properties across the stream channel rather than along the channel. If wells are spaced close enough, the anisotropy may be captured in the linear interpolation between wells. If wells are widely space, however, the intersection of wells with stream channels may not capture the anisotropic correlation. In this later case, reservoir properties that are more correlated in one direction than another can be generated if the spectral density in Equation 3.2 is modified by adding the parameter, s_x, as follows:

$$S(\omega_x,\omega_z) = A_{SD}\left(\frac{\delta(\omega_x)}{\omega_z^{\alpha}} + \frac{B_{SD}}{(s_x\omega_x^2 + \omega_z^2)^{\beta}}\right) \tag{4.1}$$

Changing the value of s_x will stretch the distribution of splotches in the x or the z direction, depending on whether it is greater than or less than one, respectively.

4.4.3 *Cross Section With Many Units*

We have discussed previously that dividing and combining geologic units may be necessary to obtain a fGn distribution. Geologic units can also be distinguished based on their B_{SD} value in Equation 3.2. B_{SD} determines the amount of layering in a distribution. Two geologic units may, together, be fGn, but one can be more continuous laterally than the other (e.g., the core photo in Figure 3.40). If this is the case, the two units should be divided into separate units to build the cross sections. If this is not done, the layering character of the two units will be mixed together resulting in an incorrect connectivity for both units.

Geologic units may also need to be divided, because they are believed to be caused by completely uncorrelated geological events. The distributions generated using Equation 3.2 have a built-in correlation in the vertical direction. The splotches that are added are isotropic. If a major unconformity exists between two units, the vertical continuity is interrupted at the boundary. In order to reproduce this in the generated distribution, the unit should be divided at the unconformity so that the vertical correlation will be broken at this depth.

Plate 3 shows a section of the earth that has been generated using six different fractal cross sections. The entire thickness of the cross section is about 200 meters. The depositional environments range from intertidal mud flats to shallow marine shelf

deposits. The well separation is about 2 kilometers. Notice the differing character of each unit. It's important that the layers be isolated because, if these units had been mixed together into a single cross section, the character of the individual units would have been lost.

In Plate 3, dark blue is low porosity-poor quality reservoir rock, red is high porosity-good quality reservoir rock. Note that the top zone has uniformly low porosity. The next zone is fairly homogeneous, high porosity. The third and fourth zones are a mixture with streaks of low porosity layers found throughout the zones. The fifth zone is rather poor quality punctuated with isolated pockets of high porosity regions. The bottom zone is the most heterogeneous with splotchy mixtures of rather poorly connected high and low porosity regions.

Plate 4 shows a section of the earth at the same depth as Plate 3 but at right angles. The well at the right of Plate 4 is the same as the well at the left of Plate 3. Even so, there are noticeable differences in the cross sections. Note that the top two zones are of poorer quality in Plate 4 compared to the same zones in Plate 3. The third zone is less layered in Plate 4. The fifth and sixth zones contain more of the lowest porosity regions in Plate 4 than in Plate 3.

These same qualitative features were noted in all of the realizations of both cross sections. Yogi Soni (Conoco Inc.) generated the cross sections in Plates 3 and 4.

4.4.4 *Cross Section With Thin Coal Seams*

Figure 4.4 shows a cross section generated in a tidally dominated, marginal marine setting. The logs are separated by about 4,800 feet. The cross section is about 120 feet thick. In this formation, thin coal beds can be identified on the logs. These coal beds are believed to be much more laterally continous than other beds. Coal can be identified on the porosity logs with porosity values greater than 35 percent. This is much higher than the surrounding porosity values. The well logs were depth adjusted. A linear interpolation was then made between the logs. Coal beds have no moveable hydrocarbons, so for fluid flow simulations, their porosity should be treated as zero. In order to accommodate this in the generated cross section, first a template is made having 0 everywhere the interpolated porosities are greater than 35 percent and 1 everywhere the interpolated porosities are less than 35 percent.

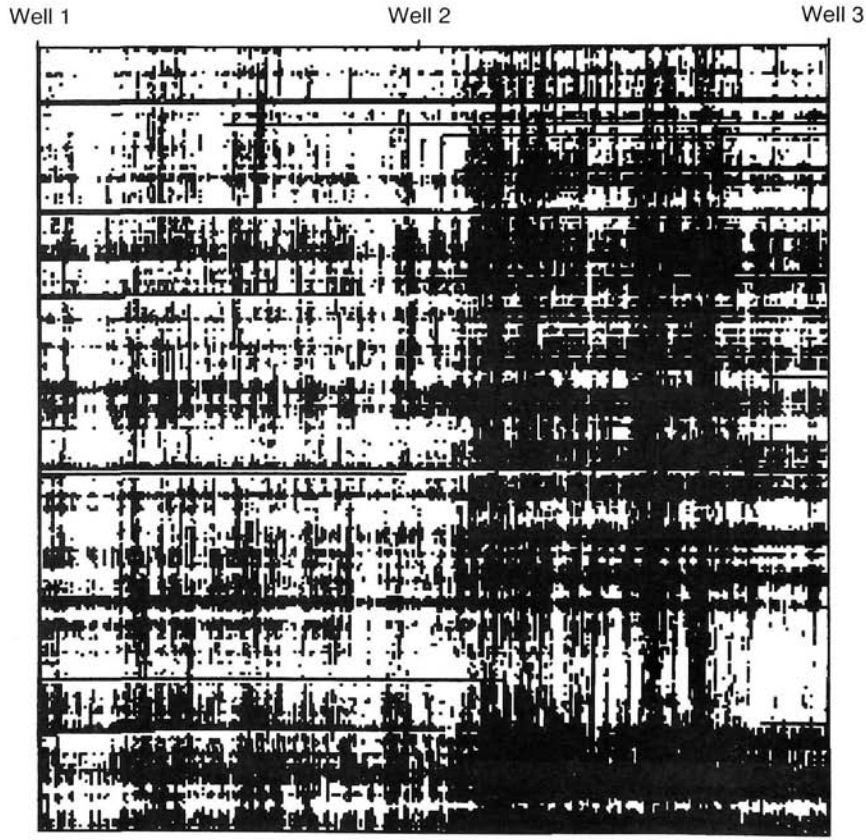


Figure 4.4. Black and white "threshold" cross section of a tidally dominated, marginal marine setting with coal beds.

A porosity distribution was generated between the wells using the fGn model. This is possible, since the small amount of coal in the formation did not appreciably damage the fGn distribution seen in the porosity logs through the formation. The coal bed distribution of 1's and 0's was then multiplied, point by point, by the fGn model distribution. This insured zero porosity wherever coal is encountered in the well log.

The final cross section is presented in Figure 4.4 by setting a threshold of 14 percent. Everywhere the porosity is less than 14 percent, the cross section is colored black. Everywhere the porosity is greater than 14 percent, the porosity is colored white. This presentation gives a visual indication of the connectivity of the reservoir with porosity greater than 14 percent. The "splotchiness" of Figure 4.4 looks to be long and thin vertically. This is actually not the case in real world coordinates. The "splotchiness" that is added was isotropic. The apparent anisotropy is produced because the vertical direction has been exaggerated (stretched) by more than 40 times its actual thickness in Figure 4.4 (generated by Bruce Reid, Conoco Inc.).

4.5 Checking Results

Once porosity distributions are generated, they should be checked. This section discusses some possible checks. Wells can be omitted and then redrilled. Statistics of the final distribution can be compared with input statistical descriptions of the well data. Pictures of the resulting distributions can be studied by the geologist to see if they are consistent with his geological model. Also, the connectivity produced in the final distributions can be compared with any production data that may reflect the actual reservoir connectivity.

The generated distributions are statistical distributions. A random number generator is used to produce many distributions consistent with the input statistics. As a result, the checks described here should be made on a number of distributions rather than just one or two.

In general, we should use all the information available to produce reservoir property distributions. However, for testing purposes, it is often useful to omit a well. The location of the omitted well can then be "logged" in a cross section containing that well location and compared to the actual well data.

Both horizontal and vertical traces can be extracted from generated distributions and statistical tests run on these traces. This essentially produces a

check that distributions have the intended statistical properties. Although both vertical and horizontal wells can be extracted from a generated cross section, the methodology described here can only input vertical wells to build the fGn distributions. How to input horizontal well data is a topic for unresolved issues.

False color pictures can be compared with geological interpretations. Some care needs to be taken in this, however. First, color scales can emphasize different things depending on the color thresholds. Second, aspect ratios of plots need to be considered. Despite these concerns, careful generation and interpretation of false color images can be an extremely good way to compare generated distributions to geological models.

The use of false color images is a science in itself. Only a few caveats will be mentioned here. The setting of color for porosity values determines the appearance of a distribution. Different color maps can produce widely different pictures. The appearance of connectivity, for example, is threshold dependent.

Automatically scaled false color images can be particularly troublesome. If a color scale is automatically generated from minimum to maximum values in the distributions, then any change in the maximum or minimum value can change the appearance of the final distribution. Even different realizations from the same data can produce different scales. The best way to avoid this problem is to use thresholds to assign colors. For example, set color thresholds at specific porosity values (e.g., 0.05, 0.10, 0.15, etc.) and use the same thresholds for all of the realizations.

Aspect ratios can also produce confusion in viewing false color pictures. There are three distinct aspect ratios. One is the aspect ratio of the grid blocks being generated. A second is the length to width ratio of the splotches generated. A third is the length to width ratio at which the cross section is plotted.

When the cross sections are generated, each grid block can be long and skinny, square, or even short and fat. This aspect ratio is determined by choosing the number of grids in each direction. In order for the aspect ratio of the grids to be unity (i.e., grids to be square), the number of grids in each direction must be proportional to the physical dimensions of the distribution being generated. Unity aspect ratio is usually best, but it may not be practical due to the number of grids required in the horizontal direction for largely spaced wells.

We must specify the aspect ratio of the splotches before generating the distributions. This second ratio is determined by the s_x parameter in Equation 4.1. If s_x is one, the noise is isotropic in real space dimensions.

The third aspect ratio is the length to width ratio at which the cross section is plotted. This has no bearing on fluid flow simulation or the "reality" of the cross section generated, but it does have a striking affect on what the eye sees. Ideally, the plotted cross section should have the actual field aspect ratio. The problem is that for most offshore reservoirs, well spacings can be very large. As a result, when cross sections are plotted in this manner, they can be 10 to 20 feet long and only a few inches high. This makes them difficult to view. Usually the vertical scale is stretched, making the vertical scale many times the horizontal scale. When viewing these, care should be taken to remember that the vertical has been grossly exaggerated. Splotches viewed on this presentation are going to look long and thin in the vertical direction, even though they may actually be isotropic in real space coordinates. Figure 4.4 illustrates this unwanted effect.

4.6 fBm Cross Sections

Most data indicate fGn rather than fBm in reservoir property distributions within the subsurface. fBm distributions were the first fractal distributions used for horizontal distributions of reservoir properties.[22] Hewett[22] justified his use of fBm using groundwater plume data.[48,49] Hewett and others produced many good field scale fluid flow simulations using fBm.[22,50-52] The reason for fBm's success in these studies may be the choice of fields studied. fBm and fGn distribution both approach a completely layered system as the variation between wells decreases with decreasing well spacing. Most of the fBm studies have been performed for closely spaced wells. As wells get farther apart, the differences between fBm and fGn distributions will increase. In this section, we will describe methods to generate fBm distributions. In Chapter 6, we will compare reservoir simulations on a field using both fBm and fGn distributions.

There are several ways of generating fBm cross sections. The first method used by Hewett[22] was midpoint displacement as described by Voss[11,53] for each depth. But since each depth interval is generated separately by this procedure, fBm is not generated in the vertical direction away from the control wells. Hewett and Behrens[51] changed their procedure to spectral methods. We will also use a spectral method to generate fBm distributions, except we will use the fast Fourier transform and not the Weierstrass-Mandelbrot[11,53] procedure. This calculational difference should introduce no differences in the distributions generated.

We use a spectral density that produces fGn vertically, but fBm horizontally. The spectral density is used to generate a noise array. This noise array is then added to a smooth map generated between well pairs. The smooth map may be from kriging, linear interpolation, or constant layer values. We chose to use linear interpolation in our examples. The noise is added by a procedure outlined by Journel and Huijbregts.[20] This method is different from the above fGn procedure. The mathematical details are in Appendix A4.3.

Figure 4.5 shows a fBm cross section generated using the same two wells that were used in Figure 4.1. Notice that Figure 4.5 is less broken up than Figure 4.1. When fBm is added to the cross section, the added "noise" is much more correlated. The effect of fGn and fBm distributions on fluid flow will be discussed in Chapter 6.

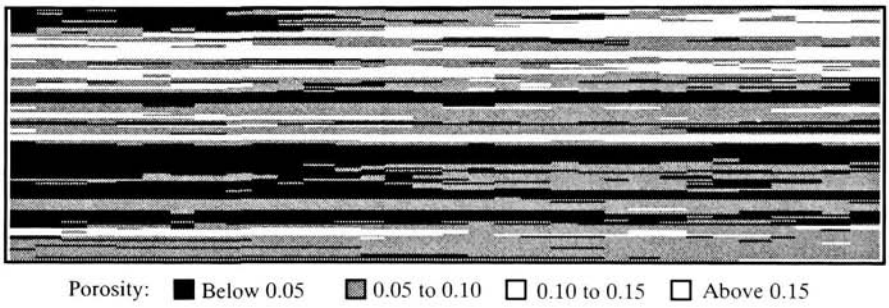


Figure 4.5. fBm cross section generated based on the same wells as used in the fGn cross section in Figure 4.1.

4.7 3D fGn Distributions

Three dimensional fGn distributions are generated in a manner very similar to fGn cross sections. The main differences are (1) the smooth map produced between wells, and (2) the spectral density equation for 3D fGn noise.

The smooth map between wells is generated either by a constant layer map or by a trilinear interpolation algorithm. These are used because they both become the same as their 2D counterparts (constant layer values and linear interpolation) between well pairs. In order to use trilinear interpolation, the wells must first be

connected by lines so that each well corresponds to the vertex of a triangle. Fortunately, any geometric arrangement of vertical wells can be triangulated.

A 3D noise array is generated using the following spectral density equation:

$$S(\omega_x,\omega_y,\omega_z) = \frac{1}{[s_x\omega_x^2 + s_y\omega_y^2 + \omega_z^2]^\beta} \tag{4.2}$$

with

$$\beta = H + 0.5 \tag{4.3}$$

and

$$S(0,0,0) = 0.0 \tag{4.4}$$

x and y are taken to be planer coordinates, z the vertical coordinate. s_x and s_y can be used like s_x in Equation 4.1 to make noise distributions anisotropic.

The noise distribution is scaled so that its standard deviation is equal to the average standard deviation of the differences of the well pairs. The noise is then added to the smooth interpolation between the wells. Lastly, the entire distribution is scaled to have the same mean and standard deviation as the input well data. The overall mean and standard deviations are constrained, because controlling only the well locations produces very large fluctuations in the mean of the entire distribution among realizations.

Figure 4.6 shows a 3D distribution of reservoir properties between 4 wells located at the corners of the distribution. This figure was generated in the same reservoir as Figures 4.1 through 4.3.

Figures 4.7 through 4.9 show areal distributions of multiple realizations at the same depth interval generated by using different random number seeds. Notice the location of the splotches varies from realization to realization, but the overall appearance of the slices looks similar. The changing locations of the splotches change the connectivity between well pairs. The effect of this on fluid flow will be discussed in Chapter 6.

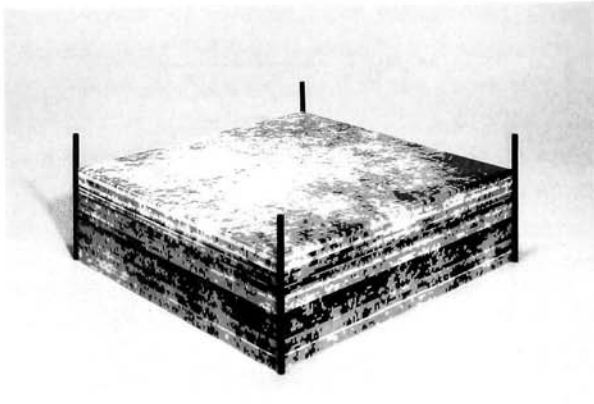


Figure 4.6. 3D fGn distribution generated in the same reservoir as Figures 4.1 through 4.3.

4.8 Large Well Spacings

Large well spacings present special problems to generating fGn distributions--especially in 3D. It is usually impossible to use square (i.e., cubic) grids in 3D when well spacings are large because of the required large number of grids. The large spacing of the wells makes the uncertainty of distributions, from a purely statistical point of view, very high. Geological input becomes even more important. We can also look for other data sets to help inform us of interwell properties. One data set is seismic data.

Nonsquare grids present a challenge for two reasons. One is that the standard deviation of data sets are usually volume dependent. The other is that permeability can also be volume dependent. The zeroth order approach is to ignore the problems and just use the well data as though the aspect ratio of the grids and the volume of the grids are the same as the rock volumes investigated by the logging tool. If this is done, cross sectional distributions are probably preferable to 3D distributions,

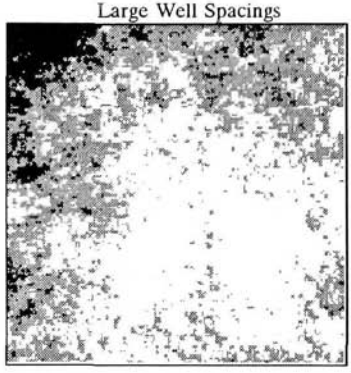


Figure 4.7. Areal slice from Realization #1 of the 3D fGn distribution.

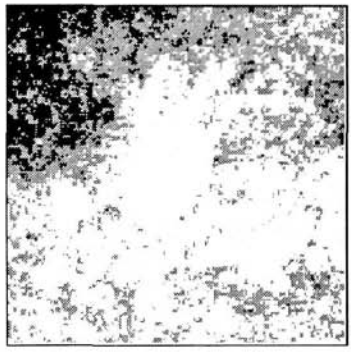


Figure 4.8. Areal slice showing same layer as Figure 4.7 but from 3D Realization #2.

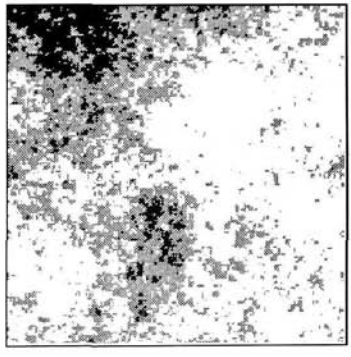


Figure 4.9. Areal slice showing same layer as Figure 4.7 but from 3D Realization #3.

because the grids can be generated closer to square. (Not as many are needed.) Another approach is to generate "representative" distributions in some way (e.g., with cross sections) and then try to scale them up to 3D blocks. In the end, however, this problem is not completely solved.

Recently efforts have been made to relate seismic data to porosity. Doyen[54] claims seismic attributes, such as acoustic impedance, correlate with porosity. Three dimensional seismic surveys can provide information over the entire reservoir,[55] but surface seismic data give information about only large scale features. Even if seismic data correlations with porosity cannot be found, geological information sometimes provides a rough outline of the large scale structure. In the next section, we show how to generate distributions that honor both the statistical data from the wells and the large features established by seismic or other data sources.

4.9 Integration of Large and Small Scale Data

The integration of well data with large scale data is a challenge. The spatial resolution of surface seismic data is typically 40 to 100 feet. Porosity well log measurements have a vertical resolution of 2 to 3 feet. Seismic surveys cover the entire interwell region, but porosity well logs sample less than one foot into the rock.

Journel and coworkers have developed indicator kriging methods that merge data at different scales. Their methods have been applied to field data with some success.[56] The methods are well documented in a recent book.[57] A discussion of these methods would take us beyond the scope of this book.

Instead, this section provides an alternate method based on Fourier transforms that is an extension of the methods described earlier in this book. We will first illustrate the method with a one dimensional data set and then extend it to 2D. 3D extension is straightforward and will not be covered here. We will use synthetic data sets to demonstrate the methods.

4.9.1 *One Dimensional Example*

Let us assume that the solid curve in Figure 4.10 corresponds to the true porosity well log. A smooth porosity curve is generated from the original curve by averaging every 5 values of the true curve and plotting these averages at the middle

of each of the 20 averaged intervals. Then, values are reassigned to a fine grid by using linear interpolation. We will let the smoothed curve represent porosity derived from seismic data or a geological model of the true distribution. We would like to reproduce the "true" porosity curve from statistical measures of the true porosity curve and the average data curve. Unfortunately this is not possible. The best we can do is generate a curve that honors the averaged curve and is statistically similar to the true curve. We will do this by generating a curve with the same mean, standard deviation, and H value as the original curve, as well as the same average values as the original curve. This can be done with Fourier methods.

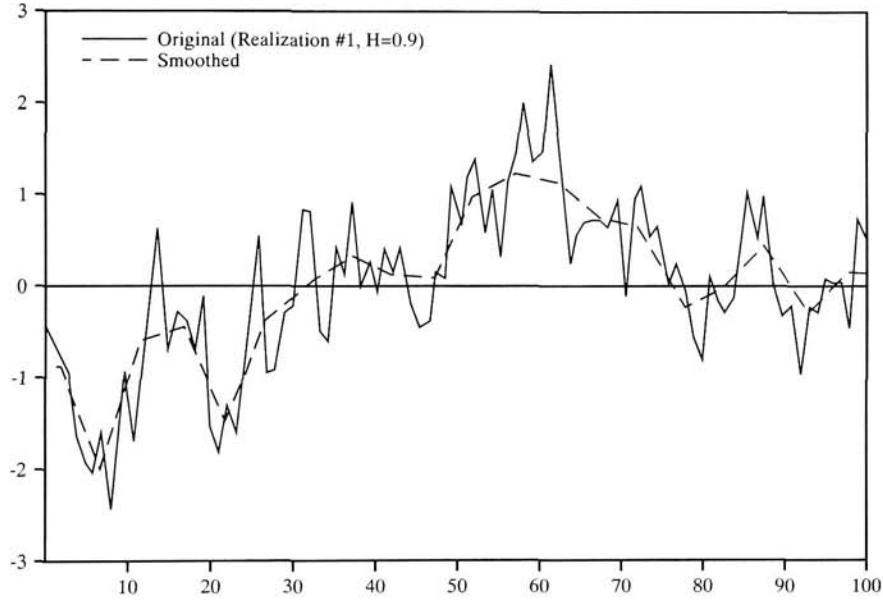


Figure 4.10. Smoothed and original traces for Realization #1.

The average curve captures the low frequency information of the original curve, but the smoothing process destroys the high frequency information. As described in Chapter 2 and Appendix A2.3, a Fourier transform of a curve produces an amplitude array and a phase array. All the information of a curve is captured in these two arrays. The resulting arrays for the original and smoothed curves are compared in Figures 4.11 and 4.12. Figure 4.11 contains the amplitude arrays.

Figure 4.12 contains the phase arrays. For a trace of N real numbers, the arrays are symmetric about the center point N/2+1. The low frequency information from the smoothed curve is very similar to the low frequency information from the original curve. Significant variations occur in the middle and high frequency values.

Appendix A3.3 pointed out that a smoothing filter multiplies the amplitude by a constant value at low frequencies (less than the frequency corresponding to the filter length). For a most accurate analysis, this multiple could be calculated and used to get even closer agreement between the low frequency amplitudes of the smoothed and the original curves in Figure 4.11. For the cases studied here, the results were considered accurate enough without this correction.

We now generate a curve with the same mean, standard deviation, and H value as the original curve. This curve would be a realization of the true porosity curve if we had no constraining information other than the fGn nature of the true curve. Figure 4.13 shows the generated curve as a dashed line. The character of the curve is similar to the original curve (the solid line in Figure 4.13), but the highs and lows of the realization do not occur at the same locations as those in the original curve.

To condition the new curve to the average of the original curve, we can extract the low frequency information captured in the smoothed curve and substitute it for the low frequency information in our realization. First, take the Fourier transform of the realization. Second, substitute the low frequency amplitudes and phases from the smoothed curve in Figure 4.10 into the corresponding low frequency positions in the arrays for the realization. The smooth curve has 20 independent values, because we averaged the original curve (100 points) over five-point intervals. In this case, we substitute values of amplitude and phase values from positions 2 to 10 and 92 to 100 inclusive in the arrays. In general, for a fine-scale distribution of N grids and a coarse grid of N_s grids, we would substitute values from 2 to $N_s/2$ and $N-N_s/2-1$ to N in the arrays. Third, take the inverse Fourier transform of the modified array.

The modified realization is shown as the dashed curve in Figure 4.14. It agrees with the true curve (solid line) except for high frequency noise that the smoothed curve does not capture. As a further verification, the same smoothing process is performed on the modified realization and compared to the true curve in

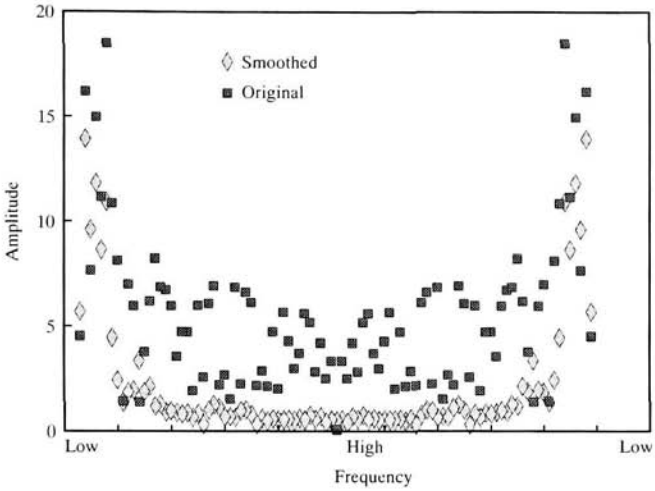


Figure 4.11. Amplitude of original and smoothed traces for Realization #1.

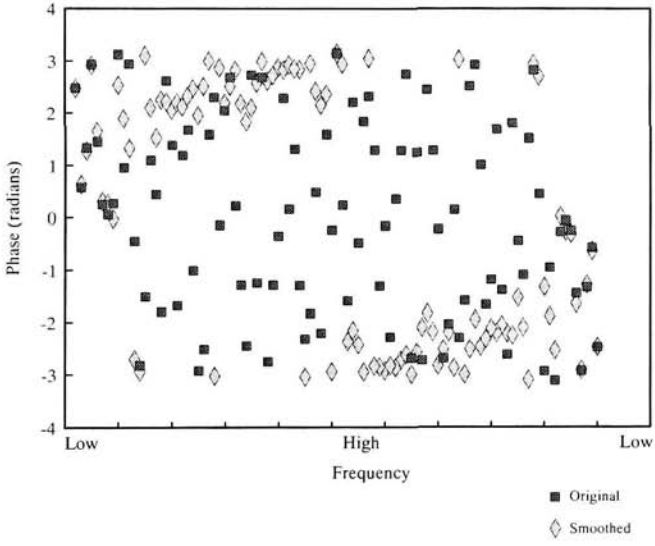


Figure 4.12. Phase of original and smoothed traces for Realization #1.

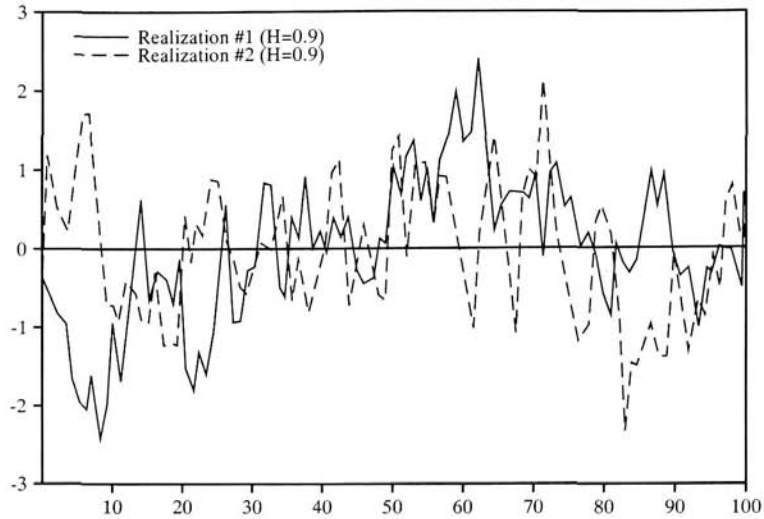


Figure 4.13. Realizations of one dimensional traces of fGn.

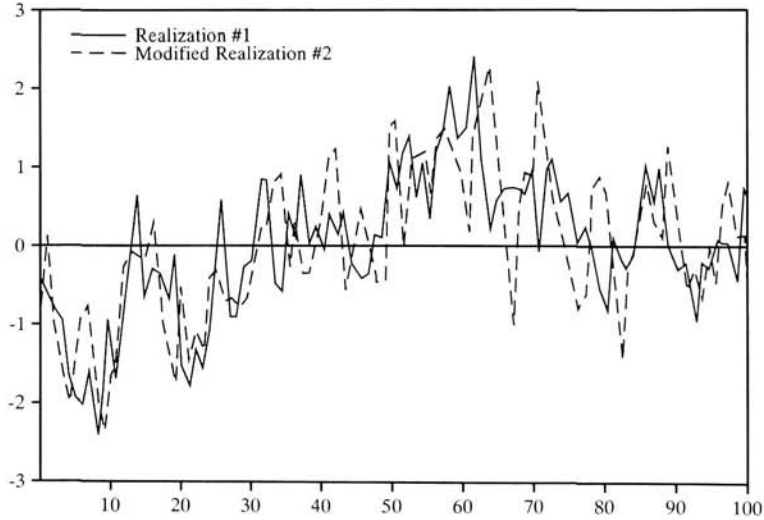


Figure 4.14. Realization #2 with low frequency information from Realization #1.

Figure 4.15. The smoothed curves match for the low frequency information. The mean of the modified realization and the original curve are both zero. The standard deviation and H agree within 5 percent.

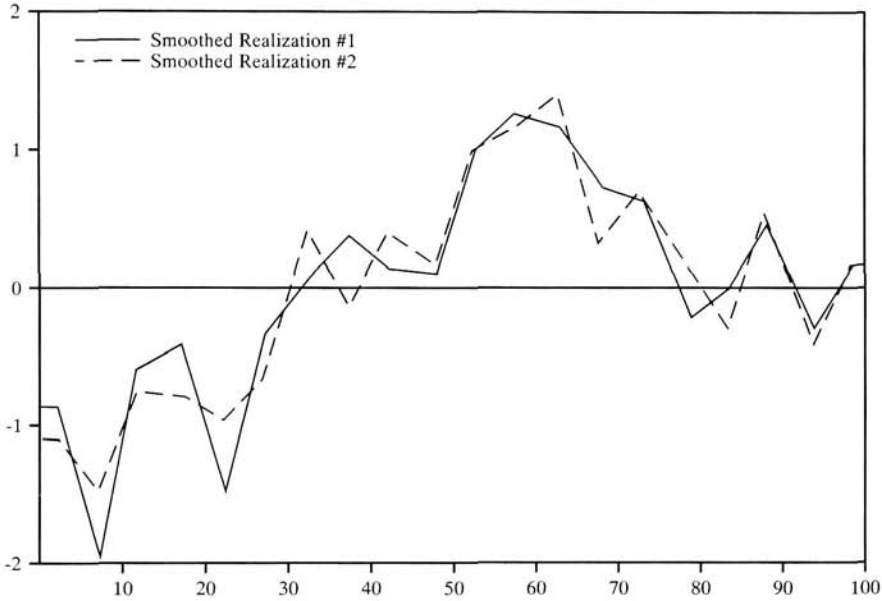


Figure 4.15. Smoothed traces of Realization #1 and modified Realization #2.

4.9.2 *Two Dimensional Example*

The above method can be extended to 2D distributions. Plates 5a and 5b are two realizations of a vertical cross section (512 x 64) generated by the methods described in Section 4.4. The only difference between realizations is the random number seed used. Realization 1 shows local porosity development in the left center of the cross section, while a similar porosity development occurs on the right side of realization 2.

We will use realization 1 to represent the "true" distribution and realization 2 to represent an approximation to the true distribution. Note that the second distribution is a poor approximation of the location of the large porosity development in the left center of the true cross section.

We apply a filtering process that is similar to the 1D case to realization 1 to get a smoothed image. A coarse grid (32 by 4) is used. Each coarse grid contains 16 x 16 fine grids. Arithmetic averaging over each 16 x 16 array provides the average value assigned to the center of the coarse grid. Then, values are reassigned to a fine grid by using bilinear interpolation. The result is the smoothed image of realization 1 shown in Plate 5c. This smoothed image represents a porosity map derived from seismic data or geological input.

By using Fourier transforms, we transplant the low frequency information from Plate 5c (seismic data) into the fine grid porosity, Plate 5b. In the modified realization (Plate 5d), the larger areas of porosity development agree with realization 1. The final result also captures the layering.

Unresolved Issues

1. Two examples in this chapter are hypothetical cases--the inclusion of data of different scales and the filling in of missing fGn data. Engineering application of these procedures is necessary to fully test these procedures.

2. All the procedures presented in this chapter for generating reservoir properties use only vertical wells. The procedures should be modified to include the use horizontal and highly deviated well data.

3. The procedure presented for generating reservoir properties does not match well data exactly. The noise that is added to the distributions is only statistically tied to the well data. A better procedure would be to use the phase arrays to condition the generated distributions to the well data directly.

4. The use of "simulation" to produce missing data values (see Appendix 4.1) can be extended to include layering information. This can be done by using the 2D covariance model of fGn (Equation A3.55) and adding an offset well to the set of known data.

5. An extension of "simulation" can use indicator kriging[57] to include seismic data in generating realizations. This same extension could be carried over into fractal distributions in the framework of multifractals.[10]

6. Can α be extracted from the log traces and β from the differences in the log traces taken in pairs, as suggested in Section 4.4.1? Will fGn result if the intermittency exponents in α and β are different? Is the assumption that the intermittency exponents for α and β equal a valid assumption?

APPENDIX A4

This appendix is divided into five parts. The first describes a technique for filling in missing data in log or core traces. The second, third, and fourth describe the generation of fGn cross sections, fBm cross sections, and 3D fGn distributions, respectively. The fifth shows the mathematics of one method of combining large and small scale data in 2D.

A4.1 Filling "Holes" in Log or Core Traces

We will begin with an fGn trace that has missing data (e.g., Figure A4.1b). This might be, for example, a set of core measurements where not all of the core was recovered. Suppose we believe the entire interval is fGn with known mean, standard deviation, and H (e.g., from an ensemble of many wells in the area). In this section, we will show one way the missing data can be "filled in" consistent with what is known.

We will use the geostatistical procedure known as "simulation."[57] This procedure produces a good result and is also a demonstration of using fGn within the framework of "standard" geostatistical tools.

Simulation begins with kriging. Kriging calculates an expected mean and standard deviation at an unknown data location from surrounding "known" data. A Gaussian distribution can be defined from the calculated mean and standard deviation. Simulation proceeds by making a random draw from this Gaussian distribution. The value drawn is placed at the location of the missing data value. This new data value is added to the other "known" data, and the process is repeated at another unknown data location. This process is repeated until all the unknown data locations are filled. A summary of the mathematics of this process follows.

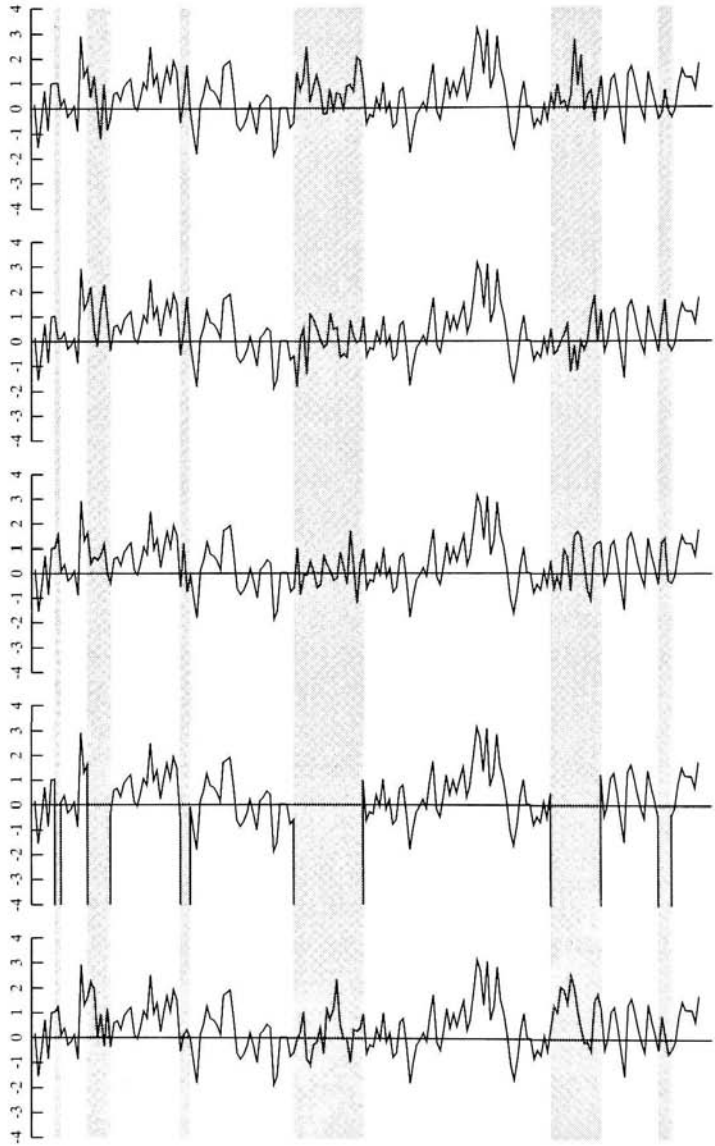


Figure A4.1. Filling "holes" in fGn data. (a) original trace, (b) original trace minus missing data, (c) Realization #1 with synthetic missing data filled in, (d) Realization #2 with synthetic missing data filled in, and (e) Realization #3 with synthetic missing data filled in.

Let the circles in Figure A4.2 represent the locations where data values are known. (Here we have illustrated the problem by showing a 2D data set. The processes are applicable to 1, 2, or 3D.) We wish to determine the mean and standard deviation at another point, marked with an x in Figure A4.2.

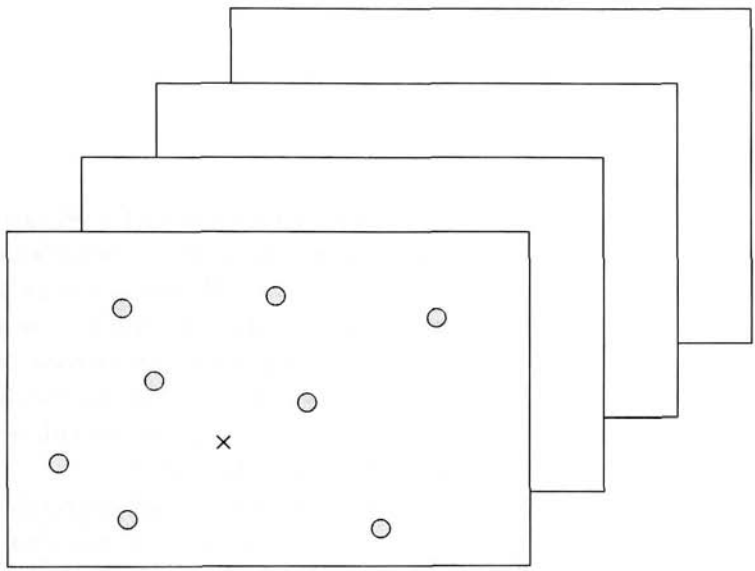


Figure A4.2. Example geometric arrangement of known (circles) and unknown (x) data locations from an ensemble of all possible distribution.

In reality, of course, the unknown value at x has only one value, but we will first consider all possible distributions consistent with our fGn model with a specific geometric arrangement of known and unknown points. From this, we will derive a set of general constraints. Next, we will generate realizations consistent with the geometric constraints and a specific set of known data values.

We express the expected value for realization k at x, $g_0'(k)$, as a weighted sum of the known data values at the circles, $g_j(k)$. That is,

$$g_0'(k) = \sum_{j=1}^{N} w_j \, g_j(k) \qquad \textbf{(A4.1)}$$

where

w_j = weight of the j^{th} point

and

N = number of known points

Notice that we assume that the weights, w_j, do not depend on the particular realization, k. The rationale behind this assumption is that closer points should have larger influence (i.e., larger weights) than farther points. How close is "close" should depend on the range of the variogram (or the range of the covariance). The spatial location and the covariance model should be sufficient to define these weights independent of the specific data values at the circles. Once the weights are calculated, we will generate a set of realizations that are consistent with both the geometric arrangement of the points and the specific data values.

The assumptions we will make are as follows. First, we will assume that, over all possible realizations, the average value at x is the same as the average value at the circle locations. This statement is a "local" stationarity assumption for the mean. Second, we will assume that the standard deviation of all possible values at the unknown data location is a minimum. Finally, we will assume that the ensemble is fGn. This last assumption is stronger than necessary for the general application of simulation. The relaxation of it can produce a rich variety of other distributions,[20] but we will restrict ourselves here to fGn.

Now, we will invoke each of our assumptions to produce a set of equations that can be solved for the w_j values. We define D(k) as the difference in the true value at x, $g_0(k)$, and the expected value at x, $g_0'(k)$. That is,

$$D(k) = g_0(k) - \sum_{j=1}^{N} w_j g_j(k) \qquad \textbf{(A4.2)}$$

The ensemble average of D(k) is

$$D = \langle D(k) \rangle = \frac{1}{N_e} \sum_{k=1}^{N_e} D(k) \tag{A4.3}$$

where N_e equals the number of realizations in the ensemble

Then,

$$D = \langle D(k) \rangle = \langle g_0(k) - \sum_{j=1}^{N} w_j g_j(k) \rangle \tag{A4.4}$$

We cannot hope that an expectation value, $g_0'(k)$, will always be exactly equal the true value, $g_0(k)$. But, over many ensembles, we wish to make the average expected value, $< g_0'(k) >$, equal to the average correct value, $\langle g_0(k) \rangle$. This can be done by requiring $D=0.0$, that is,

$$D = \langle g_0(k) \rangle - \sum_{j=1}^{N} w_j \langle g_j(k) \rangle = 0.0 \tag{A4.5}$$

Our first assumption is that the average of the values at the known locations, $\langle g_j(k) \rangle$, is the same as the average at the unknown location, $\langle g_0(k) \rangle$, or

$$\langle g_0(k) \rangle = \langle g_j(k) \rangle \tag{A4.6}$$

So, except for the special case of $\langle g_0(k) \rangle = 0.0$, Equations A4.5 and A4.6 produce the following constraint on the weights:

$$\sum_{j=1}^{N} w_j = 1 \tag{A4.7}$$

To make the standard deviation of the missing data location a minimum, we first express the standard deviation of D(k), σ_D, in terms of the weights, w_j. Now

$$\sigma_D^2 = \langle (D(k) - \langle D(k) \rangle)^2 \rangle \tag{A4.8}$$

After a bit of algebra,[17] Equation A4.8 yields

$$\sigma_D^2 = \sigma^2 + \sum_{h=1}^{N} \sum_{j=1}^{N} w_h w_j \mathrm{Cov}(g_h, g_j) - 2 \sum_{j=1}^{N} w_j \mathrm{Cov}(g_j, g_0) \tag{A4.9}$$

where

$$\sigma^2 = \langle (g_0(k) - \langle g_0(k) \rangle)^2 \rangle \tag{A4.10}$$

and the covariances are

$$\mathrm{Cov}(g_h, g_j) = \langle (g_h(k) - \langle g_h(k) \rangle)(g_j(k) - \langle g_j(k) \rangle) \rangle \tag{A4.11}$$

$$\mathrm{Cov}(g_j, g_0) = \langle (g_j(k) - \langle g_j(k) \rangle)(g_0(k) - \langle g_0(k) \rangle) \rangle \tag{A4.12}$$

We wish to minimize σ_D^2 in Equation A4.9 subject to the constraint Equation A4.7. Using a Lagrange multiplier, λ, we write

$$\sigma_D^2 = \sigma^2 + \sum_{h=1}^{N} \sum_{j=1}^{N} w_h w_j \mathrm{Cov}(g_h, g_j) - 2 \sum_{j=1}^{N} w_j \mathrm{Cov}(g_j, g_0) + 2\lambda \left(\sum_{j=1}^{N} w_j - 1 \right) \tag{A4.13}$$

The minimum is calculated by setting the partial derivative with respect to each w_j in Equation A4.13 to zero. This yields a set of N linear equations. Adding Equation A4.7 to the set of N equations yields N+1 equations with N+1 unknowns. (The unknowns are the N w_j values and λ.) The equations to be solved are the following:

$$\frac{\partial \sigma_D^2}{\partial w_j} = 0 \quad \text{for} \quad j = 1, 2, 3, \ldots, N \tag{A4.14}$$

and

$$\sum_{j=1}^{N} w_j = 1 \tag{A4.15}$$

The resulting set of equations can be expressed in matrix notation.

$$\mathbf{C} \cdot \boldsymbol{w} = \boldsymbol{D} \tag{A4.16}$$

where

$$\mathbf{C} = \begin{pmatrix} Cov(g_1,g_1) & Cov(g_1,g_2) & \dots & Cov(g_1,g_N) & 1 \\ Cov(g_2,g_1) & Cov(g_2,g_2) & \dots & Cov(g_2,g_N) & 1 \\ \dots & \dots & \dots & \dots & \dots \\ Cov(g_N,g_1) & Cov(g_N,g_2) & \dots & Cov(g_N,g_N) & 1 \\ 1 & 1 & 1 & \dots & 0 \end{pmatrix} \tag{A4.17}$$

$$\mathbf{w} = \begin{pmatrix} w_1 \\ w_2 \\ \dots \\ w_N \\ \lambda \end{pmatrix} \tag{A4.18}$$

$$\mathbf{D} = \begin{pmatrix} Cov(g_1,g_0) \\ Cov(g_2,g_0) \\ \dots \\ Cov(g_N,g_0) \\ 1 \end{pmatrix} \tag{A4.19}$$

For fGn, the covariance, $Cov(g_k,g_j)$, is only a function of separation of the data locations. For 1D, the fGn covariance equation is Equation A2.92. For 2D, it is Equation A3.55. If the mean, standard deviation, H, and the separation of the points are known, these equations uniquely determine the **C** and **D** matrices and hence the w_j values.

The solution is

$$\mathbf{w} = \boldsymbol{C}^{-1} \cdot \boldsymbol{D} \tag{A4.20}$$

The resulting N w_j values can be used to estimate $g_0'(k)$ from Equation A4.1 and the known values $g_j(k)$.

The **C** matrix can become large for large data sets. Calculating $\mathbf{C}^{-1}$ for large matrices can require too much computational time. To minimize computational time, often a set range, R, is chosen. Only known data values within this range are considered "known." The range must be large enough to include a number of known values and small enough to make the inversion of the **C** matrix reasonably rapid.

The standard deviation, σ_D, can be calculated by substituting the w_j values directly into Equation A4.9. A quicker way to calculate σ_D is to note that (see Reference No. 17)

$$\sigma_D^2 = \sigma^2 - \mathbf{w} \cdot \boldsymbol{D} \tag{A4.21}$$

We have now defined an algorithm to generate the set of weights, w_j. The w_j values are geometric constraints, because they depend only on the covariance and the location of the points. They do not depend on the explicit values of the data. We now wish to generate realizations that are consistent with our specific data as well as the weights. To do this, we calculate the estimate at x, g_0', as

$$g_0' = \sum_{j=1}^{N} w_j g_j \tag{A4.22}$$

where g_j are the specific data values at the circles. g_0' is our most probable estimate of the data value at x. For Gaussian distributions, the most probable and the mean values are the same. To calculate the standard deviation of the difference in g_0' and

the true value of $g_{0,\ \sigma_D}$, we use Equation A4.9. To evaluate σ in Equation A4.9, we assume σ is the same as the standard deviation of the entire distribution.

Next, a random number generator is used to choose a specific value from a Gaussian distribution with mean, g_0', and standard deviation, σ_D. This can be done using commercial software, e.g., IMSL[23] or the technique suggested by Voss.[11] The chosen value is assigned to the x location and is added to the set of known values. Thus, the x location becomes a circle. Another location where the value is unknown (i.e., another location, x) is chosen, and the process is repeated until all of the missing data locations are filled.

The entire process of filling in all the missing data values can be repeated any number of times. Each time, a different realization will be produced. Figures A4.1 c, d, and e show three different realizations produced by this method. For these realization, the following parameters are used:

$\bar{g}$ = 0.20 (the average value of the distribution)
σ = 0.05 (the standard deviation of the distribution)
H = 0.8 (intermittency exponent)
R = 1/5 N (range of the values used for the C matrix)
N = 200 (number of points in the entire sequence)

For the realizations, σ and $\bar{g}$ bar are calculated from the known values in Figure A4.1b. H is assumed to be known as 0.8. Note that from any single realization, it is very difficult to tell where data are added.

A4.2 fGn Cross Section Generation

To build fGn cross sections, the left and right edges of a cross section, $A_{j,\ell}$, are filled with well log data ($W_{L,\ell}$ for the left edge well log data and $W_{R,\ell}$ for the right edge well log data):

$$A_{1,\ell} = W_{L,\ell} \text{ and } A_{N_X,\ell} = W_{R,\ell}, \text{ for } \ell = 1 \text{ through } N_z \qquad \text{(A4.23)}$$

where

N_x = number of grids in the x direction (horizontal)
N_z = number of grids in the z direction (vertical)

To calculate a layered image, we either use linear interpolation between the known edges or assign a constant value at each depth. If linear interpolation is chosen, then

$$A_{j,\ell} = A_{1,\ell} + \frac{(A_{N_x,\ell} - A_{1,\ell})}{(N_x - 1)} \times (j - 1) \tag{A4.24}$$

If we choose a constant value at each depth, then

$$A_{j,\ell} = (A_{1,\ell} + A_{N_x,\ell})/2 \tag{A4.25}$$

Next, the amount of fGn noise to be added is calculated. It is assumed that the well logs have been put on depth, and therefore, differences between the left and right log values (other than an overall trend) are the result of noise. To calculate the amount of noise, first any overall trends are removed by subtracting the average of each log from each log value. This gives

$$\phi_{L,\ell} = W_{L,\ell} - \overline{W}_L \tag{A4.26}$$

where

$$\overline{W}_L = \frac{1}{N_Z} \sum_{\ell=1}^{N_z} W_{L,\ell} \tag{A4.27}$$

and

$$\phi_{R,\ell} = W_{R,\ell} - \overline{W}_R \tag{A4.28}$$

where

$$\overline{W}_R = \frac{1}{N_z} \sum_{\ell=1}^{N_z} W_{R,\ell} \tag{A4.29}$$

The variance, V, of the difference in the two logs without trends is calculated,

$$V = \frac{1}{2\,N_z} \sum_{\ell=1}^{N_z} (\phi_{L,\ell} - \phi_{R,\ell})^2 \tag{A4.30}$$

V corresponds to the variogram at lag N_x of the noise that needs to be added.

A noise array is now constructed from the appropriate spectral density, S, of fGn noise. The spectral density of the noise to be added is

$$S(\omega_x,\omega_z) = \frac{A_{SD}}{[s_x\ \omega_x^2 + \omega_z^2]^\beta} \tag{A4.31}$$

with

$$\omega_x = 2\pi f_x \tag{A4.32}$$

$$\omega_z = 2\pi f_z \tag{A4.33}$$

$$\beta = H \quad \text{where} \quad 0.5 < H < 1 \tag{A4.34}$$

Here f_x and f_z are the x and z frequencies in Fourier space. s_x can be used to produce anisotropic noise, but in all of the examples in this book, s_x has been set equal to 1.0 so that all noise added is isotropic.

The noise array, $g_{j,\ell}$, is generated with H determined from a fractal analysis of the well logs. Using the methods in Appendix A3.5, we evaluated the noise array, $g_{j,\ell}$, as a discrete inverse Fourier transform. Repeating Equation A3.35, we write

$$g_{j,\ell} = \sum_{m=1}^{N_x} \sum_{p=1}^{N_z} G_{m,p} e^{2\pi i[(j-1)(m-1)/N_x + (\ell-1)(p-1)/N_z]} \qquad \textbf{(A4.35)}$$

Here,

$$G_{m,p} = R_{m,p} e^{i\theta_{m,p}} \qquad \textbf{(A4.36)}$$

To evaluate coefficients $R_{m,p}$, we write a discrete form of Equation A4.31 as

$$R^2_{m,p} = \frac{1}{\left[s_x (m-1)^2 + E_z(p-1)^2\right]^\beta} \qquad \textbf{(A4.37)}$$

$$\text{for } \frac{N_x}{2} \geq m - 1 \geq 0 \quad \& \quad \frac{N_z}{2} \geq p - 1 \geq 0$$

$$E_z = \left(\frac{L_x N_x}{L_z N_z}\right)^2$$

except

$$R_{1,1} = 0.0 \qquad \textbf{(A4.38)}$$

In these equations, the size of the $g_{j,\ell}$ array to be generated is L_xN_x by L_zN_z, where the number of grids in each direction is N_x and N_z and the size of each grid is L_x by L_z.

We now invoke the symmetry and phase requirements that have been used previously for the core photos in Appendix A3.5 (Equations A3.42 through A3.49). These equations assign values throughout the remaining $G_{m,p}$ array.

For a most accurate distribution, only one-half to one-fourth of the original distribution should be retained,[11] but for this work, the entire array is used.

Note $g_{j,\ell}$ only needs to be generated to agree with Equation A4.31 to within a multiplicative constant, because $g_{j,\ell}$ will be rescaled to the well data before it is used. Equation A4.38 makes the average of the noise array to be equal to zero.

The desire now is to adjust the variogram of lag N_x of the generated noise array to be equal to V. An indirect approach to this produces a more stable solution than a direct one. A direct approach would be a linear scaling that would set the variogram of lag N_x equal to V, but if it is assumed that the variogram of the generated noise array has reached its sill for a lag equal to the well separation, then the variance of the noise array, V_n, is equal to the variogram of lag N_x. Since there are many more values in the variance than in the variogram of lag N_x, the variance produces a more stable estimate of the variogram of lag N_x. Although, strictly speaking, fGn never reaches a sill, the rate of change of fGn asymptotically goes to zero. For large lags, fGn changes very slowly. The variance of the array is therefore used for scaling the noise. The variance of the noise array, V_n, is calculated as follows:

$$V_n = \frac{1}{N_x N_z} \sum_{j=1}^{N_x} \sum_{\ell=1}^{N_z} (g_{j,\ell} - \bar{g})^2 \qquad \textbf{(A4.39)}$$

with

$$\bar{g} = \frac{1}{N_x N_z} \sum_{j=1}^{N_x} \sum_{\ell=1}^{N_z} g_{j,\ell} \qquad \textbf{(A4.40)}$$

Since the average value in the generated noise array is zero, the following linear scaling gives an array, $F_{j,\ell}$, with the desired amount of noise and zero mean (see Appendix A2.6):

$$F_{j,\ell} = g_{j,\ell} \sqrt{V/V_n} \qquad \textbf{(A4.41)}$$

The scaled noise array, $F_{j,\ell}$, is then added to the layered array, $A_{j,\ell}$:

$$B_{j,\ell} = A_{j,\ell} + s_{noise} F_{j,\ell} \qquad \textbf{(A4.42)}$$

By adjusting the factor s_{noise}, we can vary the amount of noise added to the layered array. In all the generated distributions in this book, s_{noise} is set to one.

Choosing linear interpolation, Equation A4.24, to form the layered array introduces trends in any horizontal trace. These trends alter the statistical structure of the fGn in any horizontal trace. The alternative of assigning a constant value at each depth (Equation A4.25) introduces no undesired trends. The sum of the layer cake array and the fGn noise array would still have fGn statistical structure. However, the resulting cross section would not honor the input well log values at the edges as well as the procedure with linear interpolation. We use linear interpolation unless noted otherwise.

With linear interpolation, $B_{j,\ell}$ in Equation A4.42 has a total variance greater than the variance of the original log data. The average of the resulting array, $B_{j,\ell}$, is unchanged from the linear interpolation array, since the average of the noise array, $F_{j,\ell}$, is zero.

One final linear scaling is made. Its purpose is to adjust the variance and average of the edges of the array to agree with the variance and average of the log data. This is done so that the edges of the final cross section will closely match the input well log data.

The linear scaling that gives the desired variance and average is the following:

$$C_{j,\ell} = \left[\frac{V_W}{V_H}\right]^{1/2} (B_{j,\ell} - \overline{B}_H) + \overline{B}_W \qquad \text{(A4.43)}$$

where

$$\overline{B}_W = (\overline{W}_L + \overline{W}_R)/2 \qquad \text{(A4.44)}$$

$$\overline{B}_H = [\,\overline{B}_L + \overline{B}_R\,]/2 \qquad \text{(A4.45)}$$

$$V_W = \left[\frac{1}{N_z}\sum_{\ell=1}^{N_z}(W_{L,\ell} - \overline{W}_L)^2 + \frac{1}{N_z}\sum_{\ell=1}^{N_z}(W_{R,\ell} - \overline{W}_R)^2\right]/2 \qquad \text{(A4.46)}$$

$$V_H = \left[\frac{1}{N_Z} \sum_{\ell=1}^{N_Z} (B_{1,\ell} - \bar{B}_L)^2 + \frac{1}{N_Z} \sum_{\ell=1}^{N_Z} (B_{N_x,\ell} - \bar{B}_R)^2 \right] / 2 \qquad \text{(A4.47)}$$

$$\bar{B}_L = \frac{1}{N_z} \sum_{\ell=1}^{N_z} B_{1,\ell} \qquad \text{(A4.48)}$$

$$\bar{B}_R = \frac{1}{N_z} \sum_{\ell=1}^{N_z} B_{N_x,\ell} \qquad \text{(A4.49)}$$

Equations A4.27 and A4.29 are used to evalute $\overline{W}_L$ and $\overline{W}_R$ in Equation A4.46.

The variance and average value of the edges of the final array, $C_{j,\ell}$, are now the same as the variance, V_W, and average, $\bar{B}_W$, of the input well log data from both wells. As a result of scaling the edges rather than the overall array average to the well log data, the overall average and variance of the final array may be greater than or less than the average and variance of the input well log data for a particular realization. This scaling is used in Figures 4.1 through 4.3. The figures are generated using porosities measured by well logs and square grids of size 2 ft by 2 ft.

An alternate linear scaling can be used to set the overall variance and mean of the cross section to match the corresponding measures of the well data. Then, instead of Equation A4.43, the final linear scaling is

$$C_{j,\ell} = \left[\frac{V_W}{V_A} \right]^{1/2} (B_{j,\ell} - \bar{B}_A) + \bar{B}_W \qquad \text{(A4.50)}$$

where

$$V_A = \frac{1}{N_x N_z} \sum_{j=1}^{N_x} \sum_{\ell=1}^{N_z} (B_{j,\ell} - \bar{B}_A)^2 \qquad \text{(A4.51)}$$

$$\bar{B}_A = \frac{1}{N_x N_z} \sum_{j=1}^{N_x} \sum_{\ell=1}^{N_z} B_{j,\ell} \qquad \text{(A4.52)}$$

A4.3 fBm Cross Section Generation

Hewett and Behrens[51] describe the method to build fBm cross sections. Following their method, the left and right edges of a cross section, $A_{j,\ell}$, are filled with well log data as shown in Equation A4.23. A linear interpolation is made between the edges of the array as shown in Equation A4.24.

fBm noise is now generated. This noise is actually fBm in the horizontal direction and fGn in the vertical direction. The noise is generated from an equation for the spectral density similar to the way fGn noise was generated. As a discrete inverse Fourier transform, the noise array, $g_{j,\ell}$, is generated in the same manner as fGn (Equations A4.35 through A4.38) except

$$R^2_{m,p} = \frac{1}{[(m-1)^2 + \hat{E}_z(p-1)^2]^{\beta}\,(m-1)^2} \qquad \text{(A4.53)}$$

with

$$\hat{E}_z = N_x/N_z \qquad \text{(A4.54)}$$

and

$$\beta = H + 1 \quad \text{where} \quad 0.5 < H < 1 \qquad \text{(A4.55)}$$

Note that this aspect ratio, $\hat{E}_z$, is based on the number of grids, not the physical dimensions of the cross section. As with fGn, $g_{j,\ell}$ only needs to be generated to within a multiplicative constant, because $g_{j,\ell}$ will be rescaled to the well data before it is used.

The noise array, $g_{j,\ell}$, is now scaled so that the total variance of the noise array, V_n, matches the variance of the well logs, V_w. This is done by the following linear scaling of the noise array:

$$F_{j,\ell} = \left[\frac{V_W}{V_n}\right]^{1/2} g_{j,\ell} \tag{A4.56}$$

where

$$V_n = \frac{1}{N_x N_z} \sum_{j=1}^{N_x} \sum_{\ell=1}^{N_z} (g_{j,\ell} - \bar{g})^2 \tag{A4.57}$$

$$\bar{g} = \frac{1}{N_x N_z} \sum_{j=1}^{N_x} \sum_{\ell=1}^{N_z} g_{j,\ell} \tag{A4.58}$$

and V_W is defined by Equation A4.46.

The next step is to add the noise array, $g_{j,\ell}$, to the linear interpolation array, $A_{j,\ell}$, in such a way that the edges of the final array are similar to the original log data. This is done by a procedure outlined by Journel and Huijbregts.[20] First, a linear interpolation between the edges of the noise array is made,

$$H_{j,\ell} = F_{1,\ell} + \frac{(F_{N_x,\ell} - F_{1,\ell})}{(N_x - 1)} \times (j - 1) \tag{A4.59}$$

This array, $H_{j,\ell}$, is subtracted from the original noise array so that the edges of the new noise array are zero:

$$\psi_{j,\ell} = F_{j,\ell} - H_{j,\ell} \tag{A4.60}$$

The resulting array, $\psi_{j,\ell}$, is now added to the linear interpolation array, $A_{j,\ell}$,

$$B_{j,\ell} = A_{j,\ell} + \psi_{j,\ell} \tag{A4.61}$$

As with fGn, one final linear scaling is made. The purpose is to adjust the variance of the final array to the variance of the well logs while preserving the average of the

edges of the array to the average of the well log data. The linear scaling which gives the desired result is the following:

$$C_{j,\ell} = \left[\frac{V_W}{V_H}\right]^{1/2} (B_{j,\ell} - \overline{B}_H) + \overline{B}_W \qquad \textbf{(A4.62)}$$

where

$$V_H = \frac{1}{N_x N_z} \sum_{j=1}^{N_x} \sum_{\ell=1}^{N_z} (B_{j,\ell} - \overline{B}_A)^2 \qquad \textbf{(A4.63)}$$

$$\overline{B}_A = \frac{1}{N_x N_z} \sum_{j=1}^{N_x} \sum_{\ell=1}^{N_z} B_{j,\ell} \qquad \textbf{(A4.64)}$$

$$\overline{B}_H = \left[\frac{1}{N_z} \sum_{\ell=1}^{N_z} B_{1,\ell} + \frac{1}{N_z} \sum_{\ell=1}^{N_z} B_{N_x,\ell}\right]/2 \qquad \textbf{(A4.65)}$$

$$\overline{B}_W = \left[\, \overline{W}_L + \overline{W}_R \,\right]/2 \qquad \textbf{(A4.66)}$$

and V_W is defined by Equation A4.46. $\overline{W}_L$ and $\overline{W}_R$ are defined by Equations A4.27 and A4.29, respectively.

For the fBm cross section in Figure 4.5, the grids are generated from porosity log data at the scale they are to be used (here 2 ft by 16 ft) in reservoir simulations (Chapter 6).

A4.4 The Generation of 3D Distributions

A 3D fGn distribution is generated similar to a 2D fGn distribution (Appendix A4.2). First, the program interpolates between well data. Second, fGn is generated and scaled based on the differences between the well log data sets. Third, fGn is added to the interpolated well data. Lastly, the final array is scaled to the well data.

First, a trilinear interpolation of the well data, $W_{q,\ell}$, is made at each depth. The results are assigned to the grid values, $A_{j,k,\ell}$. This is done by locating each grid point within a triangle of wells. Once it is determined that the point to be evaluated is interior to three wells, the value of the point is assigned as a weighted average of the three bounding well values. The weights are assigned proportional to the relative area of each triangular section opposite the corresponding well location. (See Figure A4.3).

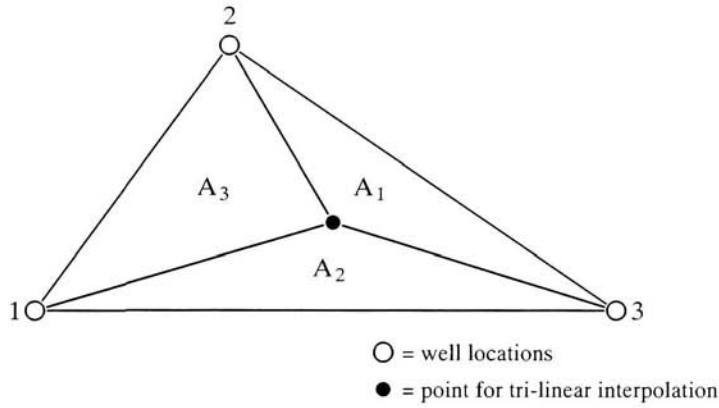


Figure A4.3. Weighting scheme for trilinear interpolation.

The calculation of the weights and whether or not a particular point is bounded by 3 wells is based on cross products. The area of a triangle, A_Δ, bounded by two vectors, $\vec{v}_1$ and $\vec{v}_2$, is

$$A_\Delta = (\vec{v}_1 \times \vec{v}_2)/2 \qquad \textbf{(A4.67)}$$

If the point (x,y) is the origin and (x_1,y_1) and (x_2,y_2) are the end points of $\vec{v}_1$ and $\vec{v}_2$, respectively,

$$A_\Delta = [(x_1 - x)(y_2 - y) - (x_2 - x)(y_1 - y)]/2 \qquad \textbf{(A4.68)}$$

Let q_1, q_2, and q_3 be the three wells that define the triangle that contains the grid point, (x,y).

If the coordinates of the three bounding wells are (x_1,y_1), (x_2,y_2), and (x_3,y_3), then the area of the entire triangle bounded by the three wells is

$$D_\Delta = [(x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1)]/2 \qquad \textbf{(A4.69)}$$

The weights needed, A_1, A_2, and A_3, are calculated as follows:

$$w_1 = [(x_1 - x)(y_2 - y) - (x_2 - x)(y_1 - y)]/D_\Delta \qquad \textbf{(A4.70)}$$

$$w_2 = [(x_2 - x)(y_3 - y) - (x_3 - x)(y_2 - y)]/D_\Delta \qquad \textbf{(A4.71)}$$

and

$$w_3 = [(x_3 - x)(y_1 - y) - (x_1 - x)(y_3 - y)]/D_\Delta \qquad \textbf{(A4.72)}$$

The value assigned to the grid point, $A_{j,k,\ell}$, is

$$A_{j,k,\ell} = w_1 W_{q_1,\ell} + w_2 W_{q_2,\ell} + w_3 W_{q_3,\ell} \qquad \textbf{(A4.73)}$$

for all values of j and k. $W_{q_1,\ell}$ is the well data from well number q_1 at depth index ℓ. In these calculations, the x and y values are all in well units, e.g., feet.

If the point is interior to the triangle, w_1, w_2, and w_3 are all positive. In order to allow for round off, this is checked by insuring that each w_i is greater than or equal to ϵ, where

$$\epsilon = \frac{-1}{10N_xN_y} \tag{A4.74}$$

Here, N_x and N_y are the number of grids to be generated in the x and y directions.

The amount of fGn noise to be added is calculated as follows. It is assumed that the well logs have been put on depth, and therefore, differences between the well log values (other than overall trends) are the result of noise. To calculate the amount of noise, first any overall trends are removed by subtracting the average of each log from each log value. This gives:

$$\phi_{q,\ell} = W_{q,\ell} - \overline{W}_q \tag{A4.75}$$

where

$$\overline{W}_q = \frac{1}{N_z} \sum_{\ell=1}^{N_z} W_{q,\ell} \tag{A4.76}$$

Here,

N_z = number of grids in the z direction.

The number of possible pairs, N_p, of N_W wells taken two at a time is

$$N_p = \frac{N_W!}{2\,(N_W - 2)!} \tag{A4.77}$$

so that the average variance of all well pairs, V, can be expressed as

$$V = \frac{1}{2N_zN_p} \sum_{q=1}^{N_p} \sum_{\ell=1}^{N_z} (\phi_{q,\ell} - \phi_{q+1,\ell})^2 \tag{A4.78}$$

Here this sum is taken only over well pairs that are different. If two wells are identical, the resulting zero variance for that well pair is not averaged with the nonzero values. This is done so that a set of 4 wells with three of them being

identical will produce the same value of V that is calculated by the 2D procedure for a cross section between the two different wells (Appendix A4.2). V corresponds to the variogram at a lag equal to the average well spacing. This is assumed to be the total variance of the noise that needs to be added.

A noise array is now constructed from the appropriate spectral density, S, of fGn noise. The spectral density of the noise to be added is:

$$S(\omega_x,\omega_y,\omega_z) = \frac{1}{[s_x \omega_x^2 + s_y \omega_y^2 + \omega_z^2]^{\beta}} \qquad \text{(A4.79)}$$

with

$$\omega_x = 2\pi f_x \qquad \text{(A4.80)}$$

$$\omega_y = 2\pi f_y \qquad \text{(A4.81)}$$

$$\omega_z = 2\pi f_z \qquad \text{(A4.82)}$$

Here f_x, f_y, and f_z are the x, y, and z frequencies in Fourier space.

The noise array, $g_{j,k,l}$, is generated with β determined from a fractal analysis of the well logs (β=H+0.5). s_x and s_y are parameters that can stretch the x and y directions. They might be thought of as scaling the x and y coordinate axis relative to the z axis. In all of the examples in this book, s_x=s_y=1.0, which corresponds to isotropic noise.

The noise array needs to be generated to agree with Equation A4.79 only to a multiplicative constant, because it will be rescaled to the well data before it is used. In the Fourier transform, the average value of the noise array is made to be zero by defining S(0,0,0) to be:

$$S(0,0,0) = 0.0 \qquad \text{(A4.83)}$$

The noise array, g, is generated from a complex frequency array, G. The frequency array is made up of an amplitude, R, and phase array, θ. R and θ are generated from the spectral density, S, as follows:

$$R_{m,n,p} = +\sqrt{S_{m,n,p}} \tag{A4.84}$$

where

$$S_{m,n,p} = \frac{1}{(s_x(m-1)^2 + s_y E_y(n-1)^2 + E_z(p-1)^2)^{\beta}} \tag{A4.85}$$

where,

$$\beta = H + \frac{1}{2} \tag{A4.86}$$

$$E_y = \left(\frac{L_x N_x}{L_y N_y}\right)^2 \tag{A4.87}$$

$$E_z = \left(\frac{L_x N_x}{L_z N_z}\right)^2 \tag{A4.88}$$

For

$$0 \le m - 1 \le \frac{N_x}{2} \tag{A4.89}$$

$$0 \le n - 1 \le \frac{N_y}{2} \tag{A4.90}$$

and

$$0 \le p - 1 \le \frac{N_z}{2} \tag{A4.91}$$

except

$$R_{1,1,1} = 0 \tag{A4.92}$$

In these equations, the size of the g array to be generated is L_xN_x by L_yN_y by L_zN_z, where the number of grids in each direction is N_x, N_y, N_z and the size of each grid is L_x by L_y by L_z.

In order for g to be real, the R array is required to have the following symmetries:

$$R_{1,N_y-n+2,p} = R_{1,n,p} \tag{A4.93}$$

and

$$R_{\frac{N_x}{2}+1,N_y-n+2,p} = R_{\frac{N_x}{2}+1,n,p} \tag{A4.94}$$

for

$$1 < n \le \frac{N_y}{2} \quad \& \quad 1 < p \le \frac{N_z}{2}. \tag{A4.95}$$

$$R_{N_x-m+2,1,p} = R_{m,1,p} \tag{A4.96}$$

and

$$R_{N_x-m+2,\frac{N_y}{2}+1,p} = R_{m,\frac{N_y}{2}+1,p} \tag{A4.97}$$

for

$$1 < m \leq \frac{N_x}{2} \quad \& \quad 1 < p \leq \frac{N_z}{2}. \tag{A4.98}$$

$$R_{m,N_y-n+2,1} = R_{m,n,1} \tag{A4.99}$$

and

$$R_{m,N_y-n+2,\frac{N_z}{2}+1} = R_{m,n,\frac{N_z}{2}+1} \tag{A4.100}$$

for

$$1 < m \leq \frac{N_x}{2} \quad \& \quad 1 < n \leq \frac{N_y}{2}. \tag{A4.101}$$

$$R_{m,N_y-n+2,p} = R_{m,n,p} \tag{A4.102}$$

for

$$1 < m \leq \frac{N_x}{2} \quad \& \quad 1 < n \leq \frac{N_y}{2} \quad \& \quad 1 < p \leq \frac{N_z}{2} \tag{A4.103}$$

$$R_{N_x-m+2,N_y-n+2,p} = R_{m,n,p} \tag{A4.104}$$

$$1 < m \leq \frac{N_x}{2} \quad \& \quad 1 < n \leq N_y \quad \& \quad 1 < p \leq \frac{N_z}{2}. \tag{A4.105}$$

The amplitude, R, and phase arrays, θ, are combined to produce an array, G, of the noise in frequency space as follows:

$$G_{m,n,p} = R_{m,n,p}\cos(\theta_{m,n,p}) + i[R_{m,n,p}\sin(\theta_{m,n,p})] \quad \text{(A4.106)}$$

The phase array, θ, is set to a random number between $-\pi$ and $+\pi$ such that the following symmetry requirements are preserved:

$$G_{1,N_y-n+2,\frac{N_z}{2}+1} = G^*_{1,n,\frac{N_z}{2}+1} \quad \text{for} \quad 1 < n \le \frac{N_y}{2} \quad \text{(A4.107)}$$

$$G_{1,N_y-n+2,1} = G^*_{1,n,1} \quad \text{for} \quad 1 < n \le \frac{N_y}{2} \quad \text{(A4.108)}$$

$$G_{\frac{N_x}{2}+1,N_y-n+2,\frac{N_z}{2}+1} = G^*_{\frac{N_x}{2}+1,n,\frac{N_z}{2}+1} \quad \text{for} \quad 1 < n \le \frac{N_y}{2} \quad \text{(A4.109)}$$

$$G_{1,N_y-n+2,N_z-p+2} = G^*_{1,n,p} \quad \text{for} \quad 1 < p \le \frac{N_z}{2} \;\; \& \;\; 1 < n \le N_y \quad \text{(A4.110)}$$

$$G_{1,1,N_z-p+2} = G^*_{1,1,p} \quad \text{for} \quad 1 < p \le \frac{N_z}{2} \quad \text{(A4.111)}$$

$$G_{N_x-m+2,1,\frac{N_z}{2}+1} = G^*_{m,1,\frac{N_z}{2}+1} \quad \text{for} \quad 1 < m \le \frac{N_x}{2} \quad \text{(A4.112)}$$

$$G_{N_x-m+2,1,1} = G^*_{m,1,1} \quad \text{for} \quad 1 < m \le \frac{N_x}{2} \quad \text{(A4.113)}$$

$$G_{\frac{N_x}{2}+1,N_y-n+2,1} = G^*_{\frac{N_x}{2}+1,n,1} \quad \text{for} \quad 1 < n \le \frac{N_y}{2} \quad \text{(A4.114)}$$

$$G_{N_x-m+2,N_y-n+2,1} = G^*_{m,n,1} \text{ for } 1 < m \le \frac{N_x}{2} \ \& \ 1 < n \le N_y \quad \textbf{(A4.115)}$$

$$G_{N_x-m+2,1,N_z-p+2} = G^*_{m,1,p} \text{ for } 1 < p \le \frac{N_z}{2} \ \& \ 1 < m \le N_x \quad \textbf{(A4.116)}$$

$$G_{N_x-m+2,N_y-n+2,\frac{N_z}{2}+1} = G^*_{m,n,\frac{N_z}{2}+1} \text{ for } 1 < m \le \frac{N_x}{2} \ \& \ 1 < n \le N_y \quad \textbf{(A4.117)}$$

$$G_{N_x-m+2,N_y-n+2,N_z-p+2} = G^*_{m,n,p} \text{ for } 1 < m \le N_x \ \& \ 1 < n \le N_y \ \& \ 1 < p \le \frac{N_z}{2} \quad \textbf{(A4.118)}$$

Note that these require many values in the θ array to be zero. In particular, θ=0 for the following conditions:

$$\theta_{1,1,1} = 0 \quad \textbf{(A4.119)}$$

$$\theta_{\frac{N_x}{2}+1,1,1} = 0 \quad \textbf{(A4.120)}$$

$$\theta_{1,\frac{N_y}{2}+1,1} = 0 \quad \textbf{(A4.121)}$$

$$\theta_{1,1,\frac{N_z}{2}+1} = 0 \quad \textbf{(A4.122)}$$

$$\theta_{1,\frac{N_y}{2}+1,\frac{N_z}{2}+1} = 0 \quad \textbf{(A4.123)}$$

$$\theta_{\frac{N_x}{2}+1,\,1,\,\frac{N_z}{2}+1} = 0 \qquad \text{(A4.124)}$$

$$\theta_{\frac{N_x}{2}+1,\,\frac{N_y}{2}+1,\,1} = 0 \qquad \text{(A4.125)}$$

and

$$\theta_{\frac{N_x}{2}+1,\,\frac{N_y}{2}+1,\,\frac{N_z}{2}+1} = 0 \qquad \text{(A4.126)}$$

An inverse 3D Fourier transform is then taken using a fast Fourier transform algorithm to produce the noise array, g, from G:

$$g(x,y,z) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} G(\omega_x,\omega_y,\omega_z)e^{2\pi i(\omega_x x+\omega_y y+\omega_z z)}d\omega_x d\omega_y d\omega_z \qquad \text{(A4.127)}$$

The inverse Fourier transform is carried out numerically by using a 1D fast Fourier transform (FFT) from IMSL[23] three times in sequence. The 1D FFT is

$$\xi(x) = \int_{-\infty}^{+\infty} \Xi(\omega)e^{2\pi i\omega x}d\omega \approx \int_{\frac{-\Omega_L}{2}}^{\frac{+\Omega_L}{2}} \Xi(\omega)e^{2\pi i\omega x}d\omega \qquad \text{(A4.128)}$$

In discrete space, this is

$$\xi_j = \frac{\Omega_L}{N}\sum_{m=1}^{N} e^{2\pi i(j-1)(m-1)/N}\,\Xi_m \qquad \text{(A4.129)}$$

where $\Omega_L = 1/L$ and L is the size of the grids.

The 3D transform is carried out as repeated 1D transforms, giving

$$g_{j,k,\ell} = \frac{\Omega_N}{N_z}\sum_{p=1}^{N_z}\left[\frac{\Omega_M}{N_y}\sum_{n=1}^{N_y}\left[\frac{\Omega_L}{N_x}\sum_{m=1}^{N_x}\left[G_{m,n,p}\, e^{2\pi i\frac{(j-1)(m-1)}{N_x}}\right]e^{2\pi i\frac{(k-1)(n-1)}{N_y}}\right]e^{2\pi i\frac{(\ell-1)(p-1)}{N_z}}\right] \quad \text{(A4.130)}$$

The inverse Fourier transform assumes a periodic sequence. Our sequence is not periodic. A careful set of statistical tests will show that the spectral density of g deviates from the desired spectral density, Equation A4.79, at the longest lags. This bias can be removed if sequences longer than the needed one are generated and only the size of the sequence needed is used. (For example, Saupe[42] recommends generating a sequence two to four times larger than needed.) In the interest of saving computer time this was not done. The error introduced is that the noise added is more correlated between opposite faces of the volume generated than should be. This is not readily apparent visually, and we have noticed no problem in our simulations due to this approximation. As a result, we only generate sequences of the needed length.

The variance of the noise array, g, is

$$V_n = \frac{1}{N_x N_y N_z}\sum_{j=1}^{N_x}\sum_{k=1}^{N_y}\sum_{\ell=1}^{N_z}(g_{j,k,\ell} - \bar{g})^2 \quad \text{(A4.131)}$$

with

$$\bar{g} = \frac{1}{N_x N_y N_z}\sum_{j=1}^{N_x}\sum_{k=1}^{N_y}\sum_{\ell=1}^{N_z} g_{j,k,\ell} \quad \text{(A4.132)}$$

Since the average value in the generated noise array is zero, the following linear scaling gives an array, $G_{i,j,k}$, with the desired amount of noise and zero mean:

$$F_{j,k,\ell} = g_{j,k,\ell}\sqrt{\frac{V}{V_n}} \quad \text{(A4.133)}$$

where Equation A4.78 is used to evaluate V.

The scaled noise array, $F_{j,k,\ell}$, is multiplied by a user-defined scaling factor, s_{noise}, and is then added to the trilinear interpolation array, $A_{j,k,\ell}$,

$$B_{j,k,\ell} = A_{j,k,\ell} + s_{noise}\, F_{j,k,\ell} \tag{A4.134}$$

The inclusion of the trilinear interpolation array introduces trends in any horizontal trace. These trends alter the statistical structure of the fGn in any horizontal trace. An alternative method is to replace the trilinear interpolation array with a constant value at each depth. Then, the porosity value at a given depth would be the average of the well values at the given depth. The sum of the constant depth value array and the fGn noise array would still have fGn statistical structure. However, the resulting cross section would not honor the input well log values at the well grid locations as closely as the procedure with linear interpolation.

The array, $B_{j,k,\ell}$ has a total variance greater than the variance of the original log data. The average of the resulting array, $B_{j,k,\ell}$, is unchanged, since the average of the noise array, $F_{j,k,\ell}$, is zero.

One final linear scaling is made. Its purpose is to adjust the final variance and average of the array. Two different scaling methods are possible. One scaling method matches the mean and variance of the generated distribution at only the well locations to the mean and variance of the original log values. In the second method, the mean and overall variance of the generated distribution are matched to the corresponding values in the original well data or user specified values. If the final distribution is chosen to agree at the well locations, the following transformation is made.

$$C_{j,k,\ell} = \left[\frac{V_W}{V_H}\right]^{\frac{1}{2}} (B_{j,k,\ell} - \overline{B}_H) + \overline{B}_W \tag{A4.135}$$

where

$$\overline{B}_W = \frac{1}{N_W} \sum_{q=1}^{N_W} \overline{W}_q \tag{A4.136}$$

$$V_W = \frac{1}{N_W N_z} \sum_{q=1}^{N_w} \sum_{\ell=1}^{N_z} (W_{q,\ell} - \overline{W}_q)^2 \tag{A4.137}$$

$$V_H = \frac{1}{N_W N_z} \sum_{q=1}^{N_W} \sum_{\ell=1}^{N_z} (B_{j(q),k(q),\ell} - \bar{B}_q)^2 \qquad \text{(A4.138)}$$

$$\bar{B}_q = \frac{1}{N_z} \sum_{\ell=1}^{N_z} B_{j(q),k(q),\ell} \qquad \text{(A4.139)}$$

$$\bar{B}_H = \frac{1}{N_W N_z} \sum_{q=1}^{N_W} \sum_{\ell=1}^{N_z} B_{j(q),k(q),\ell} \qquad \text{(A4.140)}$$

The variance and average values of the final array at the well locations, $C_{j,k,\ell}$, are now the same as the variance, V_L, and average, B_W, of the input well log data. The well locations, rather than the overall array, average to the well log data. As a result, the overall average and variance of the final array may be greater than or less than the average and variance of the input well log data for a particular realization.

If the overall variance and mean are chosen to match the well values, the final scaling is as follows:

$$C_{j,k,\ell} = \left[\frac{V_W}{V_A}\right]^{1/2} (B_{j,k,\ell} - \bar{B}_A) + \bar{B}_W \qquad \text{(A4.141)}$$

where

$$V_A = \frac{1}{N_x N_y N_z} \sum_{j=1}^{N_x} \sum_{k=1}^{N_y} \sum_{\ell=1}^{N_z} (B_{j,k,\ell} - \bar{B}_A)^2 \qquad \text{(A4.142)}$$

$$\bar{B}_A = \frac{1}{N_x N_y N_z} \sum_{j=1}^{N_x} \sum_{k=1}^{N_y} \sum_{\ell=1}^{N_z} (B_{j,k,\ell}) \qquad \text{(A4.143)}$$

If the overall variance and mean are chosen to match user-specified values, V_W and $\bar{B}_W$ become the user input variance and mean, respectively.

A4.5 Combining Small and Large Scale Data in 2D

Small scale (e.g., well) and large scale (e.g., seismic) data are combined through manipulation of Fourier transforms. Section 4.9.1 describes the method in 1D. The details for a similar manipulation for 2D vertical cross sections are given below.

Consider a vertical cross section based on small scale data (Plate 5b) with grids N_x by N_z. Let the cross section of the large scale data (Plate 5c) contain N_{xs} by N_{zs} independent values. The large scale data is bilinearly interpolated onto the small scale grid. Bilinear interpolation calculates the value of a point within a set of 4 known values. Let the unknown point, g(x,y), lie within the known values, $g(x_1,z_1)$, $g(x_2,z_1)$, $g(x_1,z_2)$, and $g(x_2,z_2)$, in Figure A4.4. That is,

$$x_1 < x < x_2 \tag{A4.144}$$

and

$$z_1 < z < z_2 \tag{A4.145}$$

Then, the value of g(x,y) is calculated as

$$g(x,z) = (1-a)(1-b)\,g(x_1,z_1) + a(1-b)\,g(x_2,z_1) + (1-a)\,b\,g(x_1,z_2) + ab\,g(x_2,z_2) \tag{A4.146}$$

with

$$a = \frac{x - x_1}{x_2 - x_1} \tag{A4.147}$$

and

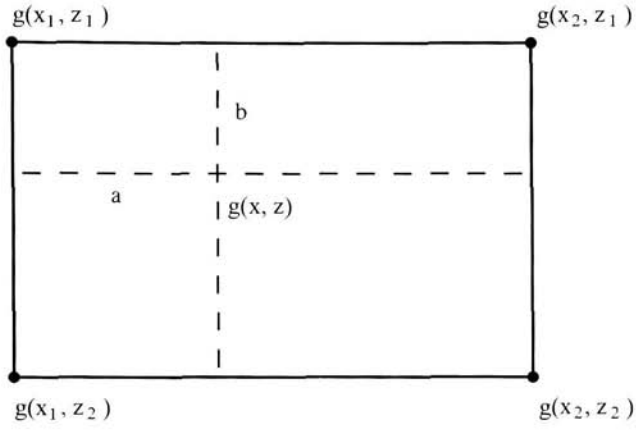


Figure A4.4. Location of points used in the bilinear interpolation formula.

$$b = \frac{z - z_1}{z_2 - z_1} \qquad \textbf{(A4.148)}$$

This process allows the calculation of a small scale grid of values from the larger scale data. The Fourier transform of the result is taken and represented by the array, $G_{m,p}$. The Fourier transform of the cross section based on small scale data (well data) is stored in the array, $F_{m,p}$. The low frequency information in the $G_{m,p}$ array is then substituted for the low frequency information in the $F_{m,p}$ array to produce the final Fourier array, $\Xi_{m,p}$. This is done as follows:

In the upper left quadrant of the arrays

$$\Xi_{m,p} = G_{m,p} \quad \text{for } 2 \le m \le N_x/2 \text{ and } 1 \le p \le N_z/2 \qquad \textbf{(A4.149)}$$

In the lower left quadrant

$$\Xi_{m,p} = G_{m,p} \quad \text{for } 2 \le m \le N_{xs}/2 \text{ and } N_z + 2 - N_{zs}/2 \le p \le N_z \qquad \textbf{(A4.150)}$$

In the upper right quadrant

$$\Xi_{m,p} = G_{m,p} \quad \text{for } N_x + 2 - N_{xs}/2 \le m \le N_x \text{ and } 1 \le p \le N_{zs}/2 \qquad \textbf{(A4.151)}$$

In the lower right quadrant

$$\Xi_{m,p} = G_{m,p} \quad \text{for } N_x + 2 - N_{xs}/2 \le m \le N_x \text{ and } N_z + 2 - N_{zs}/2 \le p \le N_z \qquad \textbf{(A4.152)}$$

and

$$\Xi_{m,p} = F_{m,p} \qquad \textbf{(A4.153)}$$

for all other values of m and p. Applying the inverse Fourier transform to the $\Xi_{m,p}$ array gives a distribution that honors both the small scale and large scale data (Plate 5d).

CHAPTER 5

PERMEABILITY DISTRIBUTIONS

Permeability quantifies the ease with which fluids flow through porous media. That is, given a pressure drop across a piece of rock of a given size and shape, how fast will the fluid flow through the rock? This relationship is given by Darcy's equation[58,59] for the case of linear flow that is illustrated in Figure 5.1,

$$q = \frac{kA}{\mu} \frac{[p_{in} - p_{out}]}{L} \tag{5.1}$$

where

k = permeability
A = cross sectional area
μ = fluid viscosity
p = pressure
q = volumetric fluid flow
L = length of rock piece

Measurement of permeability is a straightforward experiment. But to predict fluid flow at the reservoir scale, these core measurements must be scaled up in some way to the field scale. If reservoirs were homogeneous, this would be a trivial problem. The measured permeability would be the effective permeability for any volume of the reservoir. We have found in previous sections, however, that the distribution of rock properties at the reservoir scale can be far from homogeneous.

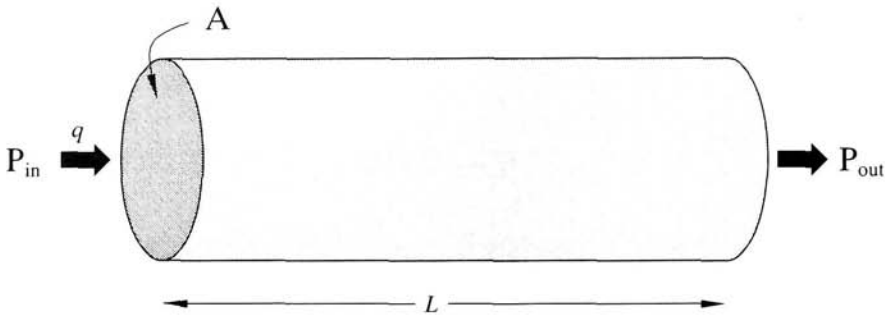


Figure 5.1. Fluid flow experiments on cores to define Darcy's equation.

The question addressed here is "how can the effective permeability of heterogeneous media be calculated?" Some surprising findings result from a study of this rather simple question. We will limit our study to fGn distributions described by Equation 3.2, but much of what is found can be applied to any correlated distribution.

5.1 Generation of Permeability Distributions

Two ways are presented to generate distributions of permeabilities on a grid network between wells. The first method is to generate fGn distributions of porosity on a grid as described in Chapter 4 and then infer permeabilities for each grid from the porosity of each grid. The second method is to generate permeability distributions directly using the fGn model introduced in Chapter 4. We have used almost exclusively the first procedure. Neither procedure is ideal. The first requires a porosity-permeability relationship at the grid block scale. The second requires continuous permeability measurements at each grid corresponding to a well location.

In order to generate a permeability distribution from the first method, a porosity-permeability relationship is needed. Porosity-permeability relationships have been used quite commonly in the petroleum industry. Theoretically, there should be no correlation between porosity and permeability. For example, billiard balls and BB's stacked in a face centered cubic array both have the same porosity but very different permeabilities.

Despite the lack of correlation in general between porosity and permeability, nature has been kind to us in many reservoirs in that the depositional process often produces at least rough porosity-permeability correlations. Examples[60,61,62] for two reservoirs are shown in Figures 5.2 and 5.3. If a porosity distribution is generated from porosity logs as described in Chapter 4, we can use one of these transforms to infer permeability at each location at which we have estimated porosity. Although there is a clear trend in both figures, a close examination reveals that permeability for a particular porosity can vary more than an order of magnitude.

The use of porosity-permeability correlations is not very satisfactory for three reasons. First, the large error bars in correlations like Figures 5.2 and 5.3 yield large uncertainties. Second, porosity-permeability relationships are not available in all reservoirs. Third, porosity-permeability relationships are generated at the core scale but are applied to the grid block scale. (We will return to this last problem in Section 5.5.6.) Nevertheless, we most often proceed along this path.

In applying the method, we generate an fGn distribution of porosities between wells (in 2D or 3D). Next, a porosity to permeability relationship is derived. For Figure 5.2, for example, a straight line is fit to the data, and the resulting porosity-permeability relationship is

$$k = a10^{b\phi} \tag{5.2}$$

where

ϕ = porosity (fraction)
k = permeability
a,b = fitting parameters ($a=1.137 \times 10^{-4}$ and $b=25.3$ for Figure 5.2)

Given a value of porosity for each grid, a corresponding value of k can be estimated from Equation 5.2. If desired, Equation 5.2 can be modified to include the uncertainty in the data. This is done by adding a Gaussian noise term, $f_g(\sigma)$, so that

$$k = a10^{b\phi + f_g(\sigma)} \tag{5.3}$$

where

$f_g(\sigma)$ = a value drawn at random from a Gaussian distribution of mean 0 and standard deviation σ.

A most general form of $f_g(\sigma)$ would allow nonGaussian distributions and σ to be a function of ϕ, but these enhancements are often not justified based on the data available. Figures 5.4 and 5.5 show computer-generated porosity-permeability relationships generated using Equation 5.3 with a and b estimated from Figure 5.2 and σ=0.8. These figures are surprisingly similar to Figure 5.2. Notice the apparent clustering of points. This clustering is due to random fluctuations in the points chosen rather than some underlying representation. Because of the similarity of Figure 5.2 to Figures 5.4 and 5.5, no further enhancement is made to Equation 5.3.

Several other porosity-permeability transforms may be useful in specific circumstances. One is to add nonlinearity to the porosity-permeability transform by expressing the exponent as a power law expansion in ϕ. For example,

$$k = a10^{b_n\phi^n+b_{n-1}\phi^{n-1}+\ldots+b_1\phi+f_g(\sigma)} \tag{5.4}$$

Another is to introduce maximum and minimum cutoffs to the porosity-permeability transform. The cutoffs would set the permeability to a fixed value above or below a given porosity. That is, a cutoff could be used so that all permeabilities above the cutoff are assigned one value, while all those below are assigned another value.

Another useful porosity-permeability transform is

$$k = a\phi^b \tag{5.5}$$

where a and b are fitting parameters.

Once an appropriate porosity-permeability transform is chosen, it is used to assign a permeability value at each grid block corresponding to the porosity at each grid. This yields the desired permeability distribution.

Should we desire to generate permeability distributions directly, without using a porosity-permeability transform, we must have continuous vertical permeability measurements at each well location to use the techniques in Chapter 4. The oil industry has a number of sophisticated logging tools to measure porosity, water saturation, and lithology, but to date, we do not have downhole tools to measure

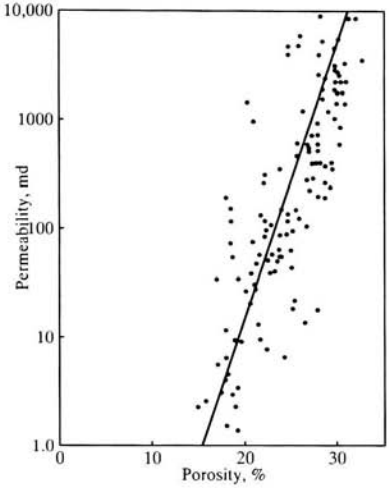


Figure 5.2. Porosity-permeability relationship for Jay field.[60,61]

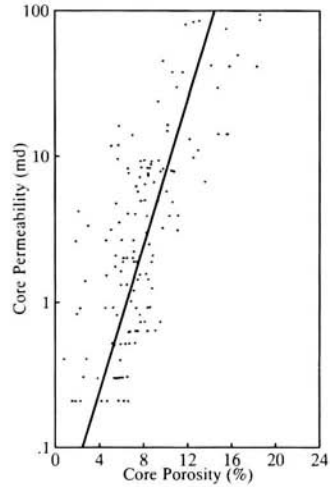


Figure 5.3. Porosity-permeability relationship for Means field.[60,62]

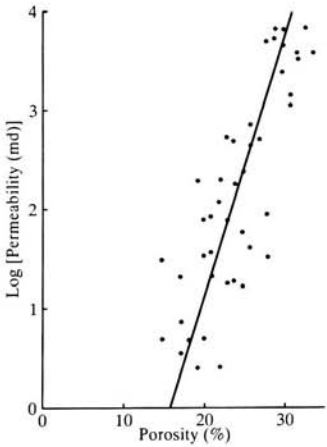


Figure 5.4. Computer-generated porosity-permeability distribution using Equation 5.3 with $a=1.137 \times 10^{-4}$, $b=25.3$, and $\sigma=0.80$.

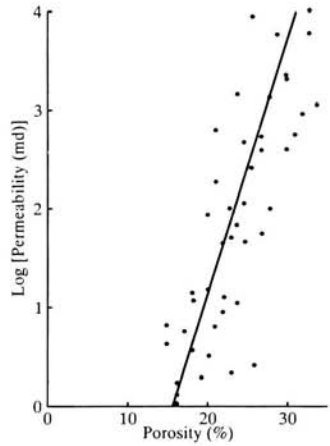


Figure 5.5. Computer-generated porosity-permeability distribution using Equation 5.3 with $a=1.137 \times 10^{-4}$, $b=25.3$, and $\sigma=0.80$.

permeability. To obtain rock permeabilities, cylindrical portions of rock (called cores) are extracted from the reservoir and brought to the surface. Permeabilities can be directly measured on these rock pieces. Usually these data are not continuous through a formation. Some pieces of rock tend to break; others disintegrate in the drilling fluid and are lost. Many times we do not even attempt to core all productive intervals. The result is a discontinuous data set.

A method of filling in the missing data has already been described in Appendix A4.1, "Filling in Holes in Data." In order to use this, a fGn distribution is required. Permeability data are often log-normal and therefore must first be transformed (e.g., by taking the logarithm of the permeabilities) in order to produce a Gaussian distribution. Also, filling the data with a statistical procedure only produces an estimate of the missing data. It should not be used if large portions of the data interval are missing.

5.2 Scaling or Averaging Reservoir Properties

In principle, once porosity and permeability distributions are generated, fluid flow simulations can be performed. In practice, the number of grids (or nodes) produced is too large for typical fluid flow simulations. In Chapter 4, we pointed out that it is important to generate fGn distributions with grids near the rock volume corresponding to the volume of rock examined by the measuring device. Usually, the input data are porosity well logs. If this is the case, at most only a few cubic feet of the reservoir are sampled by each measurement. Core measurements sample even less. On the other hand, well spacing ranges from hundreds of feet to thousands of feet. The desired fine-scale distribution produces too many grids to fit into a conventional reservoir simulator. One solution is to "scale up" or average reservoir property distributions to produce fewer grids prior to reservoir simulations.

The first-order approximation for scaling porosity from a fine scale to a coarse scale is straightforward. Porosity is defined as void volume divided by bulk volume. To calculate the average porosity of grids of equal size, we simply add the porosities of each grid and divide by the number of grids. That is simply an arithmetic average. This procedure is only a first-order approximation, because the porosity actually needed in fluid flow simulations is the connected porosity rather than the total

porosity. The connected porosity can differ from the total porosity if impermeabilities in the rock isolate void volumes so that they cannot contribute to flow. Even experimental measurements of connected versus total pore volume can present difficult problems for laboratory measurements and logging tools. Because of the difficulty of this problem, we will be satisfied with the first-order approximation for averaging porosity.

The first-order approximation for scaling permeability is not so straightforward. Average permeability depends not only on the permeability measurements of each grid, but also upon how the grids are put together to form a larger grid. The connectivity cannot be ignored even for the first-order approximation. (See Figure 2.4 and the associated discussion in Section 2.2.) Sections 5.4 and 5.5 will be devoted to the first-order calculation of average permeability.

5.3 Layers and Random Media

In this section, we will discuss effective permeability for perfectly layered and perfectly random distributions. These heterogeneities are very familiar and are included here only for completeness.

For perfectly layered distributions (with no cross flow between layers), Darcy's law can be applied to each layer separately. The pressure drop is the same across each layer. With this geometry, the average or effective permeability,[63] k_{eff}, is

$$k_{eff} = \frac{1}{N} \sum_{j=1}^{N} k_j \tag{5.6}$$

where

k_j = permeability of j^{th} layer
N = number of layers

This is often called the arithmetic average of permeability by reservoir engineers.

For flow perpendicular to the layers, an identical approach yields[63]

$$\frac{1}{k_{eff}} = \frac{1}{N} \sum_{j=1}^{N} \frac{1}{k_j} \tag{5.7}$$

This is called the harmonic average.

For completely random distributions, early flow calculations yielded a fairly good approximation to the effective permeability to be[64]

$$k_{eff} = (k_1 k_2 k_3 \ldots k_N)^{\frac{1}{N}} = \left(\prod_{j=1}^{N} k_j \right)^{\frac{1}{N}} \tag{5.8}$$

This is called the geometric average.

Reservoir engineers often choose among these averages in field studies. The results have not always been satisfactory. A number of questions remain about which average to use under which circumstances.

5.4 The Renormalization Algorithm--FRACTAM

Reservoir distributions are not perfectly layered nor perfectly random. Given a particular distribution of permeabilities, we will describe an algorithm that can calculate effective or average permeabilities for distributions that are neither perfectly layered nor perfectly random. The algorithm is based on renormalization used in percolation theory.[65]

We will limit ourselves to calculating the effective permeability for a specific set of boundary conditions. These conditions are fluid injection into one side (or face) of the distribution and fluid production on the opposite side (or face). No flow boundaries will be imposed on the other sides (or faces). This set of boundary conditions corresponds to those used in laboratory experiments on core plugs and those implicitly assumed for each grid block in reservoir simulators. In "Unresolved Issues," we will suggest ways to relax these boundary conditions.

The first step is to divide the distribution into 2^N+1 by 2^N+1 grids. We will assume we know the permeability of each grid. The permeability between each grid is estimated as the harmonic average of the permeability at adjacent grids. This procedure then produces a set of values defining the permeabilities between each grid pair. We represent this graphically as a set of tubes connecting the grid block

centers (Figure 5.6). Each node represents a grid block center, and each tube represents the permeability between grids.

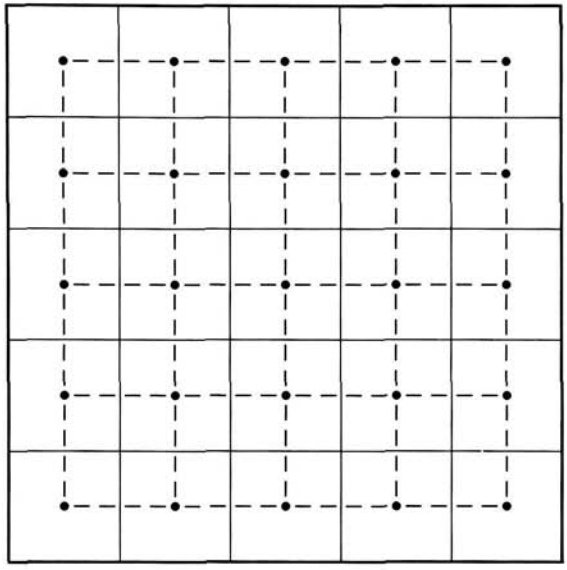


Figure 5.6. A "tube" network describing pictorially the locations of pressure values (nodes) and permeabilities between grids (tubes).

Solving for effective permeability on the tube network requires solving the set of equations in Appendix A5.1. This is normally done using sparse matrix algorithms.[66] The solution can also be done by the transfer matrix algorithm.[67] Both of these algorithms are very computationally intensive for large numbers of grids. Recently, renormalization approaches have been shown to be successful for calculating effective properties on fractal networks. King[68] was the first to publish a renormalization algorithm for estimating effective permeabilities for oil industry applications. Later, Aharony et al.[69] published another renormalization algorithm called FRACTAM. We will describe FRACTAM. We do not suggest that FRACTAM is superior to King's algorithm. We choose to describe FRACTAM only because we are more familiar with it.

FRACTAM is an abbreviation for Fast Renormalization Algorithm for Correlated Transport in Anisotropic Media. The algorithm approximates effective permeabilities on 2D and 3D grid networks. FRACTAM is almost as fast as arithmetic or harmonic averaging and has been shown to calculate effective permeabilities to within 5 percent on a wide range of permeability distributions.

The FRACTAM algorithm starts with a 2^N+1 by 2^N+1 grid network (Figure 5.6). The pressure in each grid block is assigned at the node, and the flow between blocks is assigned to the tubes. Darcy's equation, Equation 5.1, relates the pressure to the flow in each tube between node pairs.

The FRACTAM algorithm divides the corresponding tube network into small pieces, each containing 2 x 2 elements (see Figure 5.7). Each element is approximated as an H-cell in the x and z directions.[70] The effective permeability of each H-cell is calculated by assuming the pressure is the same at both inlets and both outlets. This makes the solution of flow on the small piece equivalent to a Wheatstone bridge[71] array (Figure 5.7). The Wheatstone bridge can be solved exactly (see Appendices A5.2 and A5.3). The effective permeability of each piece in the x and z directions is calculated in this manner. The result is the permeability in the x and z directions for each 2 x 2 piece.

This set of permeabilities can be thought of as a new network of permeabilities but at a larger scale than the original. This process can be repeated as many times as desired until a given grid coarseness is produced or until the effective permeability of the entire distribution is calculated (i.e., only one permeability is left in the x and z directions.).

To test the FRACTAM algorithm, we want permeability distributions that are neither perfectly layered nor random. For this reason, Aharony et al. chose permeability distributions based on a core photograph similar to Figure 1.10. Porosity is assumed to scale linearly with light intensity in the photo. Equation 5.2 is used as the permeability transformation. Actual correlations between porosity (permeability) and light intensity may or may not apply. The point of the exercise is to capture the pattern seen in the core photo. This pattern should provide more realistic permeability distributions than either purely layered or purely random cases. The distributions contain 512 x 512 grids.

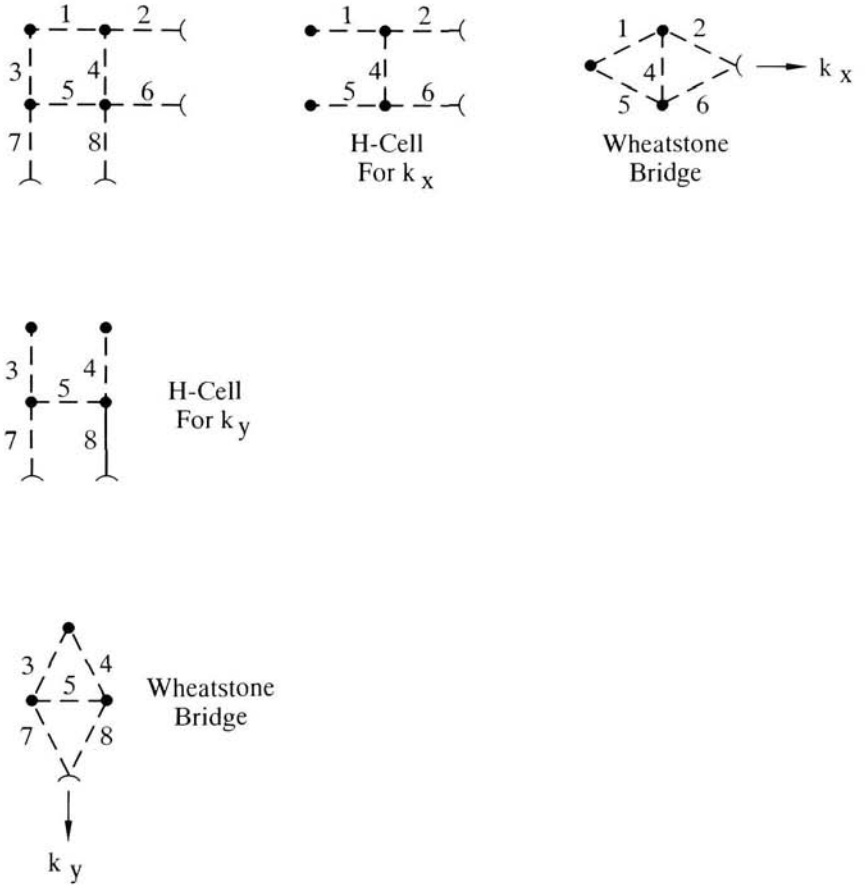


Figure 5.7. The FRACTAM algorithm in 2D.[69]

It is surprising how well the rather simple FRACTAM algorithm works. For the core photo distributions, Aharony et al. successively apply FRACTAM until single tubes are left in both the horizontal and vertical directions. Figure 5.8 shows a comparison of this and other averaging techniques. The "exact value" of permeability is based on a sparse matrix algorithm similar to that used in reservoir simulators. FRACTAM is accurate within 5 percent in all cases, which include various values for the constant a and b in the porosity-permeability transform (Equation 5.2) and various underlying distributions of porosity. All other methods have much larger errors. The techniques compared in Figure 5.8 are the arithmetic, harmonic, and geometric averages already discussed along with an upper and lower bound calculation.

The upper and lower bound calculations are made in the following way. The upper bound is calculated by setting all of the cross flow elements to infinite permeability. This is an upper bound, because the actual data will have lower permeabilities on all of these connections. The lower bound is calculated by setting all of the cross flow elements to zero permeability. This produces a lower bound because any nonzero permeable connectors can only increase the connectivity of the network over the lower bound approximation. These approximations can be useful in their own right, because the FRACTAM does not always yield the exact answer. The upper and lower bounds will bound the exact answer.

There are distributions for which the FRACTAM fails. One example is shown in Figure 5.9. Here a diagonal set of connectors is set to zero permeability. For the distribution shown, it is possible to flow fluids from one size of the grid to the other--the effective permeability is nonzero. A FRACTAM calculation yields zero permeability.

Even though there are cases where FRACTAM fails, for most distributions, the answer is within 5 percent of an exact calculation with a tremendous savings of computational time. Since our engineering estimates often contain much more uncertainty than the FRACTAM calculation, we use FRACTAM to scale permeability. In "Unresolved Issues," we suggest a potential modification to FRACTAM that may allow it to solve even more distributions.

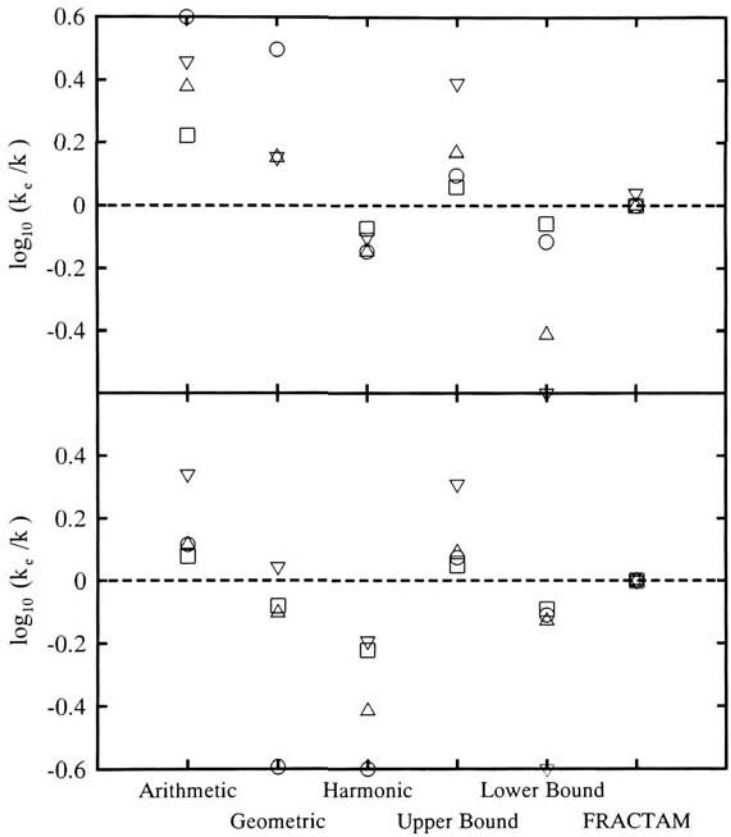


Figure 5.8. The accuracy of the various approximation schemes. k_e is the estimated permeability, and k is its exact value. The upper (lower) diagram shows results perpendicular (parallel) to the layers of the sample.[69]

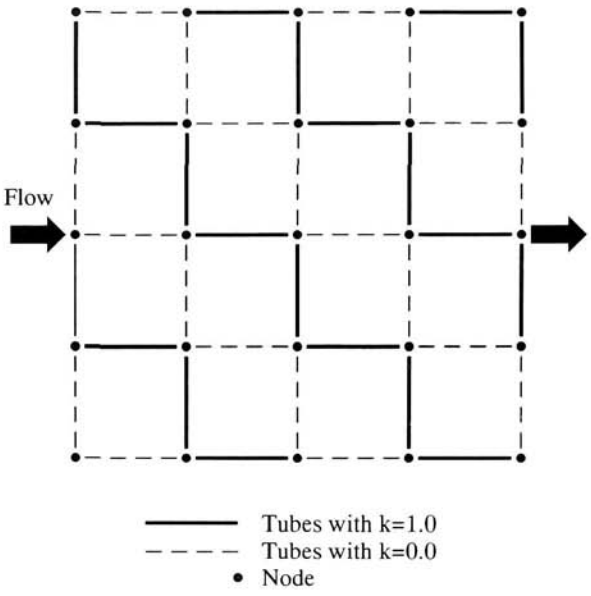


Figure 5.9. A distribution on which FRACTAM fails ($k_{true} \neq 0$, $k_{FRACTAM} = 0$).

FRACTAM is defined for scaling up grids that are 2^N+1 by 2^N+1 grids. It can be expanded to work for any grid size[33] by inserting the desired grid into a 2^N+1 by 2^N+1 array and padding the array with zero or infinite permeabilities. ("Infinite" values are approximated as 10^6*[largest permeability on original array].) Figure 5.10 shows the appropriate padding.

FRACTAM can be easily extended to 3D. A 3D equivalent to Figure 5.7 is shown in Figure 5.11. Tests in 3D show that the 3D FRACTAM algorithm is even more accurate than the 2D FRACTAM algorithm.[72] Appendices A5.2 and A5.3 describe the FRACTAM algorithm in detail.

In the next section, we use the FRACTAM algorithm to study the effective permeabilities of distributions generated using Equation 3.2. This section is not required for preparing input for reservoir simulation, but it points out a number of surprising properties of permeability distributions in heterogeneous media.

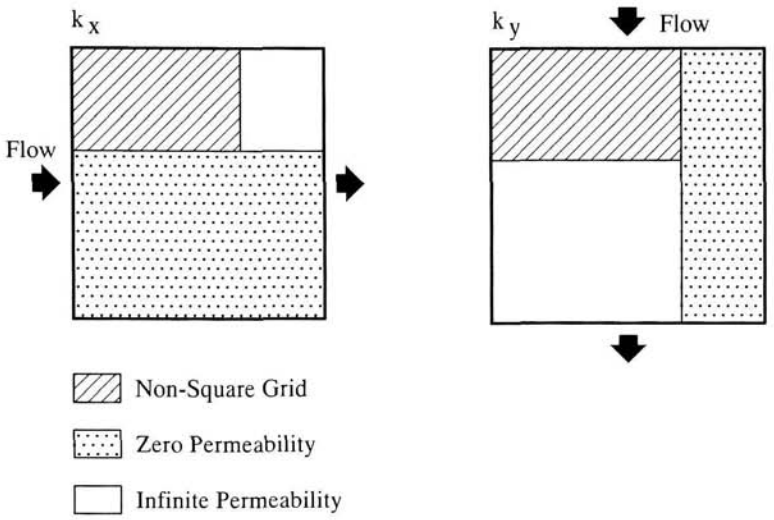


Figure 5.10. Grid for applying FRACTAM to nonsquare grids: (a) horizontal permeability and (b) vertical permeability.[33]

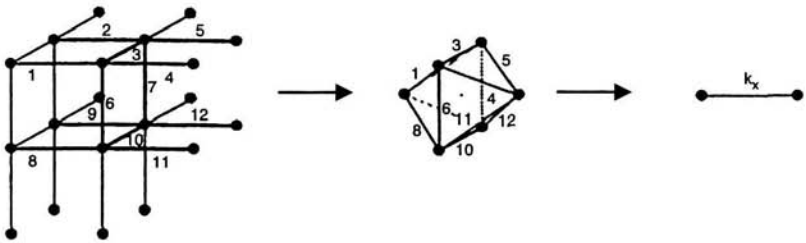


Figure 5.11. Three dimensional FRACTAM algorithm[72] shown for k_x. k_y and k_z similar.

5.5 Horizontal and Vertical Permeability Study

The following k_v-k_h study is divided into 7 parts. Each corresponds to an interesting conclusion reached in the study. The conclusions are as follows:

1. Average (or effective) vertical and horizontal permeabilities depend strongly on the spatial distribution of permeabilities.

2. There are two different ways of approaching a homogeneous distribution. One way is by decreasing the correlation length to zero; the other way is by decreasing the total variance to zero.

3. The study of percolation provides insight on fluid flow in oil reservoirs.

4. Average permeability values from 2D and 3D distributions are surprisingly similar except near percolation thresholds.

5. Permeability-porosity cross plots are generally "well behaved" over a range of grid block sizes. The only exception is near percolation thresholds.

6. If grids are averaged to produce effective permeabilities of larger grids, the effective properties can depend on the shape of the larger grids.

7. The most surprising finding is that boundary conditions can affect permeability in a heterogeneous reservoir. As a result, average reservoir permeability can be dependent upon the location and number of wells!

This study is based on fGn distributions generated using Equation 3.2. Examples of the fGn distributions studied for different layering are shown in Figures 5.12a through 5.12d. Figure 5.12a has almost no layering (B_{SD}=10 in Equation 3.2). Figure 5.12d is almost completely layered (B_{SD}=0.01). Figures 5.12b and c are in between (B_{SD}=0.1 and B_{SD}=1.0, respectively). These figures give a "feel" of the range of fGn variation with different layering. Unless otherwise stated, the base case parameters are set to the following values.

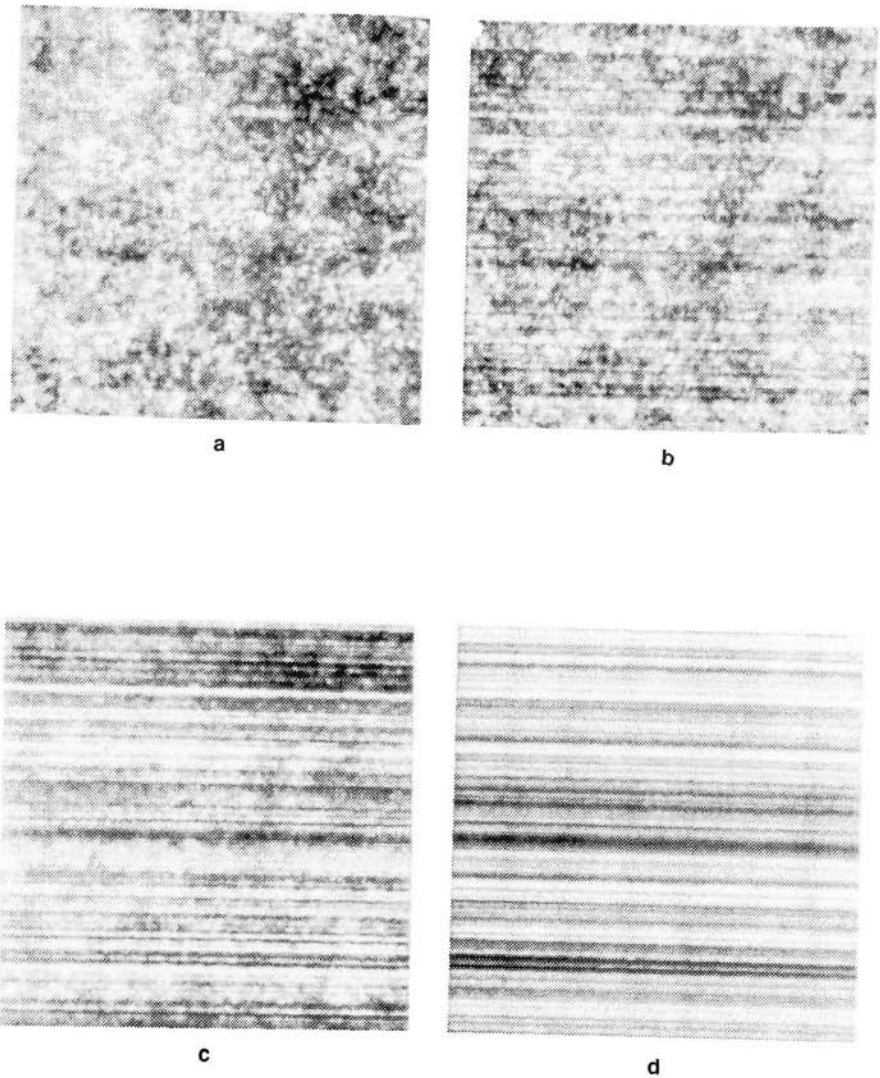


Figure 5.12. Example fGn distributions with different amounts of layering. (a) $B_{SD} = 10.0$, (b) $B_{SD} = 1.0$, (c) $B_{SD} = 0.1$, and (d) $B_{SD} = 0.01$ in Equation 3.2.

Porosity is distributed on a 400 x 400 grid with mean 0.25 and standard deviation of 0.05. A maximum of 0.50 and a minimum of 0.0 porosity is imposed. (These values are not exceeded in any of the distributions examined.) Equation 3.2 is used with $\alpha=0.6$ and $\beta=0.8$. Equation 5.2 is used to define the permeability of each grid from the porosity value (fraction) of each grid with a=1 and b=10.

5.5.1 *The Effect of Layering*

Figure 5.13 illustrates the effect of layering on the calculation of vertical and horizontal permeability. This is a variation of a study done by Carruthers.[73] Note that as layering increases (i.e., B_{SD} in Equation 3.2 decreases) k_v (k_h) approaches the arithmetic (harmonic) permeability average as expected. As layering decreases, k_v and k_h approach a value near the geometric average. The approach to the geometric average is interesting, because the nonlayered distribution is not exactly a completely random distribution. It is a "splotchy" distribution, yet both distributions yield similar average permeabilities. Figure 5.13 shows that FRACTAM is able to calculate not only the limiting cases of completely random and completely layered, but also computes permeabilities for the cases in between.

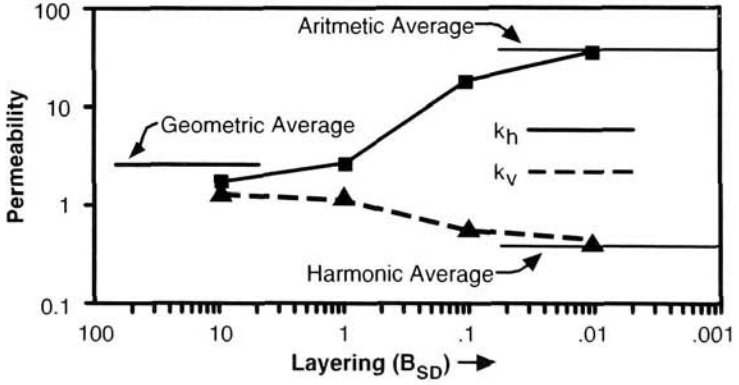


Figure 5.13. Effect of layering on effective vertical and horizontal permeabilities.

5.5.2 *Porosity-Permeability Relationships*

Figure 5.14 shows the effect of changing the porosity-permeability relationship. This is done by changing b in Equation 5.2. Figure 5.15 shows the porosity-permeability relationships used in Figure 5.14. As b is increased, the overall permeability increases. Furthermore, the difference between layered and unlayered systems increases. Differences increase because increasing b increases the range in permeability values in the cross section. Large b results in a larger difference between the highest and lowest permeability values. Conversely, small b results in small differences. In fact, if b goes to zero, the difference will go to zero. Then, all permeability values will be the same.

Note that there are two ways to approach homogeneity. One way is by removing the layers and having the splotches small compared to the "grid block" size. In this case, the correlation length goes to zero on the grid network. The other way is by having the difference between large and small values go to zero. In this case, the correlation length is unchanged, but the total variance goes to zero.

5.5.3 *Permeability Cutoffs*

Permeability cutoffs are commonly used in reservoir engineering to discriminate between reservoir and nonreservoir rock. The cutoff value is used to establish net-to-gross ratios and reservoir volume. This value has its foundation in capillary pressures. At some low permeability, it is expected that pores would be so small that capillary pressure differences between oil and water (or gas and water) would not allow the movement of the hydrocarbon (i.e., nonwetting) phase for any reasonable operating pressure. This exclusion would result in zero permeability regions within the reservoir. These zero permeability regions correspond to those grids with permeabilities below the cutoff permeability.

Some recent developments in physics, known as percolation theory,[65] are helpful in understanding the effects cutoffs can produce in heterogeneous distributions. Percolation theory is the study of flow (fluid or electrical) through a network in which a certain percent of the network bonds or nodes are set to zero (i.e., are impermeable). One question that led to this study was, "If I cut out pieces from a conducting sheet of copper at random, how much of the copper must I cut

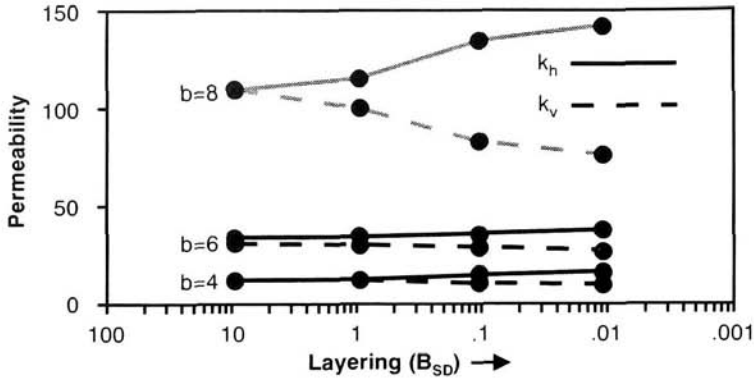


Figure 5.14. The effect of slope changes in the porosity-permeability relationship on effective permeability.

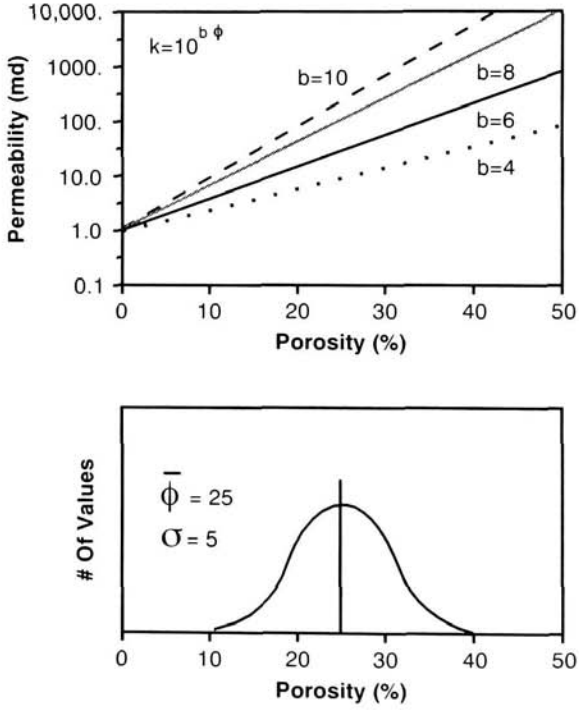


Figure 5.15. The effect of slope changes in the porosity-permeability relationship on effective permeability.

away before the sheet stops conducting?"[65] The point at which the sheet stops conducting (or starts conducting if the reverse process is used) is called the percolation threshold.

Our analogous model is how many faces of the grid blocks must one set to zero permeability before no flow will go from one edge to the opposite edge of the entire distribution. The percolation threshold is defined in this book as the minimum percent of open faces required for flow.

Note that not all the faces have to be conducting before the overall distribution starts conducting. Only enough faces must be conducting so that a continuous line of conducting faces can be traced from the left side to the right size of a cross section. Publications in the physics literature[65,74,75] have determined that for 2D square lattices, 50 percent of the faces must be open for a totally random distribution. For 3D cubic lattices,[74,75] only about 24.7 percent of the faces must be open. Thus, the percolation thresholds for 2D and 3D are 50 percent and 24.7 percent, respectively.

Figure 5.16 shows the effective permeability as a function of number of bonds remaining when the only permeabilities are randomly selected to be 1 or 0. Notice that with all the bonds open, the permeability is 1. As bonds are closed, the permeability drops until it reaches zero for 50 percent or 24.7 percent of the bonds open for 2D and 3D distributions, respectively.

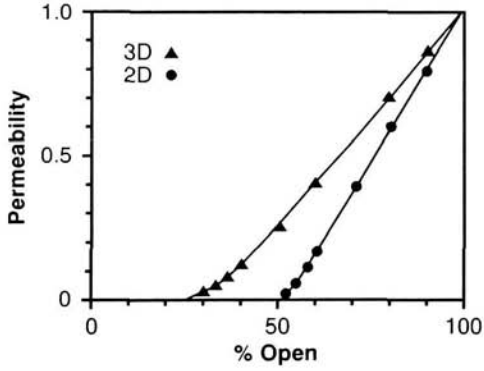


Figure 5.16. Percolation plot for 2D and 3D random distributions.

We choose to study a log normal distribution of permeability instead of ones and zeros as a model for reservoir permeability distributions. Even so, the same type of percolation response is observed for the continuous, unlayered case ($B_{SD}=10$) as seen in Figure 5.17. That is, as the cutoff value is increased (more and more of the grid faces set to zero), the permeability drops. Finally, by the time the number of grids above the cutoff decreases below 50 percent, the average permeability is zero.

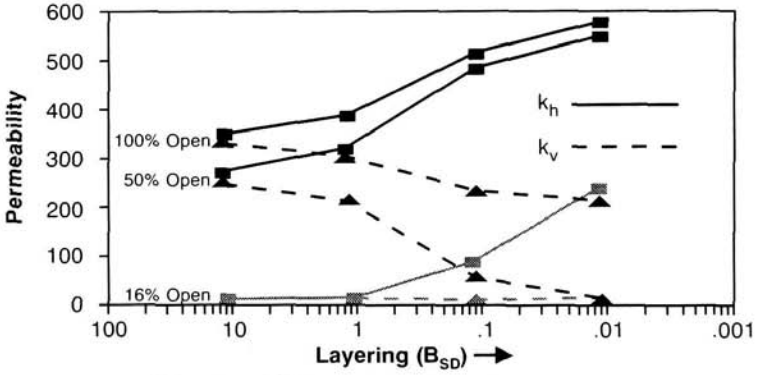


Figure 5.17. Effect of thresholds on k_v and k_h.

For layered distributions, the percolation threshold is much higher in the vertical direction than in the horizontal direction. This effect is seen in Figure 5.17 with 16 percent of the bands open, the vertical permeability is zero, but the horizontal permeability is nonzero for $B_{SD}=0.1$. That is, 16 percent is below the vertical permeability threshold but above the horizontal permeability threshold for $B_{SD}=.01$. The reason for this difference is the layering. As the cutoff is increased, a conduit for flow in the horizontal direction is maintained through highly correlated, high permeability layers. Simultaneously, some layers of horizontal permeabilities are set to zero. These act as barriers to vertical flow. Thus, vertical permeability approaches zero very quickly when cutoffs are increased in layered distributions.

This result is a good thing. Without it, most oil reservoirs would not exist. Zero vertical permeability often produces a seal for an oil or gas reservoir. At the same time, nonzero horizontal permeability permits production of the reservoir by allowing oil to drain in the horizontal direction toward production wells.

5.5.4 *2D and 3D Distributions*

Studies have been made for both 2D and 3D distributions. The resulting permeabilities are usually similar except near percolation thresholds. In most cases, a 2D distribution should give a fairly good estimate of the permeability of a 3D distribution.

A direct comparison between 2D and 3D fGn distributions can be seen in Figure 5.18. In this figure, the 2D and 3D results are superimposed for the case with no cutoffs. Small differences in 2D and 3D effective permeabilities may seem surprising at first, but a little reflection explains why the similarity is so great between 2D and 3D.

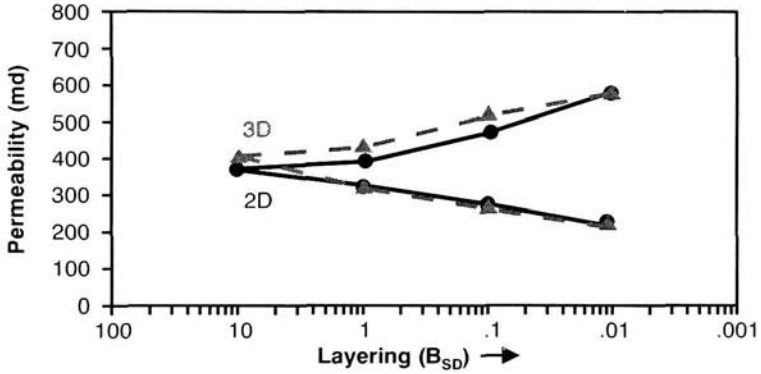


Figure 5.18. The difference in 2D and 3D permeabilities away from the 2D percolation threshold.

First, there should be no difference between 2D and 3D results for a perfectly layered distribution. Both yield the arithmetic and harmonic average in the horizontal and vertical directions, respectively. On the other extreme, Figure 5.16 shows the difference for 2D and 3D distributions for completely random distributions. While the permeabilities are not exactly the same, the differences are not great until the effective permeability of the 2D distribution approaches zero. Then, of course, any number is large compared to zero. The 3D permeabilities are different until the 3D distribution reaches its percolation threshold, where the permeability of both 2D and 3D is zero. Thus, significant differences are confined to the region between the 2D and 3D percolation thresholds.

This result implies that for most cases studied in this book, a 2D distribution should give a fairly good estimate of the permeability of a 3D distribution. This is indeed confirmed by Figure 5.18.

5.5.5 *Grid Block and Porosity-Permeability Crossplots*

What happens to the porosity-permeability relationship if grids are averaged into larger grids? This question is important for two reasons. First, porosity-permeability relationships are measured at the core scale but are applied to the grid block scale. Is this valid? Second, can the porosity-permeability relationship be scaled so that a coarser grid can be produced directly (without first generating a fine-scale distribution) using a rescaled porosity-permeability relationship? This becomes more important with large well distances, because the number of blocks at the small scale becomes very large.

We find porosity-permeability cross plots surprisingly well behaved. For nonlayered distributions, virtually no change occurs in the porosity-permeability transform relationship for larger grids. For layered distributions, the slope of the transform is unchanged, but the intercept for k_h increases while the intercept for k_v decreases. With thresholds, the results are more complicated. For distributions with thresholds, both the slope and intercept, and at times even the linearity between porosity and log (permeability), changes as the grid size increases.

For nonlayered distributions, the input porosity-permeability cross plots are preserved at all scales. For example, Figure 5.19 shows k_v and k_h for 3 coarse grid sizes (2 x 2, 5 x 5, and 10 x 10) for the 2D base case with the $B_{SD}=10$ parameter set. (In Figure 5.19, 2 x 2 means that 4 fine grids were averaged to produce a coarser grid.) Superimposed on each one of the cross plots is a solid reference line that corresponds to the input porosity-permeability transform.

For layered distributions, the slope of the porosity-permeability cross plot is relatively constant, but the intercept changes. Note that something must change in the layered systems at coarser scales, because the input k_h equals the input k_v at the smallest scale. At the coarser scale, however, k_h is greater than k_v for layered systems. (See, for example, Figure 5.13.) Figure 5.20 shows an example of the porosity-permeability transform evaluated for different grid sizes with layering. These figures are from the 2D base case parameter set with $B_{SD}=0.1$, but are typical of all the parameter sets studied.

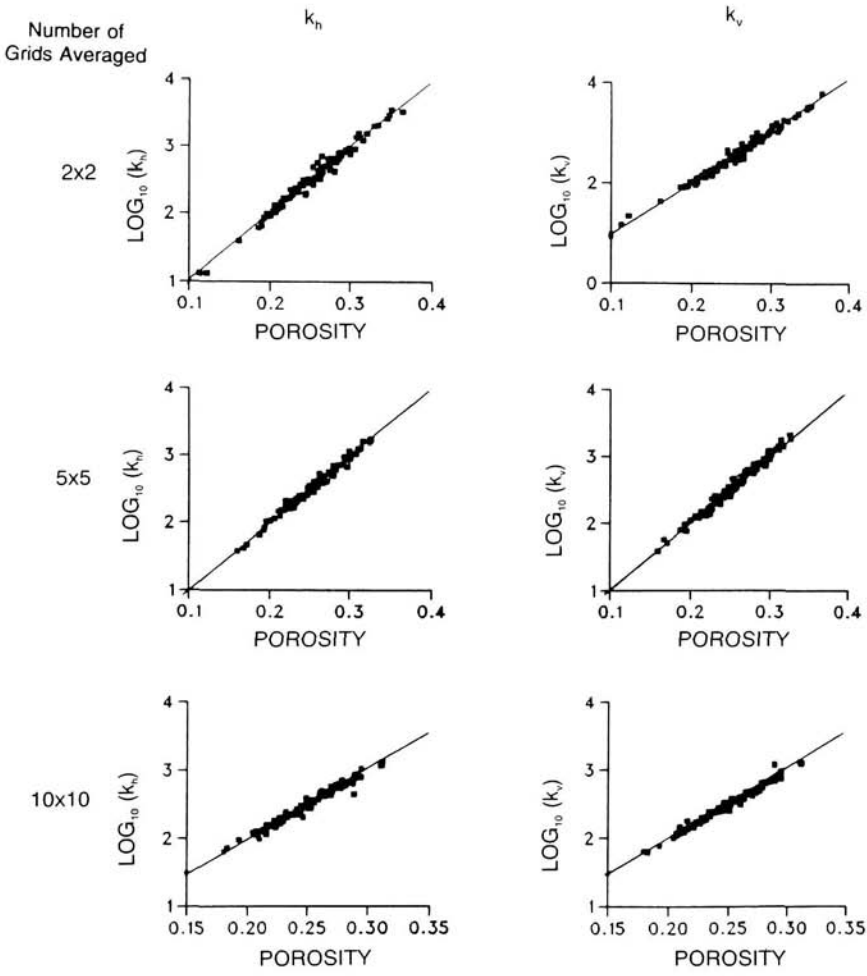


Figure 5.19. The effect of grid size on porosity-permeability relationships without layering.

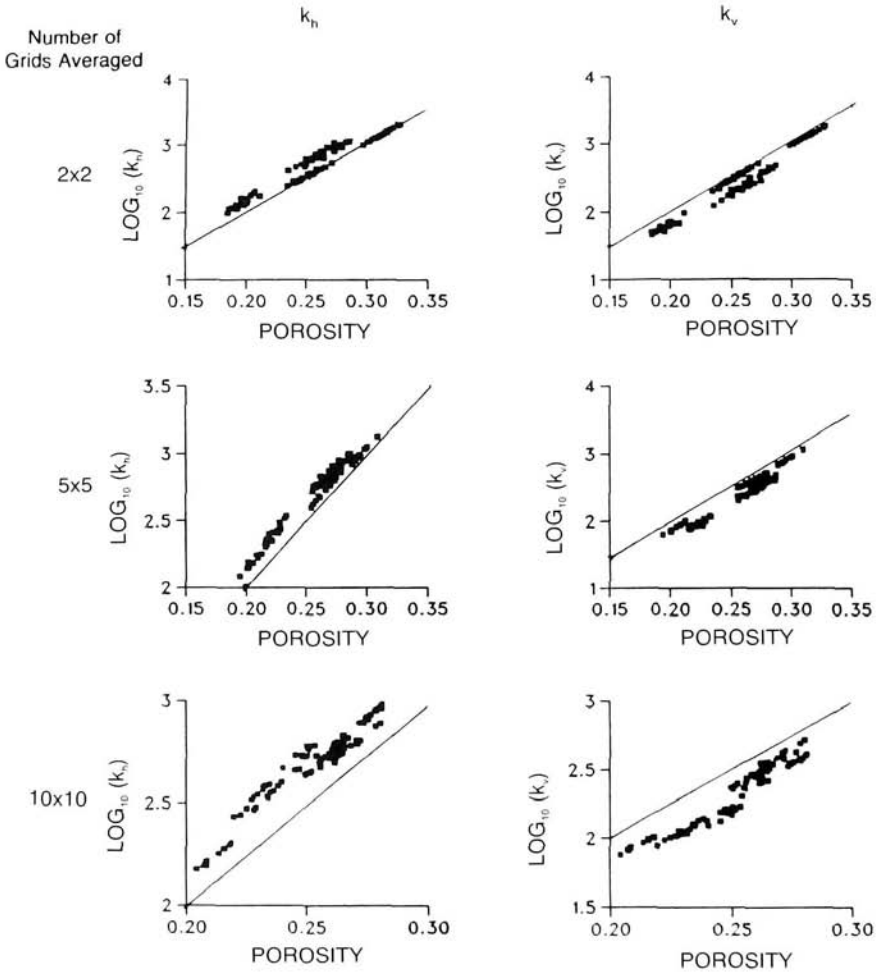


Figure 5.20. The effect of grid size on porosity-permeability relationships with layering.

For cutoffs, especially near the percolation threshold, the porosity-permeability transform often changes. Usually, both the slope and intercept changed (e.g., Figure 5.21). For some threshold values, the porosity-permeability relationship is no longer linear (e.g., Figure 5.22, k_h for less than 50 percent of the grids open to flow using the base case distribution). So far, we have not identified any overall trend in the changes produced using cutoffs.

5.5.6 *Grid Block Shapes and Effective Permeabilities*

FRACTAM was used in Section 5.5.5 to coarsen a fine grid into a coarser grid for use in reservoir simulation. Care must be taken in how this coarsening process is performed, however. Two limiting cases provide some insight.

Suppose a fine-scale grid of 100 by 100 (N_x by N_z—"x" horizontal, "z" vertical) is coarsened up to a grid that is 100 by 1. This would be accomplished by taking the arithmetic average of the permeability in each column. If the 100 x 1 grid is now put into a reservoir simulator, the effective permeability in the simulation will be the harmonic average of the coarse grids. The effective permeability would then be the arithmetic average followed by the harmonic average. Now let's do the reverse. First, coarsen the 100 by 100 grid to a 1 by 100 grid. This requires the harmonic average. Now, complete the process as before by taking the arithmetic average. These results are not the same, and neither result is the correct average permeability of the distribution.

Wilson[76] expanded this result by studying many different coarsening processes. He concluded that the calculated permeability is "path dependent." That is, the calculated average depends on exactly what path is taken (i.e., in what order and to what geometry are the grids resized). Wilson did his study using FRACTAM, but it is clear from the simple example above that this result would have occurred no matter what averaging technique was used (e.g., a reservoir simulator would have yielded the same result.) What has happened is that artificial boundary conditions have been imposed in order to find the "average" permeability of each grid block.

At each level of coarsening, no flow is allowed in the vertical direction when the horizontal permeability is calculated for each block. This is necessary in order to define horizontal permeability for each block in a meaningful way. In reality, however, fluid can flow out of one block, say down, through part of an adjacent block, and then back up and along the first block (see Figure 5.23). In this way, the

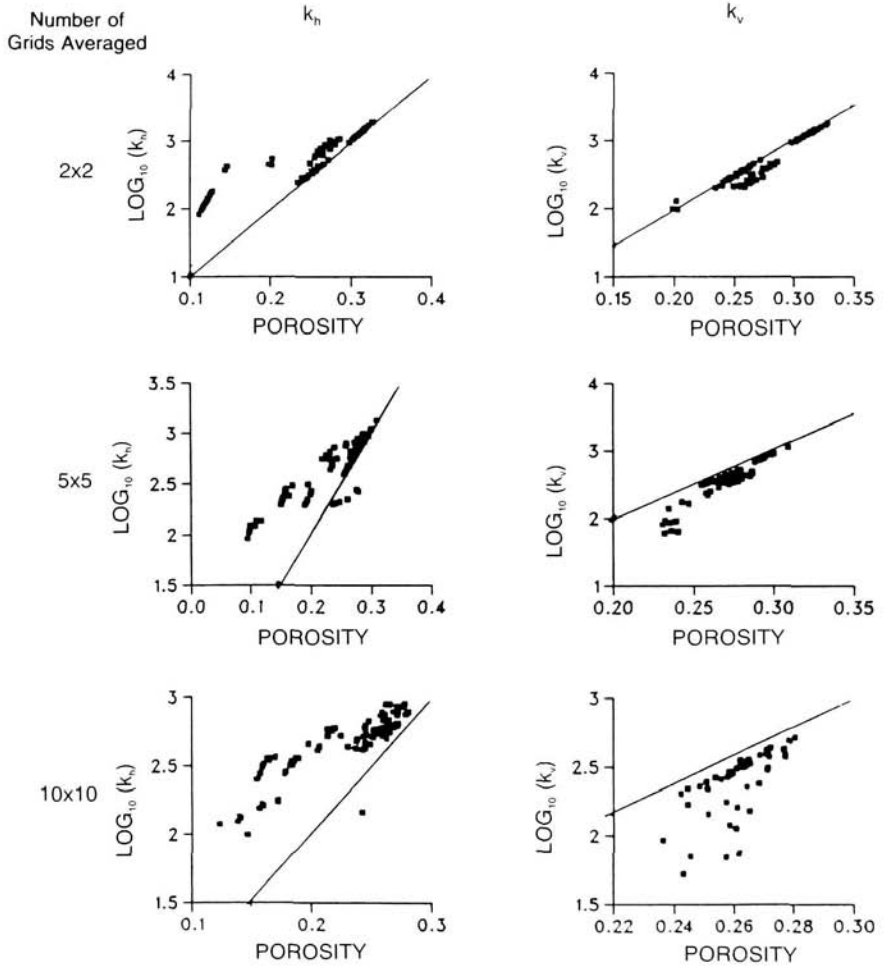


Figure 5.21. The effect of grid size on porosity-permeability relationships with layering and thresholds.

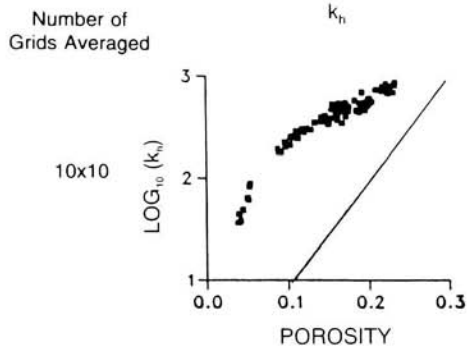


Figure 5.22. The effect of grid size on porosity-permeability relationships with layering, near a percolation threshold.

fluid can bypass a low permeability area (a block) in the middle of the reservoir. The effective horizontal permeability of this block if it is isolated, however, may be very small, while the effective horizontal permeability of the block imbedded in the reservoir may be large.

This result is an artifact of the way the intermediary grid is defined. It is not a "real" property of the reservoir, but just a computational nuisance. The moral of this study is that care should be taken whenever the aspect ratio of each coarse grid block varies greatly from the aspect ratio of the entire grid. For results acceptable within engineering accuracy, exactly how different the aspect ratio of the grids can differ from the aspect ratio of the entire distribution, we do not know.

It is disturbing that the geometry of intermediate grids can affect the calculation of the overall permeability, but this is not nearly as disturbing as the finding in the following section.

5.5.7 *Permeability as a Function of Distance*

A study has been made of permeability as a function of distance from injector to producer in a heterogeneous distribution. We use a distribution defined by the 2D, base case parameter set for B_{SD}=10, 1, 0.1, and 0.01. An ensemble of ten fine-

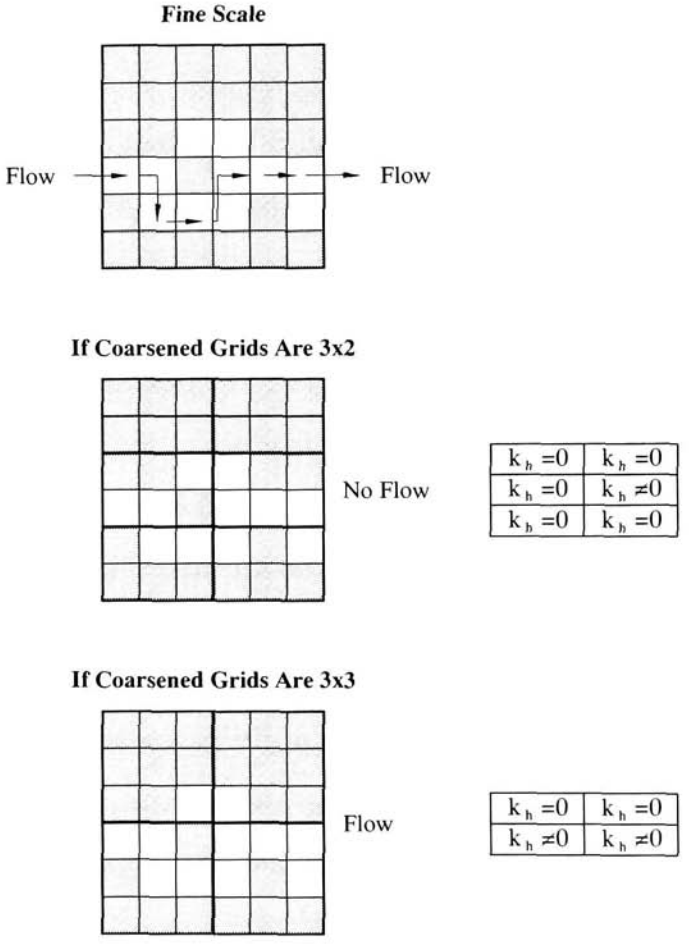


Figure 5.23. The choice of grid coarsening can affect the effective permeability calculation.

scale grids are sliced into vertical pieces of size x = 4, 10, 20, and 40 and z = 400. The average horizontal permeability of each ensemble is then calculated and plotted versus x in Figure 5.24.

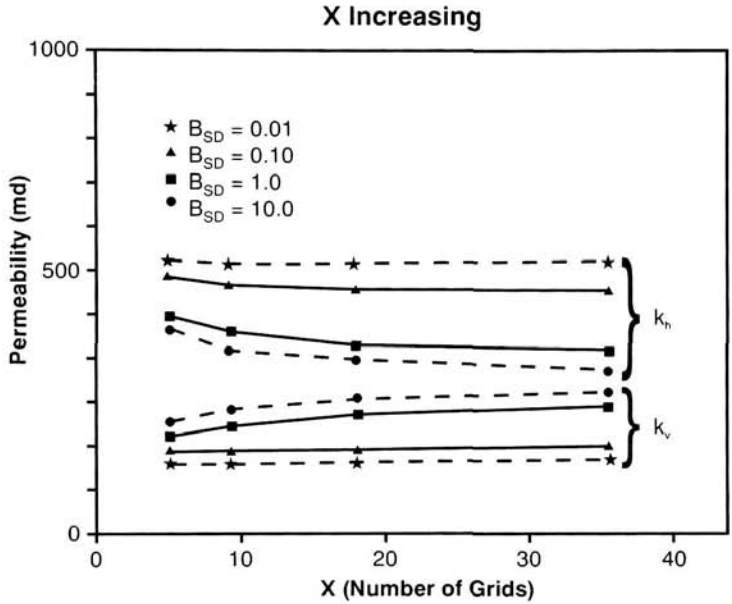


Figure 5.24. Effect of grid size on k_v and k_h.

Note that the case with virtually no layering (B_{SD}=0.01) is not very interesting. k_h and k_v are essentially identical and independent of x. But as layering increases ($B_{SD} \rightarrow 10.0$), the horizontal (vertical) permeability decreases (increases) with increasing x! This is not an artifact of FRACTAM.

This result is extremely disturbing. We have been taught to believe that permeability is a rock property. The permeability of a piece of rock is the same, no matter where it is placed in the reservoir. This is true for homogeneous distributions, but this is not true for heterogeneous ones. Permeability that depends on position can be demonstrated with a simple example.

Note that in the earlier results, we found that very layered distributions produced an average horizontal permeability equal to the arithmetic average. This

is as expected. The average permeability of a completely unlayered system approached the geometric average.

Now consider the permeability of a random distribution, Figure 5.25. If only the first column of grid blocks is used, the average flow must be the arithmetic average, since for a single column of blocks, the "effective" formation is perfectly layered. We know, however, that if we include more of the blocks in the x direction, the answer must eventually approach the geometric average, so the average for x=1 is arithmetic and the average for x="large number" is geometric. A plot of average versus x would begin at the arithmetic average and approach the geometric average. Hence, the effective permeability is a function of x up to some "large number."

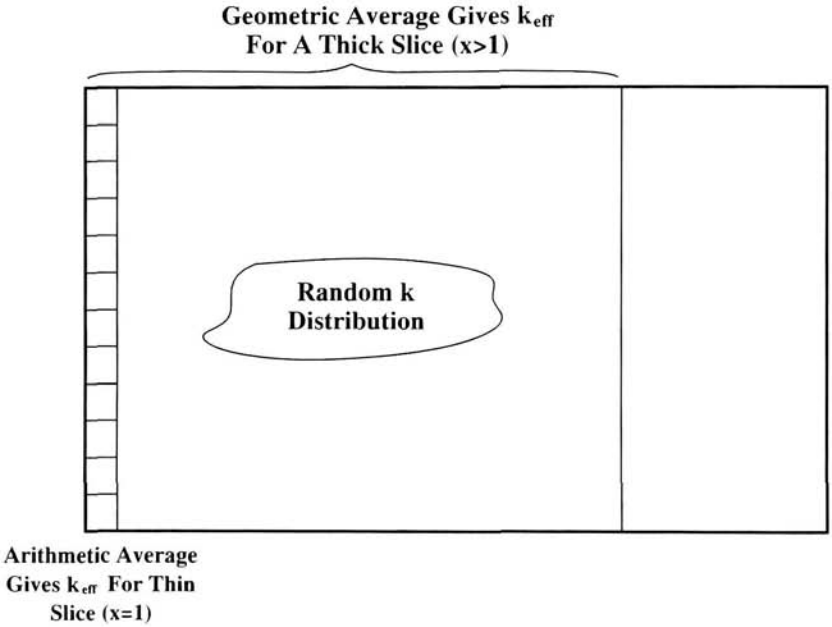


Figure 5.25. Effect of grid size on effective permeability (k_{eff}).

The results plotted in Figure 5.24 simply help quantify this "large number." From Figure 5.24, we can see that for the layered distribution studied (B_{SD}=0.01), this "large number" is less than 10. For an unlayered distribution (B_{SD}=10), this number is larger than 40. The volume of rock required before the permeability stops changing with size has been called the representative elemental volume (REV) in the

literature.[77] The problem identified here is that in some distributions, this volume can be very large, e.g., larger than grid blocks in reservoir simulation or larger than the volume of rock drained by a single well.

These results are in agreement with pressure transient tests discussed in Chapter 7. In Chapter 7, we find that some pressure transient tests can be interpreted more accurately by assuming that permeability varies around a producer (or injector) as a function of the distance away from a well to a negative fractional power. It seems that, unfortunately, for heterogeneous reservoirs, we are going to have to come to terms with the idea that permeability is not just a rock property, but that permeability can also depend on the boundary conditions used to measure it.

Unresolved Issues

1. A more complete study of the FRACTAM algorithm should be made to find out exactly when and where it fails and how badly. Since cross bedding is a problem for FRACTAM, an extension of this algorithm to include the tensor character of permeability would probably yield a more accurate and complete algorithm.

2. More research is needed on the effect of boundary conditions on effective permeabilities in heterogeneous distributions. When are these effects important and when can they be ignored?

3. A better way to distribute permeabilities on a fine grid is needed. Both direct generation and porosity-permeability transform methods presented here leave much to be desired.

4. More study is needed on the effect of grid size on porosity-permeability relationships, especially near percolation thresholds.

APPENDIX A5

This appendix is divided into three parts. The first defines the equations to be solved for a grid network model of flow through porous media. The second gives

the equations needed for the FRACTAM algorithm. The third lists the computer code for the FRACTAM algorithm in 2D and 3D.

A5.1 Equations for Solving for k_{eff}

This appendix describes the required equations to estimate k_{eff} on an areal grid network. We will assume a 2D areal grid network made up of rectangular grids of size L_x by L_y by L_z (Figure A5.1a). We assign x and z permeabilities to a node at the center of each grid. The following derivation can easily be expanded to 3D. We convert from an N by N grid network to an N-1 by N tube network, shown in Figure A5.1b, to solve for flow in the x direction. In the tube network, we assume a common source at the inlet and outlet illustrated by vertical bars on each side of the grid. An analogous tube network is applied for flow in the z direction but is not shown.

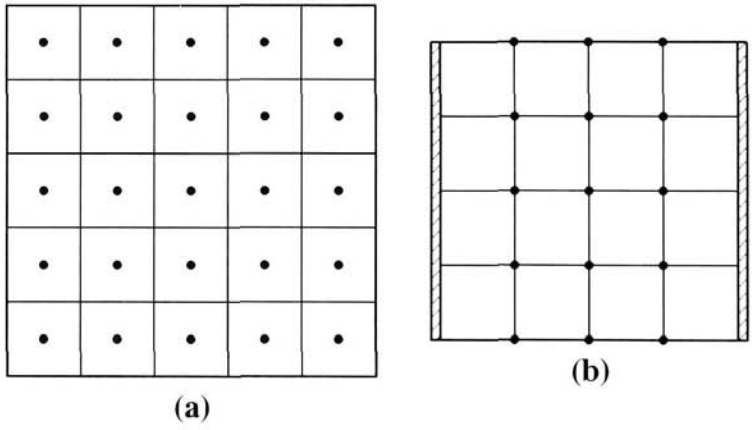


Figure A5.1 Grid used to derive equations, (a) grid network and (b) tube network.

Two types of equations are needed to solve for single phase, incompressible fluid flow in the x direction on a tube network. The first is the flow in the tubes. The second is the conservation of mass at the nodes. In order to get the flow in the tubes, we need to know the permeability of each tube. The tube permeabilities are assigned from the permeabilities at the nodes at either end of the tube using Equation 5.7. Thus

$$\frac{1}{k_x(j,\ell\text{-}j+1,\ell)} = \frac{1}{2}\left[\frac{1}{k_x(j,\ell)} + \frac{1}{k_x(j+1,\ell)}\right] \tag{A5.1}$$

Darcy's equation (Equation 5.1) for each tube can be simplified by defining the conductivity, $\hat{g}_x(j,\ell)$, between nodes (j,ℓ) and $(j+1,\ell)$ as

$$\hat{g}_x(j,\ell) = \frac{2k_x(j,\ell)k_x(j+1,\ell)L_yL_z}{\mu[k_x(j,\ell) + k_x(j+1,\ell)]L_x} \tag{A5.2}$$

The conductivity of the tube between nodes (j,ℓ) and $(j,\ell+1)$ is

$$\hat{g}_z(j,\ell) = \frac{2k_z(j,\ell)k_z(j,\ell+1)L_xL_y}{\mu[k_z(j,\ell) + k_z(j,\ell+1)]L_z} \tag{A5.3}$$

Darcy's equation can now be written in the x direction as

$$q_x(j,\ell) = \hat{g}_x(j,\ell)[p(j,\ell) - p(j+1,\ell)] \quad \text{for} \quad j = 1, N-1 \text{ and } \ell = 1, N \tag{A5.4}$$

In the z direction,

$$q_z(j,\ell) = \hat{g}_z(j,\ell)[p(j,\ell) - p(j,\ell+1)] \quad \text{for} \quad j = 2, N-1 \text{ and } \ell = 1, N-1 \tag{A5.5}$$

Equations A5.4 and A5.5 are a set of $2(N-1)^2$ equations.

For incompressible flow, the sum of the flow into each node must equal the flow out of each node. This produces $(N-2)N$ equations,

$$q_x(j,\ell) + q_z(j,\ell) = q_x(j-1,\ell) + q_z(j,\ell-1) \quad \text{for} \quad j = 2,\ N-1 \ \text{and} \ \ell = 1,\ N \qquad \text{(A5.6)}$$

where

$$q_z(j,0) = 0.0 \quad \text{for} \quad j = 1,\ N \qquad \text{(A5.7)}$$

and

$$q_z(j,N) = 0.0 \quad \text{for} \quad j = 1,\ N \qquad \text{(A5.8)}$$

We assume a common source at all inlet and a common sink at all outlet pressures. The inlet and outlet pressures are assumed known to be p_{in} and p_{out}, respectively. That is,

$$p(1,\ell) = p_{in} \quad \text{for} \quad \ell = 1,\ N \qquad \text{(A5.9)}$$

$$p(N,\ell) = p_{out} \quad \text{for} \quad \ell = 1,\ N \qquad \text{(A5.10)}$$

All nodal permeabilities are assumed to be known, and therefore, all $\hat{g}_x(j,\ell)$ and $\hat{g}_z(j,\ell)$ are known. Thus, we have $3N^2$-6N+2 unknowns $(2(N-1)^2$ velocities and N(N-2) pressures). Equations A5.4, A5.5, and A5.6 provide $3N^2$-6N+2 equations. Matrix inversion algorithms can solve for the unknowns exactly.[66]

A5.2 FRACTAM Equations

The flow equations can be solved exactly, but it is a time-consuming process. A faster approach is to use the FRACTAM algorithm.

The following FRACTAM equations apply for the Wheatstone bridge (see Figure 5.7) that estimates k_x. With the appropriate changes to the indices, the equations apply also to k_z. The effective conductivity in the x direction is

$$\frac{1}{\hat{g}_x} = \frac{1}{\hat{g}_c} + \frac{1}{(\hat{g}_d + \hat{g}_e)} \tag{A5.11}$$

with

$$\frac{1}{\hat{g}_d} = \frac{1}{\hat{g}_a} + \frac{1}{\hat{g}_2} \tag{A5.12}$$

$$\frac{1}{\hat{g}_e} = \frac{1}{\hat{g}_b} + \frac{1}{\hat{g}_6} \tag{A5.13}$$

$$\hat{g}_a = \hat{g}_1\hat{g}_4\left[\frac{1}{\hat{g}_1} + \frac{1}{\hat{g}_4} + \frac{1}{\hat{g}_5}\right] \tag{A5.14}$$

$$\hat{g}_b = \hat{g}_4\hat{g}_5\left[\frac{1}{\hat{g}_1} + \frac{1}{\hat{g}_4} + \frac{1}{\hat{g}_5}\right] \tag{A5.15}$$

and

$$\hat{g}_c = \hat{g}_5\hat{g}_1\left[\frac{1}{\hat{g}_1} + \frac{1}{\hat{g}_4} + \frac{1}{\hat{g}_5}\right] \tag{A5.16}$$

These equations are based on a transfer matrix algorithm (Derrida et al.).[67] These equations can be verified by back substitution.

This set of equations is repeated for each H cell in the x and z directions over the entire grid. When completed, a new grid is made from the $\hat{g}_x$ and $\hat{g}_z$ calculated for each grid. This can be stopped at any scale or repeated until a single $\hat{g}_x$ and $\hat{g}_z$ result. The effective permeabilities in the x and z directions can then be calculated using

$$k_x = \hat{g}_x \frac{n_x \bar{L}_x}{n_y L_y n_z L_z} \tag{A5.17}$$

and

$$k_z = \hat{g}_z \frac{n_z \bar{L}_z}{n_x L_x n_y L_y} \tag{A5.18}$$

where n_x, n_y, and n_z are the number of grids in each direction that have been rescaled.

A5.3 FRACTAM

The following 2D algorithm evaluates the conductance in the x direction given in Figure 5.7 and Equations A5.11-A5.15 for square grids. The answer is stored in variable g2d. The vector g(j) contains the local conductance following the number scheme in Figure 5.7.

```
eps=1.0e-14
a22=g(6)+g(4)+g(5)
if(a22.lt.eps) then
   c=0.0
  else
    c=1./a22
endif
a11=g(6)+g(2)-(g(6)*g(6)*c)
a31=-g(2) - (g(4)*g(6)*c)
a33=g(2)+g(4)+g(1)-(g(4)*g(4)*c)
if(a33.lt.eps) then
   c=0.0
  else
   c=1./a33
endif
g2d=a11-a31*a31*c
```

The following 3D algorithm evaluates the conductance in the x direction in Figure 5.11. The answer is stored in variable g3d. The vector g(i) contains the local conductance following the number scheme in Figure 5.11.

```
eps=1.e-14
b22=g(1)+g(3)+g(4)
if(b22.lt.eps)then
   c=0.0
 else
    c=1./(1.+(g(6)/b22))/b22
endif
a11=g(1)+g(2)-(g(1)*g(1)*c)
a21=-g(1)+(b22*g(1)*c)
a22=b22-(b22*b22*c)
a31=-g(2)-(g(3)*g(1)*c)
a32=-g(3)+(g(3)*b22*c)
a33=(g(2)+g(3)+g(5))-(g(3)*g(3)*c)
if (a33.lt.eps) then
   c=0.0
  else
   c=1./(1.+(g(7)/a33))/a33
endif
a11=a11-(a31*a31*c)+g(8)+g(9)
a21=a21-(a32*a31*c)-g(8)
a22=a22-(a32*a32*c)+g(8)+g(10)+g(11)
a31=a31-(a33*a31*c)-g(9)
a32=a32-(a33*a32*c)-g(10)
a33=a33-(a33*a33*c)+g(9)+g(10)+g(12)
if(a22.lt.eps) then
   c=0.0
  else
   c=1./a22
endif
a11=a11-(a21*a21*c)
a31=a31-(a32*a21*c)
```

```
a33=a33-(a32*a32*c)
if(a33.lt.eqs) then
   c=0.0
  else
   c=1./a33
endif
g3d=a11-a31*a31*c
```

CHAPTER 6

RESERVOIR SIMULATIONS

We now describe ways of using the porosity and permeability distributions generated in Chapters 4 and 5 in multiphase flow calculations. The modeling of the flow of oil, water, and gas through reservoirs is called reservoir simulation.[78] The simulations are used to predict oil and gas recovery rates, which can be used in an economic evaluation of a field project. Simulations also assist the engineer in deciding a development strategy. This strategy may include injecting water or other fluids to enhance oil recovery. At the very least, the engineer and geoscientists must determine the required number of wells and their locations.

Reservoir simulation emerged in the 1960's as a practical tool as computer technology developed. Prior to computer simulations, the reservoir was treated on a gross average basis in a tank model,[79] but the tank models cannot adequately account for the variations in reservoir properties in space. Reservoir simulations divide the reservoir into blocks and apply equations for flow in porous media to each block. Then in principle, each block can have its own set of properties that correspond to that volume of the reservoir. Yet, the blocks are interdependent because oil, water, and gas can flow from one block into its neighboring blocks. The flow equations between blocks can be solved by finite difference techniques,[78] which are commonly applied to engineering problems.

The technology of reservoir simulation is constantly being improved and advanced. Scientists and engineers continue to propose new methods to simulate more complex recovery schemes along with better models of the physical/chemical processes. In this chapter, our emphasis is on incorporating reservoir heterogeneity into reservoir simulations and not complex recovery mechanisms. Therefore, we will

deal with only one basic, but very common, recovery process known as waterflooding. In the United States, as much as half of the current oil production is the result of water injection.[60]

6.1 Fluid Permeability With Two Flowing Phases

To model waterflooding, we must describe the flow of two immiscible phases (oil and water) through porous rock. The permeability of each flowing phase is estimated from experimental data by assuming Darcy's law applies to each phase.[59,60] Then, the effective permeability of each phase is assumed to depend only on the saturation of that phase. The saturation of the phase is defined as the volume occupied by the phase divided by the total pore volume. The effective permeability ranges from zero at the minimum saturation of the phase to a maximum when the second phase is immobile. Correlations between effective permeability and saturation based on experimental data form the basis for modeling the flow of two or more phases.

Based on the hypothesis that Darcy's law applies to each phase, we write the volumetric flow rate of oil (q_o) and water (q_w) in terms of the pressure in the respective phase, p_o and p_w. Darcy's law states that the flow rate depends on the pressure drop, viscosity (μ_o or μ_w), and permeability,

$$q_o = \frac{k_{ro} k A}{\mu_o} \frac{dp_o}{dx} \tag{6.1}$$

$$q_w = \frac{k_{rw} k A}{\mu_w} \frac{dp_w}{dx} \tag{6.2}$$

Here k_{ro} and k_{rw} represent the relative permeability of oil and water, respectively. The relative permeability is defined as the effective permeability of a phase to a base permeability, k. Often the base permeability is chosen to be the effective permeability of oil at the residual water saturation. Then, the oil relative permeability curve starts at unity as shown in Figure 6.1. Note the relative permeability curve for each phase is assumed to be a function of only the phase

saturation. Since the curves in Figure 6.1 apply for a two-phase system, only one-phase saturation needs to be specified.

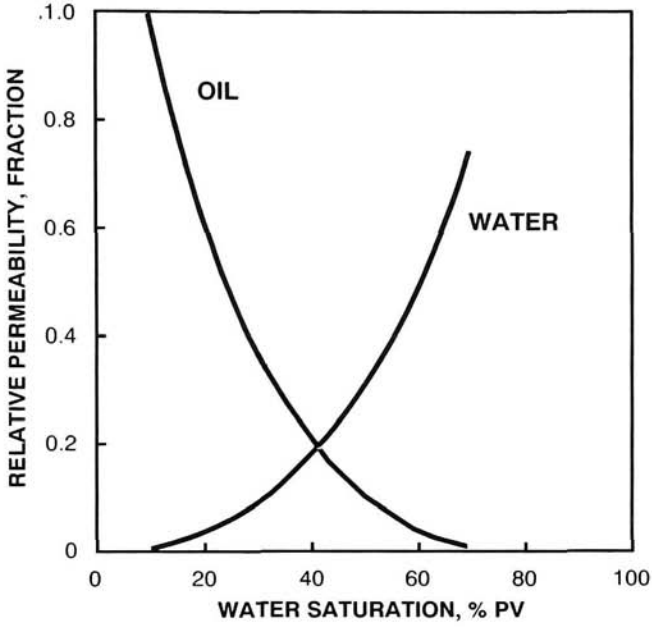


Figure 6.1. Water-oil relative permeability curves.

In Equations 6.1 and 6.2, the phase pressures for oil (p_o) and water (p_w) are related through capillary pressure effects.[59,60] The concept of capillary pressure in porous rocks is based on the observation of capillary effects in small tubes. In a standard physics experiment, when a small glass tube is inserted into a container of water, the water inside the tube rises above the baseline water level. The preferential wetting of the glass by water induces surface forces at the air/water interface. For small tube diameters, the induced surface forces extend over the entire interface and cause a significant pressure difference between the two fluid phases across the interface.

Similar to the capillary tube, pores in reservoir rocks are small, but the complex pore structure in rocks generates fluid interfaces that are far more complex than seen in the above experiment. To have a working definition, we set the water-oil capillary pressure as the pressure in the oil phase minus the pressure in the water

phase, $p_c=p_o-p_w$. Experimental capillary pressure measurements are usually plotted versus water saturation.

The assumptions that relative permeabilities and capillary pressures depend only on the water saturation greatly simplify the problem but should be viewed as working approximations. In experiments, the relative permeability of a phase usually depends on a number of other variables including the path that was followed to reach the saturation. This path dependence, or hysteresis, also applies to capillary pressure measurements. Still, the above description provides an engineering framework to model oil and water flow in reservoir rocks.

6.2 Scaling Up Reservoir Properties

Relative permeability measurements in the laboratory usually handle small rock samples with dimensions in inches. On the other hand, a reservoir simulation often covers an area extending over many square miles. Generally, computational time limits simulations to no more than 100,000 grid blocks. Thus, the grid dimensions are often hundreds of feet. We cannot expect the relative permeability measurements based on small samples to always carry over to these much larger scales.

Another problem is the porosity and permeability distributions between wells are unknown. Data at vertical wells provide measures of vertical heterogeneity but little lateral information. Since rocks properties change more rapidly vertically than laterally, the engineer often constructs models of stacked layers to treat the vertical heterogeneity. Properties are often uniform or mapped with smooth contours within a given layer, because we have so little horizontal information.

Both the great jump in scale and the simple layer cake model often lead to inadequate agreement between simulation results and field performance data. Then, the engineer goes through a history matching process to align the simulation results with field production data. Often relative permeability curves are changed from their laboratory values, because the scaling up problem makes them suspect. After the engineer obtains a sufficiently good match, he runs the simulation to predict future reservoir performance.

Fractal distributions like those generated in Chapter 4 and 5 provide more realistic models for permeability and porosity distributions. However, the scaling up problem to large grid sizes remains. Chapter 5 deals with the scaling up problem for

single phase flow. But the scale up problem for reservoir simulations must handle multiple phases.

The present chapter compares three different approaches to this scale up problem for waterfloods. Two of the methods start with waterflood displacements on 2D vertical cross sections. The first method scales up these 2D flow displacements to an areal streamtube model. The second method uses pseudorelative permeabilities to scale up flow to a coarser areal grid. The third method depends solely on the FRACTAM algorithm for single-phase flow to coarsen 3D distributions. Then, the coarser distributions are placed into a reservoir simulator. No single method is best for all applications. Instead, the particular field application dictates which method to choose.

We use a mature waterflood in southeastern New Mexico as a field case study for all three methods. In this field, the majority of development drilling occurred in the early 1940's. Wells produce from Grayburg dolomitic sands and San Andres dolomite pay zones at depths ranging from 3,600 to 4,300 feet (Figure 6.2). The primary production mechanism of the reservoir was solution gas drive.

Waterflood operations on five-spot patterns were phased in during the mid to late 1960's. The original 40-acre well spacing was infilled to 20 acres to form inverted nine-spot patterns in the early 1970's. Operations of the inverted nine-spot patterns generally remained unchanged for nearly 20 years. This long period of nearly constant operating conditions supports our use of a typical well pattern element to represent an entire section (1 square mile). We focus on the waterflood performance in two separate sections which we will denote Area A and Area B.

The porosity and permeability distributions in Chapters 4 and 5 are generated by a stochastic model. Therefore, the flow simulations are based on multiple realizations of porosity and permeability, each consistent with the same input data. For costly flow simulations, we will want to reduce the number of required simulations. Near the end of this chapter, we discuss an approach to rank the realizations and then select only a handful for comprehensive simulations.

6.3 Hybrid Finite Difference/Streamtube Method

Our first approach to waterflood simulations is a hybrid of finite difference and streamtube models.[80] Historically, both models have been used independently to simulate waterflood performance. As outlined at the beginning of this chapter, the

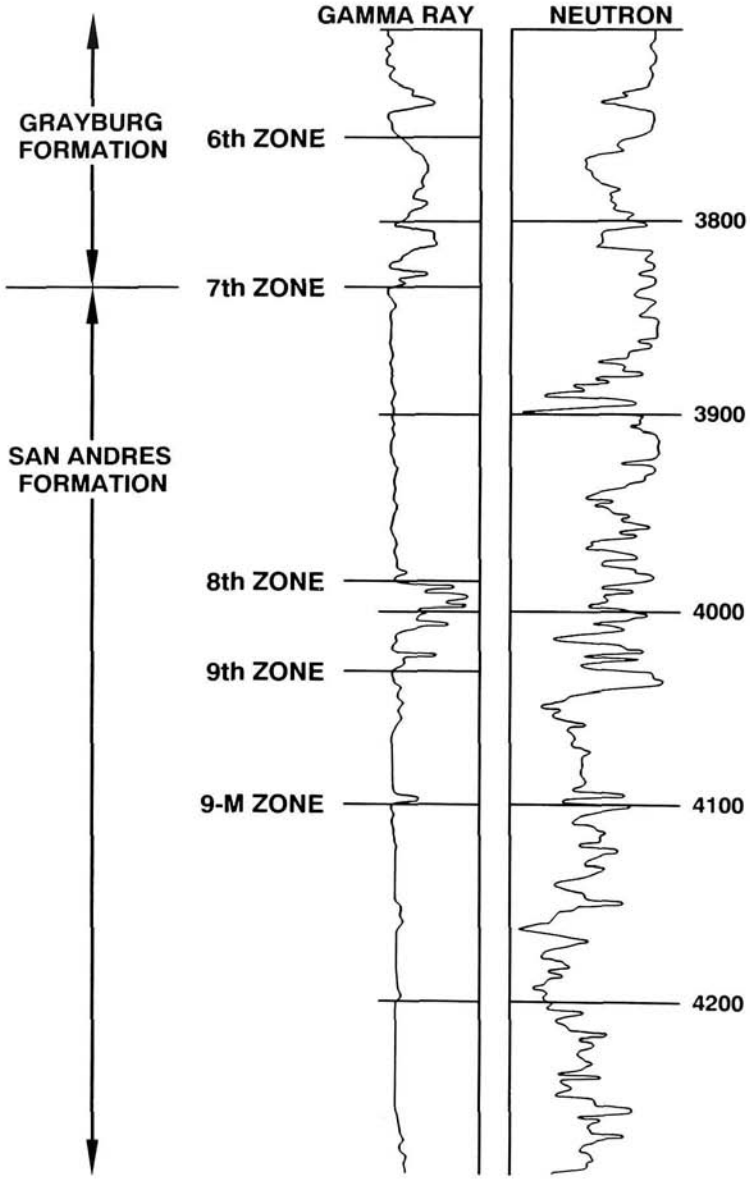


Figure 6.2. Typical porosity log from field case study.

finite difference model is based on dividing the reservoir into grid blocks. The streamtube approach is based on the concept of a streamline. Streamlines are paths followed by fluid particles as they move away from an injection well to a production well. A set of streamlines is shown in Figure 6.3 for an inverted nine-spot pattern. An inverted nine-spot pattern consists of a central injection well surrounded by nine producers. Figure 6.3 shows a symmetric element that is 1/8 of a pattern. A streamtube is the region bounded by two streamlines. The streamtube model simulates the displacement of oil by water along each streamtube. Hewett and Behrens[80] have combined the finite difference and streamtube approaches in a hybrid method that incorporates 2D fractal distributions. We will refer to their method as the streamtube method.

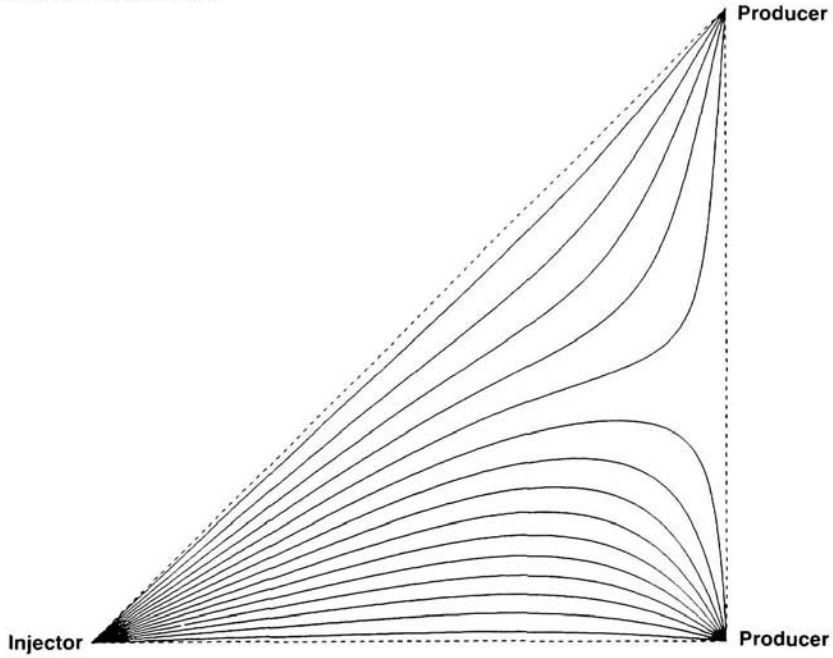


Figure 6.3. Streamlines for a one-eighth of an inverted nine-spot pattern with equal rates at producers.

The streamtube method starts with a 2D vertical cross section with fine grids (Figure 4.1). Equation 5.2 transforms the porosity distribution in Figure 4.1 into a corresponding permeability distribution. The FRACTAM algorithm, described in Chapter 5, is used to reduce the number of grids to fit into a reservoir simulator. In

our example, we use the FRACTAM algorithm to reduce the number of grids in the permeability distribution corresponding to Figure 4.1 and get the permeability distribution shown in Figure 6.4. Given a cross section, the streamtube method contains four major steps.[80]

1. Place the cross section into a conventional reservoir simulator and perform displacement calculations.

2. Determine fractional flows and mobilities from the reservoir simulation performed on the cross section.

3. Generate streamtubes for the field project.

4. Perform displacement calculations on the streamtubes using the fractional flows and mobilities from the cross-sectional simulation.

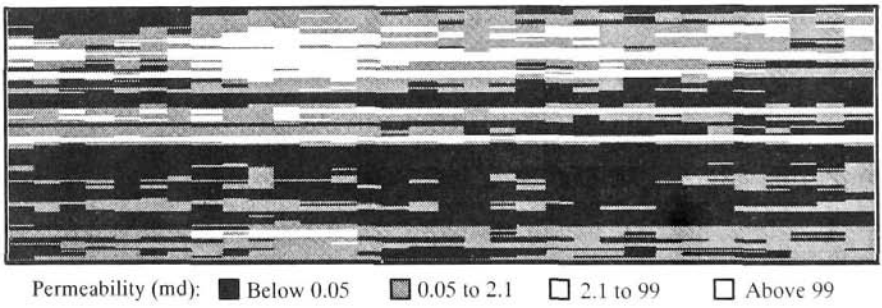


Figure 6.4. Fractal distribution of permeability for Area A based on fGn in the horizontal direction (H=0.87).

First, we will discuss some limitations of the streamtube method. Next, we will address each of the above four steps in more detail. The streamtube method is then applied to our field case. The simulated results are compared to field production data. Finally, we test the sensitivity of various parameters used to generate the cross sectional distributions of reservoir properties.

6.3.1 *Limitations*

The streamtube method circumvents the need for a full 3D model, but the 2D vertical cross section must capture all the significant physical mechanisms. For example, the hybrid method cannot handle gravity override in steeply dipping beds, unless the well pattern (injectors and producers) aligns the dominant flow direction along the axis of dip. With such alignment, 2D cross sections parallel to the axis of dip may capture gravity effects. If the injector/producer arrangement produces significant flows not aligned with the dip axis, no single cross section can capture the resulting 3D flow patterns. As another example, 2D cross sections cannot capture 3D flow patterns generated by a strong bottom water drive interacting with the areal flow patterns from injection/production wells.

Furthermore, the initial saturation distribution in the 2D cross section must adequately represent the 3D saturation distributions. A set of fractional flow and mobility curves from Step 2 above is only valid for the initial saturation distribution used in their derivation.

6.3.2 *Reservoir Simulations on Cross Sections*

Based on the interpolated porosity cross section, a conventional reservoir simulator (finite difference model) calculates the interwell flow in the reservoir. Careful thought should be given to the fluid and rock properties that are required as input into the simulator.

For our field case, laboratory relative permeabilities are available from core experiments for the Grayburg 6th and San Andres 9th Massive zones. The curves for each zone are grouped together. Each set of curves is rescaled to the formation initial water and residual oil saturation by linear scaling the total saturation change during the measurement. For the Grayburg 6th Zone, eight out of the ten curves fall in fairly narrow bands as shown in Figures 6.5 and 6.6. One set of oil and water curves in the middle of these bands is chosen as representative. These representative curves are from Core Plug No. 8 in Figures 6.5 and 6.6. The chosen curves are assigned to each grid block in the Grayburg portion of Figure 6.4. The same procedure is applied to the San Andres curves.

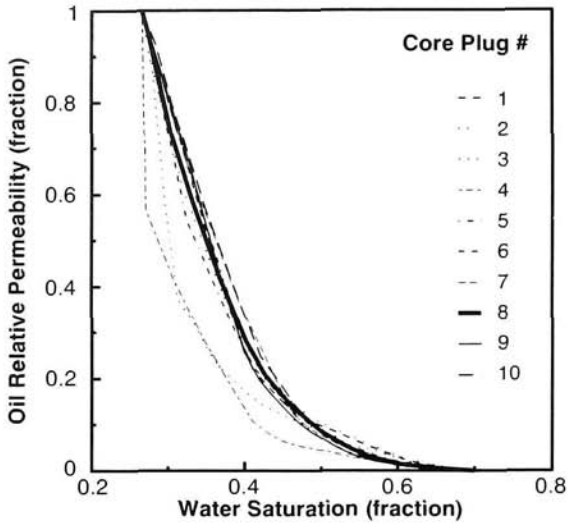


Figure 6.5. Grayburg oil relative permeability curves.

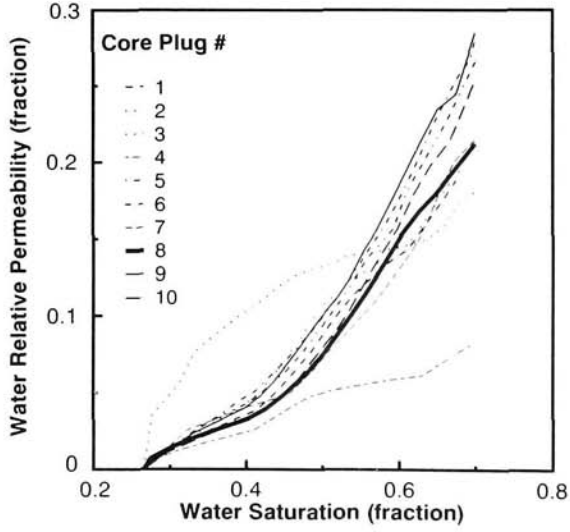


Figure 6.6. Grayburg water relative permeability curves.

The cross-sectional simulations treat the waterflood period but not the primary production period. All pressures in the cross section are maintained above the bubble point pressure to prevent any nonzero gas saturation. A single injection well is opened to every grid block on the extreme left of the cross section (Figure 6.4). The injection rate is controlled by the specified maximum injection rate. A single producer is opened to every grid block on the extreme right. The bottom hole pressure of the producer remains near the initial pressure throughout the simulation. Gravity effects are considered negligible and are not included, because the difference in oil and water densities is small. Furthermore, dense dolomite layers that separate the permeable rock zones extend over large areas and block gravity override effects. The initial water saturation is adjusted to give an initial water cut that compares with field data.

Following Hewett and Behrens,[80] we vary the transverse thickness of the grids with streamwise position. The idea is to model the shape of a stream channel between the wells. Obviously, the choice of a typical streamtube geometry is somewhat arbitrary. The purpose of using a variable width cross section is to account for the converging and diverging flows near the wells. Most simulators generate records of grid block data for subsequent post-processing. Arrays of pressure and water saturation are saved along with production/injection rates. Water saturations late in the waterflood for one cross-sectional simulation are shown in Figure 6.7.

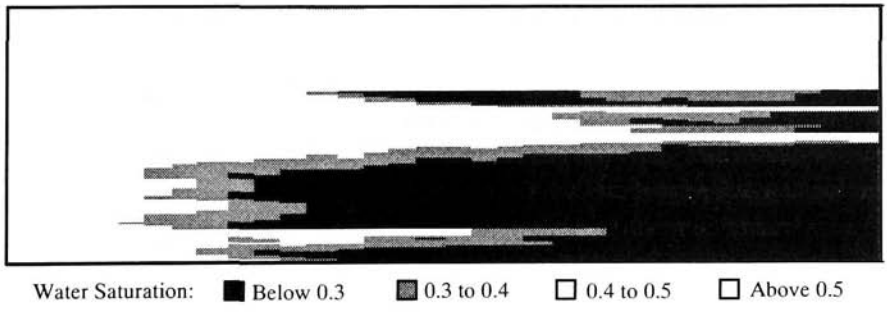


Figure 6.7. Water saturations late in the waterflood.

6.3.3 *Fractional Flow and Mobility Calculations*

The results of the cross-sectional simulation are fed into a streamtube model to calculate areal fluid displacements in the field. As an intermediate step, the output from the simulator must be reduced to an equivalent single layer solution. This requires the calculation of the fractional flow of water and oil (reservoir conditions) at the producer as a function of the cross-sectional pore volumes (fraction) injected. The output includes a fractional flow curve like the one shown in Figure 6.8.

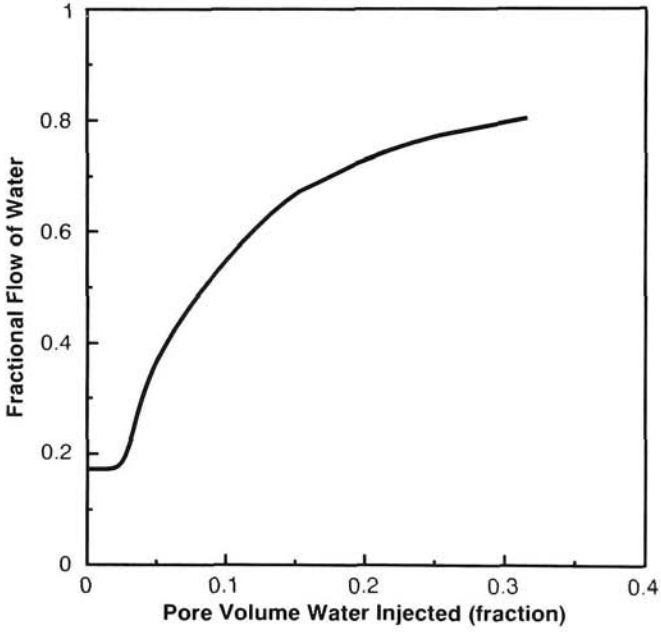


Figure 6.8. Fractional flow curve at outlet of vertical cross section.

The local mobilities for an equivalent single layer are obtained by integration in the vertical direction at fixed streamwise position x. Details are given in Appendix A6.2.1. After evaluation of the proper local mobility at each streamwise position, the entire array of mobilities is stored.

Instead of the streamwise position, a more appropriate coordinate is the fractional cumulative pore volume at x. Let PV(x) be the cumulative pore volume between the injection point and the local streamwise position and L be the total streamwise length of the cross section. Then, at any streamwise position x, the fractional cumulative pore volume is PV(x)/PV(L). At selected time steps, the local mobilities along the cross section are stored as a function of PV(x)/PV(L). This gives a two dimensional table for mobilities as a function of both Q(t)/PV(L) and PV(x)/PV(L), where Q(t) is the cumulative injected reservoir volume.

The fractional flow and mobilities for the equivalent single layer serve as input data for the displacement calculations on the streamtubes in Section 6.3.5.

6.3.4 *Generation of Streamtubes*

Fluid displacement in the field well pattern is calculated by incorporating the above fractional flow and mobility correlations into an areal streamtube model. A single-layer model is used, and we rely on the correlations to account for reservoir heterogeneity.

The first step is the generation of streamlines for the field well pattern. For equal fluid mobilities and a homogeneous, infinite reservoir, the generation of streamlines is a straightforward calculation as outlined by LeBlanc and Caudle[81] (see Appendix A6.2.2.) With their technique, one can track the movement of a particle from an injector to a producer to generate a streamline. If n particles spaced evenly around an injector are used to generate streamlines, then each of these streamlines is the center of a streamtube that carries 1/n times the well injection rate. A set of streamlines for the inverted nine-spot pattern is shown in Figure 6.3. Only one-eighth of a pattern needs to be considered due to symmetry. These streamlines correspond to equal mobility fluids and a single homogeneous reservoir layer.

For nonunity mobility ratios, the velocity field continuously changes. One approach to handling this difficulty is to recalculate the streamlines as the displacement proceeds. However, Martin and Wegner[82] found that a much simpler approach is valid for mobility ratios from 0.1 to 100. The effects of the nonunity mobility ratio can be approximated by relaxing the requirement of equal volume flow through each streamtube. Instead, the fluid injected into each streamtube is allocated according to the total mobility of the fluids along the tube. This method is also used by LeBlanc and Caudle.[81]

In our field case, we need streamlines for a pattern element from repeated inverted nine-spot patterns. The streamlines are generated by considering a large number of regular patterns in an infinite reservoir. The streamlines are taken from the middle pattern where the distortion from edge effects is considered negligible.

The method can be generalized to include other regular patterns or even an arbitrary well arrangement with reservoir boundaries specified. In generating the streamlines, we assume the well geometries and relative rates are unchanged during the waterflood life. Martin and Wegner[82] have relaxed this assumption by updating the streamtubes during the displacement calculations from instantaneous saturation and mobility distributions.

6.3.5 *Streamtube Displacement Calculation*

The next step is a displacement calculation. Each streamtube is divided into segments along the flow direction. The fractional flow curve (Figure 6.8) is applied at the producer end of each streamtube. The 2D mobility table described in Section 6.3.3 is used to assign mobilities to each segment along the streamtube. Each segment has its own PV(x). The fractional flows and mobilities are updated at each time step.

The total resistance, R_j, along the jth streamtube is calculated by a line integration taking into account the local total mobility and streamtube width. (See Appendix A6.2.3.) The resistances are summed for all the streamtubes that begin at a specified injector. The sum is denoted by R_T. For a specified rate at the injector, the fraction of fluid allocated to the jth streamtube is proportional to the sum of resistances, R_T divided by the resistance of the jth streamtube, R_j. This method of allocating injected water among the streamtubes is valid, if the same pressure drop occurs between the injector and all producers that receive its support. For many waterfloods this assumption is reasonable.

At each time step, the displacement calculation follows these steps.

1. The fluid rate at each injector is specified as an input parameter.

2. The total resistance in each streamtube is determined by line integration.

3. The injected fluid is allocated among the streamtubes inversely proportional to the total resistance of each streamtube.

4. The incremental pore volume injected for each stream tube is the injected volume divided by the volume of the tube. A corresponding volume is produced.

5. For each streamtube, the cumulative pore volume injected determines the fractional phases of the produced fluid from the correlation in Figure 6.8.

6. The contributions of all tubes terminating at a producer are added to give the producer's oil and water rates at that time step.

The above steps are repeated at each time step.

6.3.6 *Comparison With Field Production Data*

In our field case, the fluid displacement calculations are performed without adjusting any available parameters except initial water saturations. The initial water cuts are as large as 35 percent at the start of the waterflood. Evidently, these high initial water cuts are due to water transition zones with some possible water encroachment from edges and underlying aquifers. These initially high water cuts made it necessary to adjust the initial water saturations in the reservoirs to produce correspondingly high initial water cuts in the simulations. From the input relative permeability curves and fluid viscosities, the water cut can be calculated for a given water saturation.[59] The initial water cut is matched by assuming it is constant with depth along the wellbore.

The streamtube method predicts water cuts as a function of pore volumes of water injected. The method does not predict rates of injection or production. In the field case, there is a one-to-one correspondence between cumulative produced volumes and time. The time coordinates in the following graphs are obtained by simply honoring historical cumulative produced total fluid volumes.

The statistical model can generate an unlimited number of realizations. For Area A, flow simulations on 20 fGn realizations give a range of water cut curves.[46] The shaded region in Figure 6.9 covers the maximum range of all 20 realizations. The chosen water saturations give an initial water fractional flow of 12 percent. This

initial condition forces the simulation to agree with the field data for the first year. The agreement between the simulation and field data at later years occurs without any further adjustments.

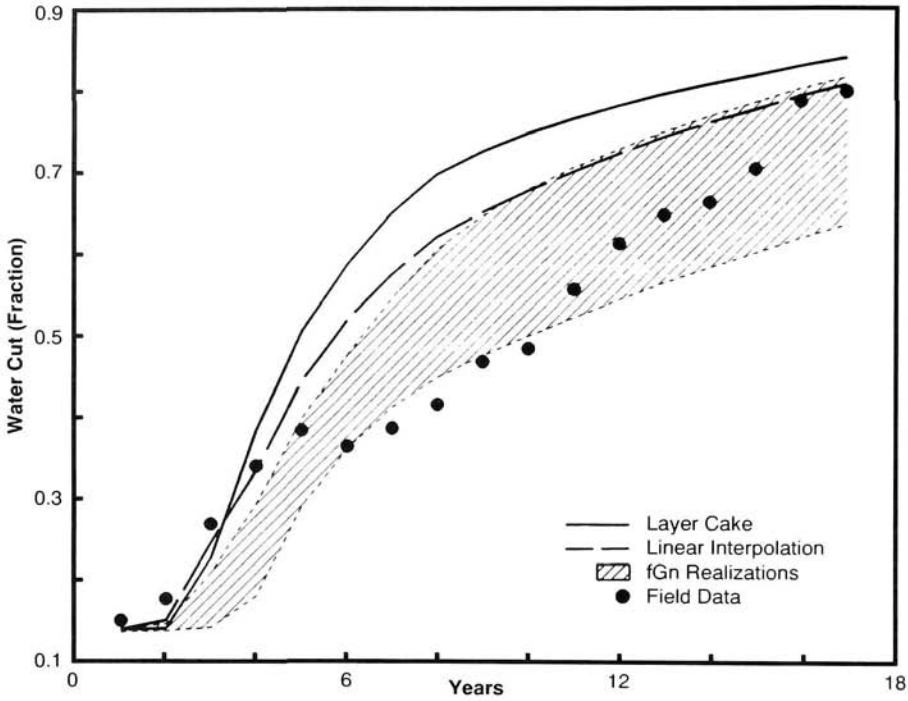


Figure 6.9. Simulated water cuts for 20 fGn realizations for Area A (higher S_{wi}).

For reference, water cut curves are also shown for linear interpolation (cross section is shown in Figure 6.10) and layer cake cross sections. The linear interpolation distribution is created by linearly interpolating between the well data on the left and the well data on the right of the cross section at each depth. The layer cake distribution is created by taking the porosity values from the well at the right-hand edge and extending them across the cross section. For both linear interpolation and layer cake distributions, the simulated water cut curve is above the field data (Figure 6.9).

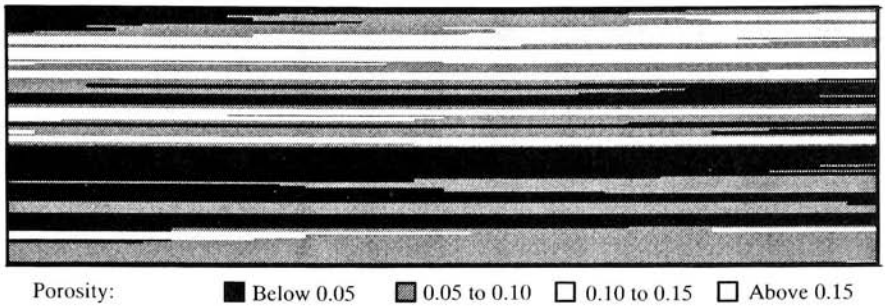


Figure 6.10. Linear interpolation of porosity for Area A (coarse grid).

The frequency distribution of cumulative oil production is displayed by the lightly shaded bars in Figure 6.11. For a 10-acre pattern element, the cumulative oil production ranges from 109 to 135 MSTB with a mean of 122 MSTB. The field oil production is 124 MSTB. For the layer cake case, the cumulative oil production is 81 MSTB, substantially less than all fGn cases and the field production. The linear interpolation case has a cumulative oil production of 104 MSTB. This is slightly less than the range of all the fGn cases and significantly less than the field production.

A set of earlier studies[22,50-52] on other reservoirs are based on fBm cross sections like the one shown in Figure 4.5. We can also apply fBm cross sections to our case study and compare fGn and fBm results. The only difference in the procedure for fBm cross sections is the generation of the cross section (Appendix A4.3).

In applying fBm cross sections, we adjust the initial water saturation to give the best match of the entire field water cut curve. To get a reasonable match to the entire curve, the simulated initial water cut had to be lower than the initial field data (Figure 6.12). The chosen water saturations give an initial water fractional flow of 7 percent in both the Grayburg and San Andres formations. We then repeated the linear interpolation and layer cake studies using a 7 percent initial water saturation. As shown in Figure 6.12, the linear interpolation case lies within the band corresponding to fBm cross sections. The layer cake case is above most of the fBm realizations and most of the field data values.

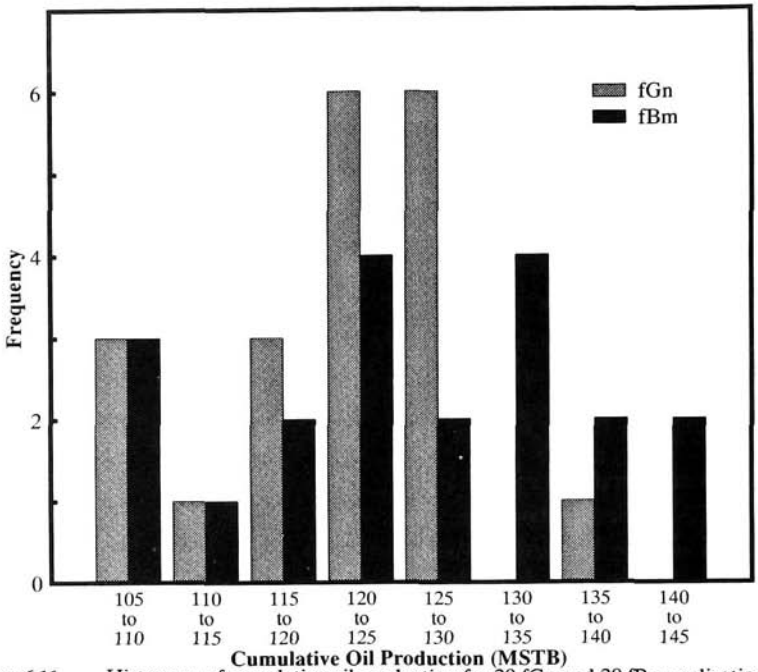


Figure 6.11. Histogram of cumulative oil production for 20 fGn and 20 fBm realizations (10-acre pattern element in Area A).

Flow simulations on 20 fBm type cross sections give a broader cumulative oil production histogram (Figure 6.11) than 20 fGn cross sections. The cumulative oil production for the fBm cases range between 105 to 143 MSTB with a mean of 125 MSTB. The corresponding linear interpolation case gives a cumulative oil production of 120 MSTB. This value is slightly below the mean of the fBm type cross sections. The layer cake cross section gives the lowest cumulative oil production of 95 MSTB.

Flow simulations for Area B give similar agreement with field water cut data for both fGn and fBm cross sections (Figure 6.13). For both cross sections, the initial water saturations are set to give an initial water fractional flow of 17 percent in all zones at reservoir conditions.

The field water cut data have considerable scatter and cannot be used to definitively say whether fBm or fGn cross sections are better. The close well spacing (20 acres per well) in our field example may diminish the differences between the two types of cross sections. As the well spacing decreases, simple linear interpolation becomes more valid and the importance of fractal noise diminishes. Still, the fGn

distributions should be more representative of reservoirs, because they are more consistent with outcrop and core photographs and horizontal well data in Chapter 3.

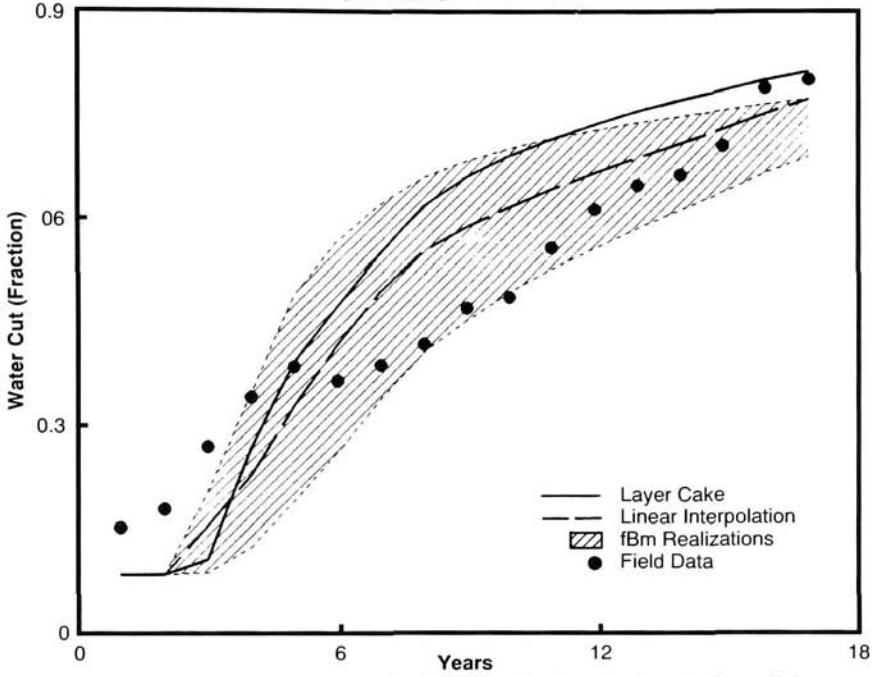


Figure 6.12. Simulated water cuts for 20 fBm realizations for Area A (lower S_{wi}).

6.3.7 *Sensitivity Studies*

All of the reservoir and fluid properties we specify in the above reservoir simulations have associated uncertainty. A complete sensitivity study would require varying all of the properties within their associated uncertainty. This process would be very time consuming. Instead, we focus on several parameters related to the generated fGn porosity and permeability distributions. These parameters are the intermittency exponent, H, the slope of the porosity-permeability transform, the ratio of vertical to horizontal permeability, and the relative permeabilility curves. In our field case, cumulative oil recovery is most sensitive to the slope of the porosity-permeability correlation.

The generation of the fGn porosity cross sections requires an input value for H. Figure 4.1 and Figure 6.14 have been generated with the same input data except

for H. In Figure 4.1 H equals 0.87, while H equals 0.6 in Figure 6.14. Even the random number seed is the same to ensure local porosity development occurs in the same location. The lower value of H in Figure 6.14 reduces the correlation length of the fractal noise. Thus, areas of local porosity development are smaller in size.

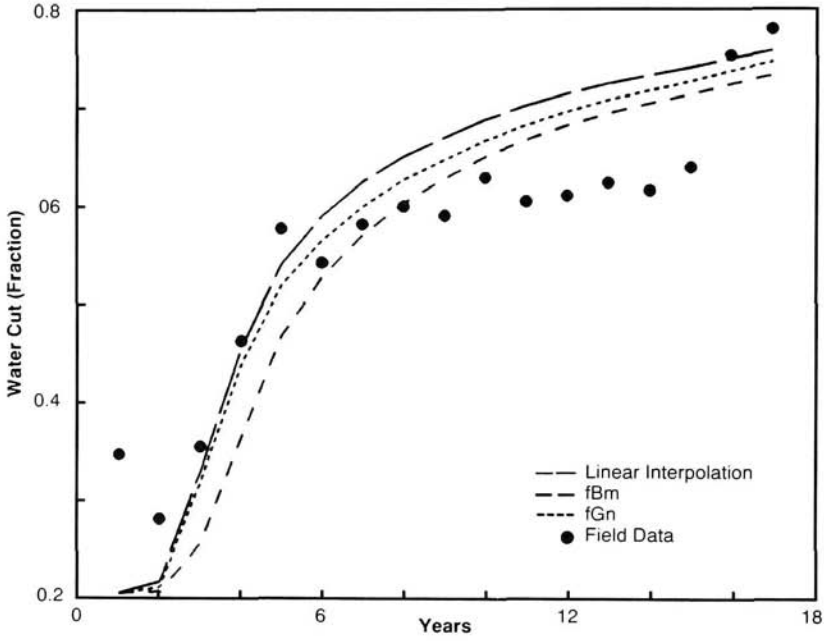


Figure 6.13. Simulated water cuts in Area B compared to field data.

We found that flow simulations in our field test case are insensitive to changes in H between 0.6 and 0.9. The cumulative oil recovery varies only 3 percent. This variation is much less than the variation among realizations in Figure 6.11. Later in Section 6.4.3, we vary the amount of fractal noise that is added to vertical cross sections. Flow simulations are much more sensitive to the amount of noise added than changes in H between 0.6 and 0.9.

Equation 5.2 converts the porosity distribution to a corresponding permeability distribution. In our field case, the range of permeabilities covers five log cycles over the corresponding porosity range of 2 percent to 17 percent. Decreasing this range to three log cycles of permeability by decreasing b in Equation 5.2 lessens the permeability contrast between high porosity zones versus low porosity zones. The

result is better vertical sweep in the flow simulations with lower water cuts and more oil recovery. The cumulative oil recovery increases by 15 percent. On the other hand, an increase in the range of permeabilities to cover six log cycles has the opposite effect and yields a 15 percent decrease in cumulative oil production.

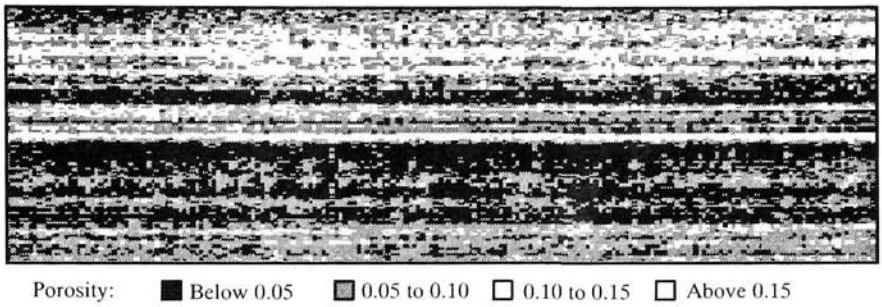


Figure 6.14. Fractal distribution of porosity for Area A based on fGn in the horizontal direction (H=0.6).

A vertical to horizontal permeability ratio of 0.1 was used in the above simulations. This ratio is just a best guess. Additional simulations show that water cuts and cumulative oil production are insensitive to changes in the permeability ratio between 0.01 and 1.0. The cumulative oil production remains within a ± 5 percent range. This insensitivity will probably not carry over to other cases with significant crossflow between layers or large gravity effects.

Relative permeability curves in the above simulations are from the middle of the bands (base case) established by core plug measurements (Figures 6.5 and 6.6). To test the sensitivity of oil recovery to the chosen set of laboratory curves, the Grayburg formation is modeled separately. We choose two sets of curves, each set representing one of the two edges of the bands in Figures 6.5 and 6.6. Since the shape of one of the water curves (#3) is unlike the others in Figure 6.6, this core plug was eliminated from the selection process. In flow simulations using the two extreme relative permeability curves, the oil recovery remains within about 10 percent of the base case result. Therefore, oil recovery is less sensitive to the choice of the relative permeability curve set than the slope of the porosity-permeability relationship.

All of the above results deal with the sensitivity of cumulative oil recovery but not flow rates. Flow rates depend on the effective permeability of the fractal distributions. The effective permeability for (single phase) linear flow in the cross section is easy to evaluate from Equation 5.1, but the permeabilities for linear flow cannot be used directly to predict injection or production rates in the field. The geometry of the flow paths is different in the field case. Still, variations in 2D permeabilities should give some indication of the corresponding 3D variations.

We found that the effective permeability can vary by a factor of 10 or more among the different types of cross sections.[46] The layer cake cross section has the highest effective permeability. The average effective permeability of 20 fGn realizations is a factor of 20 less than the layer cake permeability. The average effective permeability of 20 fBm realizations lies in between layer cake and fBm permeabilities, that is a factor of 5 less than the layer cake permeability.

The effective permeability is moderately sensitive to the value of H. The effective permeability of the cross section in Figure 6.14 (H=0.6) is about 30 percent higher than the effective permeability of Figure 4.1 (H=0.87).

6.4 Pseudorelative Permeabilities

Our second approach to waterflood simulations is using pseudorelative permeabilities. Workers at Chevron have applied the streamtube model successfully to waterfloods and CO_2 floods.[22,50-52,83] Yet, some other people view the streamtube model as inflexible, because the method cannot easily account for operational changes such as infill drilling. Pseudorelative permeability curves offer another way of mapping the simulated waterflood on the vertical cross section to an areal model.

The concept of pseudorelative permeability was developed in the 1960's to accommodate vertical sweep effects in areal simulations. In our application, the areal model of the field project has only one layer, similar to the streamtube approach. The idea is to calculate appropriate (pseudo) relative permeability curves for a single layer that account for nonuniform vertical sweep. The nonuniform sweep may occur due to gravity segregation or permeability variation. The pseudorelative permeabilities are calculated from a flow displacement on a 2D vertical cross section. As with the streamtube model approach, the 2D cross section must capture all the important physical mechanisms.

Given a cross section, the method contains three major steps.[84-86]

1. Place the cross sections into a conventional reservoir simulator and perform displacement calculations.

2. Analyze the displacement results and calculate pseudorelative permeability curves.

3. Perform displacement calculations on an areal grid using the pseudorelative permeabilities from the cross-sectional simulation.

Details on the calculations of pseudorelative permeabilities are given in Appendix A6.3 and A6.4. We start with the Kyte and Berry method[87] to calculate pseudorelative permeabilities, because their method is widely used.

We illustrate these steps in some example problems below and compare the results with the streamtube model. We start with the case of a miscible displacement of equal mobility fluids. Since the streamtubes remain unchanged during the entire displacement process, the streamtube model should give the exact solution. Next, we compare the methods for our field case. Finally, a set of sensitivity cases show the Kyte and Berry method can sometimes produce unrealistic pseudorelative permeability curves. A flux-weighted method is proposed as an alternative.

6.4.1 *Miscible Displacement*

Since the streamtube method should give the exact solution for the miscible displacement problem, we can assess the errors in the pseudorelative permeability method through a comparison of the results.

Errors in the pseudorelative permeability method are due to: (1) numerical dispersion in the finite difference model or (2) the mapping between the vertical cross-sectional results and the areal model. In the following examples, we use a sufficiently fine grid in the areal finite difference model to ensure numerical dispersion is negligible. Then, the difference in the two areal models is largely due to the mapping schemes.

The porosity distribution in Figure 6.4 (and corresponding permeability distribution) is chosen for the study. We use a conventional black-oil simulator to model a miscible flood with a tracer injected into every grid block on the extreme left of the cross section. The injection rate remains constant. A single producer is

opened to every grid block on the extreme right. At the producer, the bottom hole pressure remains at the initial pressure throughout the simulation. After the initial pressure transient, the pressure distribution remains unchanged in the cross section, because the injected fluid and initial reservoir fluids have equal mobility.

We use the Kyte and Berry method to generate a different set of pseudorelative permeability curves at each streamwise position (grid exit face) in Figure 6.4. Figure 6.15 shows the spread among the psuedorelative permeability curves. Curves from grids 6 and 22 (from the left) bound all other curves. Simple arithmetic averaging of the entire set of 32 cases produces the average curve shown as the dashed line.

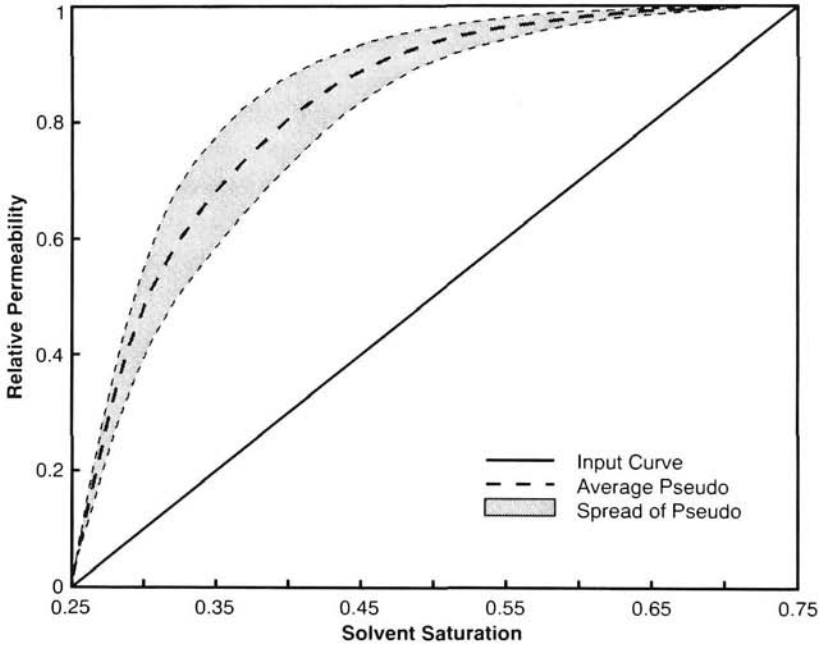


Figure 6.15. Pseudorelative permeability for the injected fluid in a miscible displacement with unit mobility.

The pseudorelative permeability curves provide a mapping from flow in the vertical cross section to an areal flow simulation. A finite difference areal model uses the pseudorelative permeabilities to simulate flow through a single layer. In particular, the relatively fine grids in Figure 6.16 handle flow in an inverted nine-spot pattern. The pattern becomes a five-spot by removing the side producer. In

Figure 6.16, the grids are parallel with the dominant flow direction between the injector and center producer (five-spot pattern). In the following simulations, we sometimes use a similar grid (not shown) that is diagonal to the dominant flow direction in the five-spot pattern.

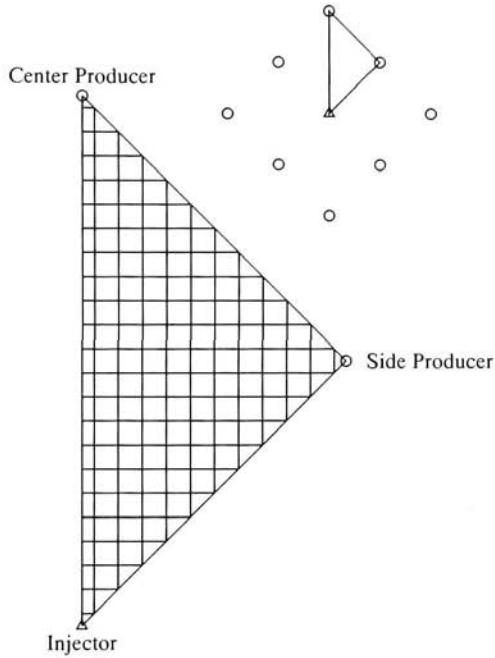


Figure 6.16. Grids that are parallel to the dominant flow direction between the injector and center producer.

The two extreme pseudorelative permeability curves in Figure 6.15 bound the responses on a five-spot pattern shown in Figure 6.17. The pseudorelative permeabilities corresponding to the highest curve in Figure 6.15 underpredict cumulative recovery by 16 percent, the largest discrepancy from the streamtube model. The average set of pseudorelative permeabilities gives a response that is very close to the streamtube simulation. In fact, the cumulative recovery is within 5 percent, which exceeds the usual accuracy of field data. In all the other simulations in this book, we use the average set of pseudorelative permeabilities.

For the relatively fine grids in Figure 6.16, grid orientation effects and numerical dispersion are small. Cumulative displaced fluid predictions are within 2

for parallel and diagonal grids. Grid orientation effects become even less for the inverted nine-spot pattern, because a single flow direction no longer dominates.

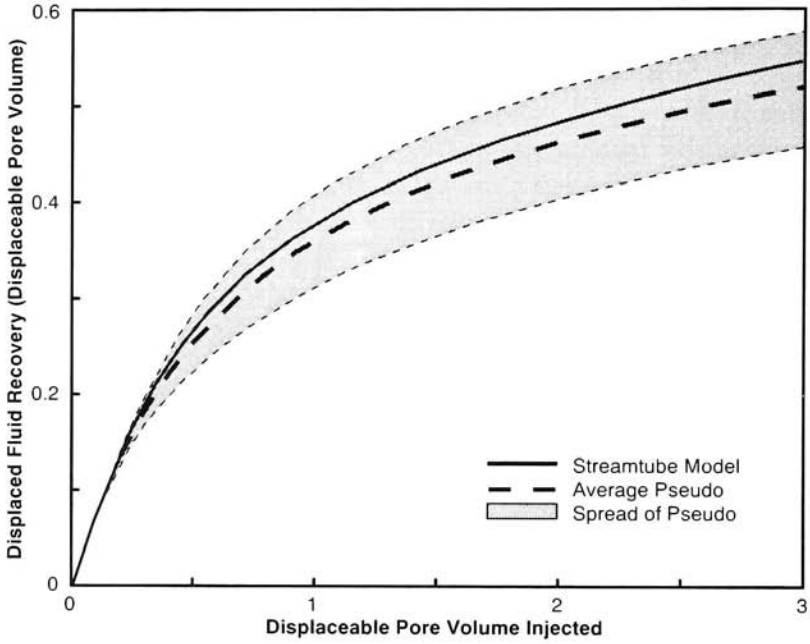


Figure 6.17. Cumulative displaced fluid recovery for miscible displacement on five-spot pattern.

6.4.2 *Comparison With Field Production Data*

Next we compare the pseudorelative permeability and streamtube method for our field test case. The results below show that two methods produce similar results. Again, we study a typical well pattern element to represent the inverted nine-spot patterns in a one-mile square area.

Recall that different sets of relative permeability curves and initial water saturations apply to the Grayburg and San Andres Formations. Different saturations add some complications in averaging pseudorelative permeabilities from different streamwise positions. In Figure 6.4, the porosity changes rapidly along a horizontal trace. Therefore, the ratio of Grayburg to San Andres pore volume among vertical columns of grids changes erratically with streamwise position. With different water

saturations in the two formations, the vertically averaged initial water saturation also varies among vertical columns of grids.

The vertically averaged saturation in each column is used to set the saturation of the beginning endpoint of the pseudorelative permeability curve for that column. Thus, the beginning endpoint saturations vary among the calculated pseudorelative permeability curves. Before finding the average psuedorelative permeability curve, the curves for each column are rescaled to the average initial and final water saturation for the entire cross section. The rescaling stretches or squeezes the curves by a linear transformation. Then, all the curves coincide to these common endpoints. The rescaling is the same one that is often applied to relative permeability curves from different core plugs.

The simulated water cuts based on the average set of pseudorelative permeabilities (Kyte and Berry) are nearly identical with the water cuts from the streamtube model (Figure 6.18). Both simulated water cut curves match the field data. The cumulative oil production values from both simulations are within 5 percent of each other.

The streamtube model results (Figure 6.18) are slightly different from those in Section 6.3. In Section 6.3, we allowed the cross-sectional width to change in the streamwise direction, similar to a streamtube between wells. On the other hand, pseudorelative permeabilities are usually based on cross sections with constant widths. For the results in Figure 6.18 and the following test cases, cross sections with constant widths are used for both the streamtube and pseudorelative permeability methods. For the streamtube model, the cumulative oil production, based on variable and constant widths, agrees within 3 percent. Therefore, for this field case, the differences are negligible.

6.4.3 *Sensitivity Studies*

Pseudorelative permeabilities produce results that are similar to the streamtube method for our field case. Are they also similar for other cases? The following set of sensitivity cases tries to answer this question. Two parameters are varied: (1) the amount of layering versus fractal noise in the distributions and (2) the mobility ratio of the waterflood. In all cases tried, the pseudorelative permeability method closely matches the streamtube method.

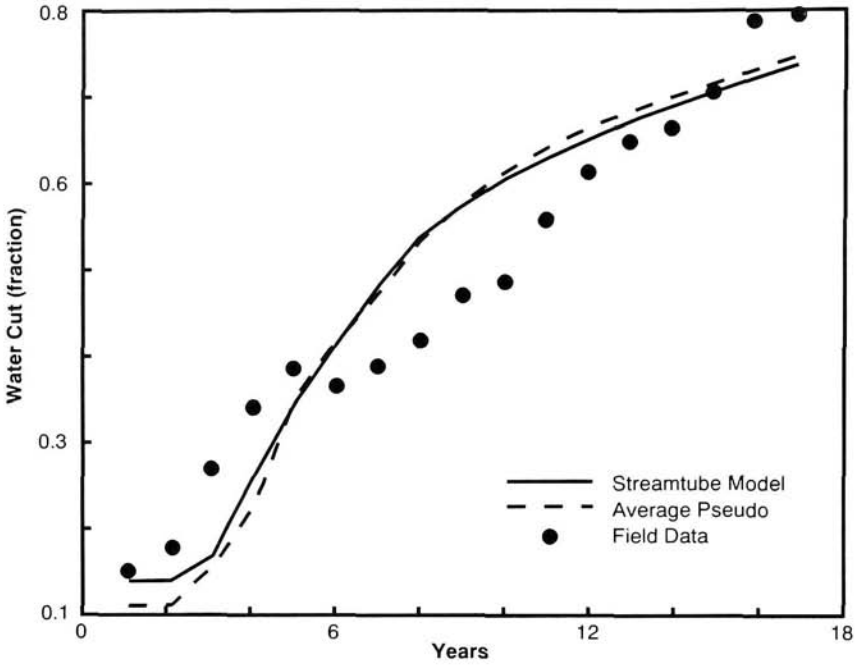


Figure 6.18. Water cuts for Area A.

The study uses three computer-generated porosity distributions that are based on Equation 3.2. Recall from Chapter 3, the value of B_{SD} determines the amount of layering. With B_{SD} set to zero, the porosity distribution becomes a layer cake as shown in Figure 6.19. As B_{SD} increases, the layers become more broken up by fractal noise. For B_{SD} equal to 1, some layering persists in Figure 6.20, but for B_{SD} equal to 10, few remnants of any layers remain, as shown in Figure 6.21. All three distributions have identical histograms. The histogram is Gaussian with a mean porosity of 9 percent and a standard deviation of 4 percent. The mean and standard deviation are typical of the Grayburg formation in our field case. For all three distributions, H equals 0.8.

Equation 5.2 transforms each porosity value in Figures 6.19 through 6.21 to a permeability value. In the transformation, a is set to 0.001 and b is set to 33.3.

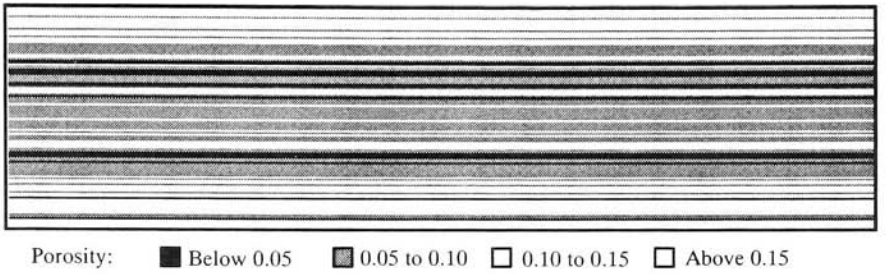


Figure 6.19. Porosity distribution with $B_{SD}=0$.

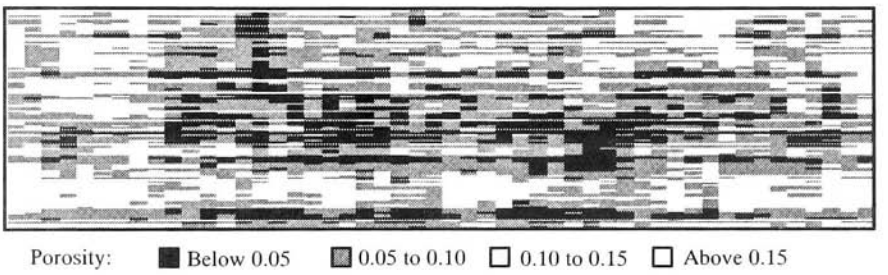


Figure 6.20. Porosity distribution with $B_{SD}=1$.

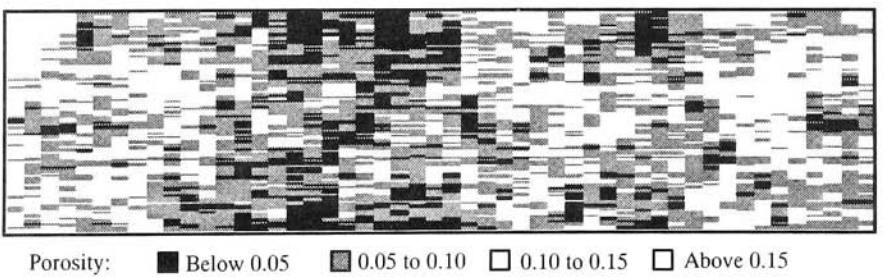


Figure 6.21. Porosity distribution with $B_{SD}=10$.

The mobility ratio is the second parameter varied in the sensitivity study. The mobility ratio is defined as

$$M = \frac{\mu_o \, k_{rw}(S_{or})}{\mu_w \, k_{ro}(S_{wi})} \tag{6.3}$$

where $k_{rw}(S_{or})$ is the relative permeability to water at residual oil and $k_{ro}(S_{wi})$ is the relative permeability to oil at initial water saturation.

We change the oil and water viscosities to get various mobility ratios. We use the relative permeability curves given by

$$k_{ro} = (1 - R)^2 \tag{6.4}$$

$$k_{rw} = 0.4 \, R^3 \tag{6.5}$$

$$R = (S_w - 0.25)/0.5 \tag{6.6}$$

where S_w ranges from 0.25 to 0.75. Martin and Wegner[82] used the same relative permeability curves in their tests of streamtube models.

For unit mobility displacements on distributions in Figures 6.19 through 6.21, the pseudorelative permeability (Kyte and Berry) and streamtube methods give nearly identical results. Cumulative oil production curves from the three different distributions are shown in Figure 6.22. For each distribution, the curves are within 4 percent for the two methods. As expected, the completely layered distribution ($B_{SD}=0$) has the lowest oil recovery. High permeability layers in Figure 6.19 channel the injected water. The oil recovery increases with the amount of added fractal noise (Figures 6.20 and 6.21). The noise breaks up these high permeability streaks giving better vertical sweep.

For low mobility ratios near 0.1 and the distribution in Figure 6.20 ($B_{SD}=1$), the Kyte and Berry method gives a water pseudorelative permeability curve that is not well behaved (Figure 6.23). Large variations occur in the water curves among streamwise positions. Appendix A6.3 gives a detailed explanation of this problem with the Kyte and Berry method. Harris and Perkins[86] report similar problems.

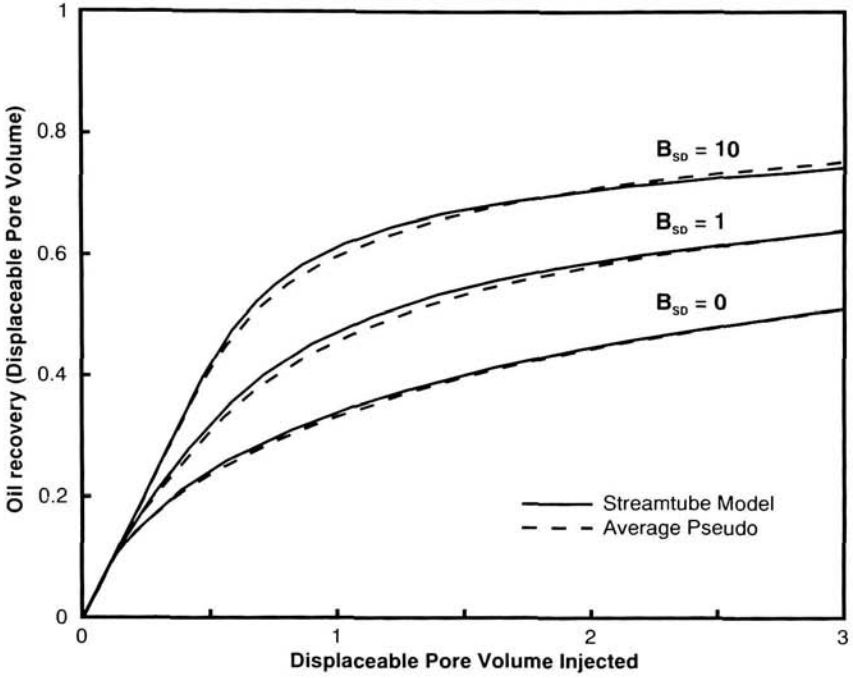


Figure 6.22. Cumulative oil production for different B_{SD} values.

Appendix A6.4 describes an alternative method[84] called the flux-weighted method to calculate pseudorelative permeabilities. For low mobility ratios, the flux-weighted method produces the curves shown in Figure 6.24. The flux-weighted method and the streamtube method produce similar cumulative oil production curves over a range of mobility ratios from 0.1 to 10. Figure 6.25 shows the results for $B_{SD}=1$.

For mobility ratios of 1 and 10, the flux-weighted method produces cumulative oil recovery curves that agree closely with the Kyte and Berry method. For mobilities less than 1, the flux-weighted method gives better-behaved results.

6.4.4 *Numerical Dispersion*

Our simulations indicate that corrections for numerical dispersion should not be included in pseudorelative permeabilities based on fractal cross sections. We get reasonable simulation results without any corrections as long as at least 7 grids are

between each injector and offset producer. This conclusion applies for waterfloods with mobility ratios between 0.1 and 10.

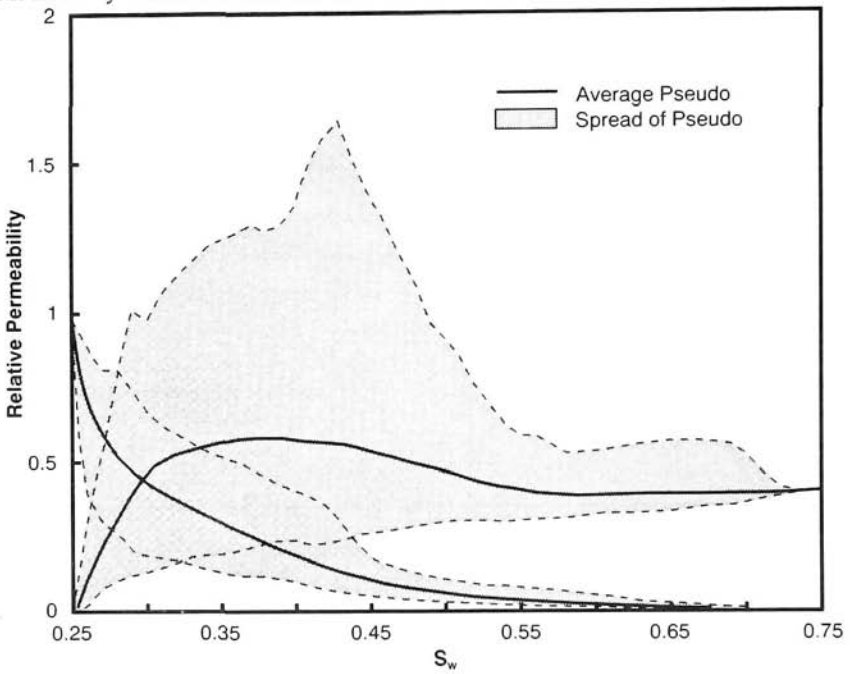


Figure 6.23. Pseudorelative permeability curves from the Kyte and Berry method for M=0.1 and B_{SD}=1.

In the past, users of the Kyte and Berry method often corrected the calculated pseudorelative permeabilities for numerical dispersion on the coarser areal grid.[85] To correct for numerical dispersion, one overlays a coarser grid (corresponding to the areal grid spacing) onto the finer cross section grid. Then, Kyte and Berry[87] demonstrate how to calculate pseudorelative permeabilities that include corrections for numerical dispersion on the coarser grid. Recently, Hewett and Behrens[85] have argued that these corrections depend on the paths between the injector and producer. Longer paths require a different correction than shorter paths. In general, we don't know the fluid paths in advance and cannot apply the proper corrections.

To get around this problem, we have used a sufficiently fine grid in the areal finite difference model that makes numerical dispersion insignificant. In all cases, the pseudorelative permeability curves originate from a vertical cross section with a finer

grid than the areal model. As the areal grid becomes coarser, numerical dispersion eventually leads to greater differences between the pseudorelative permeability and streamtube model results. If seven or more grids are between the grids containing the injector and producer, the streamtube model and pseudorelative permeability cumulative oil production agree within 10 percent.

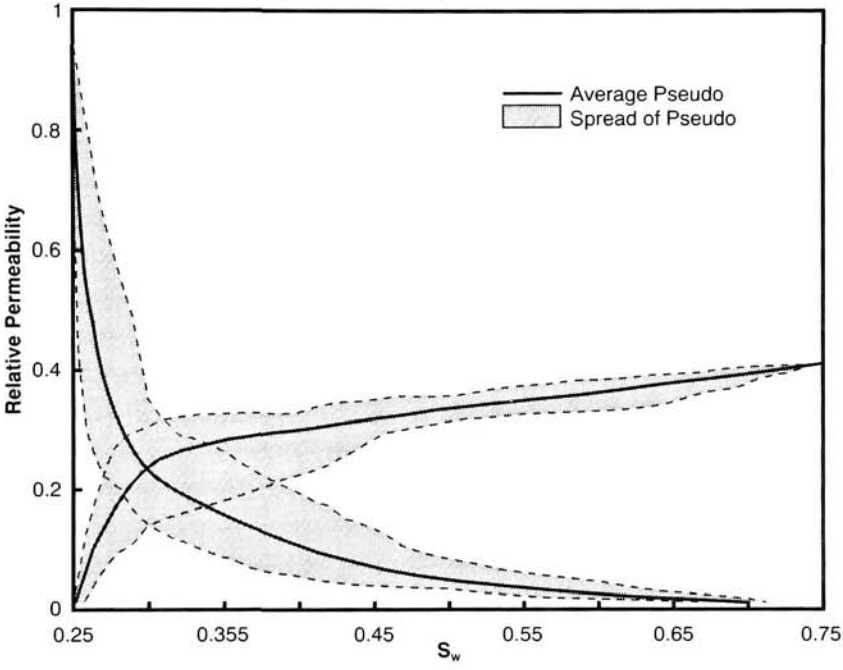


Figure 6.24. Pseudorelative permeability curves from the flux-weighted method for M=0.1 and $B_{SD}=1$.

6.5 3D Simulations

The third approach to waterflood simulations uses 3D fractal distributions. The distributions must start at sufficiently fine scale to capture the heterogeneity seen in well logs, but these fine-scale distributions contain too many grids to fit in a reservoir simulator. Therefore, we scale up to a coarser grid using arithmetic averaging for porosity and the FRACTAM algorithm for permeability.

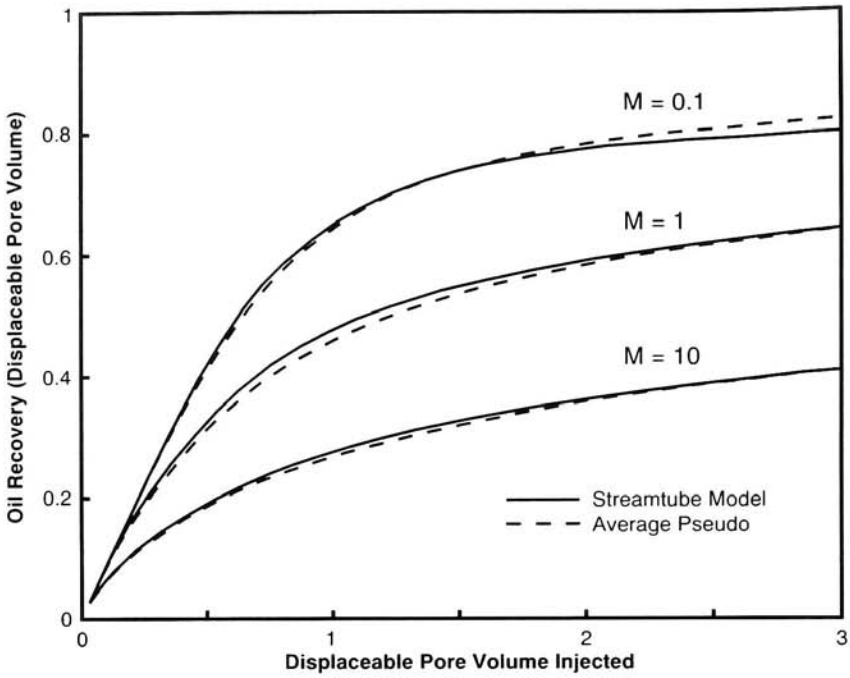


Figure 6.25. Cumulative oil production based on pseudorelative permeabilities from the flux-weighted method.

We wish to compare the results of the streamtube and pseudorelative permeability methods to the 3D simulations, but we run into a problem immediately. The first two methods used only two well logs, because they start with a single vertical cross section. For the 3D simulations, we require four wells to bound the 3D volume. In order to proceed in an orderly fashion, we first perform 3D simulations based on only the two well logs used in the other methods. Next, we perform simulations based on all four wells bounding the 3D volume under study. As explained below, we find both procedures do an acceptable job in simulating the field response. The simulated cumulative oil recovery from the 3D simulations is within 5 percent of the earlier results based on the streamtube method.[47]

6.5.1 *Fractal Distributions*

We use four wells with modern well logs that make up one-fourth of an inverted nine-spot pattern (Figure 6.26). Wells A and B have been used in the previous 2D cross section for study Area A.

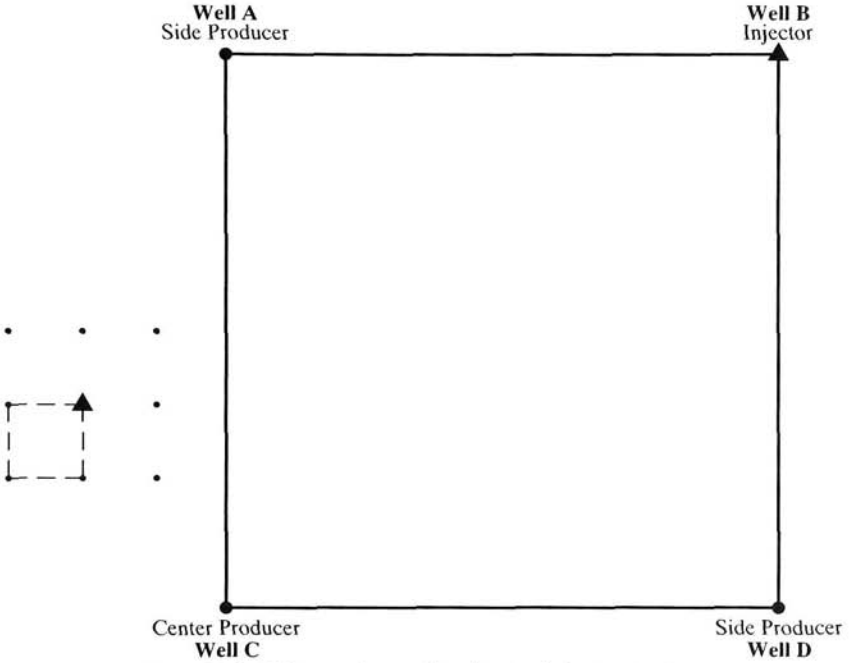


Figure 6.26. Typical 20-acre element from inverted nine-spot pattern.

For this study, the porosity logs are first averaged over 6 ft intervals. The 2 ft intervals previously used in the 2D distributions would require more than 30 million cubic grids in 3D. To reduce the number of grids, we use the averaged logs to produce 6 ft cubic grids. The method described in Section 4.7 is used to generate the 3D distributions. Each of the four wells is placed at one corner of the generated distribution. The resulting distribution is 160 x 160 areal grids and 50 vertical grids. Figure 4.6 shows a 3D porosity distribution with the top 3 grid planes sliced off to show internal heterogeneity within a productive lobe.

The number of grids is reduced further to fit into a reservoir simulator. The FRACTAM algorithm (see Section 5.4) is used to coarsen up the grid to 20 x 20 x

25 (N_x x N_y x N_z). The vertical slice in Figure 6.27 from the coarsen 3D distribution has a grid that is coarser than the grids used in the earlier 2D simulations. For this reason, the 2D study has been repeated with the streamtube method, using comparable grid sizes of the 3D simulations. The distributions in 2D and 3D are also generated on comparable grid sizes prior to any scaling up with FRACTAM.

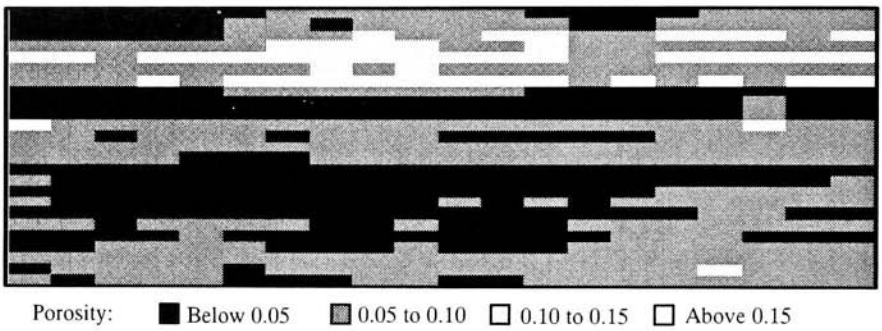


Figure 6.27. Vertical slice of 3D porosity distribution on coarser grid.

The introduction of other well logs besides Wells A and B adds some complications in making comparisons with the previous 2D cross-sectional results. Ideally, we would like to change only one thing at a time in our waterflood studies. In order to do this, we first study 3D distributions based only on the Wells A and B in Section 6.5.2. Then, Section 6.5.3 deals with distributions based on all four wells.

6.5.2 *Simulations Based on Two Well Logs*

For the 3D distributions based on only the A-B well pair, Well B still serves as the injector, but the porosity log from Well A is used as input on the remaining three corners of the 3D volume (Figure 6.26). Then, every fluid path in the 3D distribution effectively travels between the same two wells found in the cross section. In the 3D simulations, we apply a constant bottom-hole pressure at the producers. Injection rates are set by historical data.

The rates vary among producers, since preferential flow paths can exist to one producer but not the others. The first step in the generation of the 3D distribution is trilinear interpolation between values at the wells for each depth (see Section 4.7). At this stage, the pattern is balance, because the Well A well log is applied at each of the three producers. The second step is the addition of fractal noise. This noise creates the areal heterogeneity shown in Figure 4.6 and sets up preferential flow paths.

Twenty realizations give the distribution of cumulative oil recovery shown with the solid bars in Figure 6.28. Note cumulative oil recovery is given for a 20-acre pattern element, instead of the 10-acre element used in Sections 6.3 and 6.4. Twenty realizations in the 2D simulations give the distribution shown with the lighter shaded bars. The mean recovery of the 3D cases (236 MSTB) is within 5 percent of the mean recovery for the 2D cases (247 MSTB). Yet, the 3D cases give a broader range.

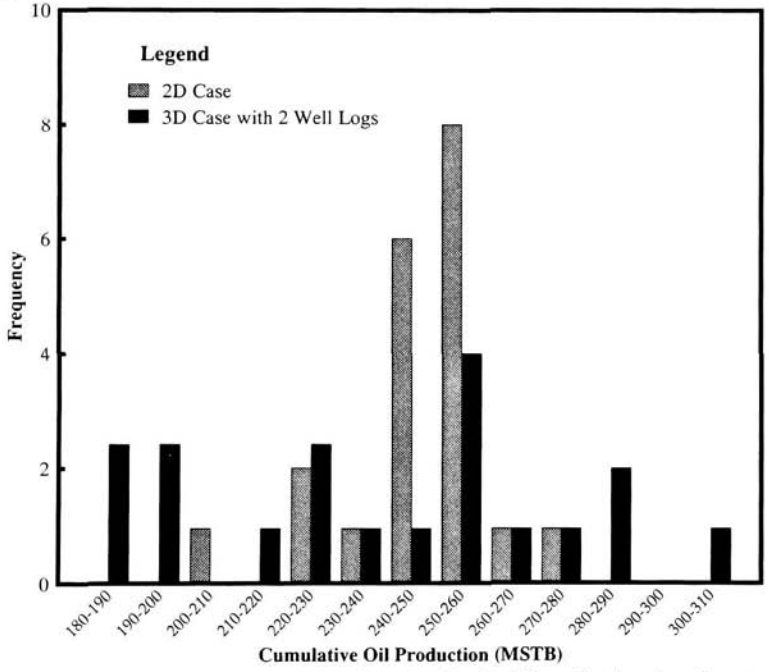


Figure 6.28. Histogram of cumulative oil production for 3D realizations based on two well logs compared to 2D results.

Areal heterogeneity in the 3D permeability distributions causes this broader range. Each producer has the same constant bottom-hole pressure. The ternary plot in Figure 6.29 confirms this interpretation. Each of the vertices in Figure 6.29 corresponds to one of the three producers in the pattern element in Figure 6.26. The relative total fluid production (RB) of each producer determines the location of each circle. If the total fluid production is equal among all three producers, the circle lies in the middle of the triangle (equal distance from all three vertices). Otherwise, the circle is placed closer to the dominant producer (with the highest total fluid production).

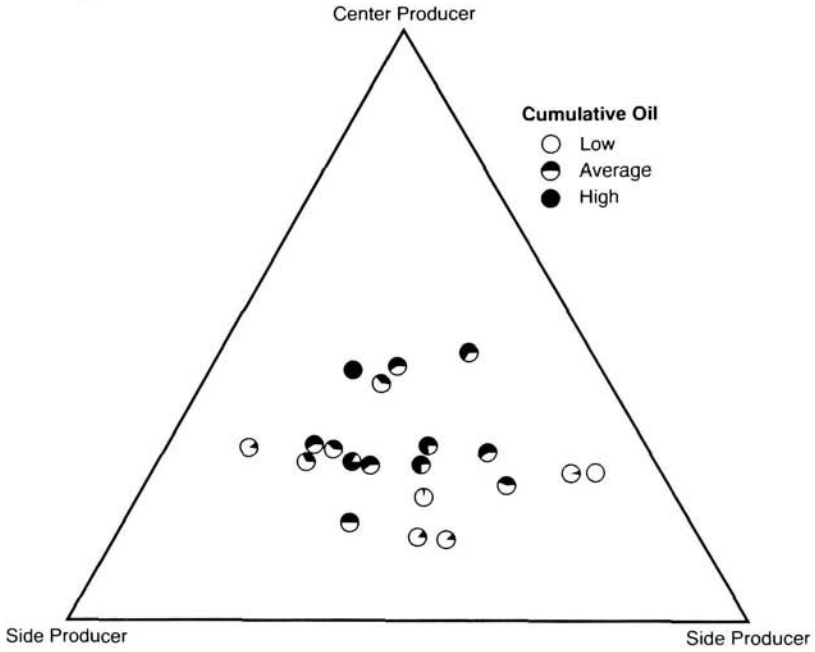


Figure 6.29. Ternary plot of cumulative oil production for 3D realizations based on two well logs.

The shading of each circle represents the cumulative oil production from a single realization. An open circle represents the lowest oil recovery among the 20 realizations, while a completely filled circle corresponds to the highest recovery. Partially shaded circles are based on a linear scale between these two extremes. Figure 6.29 shows that the greatest recoveries (more shaded circles) occur when the pattern is balanced or when the center producer dominates. If one or both of the

side producers dominate, cumulative oil recovery decreases. This trend is indicated by less shaded circles for the extreme left, right, and bottom points.

The trend of lower oil recovery whenever a side producer dominates can be explained as follows. The center producer is farther away from the injector than the side producers. If the total fluid rates of all producers are equal and the reservoir is homogeneous, then water will break through at the side producers first. After breakthrough, water will tend to channel to the side producers and bypass the remaining oil in the pattern. Increasing the center producer's rate above the side producer's rate will increase oil recovery by delaying water breakthrough at the side producers.

When the 2D cross sections are coupled to the streamtube model, the production rates of each producer are set equal. Thus, all points from the streamtube model would lie in the middle of the triangle in Figure 6.29.

6.5.3 *Simulations Based on Four Well Logs*

We now consider simulations on 3D distributions based on all four wells. The spread in oil recovery among realizations is nearly the same with either all four well logs or two well logs. Figure 6.30 shows the histograms for both cases. The spread is the same despite the fact that Well D has better porosity development than the other 3 wells.

This better porosity development makes Well D produce a greater share of the produced fluids relative to the other two producers. The same constant bottomhole pressure boundary condition at each producer allows individual well rates to vary relative to each other.

The 3D simulations with four well logs require one additional modification. The porosity/permeability correlation in Equation 5.2 works for well logs from Wells A and B where porosity values are below 16 percent. In Well D, a modification is necessary where porosities in the Grayburg 6th Zone reach 20 percent. Equation 5.2 with a equal to 0.001 and b equal to 33.3 gives permeabilities near 5,000 md for 20 percent porosity. Such high values are unrealistic, because core permeabilities remain below 100 md. Thus, a permeability ceiling of 100 md has been applied. Any grids with porosity greater than 15 percent are assigned a permeability of 100 md.

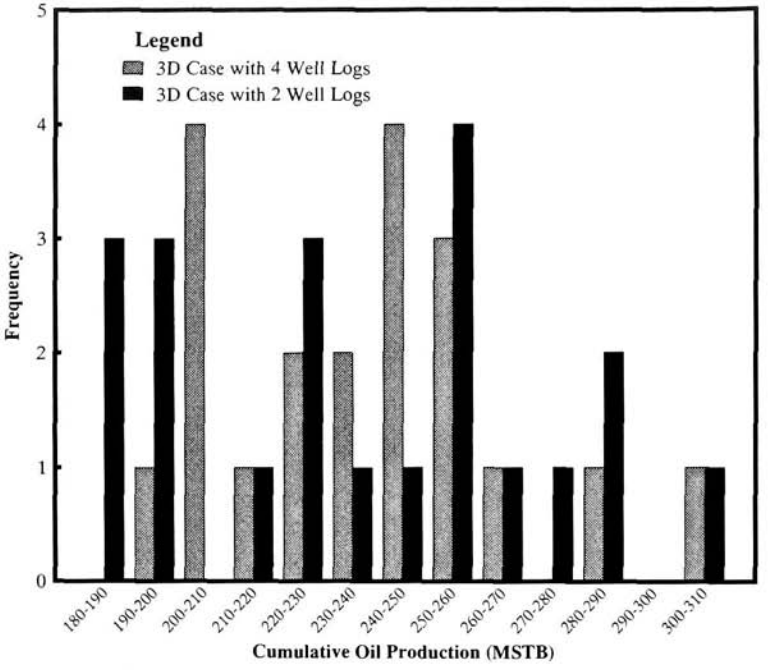


Figure 6.30. Histogram of cumulative oil production for 3D realizations based on two and four well logs.

6.6 Limitations in Scaling Up

For the above 3D flow simulations, we used the FRACTAM algorithm to reduce the number of grids to fit into a reservoir simulator. But the FRACTAM algorithm was developed for single phase flow and not multiple phases. In the following study, we see how coarse we can make the grid in our waterflood test case while maintaining tolerable errors in predicted water cuts and oil recovery.

For simplicity, we test the limitations of FRACTAM on the 2D cross-sectional permeability distribution corresponding to Figure 4.1. Prior to coarsening the grid, 20 feet of nonreservoir rock is attached to the bottom of Figure 4.1. With this addition, there are 160 grids in the vertical direction instead of 150. This allows us to coarsen the grid further without padding the sides (Figure 5.10). Figures 6.31 and 6.32 are the resulting coarser permeability distributions.

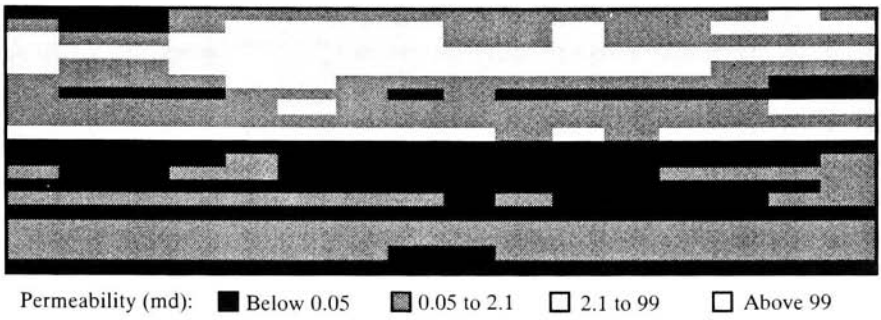


Figure 6.31. Fractal distribution of horizontal permeability with a 16 by 20 grid.

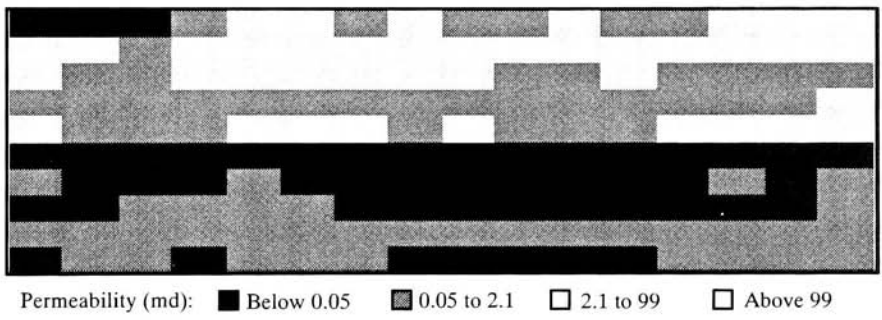


Figure 6.32. Fractal distribution of horizontal permeability with a 16 by 10 grid.

Again, we use laboratory (small scale) relative permeability curves for all grids regardless of grid size. Simulated water cuts for successively fewer grids in the horizontal direction all fall within a narrow band shown in Figure 6.33. The simulated waterflood displacement in the cross section is related to the field project by using the streamtube model in Section 6.3.

Water curves for successively small number of vertical grids are closely clustered from 20 or more layers (Figure 6.34). With 20 layers, the 8-foot thickness

of each layer is near the thickness of individual porosity lobes. With ten layers, we often average as much nonreservoir rock as reservoir rock into a grid representing an entire productive lobe. Thus, the permeability distribution becomes too homogeneous yielding lower water cuts (higher oil recovery).

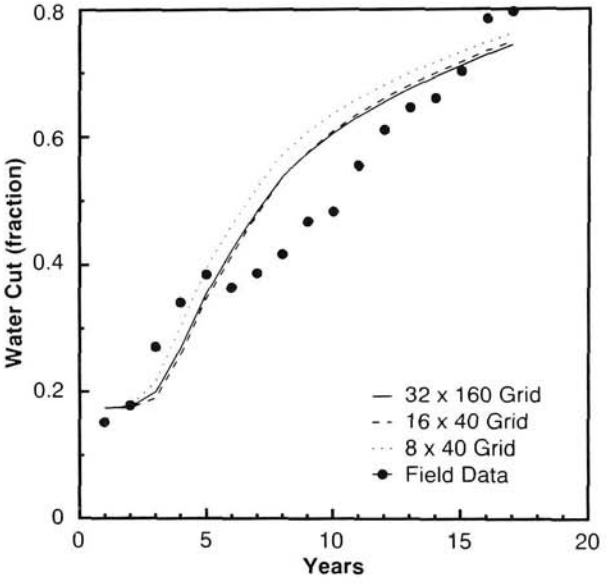


Figure 6.33. Simulated water cuts for successively fewer horizontal grids.

Therefore, scaling up with FRACTAM prior to waterflood simulations is a reasonable approach, but it has some limitations. For tolerable errors in the above example, the final coarse grid thickness cannot be larger than the thickness of individual permeability lobes.

Another approach is scaling up relative permeability curves in addition to absolute permeabilities. Kossack et al.[88] suggest one method which uses pseuedorelative permeabilities, but computational time becomes an issue for a large number of grids.

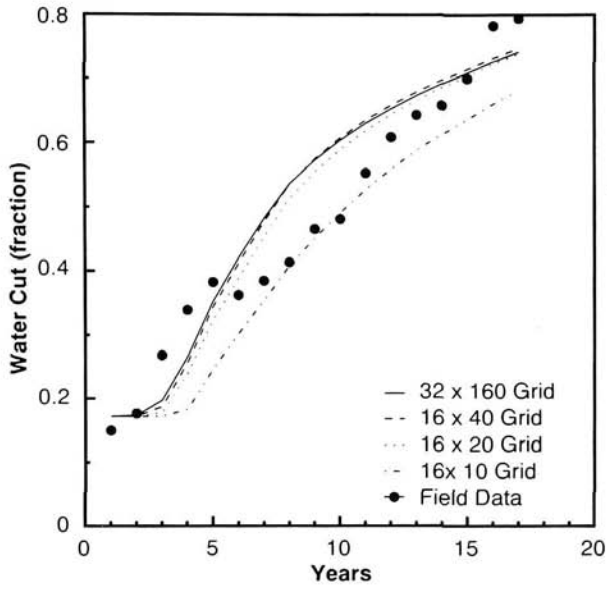


Figure 6.34. Simulated water cuts for successively fewer layers.

6.7 Ranking Realizations

In Sections 6.3 and 6.5, we performed reservoir simulations over a set of 20 realizations. The spread among realizations provides some estimate of uncertainty in oil recovery and water cuts consistent with the input data. The present study focuses on porosity and permeability distributions as a source of uncertainty. In practice, every reservoir and fluid property we specify in the reservoir simulation has an associated uncertainty (e.g., reservoir tops, faults, water saturations, etc.). We would like a set of reservoir images that capture the uncertainty of all of these parameters. At present, our ability to generate stochastic images of matrix porosity and permeability exceeds our capabilities for other parameters, such as fault and natural fracture systems. At least in the near future, our stochastic models may not always include all the important parameters.

Before proceeding further, we need to recognize some additional sources of uncertainty. In generating porosity distributions based on well logs, we usually pick one method to interpret the logs. Alternative log interpretations may be plausible

and offer other possible results. In principle, we should include other plausible log interpretations, because they represent additional uncertainty. If time and money constraints do not allow us to cover every plausible case, we should at least recognize that additional sources of uncertainty exist.

The following discussion treats parameters that can be captured in a set of stochastic images of the reservoir. If the images capture the dominant parameters, the method below offers a way to estimate the uncertainty in reservoir performance. Otherwise, the methods below provide only a subset of images, which can be used in a more complete uncertainty analysis with other important parameters.

Recall that the simulations in Sections 6.3 treat 20 porosity and permeability realizations. This direct procedure to assess uncertainty is outlined in Figure 6.35. The results of the 20 reservoir simulations are illustrated by a cumulative distribution plot instead of the histograms in Figure 6.11. The cumulative distribution plot and histogram are interchangeable. The cumulative distribution plot is generated by calculating the number of realizations that produce a particular cumulative oil recovery or less. This number divided by the total number of realizations equals the cumulative distribution. Then the cumulative distribution is plotted as a function of cumulative oil recovery.

6.7.1 *Methodology*

Ballin and Journel[89] have developed an approach that reduces the number of required reservoir simulations to assess the uncertainty. First, they rank each realization by using a fast model such as a tracer simulation. Based on this ranking, a handful of realizations are selected and placed in a reservoir simulator to estimate an approximate probability distribution for the production response. This approach circumvents the need for a large number of more comprehensive (and more costly) reservoir simulations. They tested this approach on a waterflood example and obtained favorable results. Guerillot and Morelon[90] propose a similar approach.

The first step in the Ballin-Journel approach is to rank the reservoir realizations (Figure 6.36) by using a fast simulator. The particular recovery mechanism will impact what fast simulation method to choose. For their waterflood example, Ballin and Journel chose a tracer simulation. The resulting cumulative distribution calculated from a tracer simulation will look similar to the one in Figure 6.35, but the abscissa is different. The tracer simulation gives the tracer

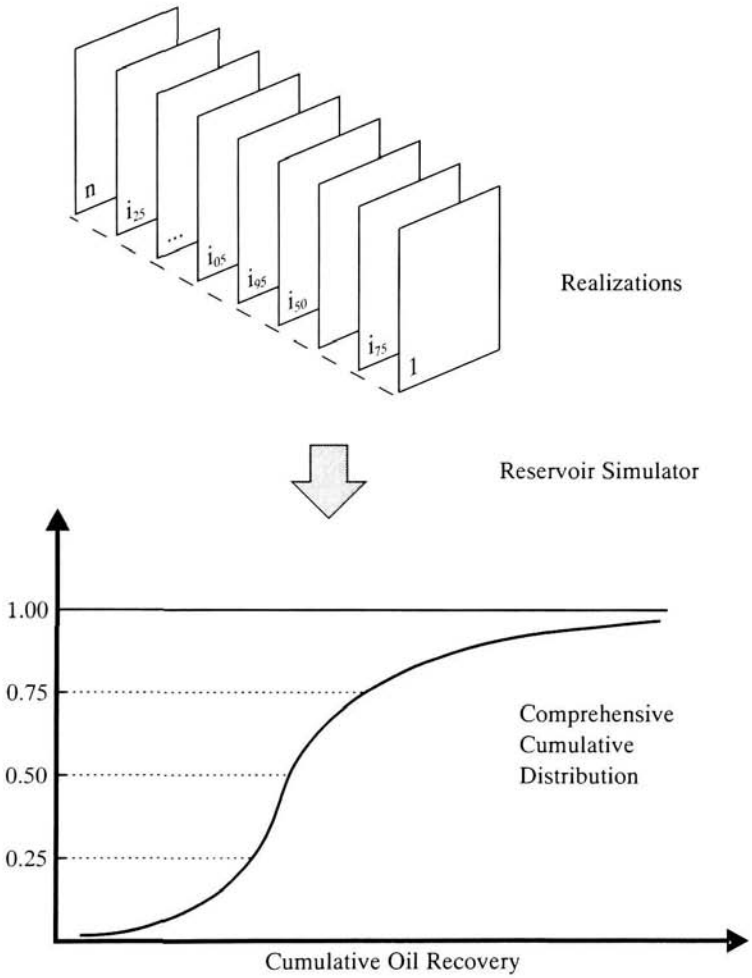


Figure 6.35. Direct procedure to generate the cumulative distribution of oil recovery.[89]

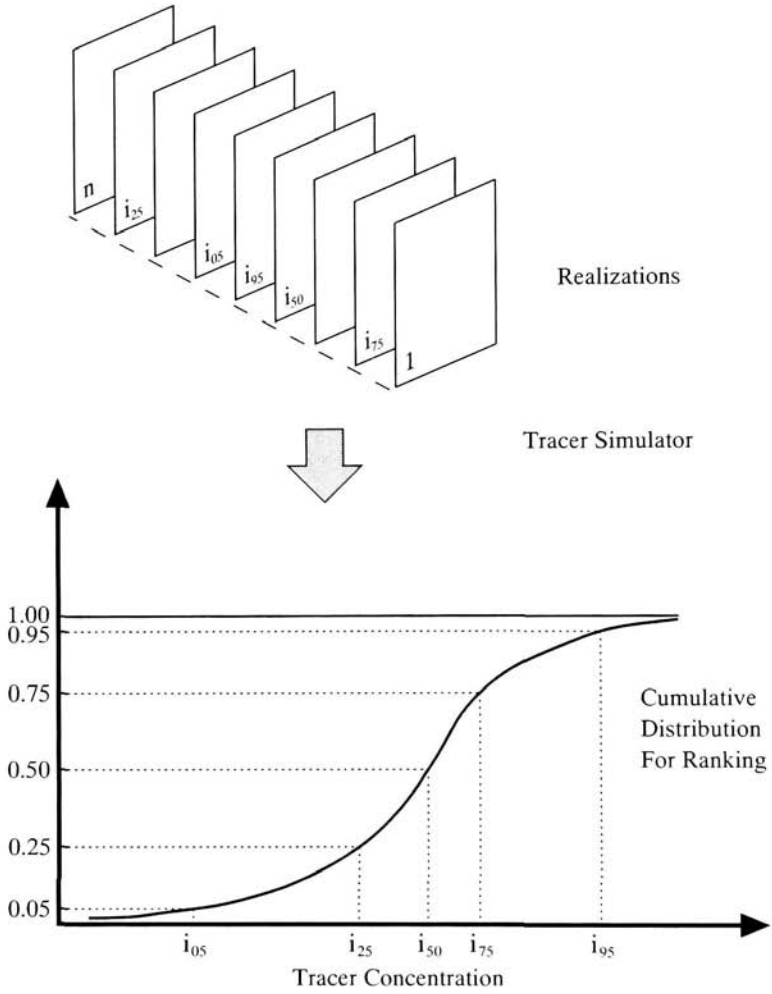


Figure 6.36. Procedure to rank realizations--first step.[89]

concentration at a producer, but nothing directly related to oil recovery. If the tracer simulation captures the most important mechanism(s), the procedure can rank the realizations according to tracer concentration at a certain time (or pore volume injected).

After completing the ranking, a small number of realizations are selected for subsequent reservoir simulations. The procedure is illustrated in Figure 6.37 for 100 previously ranked realizations. Here i_5, i_{25}, i_{50}, i_{75}, and i_{95} are the 5th, 25th, 50th, 75th, and 95th ranked realizations. The reservoir simulation is performed on only these five realizations. Each realization gives a value of oil recovery after 5 years of production. If the tracer simulation preserves the correct ordering, then these five values can be considered as the 0.05, 0.25, 0.50, 0.75, and 0.95 quantile of the oil recovery cumulative distribution. The 0.25 quantile is simply the value of oil recovery below which one-fourth of the data fall. The other quantiles have a similar meaning. Linear interpolation between these five points gives an approximate cumulative distribution curve for oil recovery.

6.7.2 *Waterflood Example*

Ballin and Journel[89] have tested this procedure on a synthetic waterflood example. The waterflood geometry is a five-spot pattern element with one layer. Permeability distributions are based on previously reported measurements from a slab of Berea sandstone.[91] Thirty-five original data values serve as conditioning data. A sequential indicator simulation algorithm[57] generates permeability realizations that honor the hard data. The porosity is constant over the 40 x 40 areal grid.

A tracer option of the ECLIPSE-100 reservoir simulator[92] serves as the tracer simulator. The full waterflood simulations are performed with the same black-oil ECLIPSE-100 simulator. The tracer simulations are 15 to 100 times faster than the corresponding waterflood simulations. The time savings depend on the relative permeability curves used for the waterflood.

To test the tracer simulation as a ranking tool, Ballin and Journel perform both the full waterflood and tracer simulations on 100 realizations. The tracer simulation starts with zero concentration within the reservoir followed by a continuous injected concentration equal to one. They find good correlation between the tracer concentration at the producer and the cumulative oil recovery from the waterflood simulation. The results in Figure 6.38 are for 3 years into the waterflood or 0.6 pore

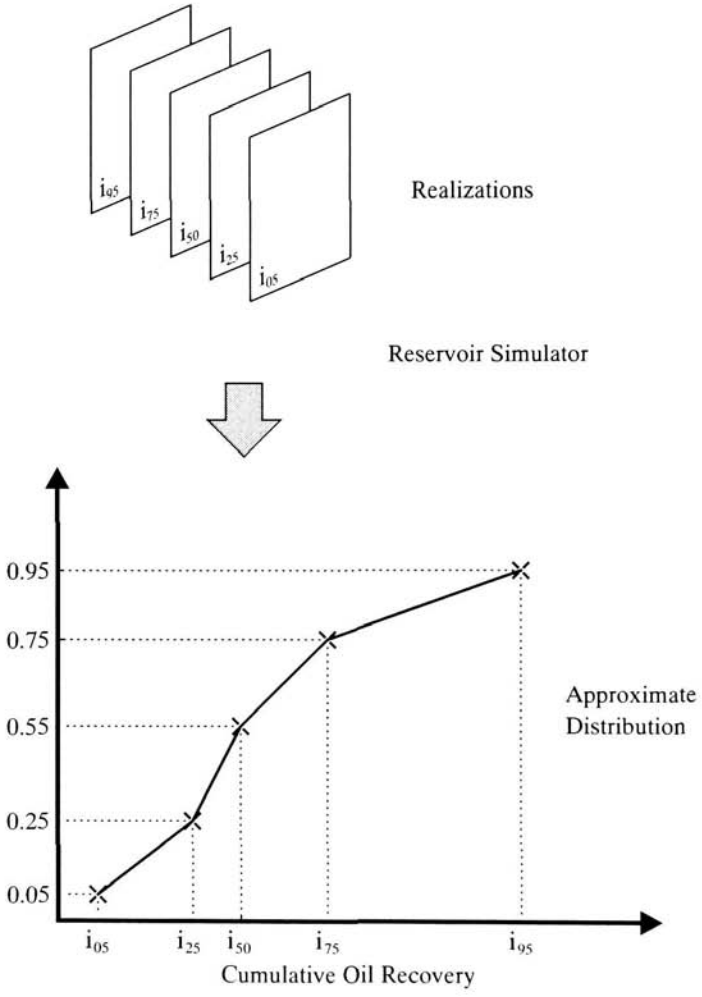


Figure 6.37. Procedure to approximate the cumulative distribution of oil recovery--second step.[89]

volume injected. The correlation is excellent. Note the ordinate is one minus the tracer concentration, not the tracer concentration.

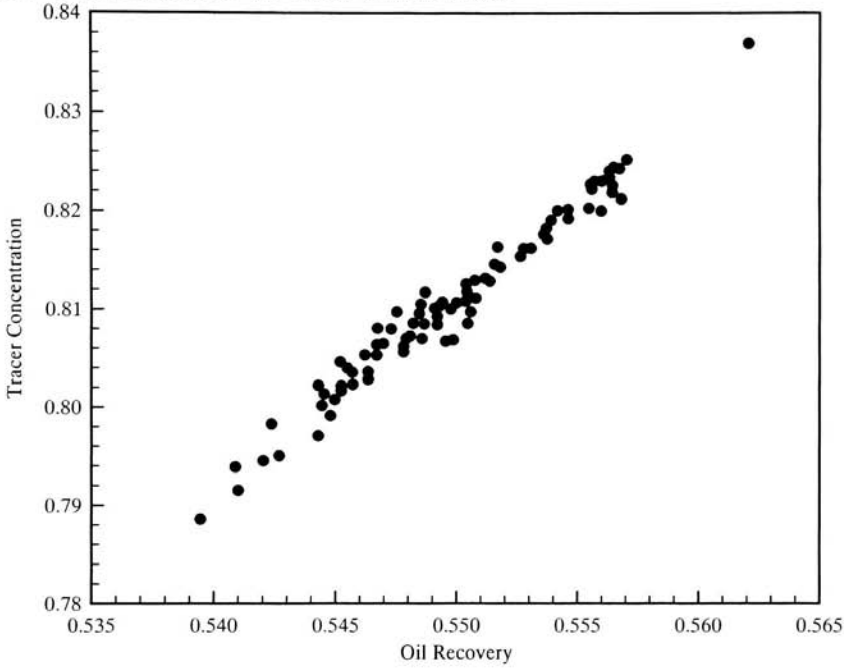


Figure 6.38. Comparison of tracer concentration with oil recovery.[89]

Next, they derive an approximate cumulative distribution curve for oil recovery based on five realizations following the outline in Figure 6.37. Figure 6.39a shows the comparison of the approximate distribution curve to the comprehensive curve, which is based on waterflood simulations over the entire set of 100 realizations. Again, the comparison is excellent. Similar results for the oil-water ratio are shown in Figure 6.39b.

These preliminary results demonstrate the potential of using a fast simulator (like a tracer model) to rank realizations. To be useful, the fast simulator must capture the most important mechanism(s). Then, geological uncertainty represented within a large set of realizations can be transferred into uncertainty in reservoir performance. The procedure requires comprehensive simulations on only a handful of realizations.

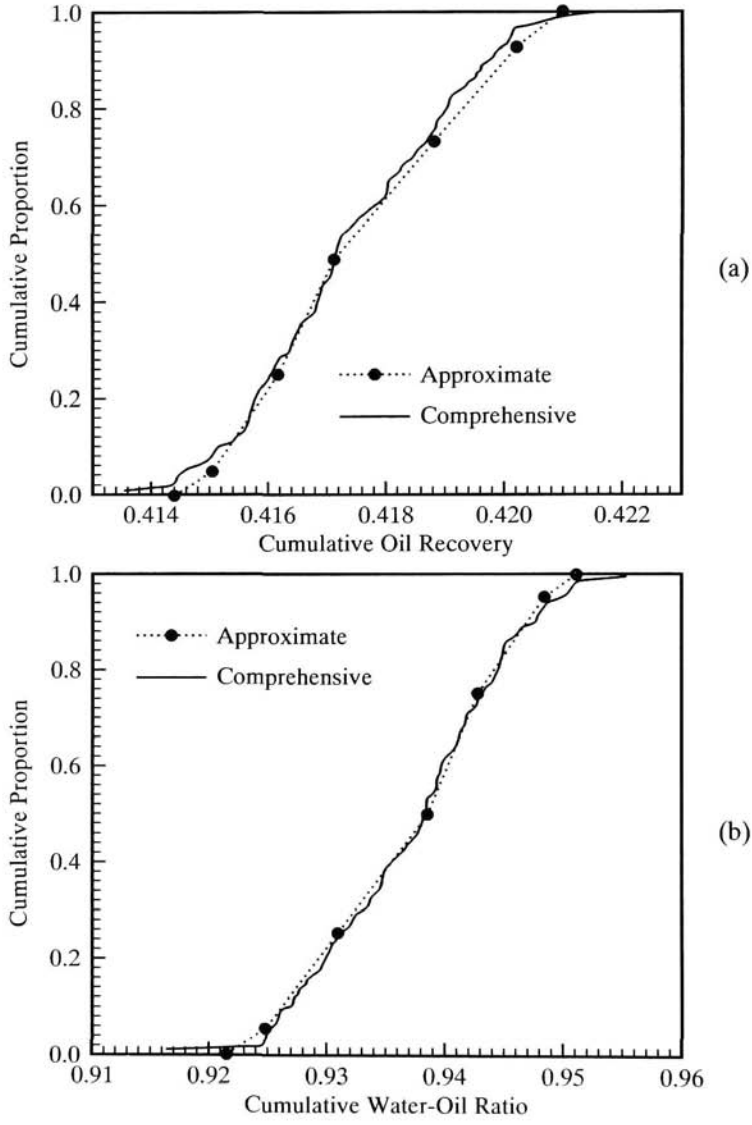


Figure 6.39. Comprehensive and approximate cumulative distributions (a) oil recovery and (b) water-oil ratio.[89]

Unresolved Issues

1. The simulations based on 3D fractal cross sections can handle gravity override effects or the bottom water drive case. Yet, some caution is in order. We have only generated and tested 3D distributions for relatively small well spacings. Our tested distributions were generated with cubic grids. In order to handle large well spacings, long, thin grids will be required. The procedure for doing this has not been fully developed or tested.

2. New screening tools need to be developed to select a handful of realizations out of a large pool for comprehensive flow simulations. The tracer simulation works well in the waterflood example by Ballin and Journel. What is an appropriate screening tool if gravity effects are important? What is appropriate for a carbon dioxide flood?

3. Scaling up relative permeability curves in addition to absolute permeability should allow coarser grids in modeling field projects. A computationally fast, yet accurate algorithm to scale up relative permeability needs to be developed.

APPENDIX A6

The streamtube (Section 6.3) and pseudorelative permeability (Section 6.4) methods both reduce the flow in a 2D vertical cross section to equivalent single-layer flow. The single-layer solution provides a mapping between the 2D cross section and an areal model for the field project. In this appendix, we describe some mathematical details for these methods. We start with a framework common to all the methods (Appendix A6.1). Then, the mappings are described for the streamtube method (Appendix A6.2), the Kyte and Berry method (Appendix A6.3), and the weighted-flux method (Appendix A6.4).

A6.1 Mappings Between 2D Cross Sections and Areal Models

The streamtube and pseudorelative permeability methods require estimates for vertically averaged fractional flows and mobilities. The relative mobility for each phase is the ratio of its relative permeability over viscosity. All the methods in this appendix share the same procedure to estimate vertically averaged fractional flows,

but each method has a different way to evaluate the vertically averaged mobilities. The following discussion gives the common expression for fractional flows. The expressions for mobilities are treated separately in the discussion of each method.

To understand the mappings, consider flow in a 2D vertical cross section with width W(x) and local height H(x) at streamwise position, x, and time, t. To reduce the flow to an equivalent single-layer solution, we must integrate over the vertical dimension, z. For incompressible fluids, the total flow volume of phase j through the cross section is calculated using Equations A6.1 and A6.2 as

$$q_j(x,t) = W(x) \int_0^{H(x)} k(x,z) \frac{k_{rj}(x,z,t)}{\mu_j} \frac{\partial \Phi_j(x,z,t)}{\partial x} dz \qquad \textbf{(A6.1)}$$

Here Φ_j is the fluid potential,

$$\Phi_j(x,z,t) = p_j(x,z,t) + \rho_j g z \qquad \textbf{(A6.2)}$$

where z is measured from an arbitrary datum. If the capillary pressure gradients are small, then

$$\frac{\partial \Phi_j(x,z,t)}{\partial x} \approx \frac{\partial \Phi}{\partial x} \qquad \textbf{(A6.3)}$$

for all phases j. Summing the contribution of both oil and water phases gives the total flow volume as

$$q(x,t) = W(x) \int_0^{H(x)} k(x,z) M_T(x,z,t) \frac{\partial \Phi(x,z,t)}{\partial x} dz \qquad \textbf{(A6.4)}$$

For the waterflood applications in Chapter 6, the total mobility $M_T(x,z,t)$ is

$$M_T(x,z,t) = \frac{k_{ro}}{\mu_o} + \frac{k_{rw}}{\mu_w} \qquad \textbf{(A6.5)}$$

Equation A6.1 gives the flow volume of phase j as $q_j(x,t)$ at position x. Then, the fractional flow of phase j across the entire cross section is

$$\bar{f}_j(x,t) = \frac{q_j(x,t)}{\sum_{k=1}^{N_p} q_k(x,t)} = \frac{q_j(x,t)}{q(x,t)} \qquad \text{(A6.6)}$$

Equation A6.6 is used by the streamtube and pseudorelative permeability methods.

A6.2 Streamtube Method

The proper allocation of the total fluid volume to each streamtube requires the total effective mobility from the cross-sectional simulation. Details are given below for these steps, which are outlined in Section 6.3.

A6.2.1 *Mobility Correlations*

The mapping for the streamtube model uses single-layer mobilities along the cross section. Hewett and Behrens[80] define the effective mobility as

$$\bar{M}_T(x,t) = \frac{\int_0^{H(x)} k(x,z)\, M_T(x,z,t)\, \frac{\partial \Phi(x,z,t)}{\partial x}\, dz}{H(x)\bar{k}(x)\frac{\partial \bar{\Phi}(x,t)}{\partial x}} \qquad \text{(A6.7)}$$

Here H(x) is the local height of the cross section with streamwise direction x. The integrand is the local mobility, $M_T(x,z,t)$, times a weighting function. The weighting function is the product of the local permeability, k(x,z), and the local pressure gradient. From Darcy's law, this product is proportional to the local streamwise flow rate. The denominator contains the vertically averaged permeability $\bar{k}(x)$ and flow potential (pressure plus gravitational effects), Φ. The vertically averaged flow potential is given by

$$\bar{\Phi}(x,t) = \frac{1}{H(x)} \int_0^{H(x)} \Phi(x,t)\, dz \qquad \text{(A6.8)}$$

To compute the vertically averaged permeability, first the harmonic average of the permeability in neighboring cells is taken at each depth. Then, the arithmetic average in the vertical direction of the previously computed harmonic averages gives $\bar{k}(x)$. From Equations A6.4 and A6.7, the expression for the flow rate is

$$q(x,t) = W(x)H(x)\bar{k}(x)\bar{M}_T(x,t)\frac{\partial\bar{\Phi}(x,t)}{\partial x} \qquad \text{(A6.9)}$$

The single-layer mobility as calculated by Equation A6.7 varies abruptly with x for the miscible displacement test case described in Section 6.4.1. Figure A6.1 is a plot of the single-phase mobility versus the streamwise coordinate x. The large scatter occurs even though $M_T(x,z,t)$ is equal to unity everywhere for the miscible displacement. The cause of the erratic behavior for the heterogeneous case becomes clear upon a closer inspection of Equation A6.7. In the numerator, the integration weights the local mobility by the local flow rate. The local flow rate is proportional to the product of permeability and flow potential gradient. A locally high permeability layer contributes to the integral according to its flow rate and not according to its local permeability. Note that the cross section contains bottlenecks where the area of high permeability streaks decrease. These bottlenecks restrict the flow rate through the high permeability streaks. Since the integrand weights according to flow rate, it does not count all the locally high permeability where the streaks bulge. On the other hand, the denominator contains the average permeability across the vertical trace.

This average permeability counts all the locally high permeabilities even if the flow is restricted by upstream or downstream bottlenecks. Therefore, a locally high value in the numerator is often not offset by the denominator.

The streamtube displacement calculations in Appendix A6.2.3 can handle this erratic behavior in single-layer mobility, but we will see later that pseudorelative permeability methods avoid this erratic behavior by using different vertically averaged mobilities.

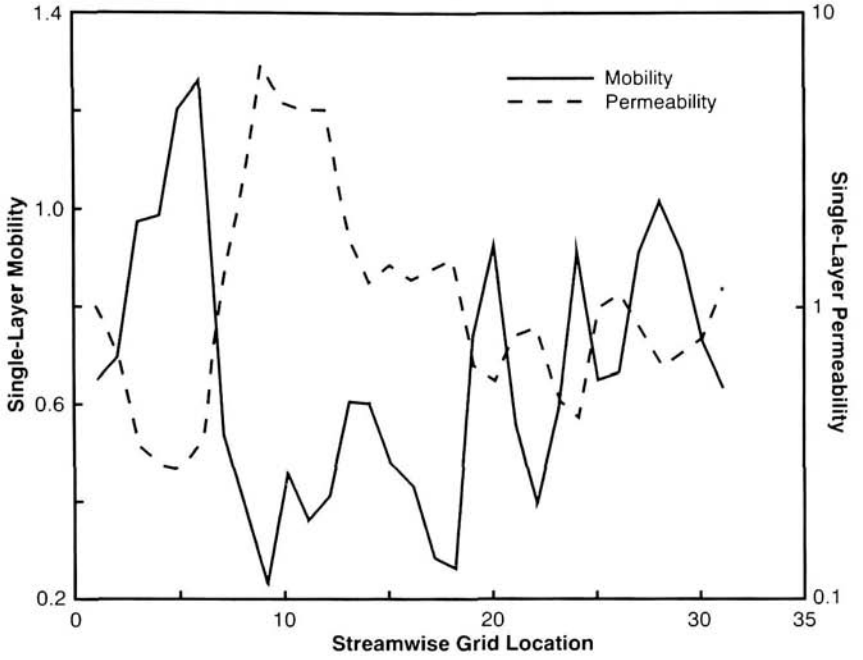


Figure A6.1. Single-layer mobility and normalized permeability versus streamwise grid location.

A6.2.2 *Generation of Streamtubes*

The generation of streamlines is a straightforward calculation as outlined by LeBlanc and Caudle.[81] Consider a single-layer reservoir with uniform porosity, ϕ, permeability, k, and thickness, h. The pressure and two-dimensional velocity components at any point (x, y) are obtained from the superposition of line sources and sink with flow rates q_j at locations (x_j,y_j). The equations are

$$\Phi(x,y) = \Phi_{av} - \frac{\mu}{4\pi kh} \sum_{j=1}^{m} q_j \ln[(x - x_j)^2 + (y - y_j)^2] \qquad \text{(A6.10)}$$

$$v_x = \frac{1}{2\pi h\phi} \sum_{j=1}^{m} q_j \frac{x - x_j}{(x - x_j)^2 + (y - y_j)^2} \qquad \text{(A6.11)}$$

$$v_y = \frac{1}{2\pi h\phi} \sum_{j=1}^{m} q_i \frac{y - y_j}{(x - x_j)^2 + (y - y_j)^2} \qquad \text{(A6.12)}$$

The symbol Φ_{av} in Equation A6.10 represents the average fluid potential in the reservoir. Using the velocity component equations, one can track the movement of a particle from an injector to a producer. To generate a family of streamlines, n particles are spaced evenly around an injector and their movement traced.

A6.2.3 *Streamtube Displacement Calculation*

The injected fluid is allocated among the streamtubes based on the relative resistance of each streamtube. Let ζ be the streamwise coordinate running along the centerline of the streamtube, beginning at the injector and ending with a value Z at the producer. The total resistance along each streamtube is

$$R(t) = \int_0^Z [\bar{k}(\zeta)\ A(\zeta)\ \bar{M}_T(\zeta,t)]^{-1}\ d\zeta \qquad \text{(A6.13)}$$

The effective mobility, $\bar{M}_T(\zeta,t)$, is evaluated by Equation A6.7. As explained in Section 6.3.3, a two dimensional mobility table is generated from the 2D cross-sectional displacement. The effective mobility in Equation A6.13 comes from this look-up table. The streamwise position in the cross section is mapped onto each streamtube through the fractional cumulative pore volume, $PV(\zeta)/PV(Z)$. We use bilinear interpolation to estimate values between table entries. The vertically averaged permeability $\bar{k}(\zeta)$ also comes from the cross-sectional results. The discussion after Equation A6.8 describes how $\bar{k}(\zeta)$ is evaluated at each streamwise location in the cross section, forming a 1D table. Then, we use linear interpolation to estimate values between table entries.

The corresponding fluid potential drop along the streamtube is

$$\Delta\bar{\Phi}(t) = q(t)\ R(t) \qquad \text{(A6.14)}$$

When the rate is specified at the injector, the fraction of fluid allocated to the jth streamtube is proportional to the ratio

$$\frac{R_T(t)}{R_j(t)}, \quad \text{where} \quad \frac{1}{R_T(t)} = \sum_j \frac{1}{R_j(t)} \tag{A6.15}$$

A6.3 Kyte and Berry Pseudorelative Permeabilities

The Kyte and Berry[87] method is often applied to generate pseudorelative permeabilities (Section 6.4). Pseudorelative permeabilities provide another mapping from the vertical cross section to an areal model of the field project. To evaluate pseudorelative permeabilities, we again calculate the fractional flow of water or oil across the entire vertical cross section, along with the single-layer mobility. Then, the pseudorelative permeability of phase j can be calculated as

$$\bar{k}_{rj}(x,t) = \mu_j \, \bar{f}_j(x,t) \, \bar{M}_T(x,t) \tag{A6.16}$$

The Kyte and Berry method uses Equation A6.6 to estimate fractional flows. In their method, the effective single-layer mobility is

$$\bar{M}_T(x,t) = \frac{\int_0^{H(x)} k(x,z) M_T(x,z,t) \, dz}{H(x) \bar{k}(x)} \tag{A6.17}$$

For the miscible displacement case with unit mobility in Section 6.4.1, Equation A6.17 produces a constant single-layer mobility of unity. This seems attractive, because we avoid the erratic behavior seen in Figure A6.1, but the Kyte and Berry method throws away information about the local bottlenecks to flow in Figure A6.1. This discarded information is important for any predictions of single-layer permeabilities or rates. Still, the Kyte and Berry method allows us to compare the shapes of the different pseudorelative permeabilities with streamwise position. This approach is reasonable, because curve shapes largely determine water cuts.

Under some conditions, the Kyte and Berry method behaves poorly as shown in Figure 6.23. To gain further insights, we perform some additional manipulations of the equations. Substituting Equations A6.1, A6.4, A6.6, and A6.17 into Equation A6.16 gives an expression for the pseudorelative permeability,

$$\bar{k}_{rj}(x,t) = \frac{\int_0^{H(x)} k(x,z)\ k_{rj}(x,z,t)\ \dfrac{\partial \Phi(x,z,t)}{\partial x}\ dz}{H(x)\ \bar{k}(x)\ \overline{\dfrac{\partial \Phi(x,t)}{\partial x}}} \qquad \textbf{(A6.18)}$$

Integration in the vertical direction gives the pseudopotential gradient in the denominator. For the integration, the weighing function is total mobility times permeability, such that

$$\overline{\frac{\partial \Phi(x,t)}{\partial x}} = \frac{\int_0^{H(x)} k(x,z)\ M_T(x,z,t)\ \dfrac{\partial \Phi(x,z,t)}{\partial x}\ dz}{\int_0^{H(x)} k(x,z)\ M_T(x,z,t)\ dz} \qquad \textbf{(A6.19)}$$

For low mobilities near 0.1, the Kyte and Berry method gives a water pseudorelative permeability curve that behaves poorly (Figure 6.23). The problem arises in Equation A6.18 where the denominator contains the product of two averages, permeability and pseudopotential gradient. The numerator contains a weighting factor that is the product of permeability and the pseudopotential gradient, but the sum of a product divided by the product of the sum can behave badly, as we saw earlier with the effective mobility in Equation A6.7.

In our applications of the Kyte and Berry method, we never compute the flow rate through the cross section. The following expression for flow rate is given for completeness. Combining Equations A6.4, A6.17, and A6.19 produces an equation for the flow rate,

$$q(x,t) = W(x)H(x)\bar{k}(x)\bar{M}_T(x,t)\overline{\frac{\partial \Phi(x,t)}{\partial x}} \qquad \textbf{(A6.20)}$$

Flow rate predictions for a single layer based on Equation A6.20 work for the cross section, but the expression does not easily carry over to the areal field model, because the pseudopotential gradient is unique to the cross section. Still, for the examples in Section 6.4 the Kyte and Berry method adequately predicts water cuts except for mobility ratios near 0.1 or lower.

A6.4 Flux-Weighted Pseudorelative Permeabilities

The following flux-weighted method to calculate pseudorelative permeabilities works for low mobility ratios near 0.1 where the Kyte and Berry method is not well behaved. The method gives results that are comparable to the Kyte and Berry method for higher mobility ratios. (See Section 6.4.3)

In the flux-weighted method,

$$\bar{M}_T(x,t) = \frac{\int_0^{H(x)} k(x,z) M_T(x,z,t) \frac{\partial \Phi(x,z,t)}{\partial x} dz}{\int_0^{H(x)} k(x,z) \frac{\partial \Phi(x,z,t)}{\partial x} dz} \qquad \text{(A6.21)}$$

The evaluations of the flux-weighted pseudorelative permeabilities in Section 6.4.3 are based on Equations A6.6, A6.16, and A6.21.

To see why the flux-weighted method avoids some of the problems experienced by the Kyte and Berry method, we now derive an explicit expression for pseudorelative permeability. Substituting Equations A6.1, A6.4, A6.6, and A6.21 into Equation A6.16 gives the flux-weighted pseudorelative permeability as

$$\bar{k}_{r\eta}(x,t) = \frac{\int_0^{H(x)} k(x,z)\ k_{r\eta}(x,z,t)\ \frac{\partial \Phi(x,z,t)}{\partial x} dz}{\int_0^{H(x)} k(x,z) \frac{\partial \Phi(x,z,t)}{\partial x} dz} \qquad \text{(A6.22)}$$

By examining Equation A6.22, we see the denominator in the flux-weighted method sums over the weighting function in the numerator. The weighting function is the local permeability times the local potential gradient. This product is the fluid flux. Unlike the Kyte and Berry method (Equation A6.18), the denominator in the flux-weighted method contains no product of sums and avoids the problems seen in Figure A6.1.

Like the Kyte and Berry method, the flux-weighted method still relies on a pseudopotential. For the flux-weighted method,

$$\overline{\frac{\partial \Phi(x,t)}{\partial x}} = \frac{\int_0^{H(x)} k(x,z) \frac{\partial \Phi(x,z,t)}{\partial x} dz}{\int_0^{H(x)} k(x,z)\, dz} \tag{A6.23}$$

Although we never calculate flow rates in our applications, Equation A6.20 gives the flow rate through the cross section.

CHAPTER 7

OTHER APPLICATIONS

The technical literature contains hundreds of papers that apply fractals to sedimentary rocks and flow through porous media. Each application treats a finite range of length scales. For example, at the small length scale end, a series of papers by Katz, Thompson, and Krohn[93-95] treat pore geometry with implications to electrical conductivity measurements. At the large scale, Barton et al.,[96] apply fractal concepts to hydrocarbon deposits in 1,600 square mile areas within the Denver and Powder River basins in the western United States. Ultimately, progress at any length scale may have consequences to reservoir engineering. Still, the present chapter (like previous chapters) remains focused on interwell length scales. We emphasize these scales, because the reservoir engineer usually deals with fluid flow problems at these scales.

This chapter covers three topics: (1) pressure transient tests, (2) faults and fractures, and (3) music from well logs. The first two topics have direct application for fluid flow between wells. The third topic is a brief diversion into the similarity of music and apparently unrelated natural phenomena.

7.1 Pressure Transient Tests

Pressure transient tests offer a way tc estimate reservoir permeability from pressure data collected in the field. These tests are the most direct means for estimating permeability at the interwell length scale. Laboratory measurements of permeability are restricted to relatively small rock samples. These laboratory

measurements must be scaled up to represent the reservoir scale. Pressure transient tests in the field circumvent this scaling up problem.

Conventional pressure transient models have been developed with the assumption that reservoirs are laterally homogeneous. Yet as we have seen in Chapter 3, core, log, and outcrop data indicate this assumption is not always justified. Still, homogeneous models are often applied to obtain an effective permeability. This approach does not work in some field cases as illustrated below. Instead, a model for a reservoir with a fractal structure matches the field data. This fractal model includes the conventional homogeneous reservoir solution as a special case.

7.1.1 *Conventional Pressure Transient Model*

A pressure buildup test is one kind of pressure transient test. In an ideal pressure buildup test, a well is shut-in after producing at a constant rate for a long time (Figure 7.1). During the shut-in period, we measure the increasing bottom-hole pressure as a function of shut-in time, Δt. From the measured pressure buildup curve and flow rate data, we can estimate the reservoir flow capacity, which is the product of the permeability times the net effective pay thickness. Often, we can estimate the net effective pay from well logs independently and solve for the permeability.

The geometry of the wellbore also influences the pressure buildup curve. In Figure 7.2, the bottom-hole pressure is measured in the wellbore near the depth of the perforations. For practical reasons, wells are often shut-in by closing a valve at the surface. Figure 7.2 illustrates a case where the sandface flow rate remains nonzero after surface shut-in. At shut-in, the wellbore pressure is less than the distant reservoir pressure. Due to this pressure difference, reservoir fluids continue to enter the wellbore through the perforations and fill the tubing/casing annulus after shut-in. This effect is called wellbore storage and distorts the early-time pressure curve. Eventually, the sandface flow rate diminishes sufficiently and wellbore storage becomes negligible. Many pressure transient models include wellbore storage effects and predict their duration.

After initial wellbore storage effects die out, the pressure change, Δp, for a homogeneous reservoir is given by[4]

$$\Delta p \sim \ln(\Delta t) \tag{7.1}$$

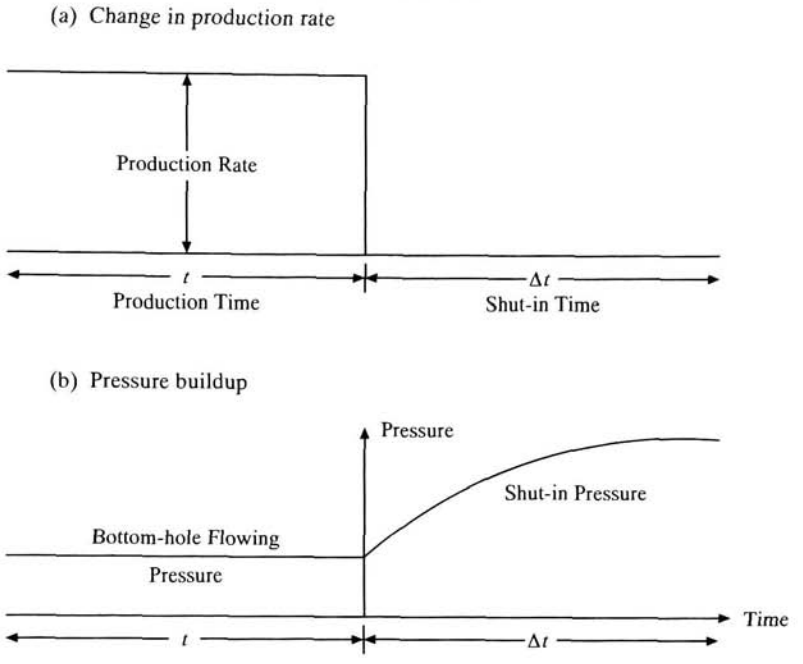


Figure 7.1. Schematic diagram of an ideal pressure buildup test.

The reservoir flow capacity is related to the slope of a Δp versus $\ln(\Delta t)$ plot. Initially, the pressure change at the wellbore reflects only the flow capacity of the near wellbore region. As the shut-in time increases, the test sees further into the reservoir. For a homogeneous reservoir, the radius of investigation scales with shut-in time as

$$r_{inv} \sim (\Delta t)^{1/2} \tag{7.2}$$

Thus, a single buildup test can be affected by the reservoir flow capacity over a range of length scales from near wellbore to the drainage radius of the producer.

Another type of pressure transient test for a producer is a drawdown test. In the ideal test, the well has been shut-in for a long period. Then, a nonzero and constant production rate is maintained. In this case, the bottom-hole pressure decreases with time. Under ideal conditions, the pressure drawdown and buildup tests give the same change in pressure except for the algebraic sign.

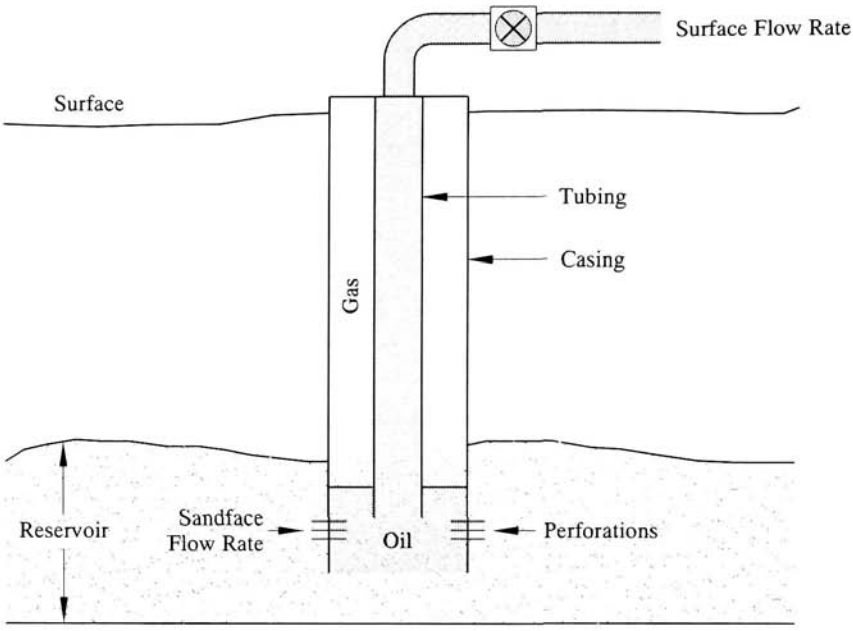


Figure 7.2. Wellbore storage in the tubing/casing annulus.

An analogous test for an injection well is a pressure falloff test. In the ideal falloff test, the injection well is shut-in after injecting at a constant rate for a long time. During the shut-in period, the bottom-hole pressure decreases according to Equation 7.1.

Over the years, many mathematical models have been developed to help interpret field tests.[97] Often theoretical pressure curves are plotted in terms of dimensionless variables as shown in Figure 7.3. The resulting plots are called type curves. Introducing dimensionless pressure, p_D, and time, t_D, allows the same curves to be applied to many different field tests. We can apply the same set of curves even though parameters such as production rate and fluid viscosity vary among different field tests. The particular curves in Figure 7.3 apply to a well that has been stimulated with a hydraulic fracture.[98] In addition to the pressure curve, it is customary in pressure transient analysis to plot the pressure derivative curve. The pressure derivative, t dp/dt, corresponds to the slope of a Δp versus $\ln(\Delta t)$ curve. Thus, the pressure derivative curve corresponds to a horizontal line when Equation 7.1 applies. This curve can be an aid in obtaining a match to field tests.[99]

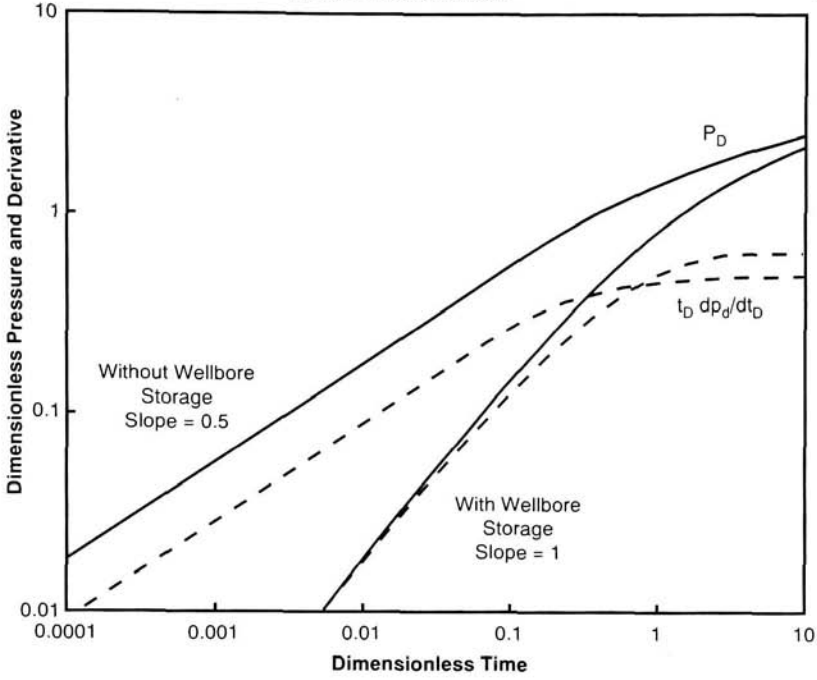


Figure 7.3. Pressure transient-type curves for a well with a vertical fracture in a homogeneous reservoir.

7.1.2 *Conventional Model Applied to Field Test*

The conventional model represented by the type curves in Figure 7.3 does not match the pressure transient field data in Figure 7.4. The field data are from a fracture stimulated well completed in the San Andres formation in New Mexico. The type curves in Figure 7.3 are typically applied to wells like this that have been hydraulically fractured. The field pressure data between 1 and 100 hours fall on a straight line with slope 0.6. The pressure derivative curve (t dp/dt) is parallel to the pressure curve. The type curves in Figure 7.3 do not show lines with similar slopes. If wellbore storage effects dominate the early time data, the type curves initially have a slope of unity. Without wellbore storage, linear flow near the vertical fracture produces an early straight line of slope one half. Straight lines with other slopes do not appear even for type curves that fall between these two extreme cases.

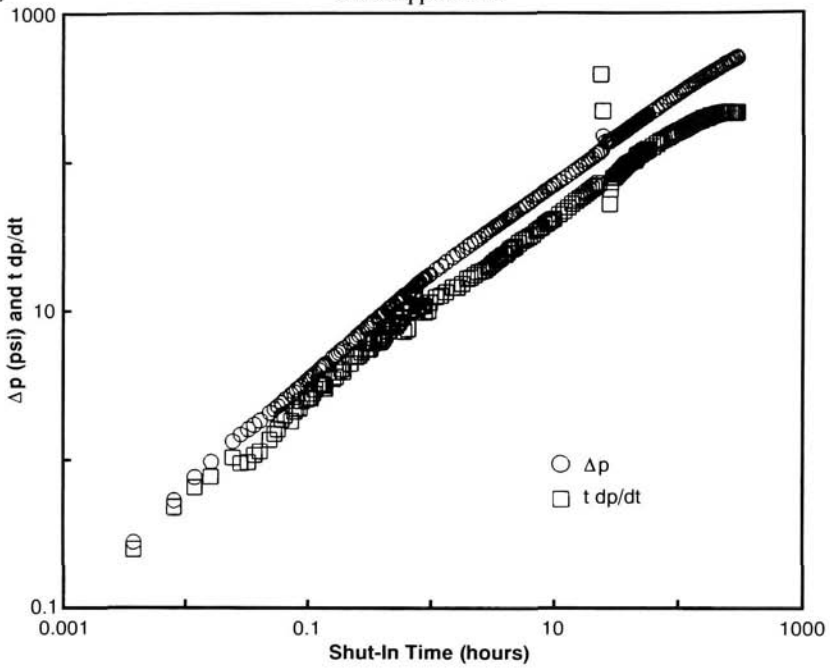


Figure 7.4. Pressure buildup field data.

As with most field tests, some unwanted events occur that distort the data. At approximately 24 hours, about 8 barrels of corrosion inhibitor were dumped into the wellbore annulus, causing a temporary jump in the bottom-hole pressure. No correction in the pressure data has been made to take out this unwanted effect. Near the end of the test, the pressure derivative curve flattens due to interference from neighboring wells.

An attempted nonlinear regression curve fit of the conventional model to the field data is shown in Figure 7.5. The measured bottom-hole pressure extends over four log cycles in time, but the conventional model can at best match only half of the data. Wellbore storage effects are taken into account by applying a method of superposition in the time coordinate. Figure 7.4 contains more field data on the pressure curve than Figure 7.5, because the computer matching technique used for the conventional model in Figure 7.5 slows down with so many field data.

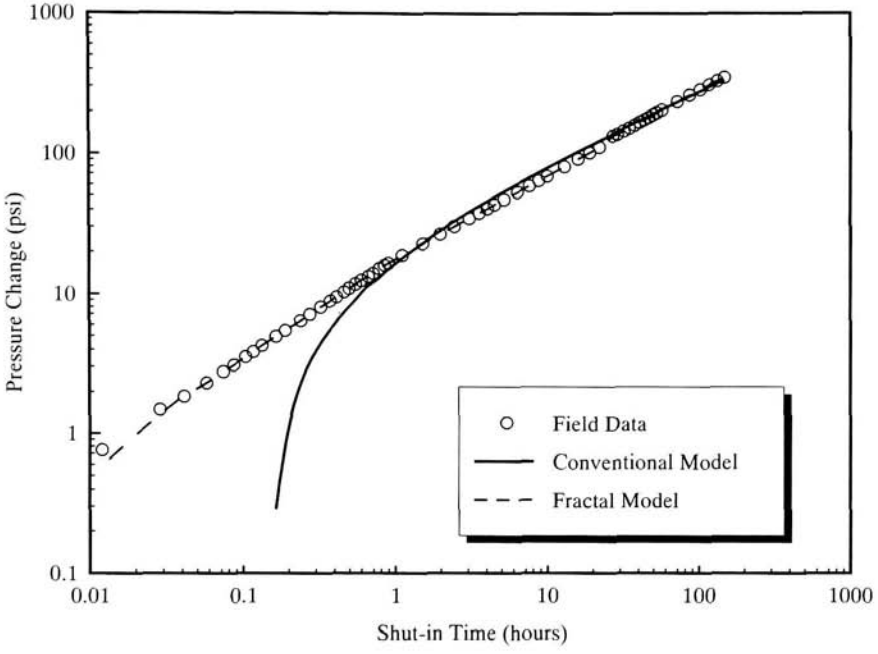


Figure 7.5. Conventional and fractal model fits to pressure buildup field data.

Many attempts have been made to get a better match with this and other homogeneous reservoir models. These attempts have introduced possible formation damage along the fracture face, non-Darcy flow within the reservoir near the wellbore, and a finite pressure drop along the fracture. The wellbore storage coefficient has also been varied over a wide range of values. The wellbore storage coefficient specifies the barrels of fluid stored in the wellbore for each psi change in bottom-hole pressure. No attempted homogeneous model match is any better than the example in Figure 7.5.

Instead, the model developed for a fractal reservoir matches the field data over four log cycles in time. This is shown as the dashed line through the data in Figure 7.5. The following sections will describe this fractal model.

7.1.3 *Fractal Permeable Network*

We suspect the conventional model fails, because permeability changes with distance from the wellbore. Then, the assumption of lateral homogeneity is invalid.

As an alternative to the homogeneous model, we consider a heterogeneous reservoir with a fractal structure. The fractal structure imposes heterogeneities at all length scales.

Core samples can serve as a model for spatial permeability variations. The thin section in Figure 7.6 is from a San Andres Formation core that has been impregnated with blue-dye epoxy. The dark color identifies the network of more permeable rock invaded by the epoxy. For the sake of discussion, consider the dark pattern as representative of the permeable network at the scale of a well pattern. The uninvaded (lighter colored rock) is treated as impermeable. Imagine a well at the upper left-hand corner. Choose a nearby point in the dark area. A nearly straight path can be found that stays within the permeable dark area back to the well. Now choose points that are successively further from the imaginary well but still in the dark area. As the distance from the wellbore increases, the path back to the well will more likely encounter larger impermeable areas to wind around. For more distant points, the path becomes more tortuous.[100] This causes the effective permeability to decrease away from the wellbore. This decreasing effective permeability with distance from the wellbore becomes the basis of the fractal pressure transient model.[101-103] Appendices A7.1, A7.2, and A7.3 provide further details.

7.1.4 *Fractal Model Applied to Field Test*

The fractal reservoir solution matches both the dimensionless pressure (p_D) and pressure derivative curves ($t_D\, dp_D/dt_D$) in Figure 7.7. The match extends over the entire four cycles of field data. The fit is made manually without a computer by simply overlaying the field data on the model type curves as explained in Appendix A7.4.

Typical uses for pressure buildup tests include estimating the flow capacity, kh, and evaluating the well completion. For the well completion, we usually want to know if the well needs a stimulation treatment or if a recent stimulation was successful. Like the conventional models, the fractal model gives estimates for kh and the equivalent wellbore radius. The equivalent wellbore radius is an evaluation of the well completion. The parameter takes into account near wellbore formation damage that decreases permeability near the wellbore. The parameter includes the effects of acid or hydraulic fracture treatments that stimulate the near wellbore region. This equivalent wellbore radius can be related to the fracture length.[97] The

effective wellbore radius is less (greater) than the actual wellbore radius if the near wellbore permeability is damaged (enhanced).

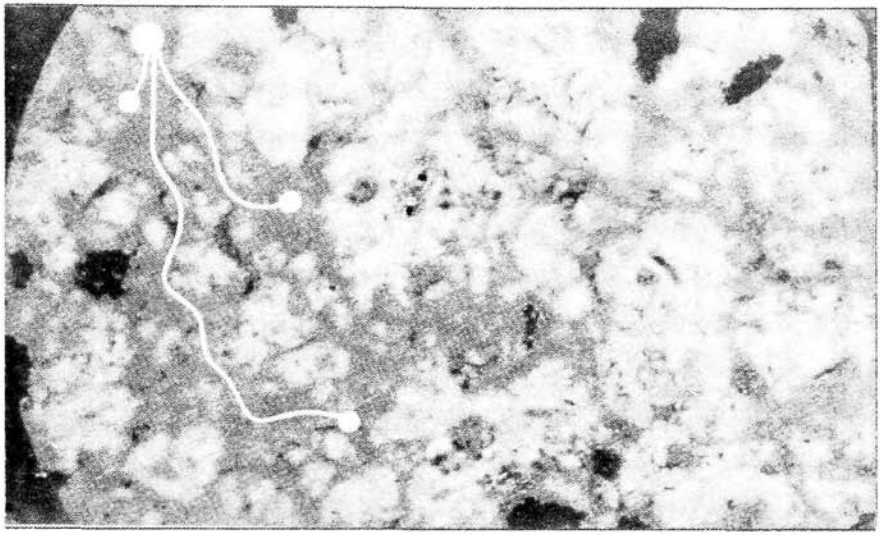


Figure 7.6. Thin section from a San Andres core impregnated by blue dye epoxy.

Appendix A7.4 demonstrates how a match between field data and the fractal reservoir solution can be used to estimate the flow capacity and the equivalent wellbore radius. The estimate of 120 md-ft for flow capacity is within the same "ball park" as an estimate based on core and fluid property data. The effect of the well's hydraulic fracture is seen as a large equivalent wellbore radius, as would be expected.

Applications of the fractal matrix model to other field tests are given in the references. Beier[102] analyzed pressure buildup tests on a producing well before and after a stimulation treatment. The fractal model shows the treatment was successful. Conventional pressure transient models are unable to interpret these field tests. Beier[104] extended the fractal model to a well with a vertical fracture and provided additional supporting field tests. Recently, Acuna et al.,[105] have applied the fractal model described by Chang and Yortsos[101] to field tests in naturally fractured reservoirs.

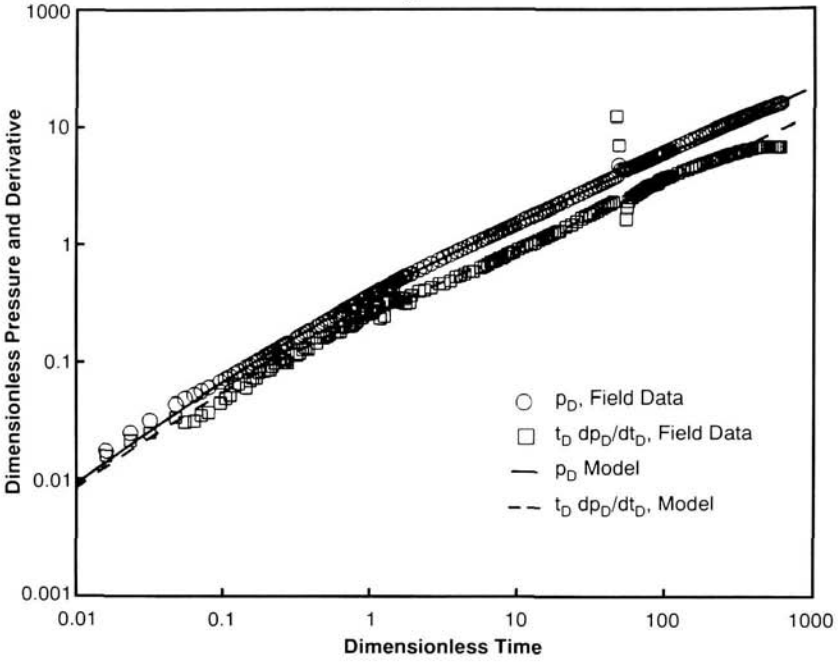


Figure 7.7. Fractal model match to pressure buildup field data.

7.2 Faults and Fractures

Faults and natural fractures often affect reservoir performance. Faults are fractures where the fracture sides have been displaced relative to one another parallel to the fracture. Faults can act as either barriers or conduits to fluid flow. As barriers, faults can serve as reservoir boundaries. Even if a fault is not sealing, its low transmissibility can impair transverse fluid flow. As conduits, faults can be flow paths into a reservoir for water in an underlying aquifer.[106]

Seismic and core data show us only a small portion of the faults and natural fractures that exist in a reservoir. Modern 3D seismic surveys can sample the entire reservoir but can resolve only large faults. Cores sample a very small portion of the reservoir and can reveal only a few of the many natural fractures. If scaling relationships for the properties of faults and fractures exist, we can use them to infer properties at intermediate scales where we have no observations.

In addition to scaling laws, we also present a simple fracture growth model below. Through this simple model, we illustrate the effects of fGn heterogeneity on a single fracture.

7.2.1 *Scaling Relationships*

In an attempt to characterize the fault population in a basin, Walsh et al.,[107] have assembled measurements of fault displacements at a wide range of length scales. The displacements are based on regional seismic data, field seismic data, and well cores from the North Sea and Troll Field. Fault displacement populations from all these sources are shown in Figure 7.8a. The displacements are expressed as heave (horizontal component). A single power-law fit (slope -0.9) is drawn to describe the relationship between fault displacements and fault density. The data span over six log cycles for fault displacements. The power-law relationship implies a fractal scaling relationship. A correction has been applied to the core data, because the core samples are on vertical transects rather than horizontal transects like the seismic data.

Walsh et al.,[107] also present fault displacement data from another offshore oil field (Figure 7.8b). The data are based on well cores and a 3D seismic survey. Displacements are expressed as fault throws (vertical component). Again, they infer a power-law relationship between the fault throw and cumulative fault density. In addition, Walsh and Watterson[108] present empirical data to demonstrate that the maximum displacements of faults are related to their lengths by a power-law relationship.

Many disciplines deal with fractures in rocks. Other correlations of fault displacements (or lengths) appear in the literature,[109,110] but few data sets span over six log cycles for displacement lengths like the data sets in Figure 7.8.

Of course, these power-law relationships do not contain all the information about the fault and fracture patterns from where they were derived. The relationships contain no information about the orientation angles and branching rules that are necessary to reconstruct patterns. Still, Heffer and Bevan[109] and Gauthier and Lake[110] have attempted to apply such scaling relationships in reservoir studies. Their applications include reservoir continuity when fault systems are present and horizontal wells intersecting fracture networks.

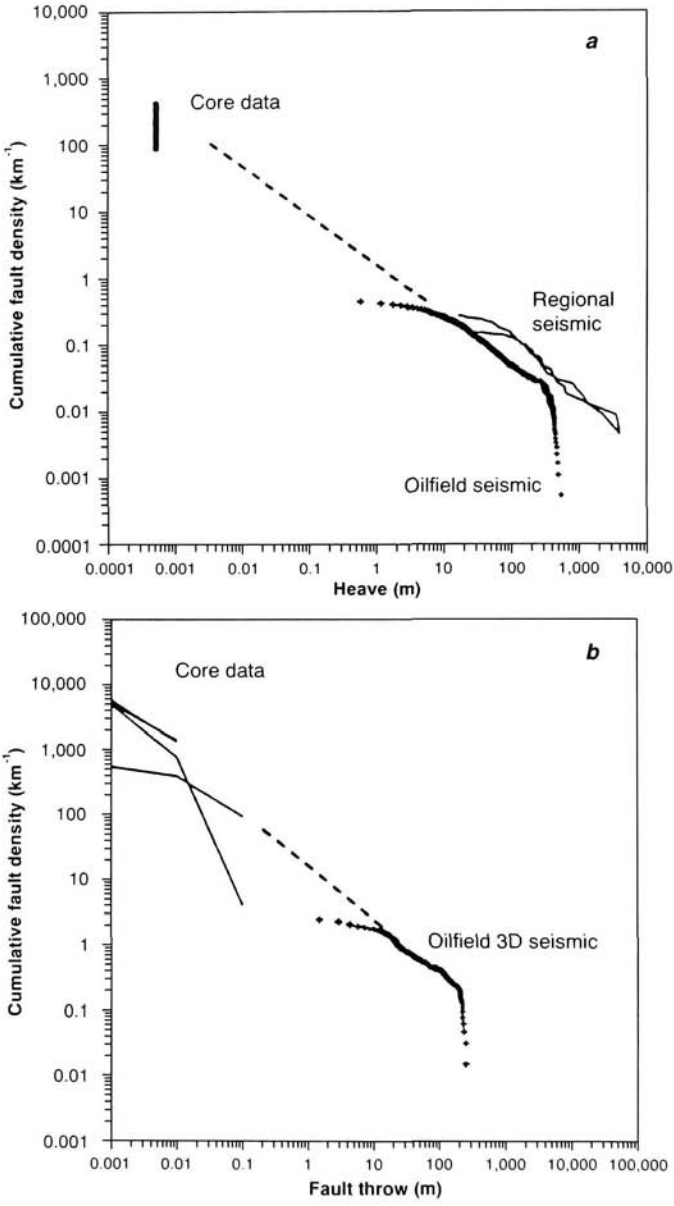


Figure 7.8. Comparison of fault displacement measurements.[107]

7.2.2 *Simple Fracture Propagation Model*

Modeling fracture growth has applications in wellbore fracture stimulation, faulting and fracturing processes in rocks, and estimation of permeabilities of individual fractures. We do not have a fundamental understanding of any of these processes yet, but progress is being made in these areas (e.g., Reference Nos. 111 and 112). In this section, we wish to suggest that rock property distributions may contribute significantly to the propagation and geometry of fractures and faults. This is done using an over simplified model of the fracturing (or cracking) process.

Many models of cracking have been proposed (e.g., Herrmann and Roux[111] and references there in). Here, we describe a simplistic model that is not designed to incorporate all of the complex dynamics of stress-strain relationships in heterogeneous media. Instead, we illustrate that fGn heterogeneity, coupled with a simple cracking model, produces cracks that look very similar to cracks in concrete (e.g., Figure 7.9) and other materials under stress. No attempt has been made to quantify this similarity. It is our hope that the use of such a simple model to produce such realistic looking cracks will encourage others to try more realistic models of cracks on fGn networks.

Figures 7.10 through 7.13 show some example results of the cracking model. These are generated by a deterministic process that begins with fGn distributions defined by Equation 3.2. The cracks are grown in the following manner:

1. The distributions generated by Equation 3.2 are linearly scaled so that each location in the distribution (j,ℓ) has a numerical value, $g(j,\ell)$, between 1 and 10.

2. A "force" term, F_c, and a "failure strength" term, B, are specified.

3. A starting crack location is chosen at the top of the distribution, and the first three grids under this location are set to 0.0 (i.e., a "nick" is made at the top of the distribution).

4. Starting from the last grid location made zero (j,ℓ), a number is assigned to each of the eight adjacent grid locations (j',ℓ') with the following formula:

$$h_{j',\ell'} = F_c \sin\theta + Bg_{j',\ell'} \tag{7.3}$$

where θ is the angle from the vertical made by a line drawn from (j,ℓ) to (j',ℓ') (see Figure 7.14). The eight surrounding grids are

$$\begin{aligned} j' &= j - 1,\ j,\ j + 1 \\ \ell' &= \ell - 1,\ \ell,\ \ell + 1 \end{aligned} \tag{7.4}$$

with the special case that

$$h_{j',\ell'} = 0 \quad \text{if } j' = j \text{ and } \ell' = \ell \tag{7.5}$$

(These equations were designed to encourage crack growth down but allow heterogeneities to interfere with the growth. The reasons for this particular formalism are described in more detail below.)

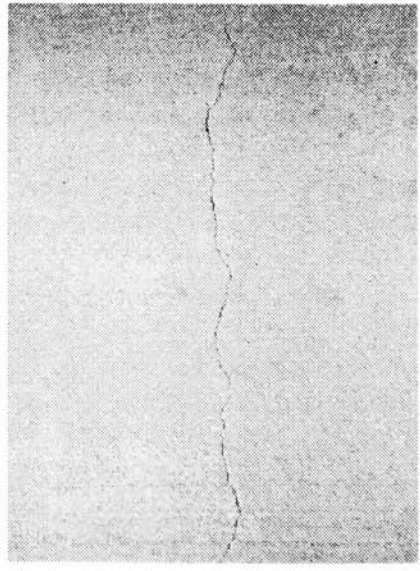


Figure 7.9. Crack in concrete sidewalk.

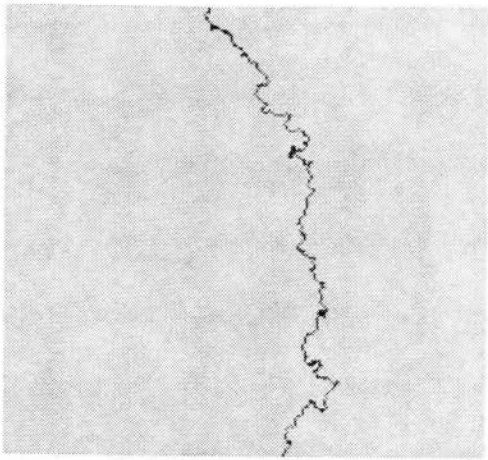


Figure 7.10. "Crack" produced in a fGn distribution with moderate stress applied in the horizontal direction.

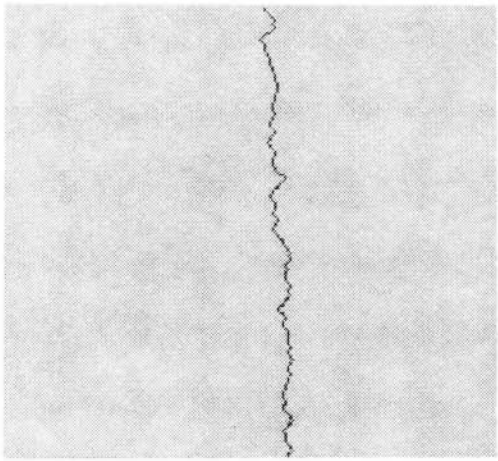


Figure 7.11. "Crack" produced with the same conditions as Figure 7.10 but with greater stress applied.

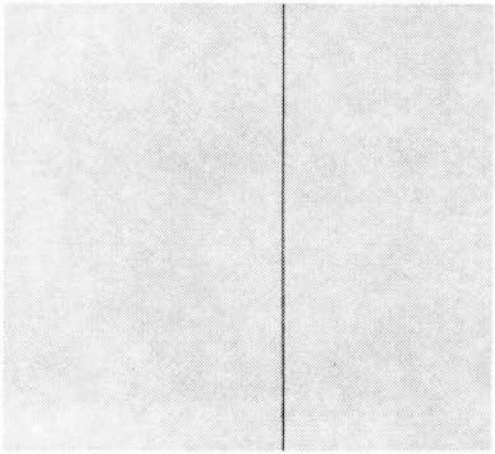


Figure 7.12. "Crack" produced with the same conditions as Figure 7.10 but with very large stress applied.

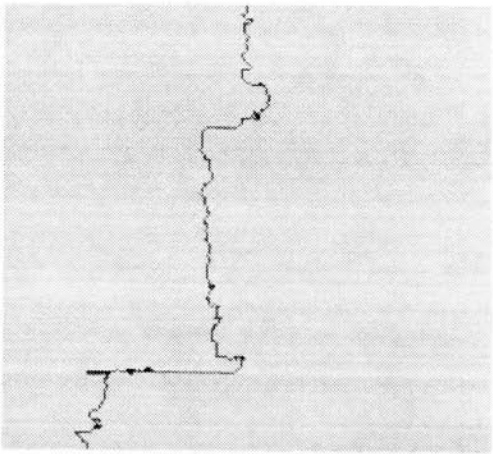


Figure 7.13. "Crack" produced in an fGn distribution with some layering and moderate stress applied in the horizontal direction.

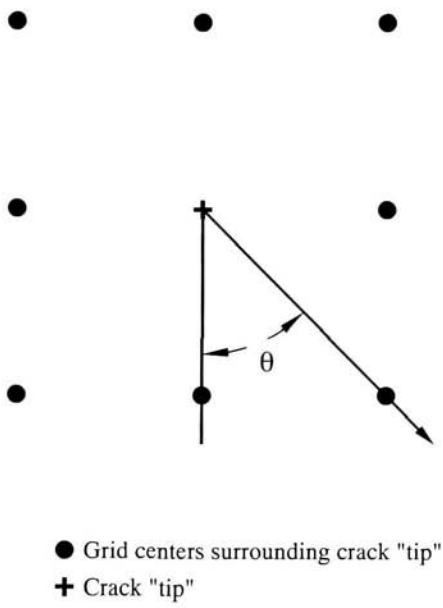


Figure 7.14. Orientation of measuring angle, θ, for Equation 7.3.

5. The grid with the largest value based on Equation 7.3 is set to zero, and Steps 4 and 5 were repeated until the crack reached the edge of the distribution.

6. Figures 7.10 through 7.13 are generated as grey scale maps from zero (black) to the maximum value of the grid (white).

The heart of the model is Equation 7.3. To understand the reason for Equation 7.3, consider a homogeneous block under horizontal stress (Figure 7.15). If a crack is initiated at the top of the block (arrow in Figure 7.15), one would expect the crack to propagate straight down in response to the stress. Equation 7.3 will force this to happen if B is very small (homogeneous distribution) or if F_c is very large (force so great that heterogeneities are negligible). An example of this is shown in Figure 7.12. For the nonhomogeneous case, Equation 7.3 tries to force propagation down, but the values of surrounding grids can cause "detours" (Figures 7.10 and 7.11). The setting of the crack location to zero prevents the crack from propagating back along itself. The idea is to model crack growth from the crack tip where stress is greatest.

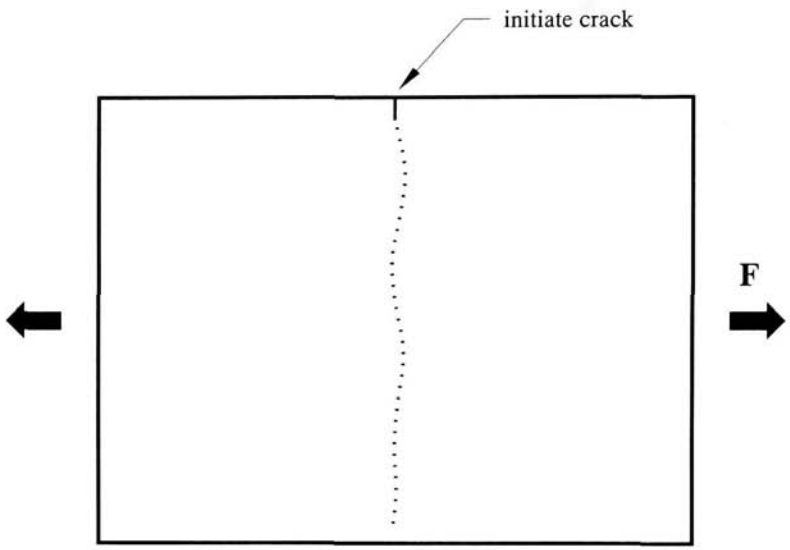


Figure 7.15. Idealized stress and crack propagation in a homogeneous media.

Figures 7.10 through 7.12 were generated with the same fGn distribution. No layering is used (i.e., $B_{SD} >> 1$ in Equation 3.2). The F_c value is changed to produce these different figures. F_c is smallest in Figure 7.10, increases in Figure 7.11, and is a very large value in Figure 7.12. Figure 7.13 has been generated using Equation 3.2 with $B_{SD} \sim 1.0$ to introduce layering. F_c and B values are chosen to allow the crack to propagate with some interference from the layers. It is interesting how the crack tends to follow layers even though the applied stress is trying to make it cross them. Such fracturing along bedding planes is common when cores are broken.

There are many limitations to this simplistic model. Stress-strain fields are not modeled on the small scale. It is assumed that the stress-strain distribution is homogeneous and that only the failure strength is determined by the heterogeneities of the distribution. The "crack" is produced by "removing" (setting to zero) the grid at the crack location rather than displacing it to one side as the crack propagates. Distortion of the media is not included. Despite all of these limitations, we believe something of the physics remains as indicated by the pictures themselves. As mentioned earlier, the hope here is not to introduce a new, improved fracture model,

but to suggest that including fGn distributions in existing models is likely to produce more realistic descriptions of crack growth.

7.3 Music From Well Logs

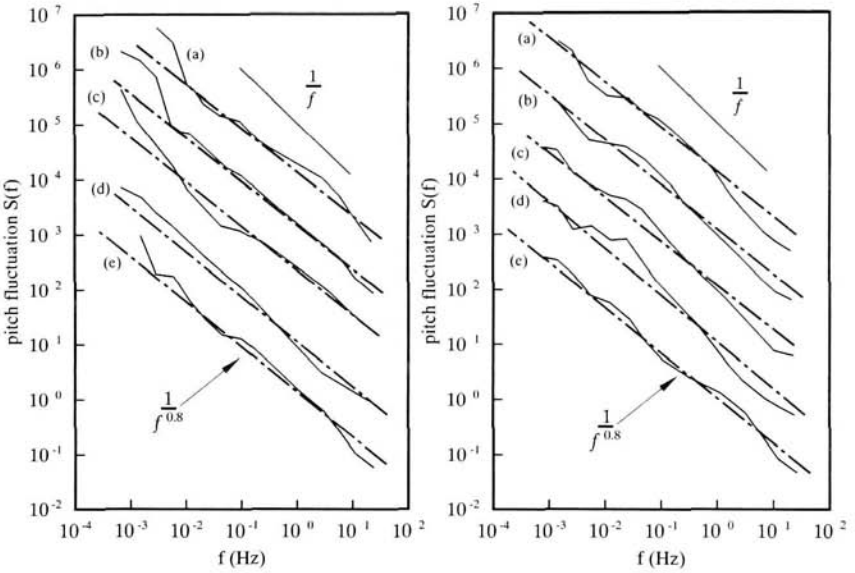
Voss and Clarke[113] discovered that many forms of music are similar to $1/f$ noise. Voss and Clarke plot the pitch fluctuations for different musical types as shown in Figure 7.16. Note that the reference line added by Voss[11] (solid line) that corresponds to $1/f$ noise could equally well be replaced by lines corresponding to $1/f^{0.8}$ (dashed lines). As illustrated in Chapter 3, well logs are also $1/f^{\beta}$ noises (fGn) with β = 2H-1 and $0.7 < H < 1.0$. In particular, H = 0.9 corresponds to $1/f^{0.8}$. Indeed, $1/f^{\beta}$ noises can be used to generate music.[113] This means that it is possible to generate music-like sounds from well log traces.

We demonstrate the technique by starting with the neutron porosity log in Figure 7.17. This log is chosen, because it is part of the fractal study of well logs in Chapter 3. As a first step, the log values are linearly scaled to frequencies corresponding to audible sounds. The result can be played directly using a microcomputer equipped with BASIC. The results are improved by separating the frequencies into bins corresponding to the notes of the C major scale. Then, rhythm is added using a simple algorithm, and the notes are transcribed to a musical score (Figure 7.18).

7.3.1 *Porosity to Frequency Map*

We start with a neutron porosity log with fGn character (Figure 7.17). The porosity values, ϕ, from the porosity log are scaled to frequency, f, so that an audible frequency corresponds to all porosity log values. Recall that one Gaussian distribution can be mapped onto another by a linear scaling of the mean and standard deviation. In particular, we use the transformation

$$f = a\,\phi + b \tag{7.6}$$



Pitch fluctuations from different musical cultures

(a) the Ba-Benzele Pygmies
(b) traditional music of Japan
(c) classical ragas of India
(d) folk songs of old Russia
(e) American blues

Pitch fluctuations in western music

(a) Medieval music up to 1300
(b) Beethoven 3rd Symphony
(c) Debussey piano works
(d) R. Strauss en Heldenlebe
(e) American blues

Figure 7.16. The $1/f$ spectra of different musical forms.[11]

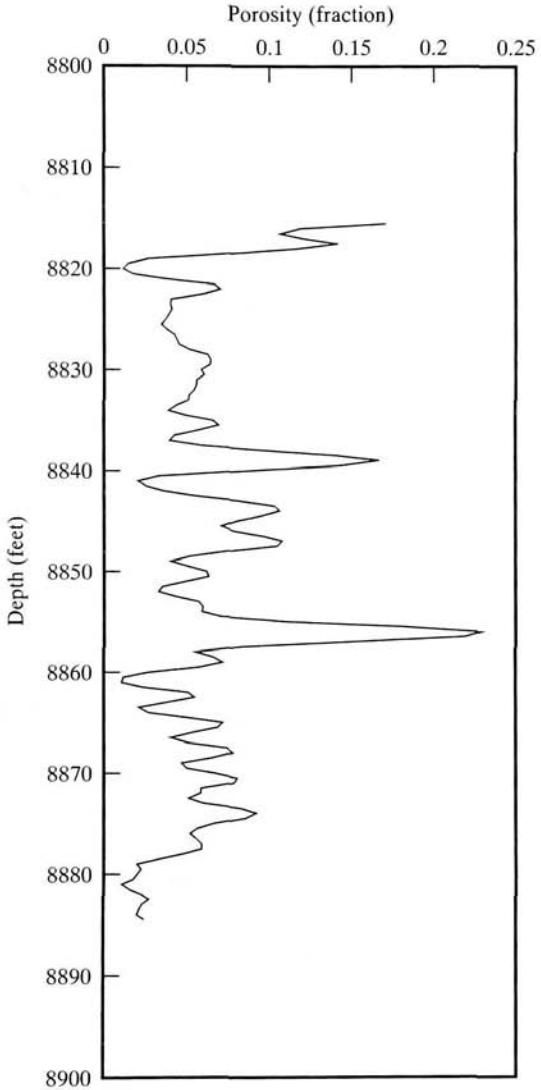


Figure 7.17. Neutron porosity well log.

Figure 7.18. Musical score.

where

$$a = 3398$$

and

$$b = 538$$

To listen to the converted frequencies, we used the BASIC programming language[114] containing a command called SOUND on a personal computer (PC). This command sends a specified frequency to the speaker in the PC for a specified duration of time. The command structure is

SOUND frequency,duration

The frequency is in Hertz (cycles per second) within a range of 37 to 32,767. The duration is in clock ticks, where there are 18.2 clock ticks per second. The frequency scaling indicated above with a duration of 3 produces a pleasant response.

7.3.2 *Frequency Corresponding to the C Major Scale*

The notes of the scale can be related to their frequency with a simple formula. This formula can be used to "bin" the frequencies from the porosity log transform into discrete notes. The binning process can be made to any key. The key of C is used for this log. Once the binning is completed, the notes can be played by sending the frequency corresponding to each note to the speaker of the PC using the SOUND command described above.

Microsoft's BASIC programming guide[114] presents a mapping from notes on a C major scale and frequency. The column in Table 7.1 entitled "known frequencies" contains these values. The notes are each given a numerical value, "Note Number," N, in Table 7.1. Then, the frequency, f, of each note for a given N is approximated using

$$f = 523.25 * (1.0595)^N \tag{7.7}$$

Table 7.1 Equation used to map notes to frequency.

Notes	Note Number	Known Frequency	Calculated Frequency	Differences	Percent Error
C	-24	130.81	130.7031	0.106814	0.081656
C#	-23		138.4800		
D	-22	146.83	146.7195	0.110413	0.075198
D#	-21		155.4494		
E	-20	164.81	164.6986	0.111358	0.067568
F	-19	174.61	174.4982	0.111789	0.064022
F#	-18		184.8808		
G	-17	196	195.8812	0.118735	0.060579
G#	-16		207.5361		
A	-15	220	219.8846	0.115396	0.052452
A#	-14		232.9677		
B	-13	246.94	246.8293	0.110681	0.044821
C	-12	261.63	261.5156	0.114337	0.043701
C#	-11		277.0758		
D	-10	293.66	293.5618	0.098142	0.033420
D#	-9		311.0287		
E	-8	329.63	329.5350	0.094999	0.028820
F	-7	349.23	349.1423	0.087666	0.025102
F#	-6		369.9163		
G	-5	392	391.9263	0.073678	0.018795
G#	-4		415.2459		
A	-3	440	439.9530	0.046928	0.010665
A#	-2		466.1302		
B	-1	493.88	493.8650	0.014969	0.003030
Middle C	0	523.25	523.25	0	0
C#	1		554.3833		
D	2	587.33	587.3691	-0.03918	-0.00667
D#	3		622.3176		
E	4	659.26	659.3455	-0.08555	-0.01297
F	5	698.46	698.5766	-0.11661	-0.01669
F#	6		740.1419		
G	7	783.99	784.1803	-0.19036	-0.02428
G#	8		830.8390		
A	9	880	880.2740	-0.27402	-0.03113
A#	10		932.6503		
B	11	987.77	988.1430	-0.37302	-0.03776
C	12	1046.6	1046.937	-0.33753	-0.03225
C#	13		1109.230		
D	14	1174.7	1175.229	-0.52951	-0.04507
D#	15		1245.155		
E	16	1318.5	1319.242	-0.74243	-0.05630
F	17	1396.9	1397.737	-0.83736	-0.05994
F#	18		1480.902		
G	19	1568	1569.016	-1.01645	-0.06482
G#	20		1662.372		
A	21	1760	1761.284	-1.28411	-0.07296
A#	22		1866.080		
B	23	1975.5	1977.112	-1.61231	-0.08161

The columns labeled "Differences" and "Percent Error" show that this formula does a good job of relating frequency values and note numbers.

The inverse of this formula maps the frequencies of the log transformation into a chromatic scale:

$$N = \text{integer} \left| \frac{\log\ (f/523.25)}{\log\ (1.0595)} \right| \qquad (7.8)$$

A better sound is produced if a map is made to a major scale rather than a chromatic scale. This is done by binning the frequency values without sharps or flats. To do this, the sharps and flats are mapped into naturals. The following transform is used:

Note In	Note Out
C	C
C#	C
D	D
D#	D
E	E
F	F
F#	F
G	G
G#	G
A	A
A#	A
B	B

Different transforms could be used to map the chromatic scale into any other major or minor key.

7.3.3 *Rhythm and Phrasing*

The transformation described so far produces one note per well log value where each well log value corresponds to 1/2 foot. The depth is mapped into time by simply setting the duration of each note in the SOUND command, but music is made up of different durations of notes and phrases. A rather simplistic map is used to determine the note duration and phrases. If the frequency values mapped from the porosity well logs decrease more than a prescribed amount (56.6 from one depth interval to the next), a SOUND command is sent to the speaker as a silent sound (frequency=32767) for a duration of 1. This "breaks up" the one note per well log value by "running together" sounds that are not separated by a silent sound and separates those that are. If consecutive sounds are the same, they sound like a single note of a longer duration. If consecutive sounds are not the same, a phrasing is apparent that lasts until the silent sound ends the phrase. The silent sound duration of 1 is short enough compared to the note duration of 3 that the silent sounds do not add any appreciable time. They sound like phrasing instead of rests.

The result is a string of numbers. Each number corresponds to one of the "Note Numbers" found in Table 7.1, except for the number used for the silent sound (-999).

7.3.4 *Musical Score*

The next step is to transcribe the above string of notes onto a musical score. Each note is mapped as an eighth note. If the same note is repeated without being separated by a silent sound, the note is considered to be longer. For example, two consecutive eighth notes, without a silent sound separating them, are written as a quarter note, and so forth. Phrasing ends at each silent sound. The resulting score is found in the Figure 7.18.

Tempo is controlled on the computer by the "duration" in each SOUND command. If each note is an eighth note, an Andante (~100 beats/minute) speed is close to a duration of 3 on the SOUND command. This duration is chosen, because it gives a "pleasant" sound. Dynamics and harmony are not available on the PC.

Obviously, the algorithm described here is not the only one possible. The porosity to frequency map parameters or even the map itself can be changed. The key, rhythm, and phrasing are all arbitrary. The parameters are adjusted to "sound good," based on a brief study. An exhaustive study has not been made.

Unresolved Issues

Pressure Transient Tests

The fractal model (Section 7.1.4) matches some field pressure transient tests in the waterflood test case in Chapter 6. An obvious question is whether simulated pressure transient tests on fGn distributions show this behavior.

Preliminary attempts at simulating a pressure drawdown on fGn cross sections produced a deviation from the homogeneous model but not as steep of a slope as observed in the field tests. Figure 7.19 shows a simulated pressure drawdown curve where the cross section in Figure 4.1 has been coarsened to a 256 x 75 grid. The simulation treats only the left half of the cross section. A producer is opened in all layers at the left edge, and either a no flow or constant pressure boundary condition is imposed on the right edge. The pressure and derivative curves in Figure 7.19 are parallel straight lines, the same trend as observed in the field. Boundary effects become important when two cases with different right-hand boundary conditions diverge.

The slope in Figure 7.19 is about 0.6, significantly greater than 0.5 for a layer cake reservoir. Still, the slope is less than observed in the field. We must compare the simulations and field tests in the same linear flow geometry simulated in the cross sections. Often wells are hydraulically fractured in the field. Then, a pressure transient test will have an early linear flow period, followed by a transition period, and then a radial flow period (Figure 7.20). The fractal model for a fractured well[104] covers each of these periods. Field tests near the wells used for the cross section have slopes of 0.74 to 0.85 in the linear flow period, significantly above the simulated result 0.6.

More work may be able to explain this discrepancy. Perhaps a finer grid is necessary or a 3D simulation is required. A larger question is whether pressure transient tests can be used to estimate parameters that are used in making the cross sections in Chapter 4.

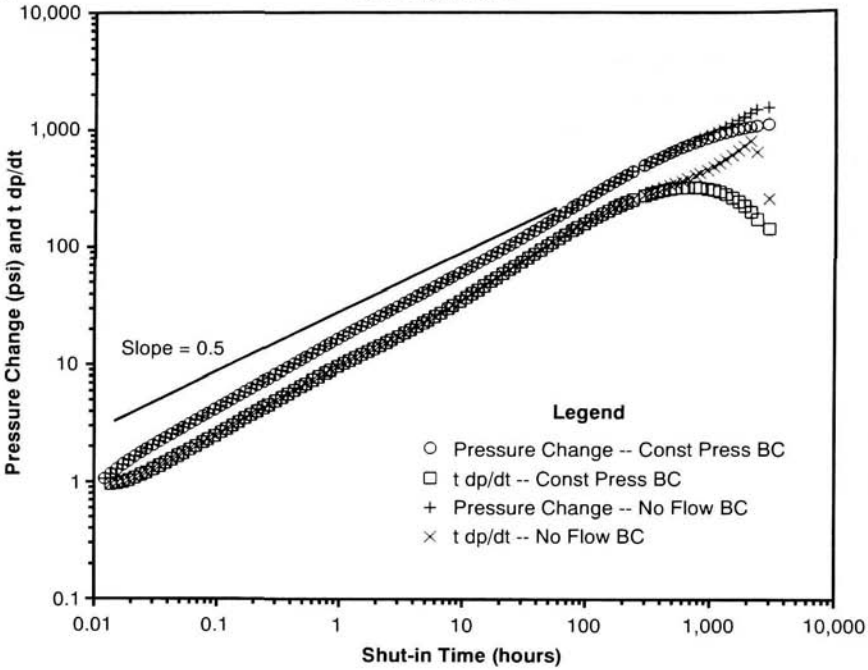


Figure 7.19. Simulated pressure drawdown on 2D vertical cross section.

Faults and Fractures

1. Modern 3D seismic surveys cannot detect relatively small faults (less than 50 feet in throw) that divide the reservoir into compartments. Techniques to stochastically model subseismic faults require orientation information in addition to the displacement distributions in Figure 7.8. Additional work in analyzing orientation of faults may lead to new techniques to stochastically model subseismic faults.

2. It would be interesting to extend the crack model to 3D and crack 3D fGn distributions generated by Equation 3.2 and then compare the surface properties of the generated crack with measured surface properties of fractured rock.

3. As mentioned in the text, the simple crack model should be replaced with more realistic crack models. Does including fGn distributions in other models produce more realistic patterns of cracks?

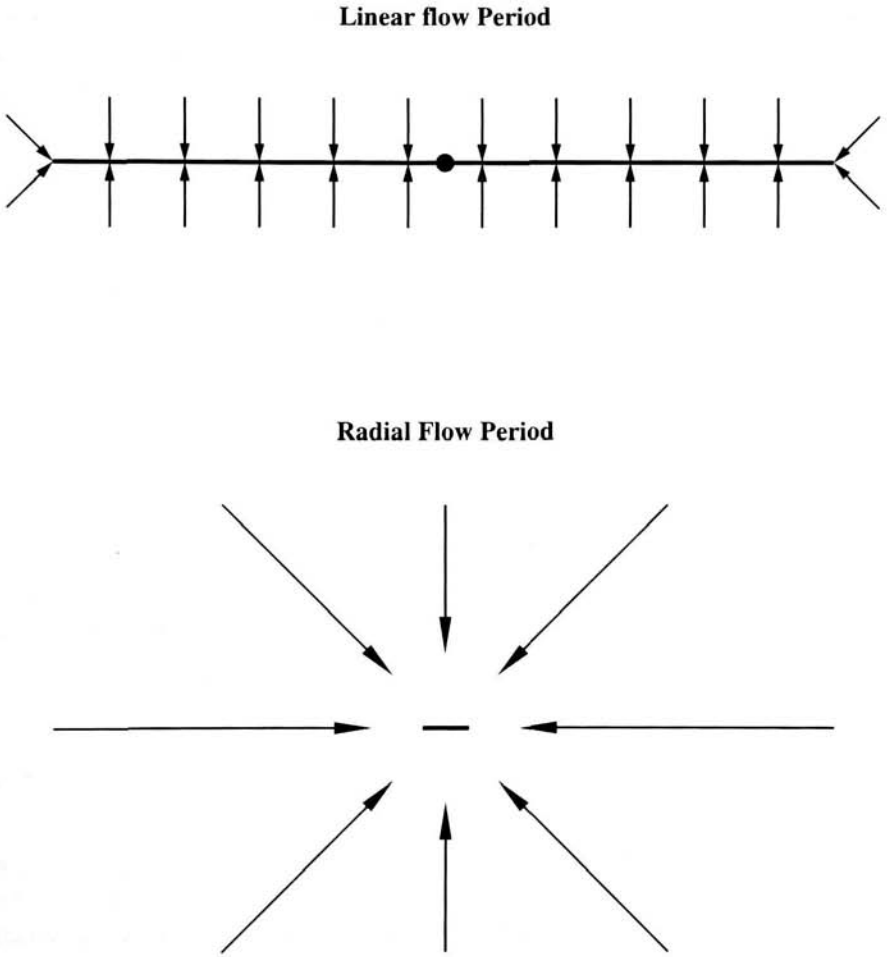


Figure 7.20 Early linear flow period and late radial flow period.

APPENDIX A7

In this Appendix, we describe the details of the fractal model for pressure transient tests. We start with some scaling relationships for the porosity and permeability on a fractal network (Appendix A7.1). We then incorporate these scaling relationships into the pressure transient model (Appendix A7.2). Typical solutions and type curves are then discussed (Appendix A7.3). Finally, we apply the model to the field test case (Appendix A7.4).

A7.1 Fractal Permeable Network

The core sample in Figure 7.6 illustrated how effective permeability can decrease with distance from the wellbore. Similar effects have also been modeled for other transport processes (Gefen et al.[115]). Like the pressure transient problem, these other processes are modeled by the diffusion equation. The link between work by Gefen et al.,[115] and our problem is the hydraulic diffusivity, η,

$$\eta = k/(\mu c_t \phi) \qquad \text{(A7.1)}$$

The hydraulic diffusivity includes the permeability, k, the fluid viscosity, μ, and the porosity, ϕ. The compressibility, c_t, accounts for contributions from both the rock and the fluid. By analogy with Gefen et al., the hydraulic diffusivity for a fractal permeable network should scale as

$$\eta(r) = k(r)/(\mu c_t \phi(r)) \sim r^{-\Theta} \qquad \text{(A7.2)}$$

where Θ is a parameter related to the topology of the network ($\Theta \geq 0$). Here, r is the distance from the wellbore. In practice, Equation A7.2 should apply over a finite range of length scales, $L_1<r<L_2$, where L_1 and L_2 are the lower and upper cutoff lengths, respectively. These cutoff lengths have not been established for any oil or gas reservoirs.

The hydraulic diffusivity contains both porosity and permeability. The porosity of an areal network scales as[101,102]

$$\phi(r) = \phi_r(r/L_r)^{-(2-d_f)} \tag{A7.3}$$

where d_f is the fractal dimension.[100] The porosity has been normalized with respect to the reference value, ϕ_r, at length scale L_r. Usually log and core data provide a reference porosity value. We expect the porosity to be nearly invariant with length scale, because oil-in-place calculations based on log and core porosities give reasonable estimates. Thus, the fractal dimension, d_f, should be 2 or slightly less. On the other hand, the shortcomings of the usual methods for averaging permeability are well known. From Equations A7.2 and A7.3, permeability scales as

$$k(r) = k_r(r/L_r)^{-(2-d_f+\Theta)} = k_r(r/L_r)^{-d_f(2/d_s-1)} \tag{A7.4}$$

where d_s is the spectral or fracton dimension.[116,117] Alexander and Orbach[117] defined the fracton or spectral dimension as

$$d_s = 2\,d_f/(2 + \Theta) \tag{A7.5}$$

As before, k_r is a reference permeability. Later, we will see that the asymptotic slope of the pressure transient curve is related to the fracton dimension. For this reason, d_s is used to describe the fractal network instead of Θ. For the areal flow cases in this chapter, $0<d_s\leq 2$.

A7.2 Fractal Reservoir Solution

Chang and Yortsos[101] have solved the pressure transient equations with porosity and permeability given by power law relationships similar to Equations A7.3 and A7.4. They directed their analysis toward naturally fractured reservoirs. With some modifications,[102] their model can also represent our problem of a reservoir with impermeable rock embedded within permeable pay zones. Below we describe the solution for a drawdown test on a circular wellbore. The fractal model has also been extended to treat a well with a vertical fracture.[104]

As usual, a diffusion equation for pressure[59] describes the flow of a single-phase, slightly compressible fluid through a reservoir. For an areally heterogeneous

reservoir, a general form of the equation applies and includes spatial variations of permeability and porosity. For radially symmetric flow in a plane, we write

$$\frac{\partial p}{\partial t} = \frac{1}{\mu c_t r\ \phi(r)} \frac{\partial}{\partial r}\left[r\ k(r) \frac{\partial p}{\partial r}\right] \tag{A7.6}$$

An equivalent wellbore radius, r_{we}, takes into account positive or negative skin as explained in Section 7.1.4. Substitution of Equations A7.3 and A7.4 into Equation A7.6 yields

$$\frac{\partial p}{\partial t} = \frac{k_{we}}{\mu c_t r \phi_{we}} \left(\frac{r}{r_{we}}\right)^{2-d_f} \frac{\partial}{\partial r}\left[r \left(\frac{r}{r_{we}}\right)^{-d_f(2/d_s-1)} \frac{\partial p}{\partial r}\right] \tag{A7.7}$$

Equation A7.7 is a generalization of the usual diffusion equation, since it applies to any permeable network of fractal dimension d_f embedded in two-dimensional Euclidean space. By setting $d_f = d_s = 2$, one obtains the usual diffusion equation. The solution of Equation A7.7 will be a smooth function for pressure. The actual pressure in the fractal network may have discontinuities due to permeability discontinuities. Still, the smooth pressure from Equation 7.7 represents a spatially averaged pressure distribution.

Before deriving the solution for a circular wellbore, we pick up the concept of an effective wellbore radius,[4] r_{we}, that is mentioned in Section 7.1.4. Replacing the reference length scale L_r by r_{we}, we rewrite Equations A7.3 and A7.4 as

$$\phi(r) = \phi_{we}(r/r_{we})^{-(2-d_f)} \tag{A7.8}$$

$$k(r) = k_{we}(r/r_{we})^{-d_f(2/d_s-1)} \tag{A7.9}$$

We need to introduce some dimensionless groups that allow us to present the solution as a single set of curves. Such groups are commonly used in pressure transient analysis. Dimensionless pressure, time, wellbore storage coefficient, and spatial coordinate are chosen as

$$p_D = 0.00708\, k_{we}\, h\, (d_f/d_s)(p_i - p) \,/\, (qB_o\mu) \tag{A7.10}$$

$$t_D = 0.000264\, k_{we}(d_f/d_s)^2 t \,/\, (\phi_{we}\mu c_t r_{we}^2) \tag{A7.11}$$

$$C_{sD} = 0.8936(d_f/d_s)C_s/(\phi_{we} c_t h r_{we}^2) \tag{A7.12}$$

$$r_D = r/r_{we} \tag{A7.13}$$

Here, h is the net effective pay (feet) of the reservoir. p_i is the initial reservoir pressure (psi), and q is the surface volume flow rate (STB/day). The formation volume factor (FVF), B_o, converts surface barrels (STB) to reservoir barrels (RB) by taking into account volume changes. The parameter C_s specifies the barrels of fluid stored in the wellbore for each psi change in bottom-hole pressure. Recall from Equation A7.1 that c_t represents compressibility. The definition of dimensionless pressure corresponds to a drawdown test. By applying the principle of superposition, one can apply the solution to a buildup test.[4]

Numerical constants appear in Equations A7.10 through A7.12, because they are written for customary oilfield units. These units appear in Table A7.1 with values for the field test.

Equation A7.7 can be written in dimensionless form as

$$(d_f/d_s)^2 r_D^{2d_f/d_s - 2} \frac{\partial p_D}{\partial t_D} = \frac{d_f - 2d_f/d_s + 1}{r_D} \frac{\partial p_D}{\partial r_D} + \frac{\partial^2 p_D}{\partial r_D^2} \tag{A7.14}$$

The boundary condition for a circular wellbore uses Darcy's equation to calculate the production rate of reservoir fluids entering the wellbore. This downhole production rate must equal the surface production rate plus the rate of wellbore storage. After writing this mass balance in dimensionless form, we get

Table A7.1 Well data and fluid properties.

Net Effective Pay, ft.	112
Porosity, Fraction	0.10
Total Compressibility, psi^{-1}	1×10^{-5}
Oil Viscosity, cp	2.46
Water Viscosity, cp	0.85
Weighted Average Viscosity, cp[a]	1.0
Weighted Average FVF, RB/STB[a]	1.00
Rates, BOPD/BWPD[b]	7/60
Pressure Measurement Method	Electronic Gauge

[a] Viscoscity and FVF for the combined flow of water and oil are weighted averages based on the oil and water production rates.
[b] Surface barrels for oil and water production rates.

$$(d_f/d_s)C_{sD}\left[\frac{dp_{pwD}}{dt_D}\right]_{r_D=1} - \left[\frac{\partial p_D}{\partial r_D}\right]_{r_D=1} = (d_f/d_s) \qquad \text{(A7.15)}$$

For an infinite reservoir, additional boundary conditions are

$$\lim_{r_D \to \infty} p_D(r_D,t_D) = 0 \qquad \text{(A7.16)}$$

$$p_D(r_D,0) = 0 \qquad \text{(A7.17)}$$

Equations A7.14 through A7.17 are similar to the dimensionless equations solved by Chang and Yortsos[101] for a network of natural fractures with no matrix participation. The only difference is the appearance of the ratio d_f/d_s in Equations A7.14 and A7.15. The equations are solved by applying a Laplace transform.[118] Let $\bar{p}_D(r_D,\tau)$ denote the Laplace transform of $p_D(r_D,t_D)$. Applying the transform to Equations A7.14 through A7.17, one obtains

$$r_D^2 \frac{d^2\tilde{p}_D}{dr_D^2} + (d_f - 2d_f/d_s + 1) r_D \frac{d\tilde{p}_D}{dr_D} - z(d_f/d_s)^2 r_D^{2d_f/d_s} \tilde{p}_D = 0 \qquad \text{(A7.18)}$$

$$\left[(d_f/d_s) C_D z \tilde{p}_D - r_D \frac{d\tilde{p}_D}{dr_D} \right]_{r_D=1} = \frac{1}{\tau} (d_f/d_s) \qquad \text{(A7.19)}$$

$$\lim_{r_D \to \infty} \tilde{p}_D(r_D,z) = 0 \qquad \text{(A7.20)}$$

$$\tilde{p}_D(r_D,0) = 0 \qquad \text{(A7.21)}$$

By a suitable change of variables, it is possible to transform Equation A7.18 into a Bessel equation. To see this, introduce the following variables

$$\tilde{p}_D = r_D^{\gamma} f(\rho) \qquad \text{(A7.22)}$$

where

$$\rho = \alpha r_D^{\beta} \qquad \text{(A7.23)}$$

Substitution of Equations A7.22 and A7.23 into Equation A7.18 gives

$$\rho^2 \frac{d^2 f}{d\rho^2} + \rho \frac{df}{d\rho} - (1 - d_s/2)^2 f - \rho^2 f = 0 \qquad \text{(A7.24)}$$

if $\alpha = \sqrt{\tau}$, $\beta = d_f/d_s$, and $\gamma = (2d_f/d_s - d_f)/2$. Equation A7.24 is Bessel's modified differential equation of order $1-d_s/2$. The general solution[118] is

$$f(\rho) = c_1 I_v(\rho) + c_2 K_v(\rho) \qquad \text{(A7.25)}$$

where $I_\nu(\rho)$ and $K_\nu(\rho)$ are modified Bessel functions of the first and second kind, respectively, and of order ν. Here $\nu = 1 - d_s/2$. The constants c_1 and c_2 are determined from the boundary conditions Equations A7.19 and A7.20. We apply the change of variables in Equations A7.22 and A7.23 to these boundary conditions. The requirement in Equation 7.20 that the pressure approaches zero far from the wellbore requires $c_1 = 0$, because $I_\nu(\rho)$ is unbounded as ρ (or r) approaches infinity. From Equations 7.19, one finds the pressure response at the wellbore to be[101,102]

$$\bar{p}_D(1,\tau) = \frac{K_\nu(\sqrt{\tau})}{\tau[C_{sD}\tau\, K_\nu(\sqrt{\tau}) + \sqrt{\tau}\, K_{\nu-1}(\sqrt{\tau})]} \qquad \textbf{(A7.26)}$$

The dimensionless pressure at the wellbore depends on d_s through the parameter ν but is independent of d_f. The actual pressure depends on d_f through the dimensionless groups (Equations A7.10 through A7.12). The inverse Laplace transform of Equation A7.26 is calculated numerically to find $p_D(1,t_D)$ using the Stehfest algorithm.[119] The resulting curves for dimensionless pressure are shown in Figures A7.1 through A7.4.

A7.3 Type Curves

The pressure drawdown solution in Appendix A7.2 models a single well (circular wellbore) in an infinite reservoir. Figure A7.1 with logarithmic coordinates gives the pressure curves for various values of the fracton dimension, d_s. For this particular set of curves, wellbore storage is set to zero. At sufficiently large times, the pressure curves ($0<d_s<2$) are straight lines with slopes $1-d_s/2$. For $d_s=2$, the conventional solution is recovered for a homogeneous reservoir.

It is customary in pressure transient analysis to also plot the pressure derivative curve, $t_D\, dp_D/dt_D$. This curve can be an aid in obtaining a match to some field tests.[99] The usual pressure derivative plot with logarithmic coordinates in Figure A7.2 also yields straight lines with slope $1-d_s/2$ at large times.

For a given value of d_s, a set of pressure responses can be generated for various values of the dimensionless wellbore storage coefficient, C_{sD}. In this sense, the conventional set of type curves for a homogeneous reservoir is a special case of

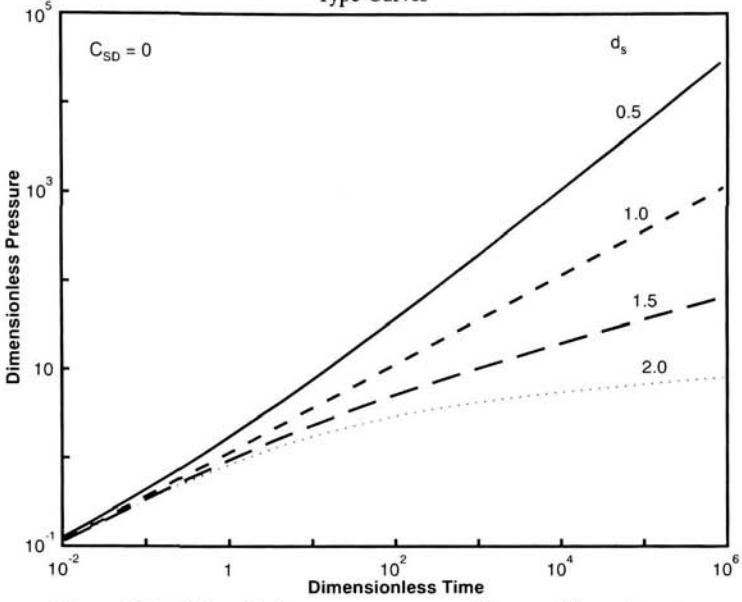


Figure A7.1. Dimensionless pressure curves with no wellbore storage.

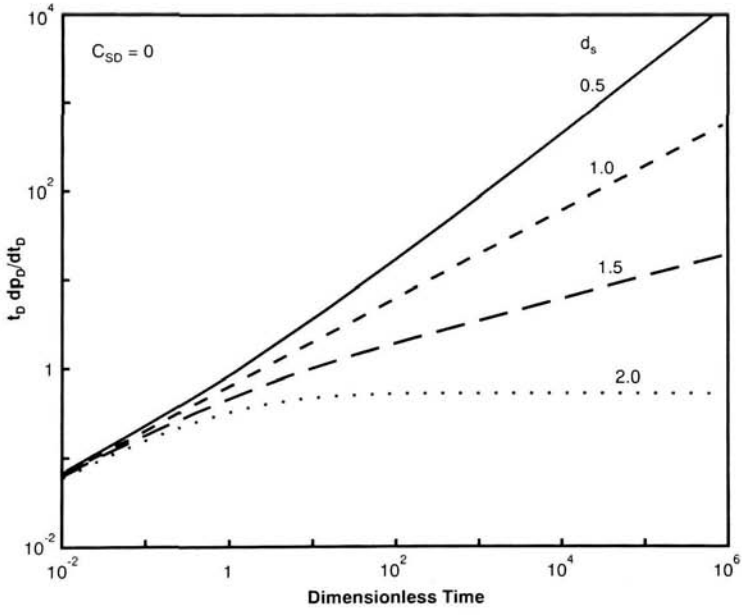


Figure A7.2. Dimensionless pressure derivative curves with no wellbore storage.

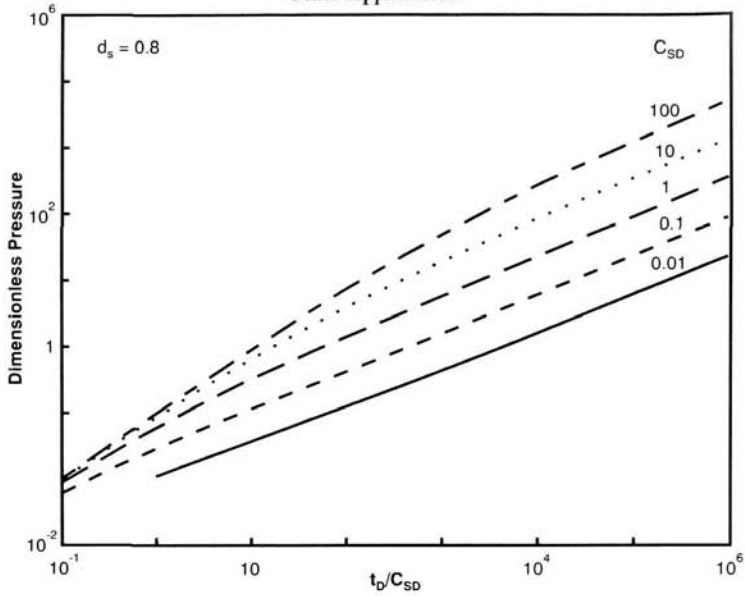


Figure A7.3. Dimensionless pressure curves for various values of C_{sD}.

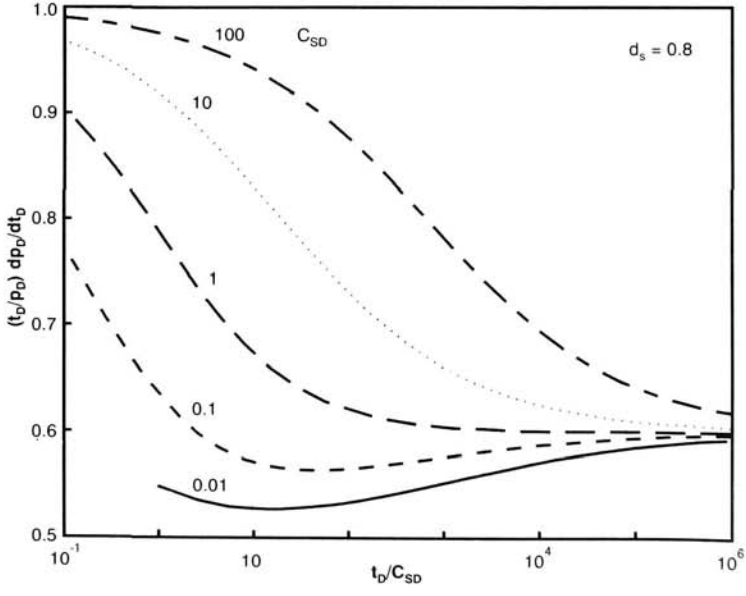


Figure A7.4 Plot of $(t_D/p_D)\ dp_D/dt_D$ for various values of C_{sD}.

the fractal reservoir solution with $d_s = 2$. A set of pressure responses for $d_s = 0.8$ in Figure A7.3 shows the usual increase in wellbore storage effects with increasing C_{sD}. Wellbore storage effects cause the early slope to be unity. The usual pressure derivative curves have similar shape and lack distinguishing characteristics. The lack of any distinguishing characteristics makes finding a unique curve to match field data difficult. For this reason, $(t_D/p_D)\ dp_D/dt_D$ curves are used in matching field data. Each $(t_D/p_D)\ dp_D/dt_D$ curve in Figure A7.4 has a specific shape that is much more pronounced. These curves represent the slopes of the pressure curves in Figure A7.3. As would be expected, for larger values of C_{sD}, $(t_D/p_D)\ dp_D/dt_D$ takes on values near unity at early times consistent with wellbore storage effects. Then, at late times, all curves approach the value of $1\text{-}d_s/2$.

A7.4 Match to Pressure Buildup Field Example

We now apply the type curves in Figure A7.1 through A7.4 to our field example. The particular curve that applies to a field test depends on two unknown parameters, the fracton dimension, d_s, and the dimensionless storage coefficient, C_{sD}. We can estimate d_s from the late time slope of the field data in Figure 7.7. The pressure and derivative curves fall on parallel lines with slope 0.60 between 1 and 150 hours. We avoid using the data after 150 hours, because the pressure derivative flattens, possibly due to interference from neighboring wells. The slope of 0.60 corresponds to $1\text{-}d_s$, with $d_s = 0.80$.

Knowing $d_s = 0.80$, we can now estimate C_{sD} from the $(t_D/p_D)\ dp_D/dt_D$ type curves in Figure A7.4. First, we replot the field data to make a (t/p) dp/dt plot. Then, the new plot is overlaid onto Figure A7.4. Since (t/p) dp/dt is dimensionless in both plots, vertical scales must overlay exactly. After adjusting the vertical scales to coincide, we move the field data in the horizontal direction for the best match. The C_{sD}=0.2 curve (not shown in Figure A7.4) matches the field data with the resulting match shown in Figure A7.5. Now we have estimated both type curve parameters and a unique type curve.

We want to determine the flow capacity and equivalent wellbore radius from the field data. Both unknowns have dimensions, but the type curves have dimensionless coordinates. To evaluate dimensional unknowns from the type curve, we need to relate the field shut-in time and pressure change to their dimensionless counterparts in the type curves.

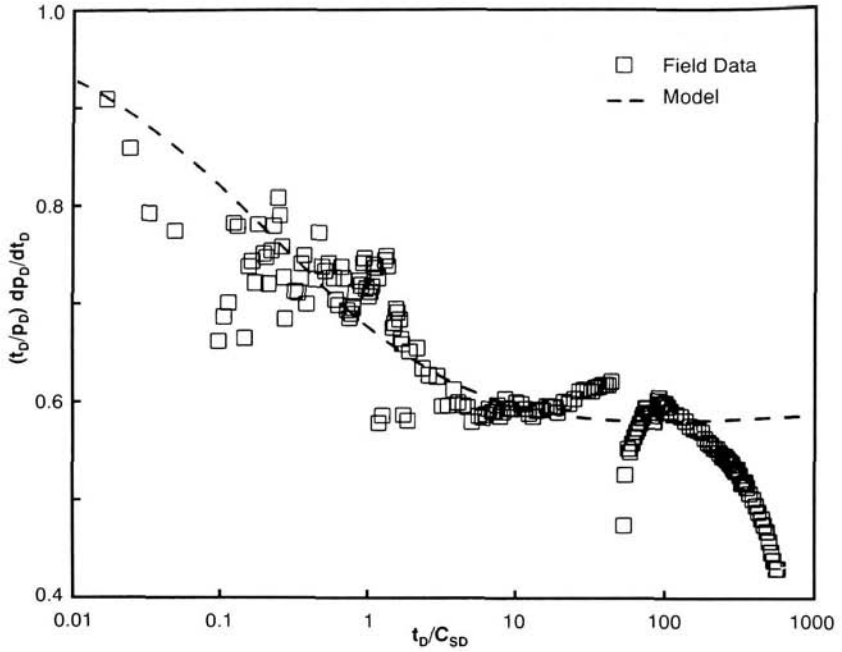


Figure A7.5. Curve match of pressure buildup on $(t_D/p_D)\ dp_D/dt_D$ plot with $C_{sD} = 0.2$.

We return to the field data match with the C_{sD}=0.2 type curve in Figure A7.5. The field data shut-in times are not shown on the horizontal scale in Figure A7.5, but they could be plotted. If we do, we find the scale for the field shut-in time, Δt, is 0.50 times the time scale, t_D/C_{sD}, on Figure A7.5. The subscript m will be used to denote the relationship (or match) between the two time scales as

$$(\Delta t)_m = 0.5\ (t_D/C_{sD})_m \qquad \textbf{(A7.27)}$$

Next, we relate the field pressure change with dimensionless pressure in the type curves. To do this, we use Figure 7.4, a plot of the dimensional pressure change and pressure derivative for the field data. We overlay Figure 7.4 on a type curve plot similar to Figure A7.3 but with the curve corresponding to C_{sD}=0.2. The time scales on the two plots are aligned with each other so that Equation A7.27 is satisfied. Then, we move the field data curve in the vertical direction for the best match. The

resulting match of the pressure curves is shown in Figure 7.7. The pressure derivative curves are also shown for completeness.

From the match in Figure 7.7, we compare the pressure scales on both the field data and the type curves. (Only the scale for the type curves is shown in Figure 7.7.) We find the scale for the field data pressure change, Δp, is 31.18 times the dimensionless pressure, p_D, scale. Using the subscript m to denote the relationship between the two pressure scales, we write

$$(\Delta p)_m = 31.18\ (p_D)_m \tag{A7.28}$$

From the above match, we can determine the flow capacity, $k_{we}h$, and the wellbore storage coefficient, C_s, as follows. Since we are analyzing a buildup instead of a drawdown test, we replace p_i-p in Equation A7.10 by Δp and t in Equation A7.11 by Δt. Combining Equations A7.10 and A7.28 and solving for $k_{we}h$ yields

$$k_{we}h = 141.2 q \mu B_o (p_D/\Delta p)_m (d_s/d_f) \tag{A7.29}$$

With the values listed in Table A7.1, d_f set equal to 2.0, and the above pressure curve match, Equation A7.29 gives $k_{we}h$ = 120 md-ft. Next, the wellbore coefficient can be estimated. Combining Equations A7.11 and A7.12 yields

$$C_s = (2.95 \times 10^{-4})(k_{we}h/\mu)(\Delta t/(t_D/C_{sD}))_m (d_f/d_s) \tag{A7.30}$$

With $k_{we}h$ = 120 md-ft and Equation A7.27, Equation A7.30 gives C_s = 0.045 RB/psi. This estimate agrees well with an independent estimate of 0.044 RB/psi based on the tubing/casing annulus dimensions and a common wellbore storage model with a rising fluid level.[120]

Finally, we estimate the equivalent wellbore radius by using Equation A7.12. First, we express the porosity, ϕ_{we}, in terms of the porosity at the wellbore scale, ϕ_w. Evaluating ϕ at the wellbore radius, r_w, in Equation A7.8 and rearranging yields

$$\phi_{we} = \phi_w (r_{we}/r_w)^{-(2-d_f)} \tag{A7.31}$$

Log and core data provide estimates of the porosity near the wellbore, ϕ_w. Combining Equation A7.31 with Equation A7.12 gives

$$r_{we}^{d_f} = 0.8936(d_f/d_s)C/(C_D \phi_w c_t h r_w^{2-d_f}) \quad \textbf{(A7.32)}$$

Values in the above paragraph and Table A7.1 give $r_{we}^{d_f} = 4500$ ft. For $d_f = 2.0$, $r_{we} = 70$ ft.

REFERENCES

1. M. Gerding, *Fundamentals of Petroleum* (Petroleum Extension Service, The Univ. of Texas at Austin, 1981).
2. J. T. Dewan, *Essentials of Modern Open-Hole Log Interpretation* (PennWell, Tulsa, Oklahoma, 1983).
3. W. D. McCain, Jr., *The Properties of Petroleum Fluids* (PennWell, Tulsa, Oklahoma, 1990).
4. C. S. Matthew and D. G. Russell, *Pressure Buildup and Flow Tests in Wells* (Society of Petroleum Engineers, Richardson, Texas, 1967).
5. J. W. Amyx, D. M. Bass, Jr., and R. L. Whiting, *Petroleum Reservoir Engineering* (McGraw-Hill, New York, 1960).
6. N. A. Anstey, *Seismic Interpretation. The Physical Aspects.* (International Human Resources Development Corp., Boston, 1977).
7. D. L. Turcotte, *Fractals and Chaos in Geology and Geophysics* (Cambridge University Press, Cambridge, 1992).
8. G. Korvin, *Fractal Models in the Earth Sciences* (Elsevier, Amsterdam, 1992)
9. B. B. Mandelbrot, *The Fractal Geometry of Nature* (W. H. Freeman and Co., New York, 1983).
10. J. Feder, *Fractals* (Plenum Press, New York, 1988).
11. R. F. Voss, in *The Science of Fractal Images*, eds. H. O. Peitgen and D. Saupe (Springer Verlag, New York, 1989), pp. 21-70.
12. M. Barnsley, *Fractals Everywhere* (Academic Press, New York, 1988).
13. H. H. Hardy, *Mathematical Geology* **24** (1992) 73-97.
14. F. F. Craig, Jr., *The Reservoir Engineering Aspects of Waterflooding* (Society of Petroleum Engineers, Richardson, Texas, 1971)

15. D. F. Morrison, *Multivariant Statistical Methods* (McGraw-Hill, New York, 1967).
16. F. Reif, *Fundamentals of Statistical and Thermal Physics* (McGraw-Hill, New York, 1965).
17. E. H. Isaaks and R. M. Strivastava, *An Introduction to Applied Geostatistics* (Oxford University Press, New York, 1989).
18. C. H. Scholz and B. B. Mandelbrot, eds., *Fractals in Geophysics* (Birkhauser Verlag, Boston, 1989).
19. J. S. Walker, *Fourier Analysis* (Oxford University Press, New York, 1988).
20. A. G. Journel and C. J. Huijbregts, *Mining Geostatistics* (Academic Press, New York, 1978).
21. M. E. Holn, *Geostatistics and Petroleum Geology* (Van Nostrand Reinhold, New York, 1988).
22. T. A. Hewett, SPE Paper 15386, presented at SPE Ann. Tech. Conf., New Orleans (1986).
23. *IMSL: Fortran Subroutine for Mathematical Applications* (IMSL Problem Solving Software Systems, Houston, 1987).
24. K. B. Oldham and J. Spanier, *The Fractional Calculus* (Academic Press, New York, 1974).
25. J. L. Jensen, D. V. Hinkley, and L. W. Lake, SPE Paper 14270, presented at SPE Ann. Tech. Conf., Las Vegas (1985).
26. G. M. Jenkins and D. G. Watts, *Spectral Analysis and Its Applications* (Holden-Day, San Francisco, 1968).
27. M. J. Lighthill, *Introduction to Fourier Analysis and Generalized Functions* (Cambridge University Press, Cambridge, 1958).
28. J. R. Garrison, Jr., W. C. Pearn, and D. U. von Ronsenberg, SPE Paper 22842, presented at SPE Ann. Tech. Conf., Dallas (1991).
29. D. L. Turcotte, Fractals **3** (1993) 491-512.
30. C. L. Farmer, in *The Mathematical Generation of Reservoir Geology*, eds. J. Fayers and P. King (Oxford University Press, New York, 1991).
31. E. Damsleth, C. Tjolsen, H. Omre, H. H. Haldorsen, *J. Petroleum Tech.* **44** (1992) 402-408.
32. B. B. Mandelbrot, verbal presentation at Fractals and Their Applications to Geology, Industrial Associates Symposium, Lamont-Doherty Geological Observatory, Palisades, New York, March 28, 1990.

33. H. H. Hardy, SPE Paper 23968, presented at SPE Permian Basin Oil and Gas Recovery Conf., Midland, Texas (1992).
34. S. E. Crane and K. M. Tubman, SPE Paper 20606, SPE Ann. Tech. Conf., New Orleans (1990).
35. *Log Interpretation Vols. 1 and 2* (Schlumberger Limited, New York, 1974).
36. H. H. Hardy, in *Reservoir Characterization III*, ed. B. Linville (PennWell, Tulsa, Oklahoma, 1993) pp. 787-797.
37. I. J. Taggart and H. A. Salisch, *APEA Journal* **31** (1991) 377-385.
38. Y. Aasum, M. G. Kelkar, and S. Gupta, *SPE Formation Evaluation* **6** (1991) 11-19.
39. W. E. Galloway, T. E. Ewing, C. M. Garrett, N. Tyler, and D. G. Bebout, *Atlas of Major Texas Oil Reservoirs* (Bureau of Economic Geology, Univ. of Texas at Austin, 1983).
40. H-H. Stolum, in *Reservoir Characterization II*, eds. L. W. Lake, H. B. Carroll, Jr., and T. C. Wesson (Academic Press, New York, 1991) pp. 579-612.
41. D. J. Goggin, M. A. Chandler, G. Kocurek, and L. W. Lake, *SPE Formation Evaluation* **7** (1992) 7-16.
42. D. Saupe, in *The Science of Fractal Images*, eds. H. O. Peitgen and D. Saupe (Springer Verlag, New York, 1988), pp. 71-136.
43. E. O. Brigham, *The Fast Fourier Transform* (Prentice-Hall, Englewood Cliffs, New Jersey, 1974).
44. M. Abramovitz and I. A. Stegun, eds., *Handbook of Mathematical Functions* (U. S. Government Print. Off., Washington, D. C., 1964) 933.
45. I. S. Gradshteyn and I. M. Ryzhile, *Table of Integrals, Series, and Products* (Academic Press, New York, 1980).
46. R. A. Beier and H. H. Hardy, in *Reservoir Characterization III*, ed. B. Linville (PennWell, Tulsa, Oklahoma, 1993) pp. 271-296.
47. R. A. Beier and H. H. Hardy, *SPE Reservoir Engineering* **9** (in press, August 1994)
48. A. Lallemand-Barres and P. Peaudecerf, *Bulletin du BRGM*, 2e Serie, Section III, No. 4 (1978).
49. J. F. Pickens and G. E. Grisak, *Water Resources Res.* **17** (1981) 1191-1211.
50. A. S. Emanuel, G. D. Alameda, R. A. Behrens, and T. A. Hewett, *SPE Reservoir Engineering* **4** (1989) 311-318.
51. T. A. Hewett and R. A. Behrens, *SPE Formation Evaluation* **5** (1990) 217-225.

52. J. L. Mathews, A. S. Emanuel, and K. A. Edwards, *J. Petroleum Technology* **41** (1989) 1136-1142.
53. R. F. Voss, in *Algorithms in Comp. Graphics*, Proceedings of the NATO A.S.I. Fund. Ilkley, U.K., 1985.
54. P. M. Doyen, *Geophysics* **53** (1988) 1263-1275.
55. P. A. Ruljtenberg, R. Buchanan, and P. Marke, *J. Petroleum Technology* **42** (1990) 22-31.
56. U. G. Araktingi, W. M. Bashore, T. T. Tran, and T. A. Hewett, in *Reservoir Characterization III*, ed. B. Linville (PennWell, Tulsa, Oklahoma, 1993) pp. 515-554.
57. C. V. Deutsch and A. G. Journel, *GSLIB: Geostatistical Software Library and User's Guide* (Oxford University Press, New York, 1993).
58. R. E. Collins, *Flow of Fluids Through Porous Materials* (Petroleum Publishing Co., Tulsa, Oklahoma, 1976).
59. L. P. Dake, *Fundamentals of Reservoir Engineering* (Elsevier, New York, 1978).
60. G. P. Willhite, *Waterflooding* (Society of Petroleum Engineers, Richardson, Texas, 1986).
61. R. M. Sneider, C. N. Tinker, and J. G. Richardson, *Reservoir Geology of Sandstones*, SPE Short Course notes presented at the Ann. Tech. Conf., New Orleans (1982).
62. C. J. George and L. H. Stiles, *J. Petroleum Technology* **30** (1978) 1547-1554.
63. B. H. Caudle, *Fundamentals of Reservoir Engineering* (Society of Petroleum Engineers, Richardson, Texas, 1967).
64. J. E. Warren and H. S. Price, *Soc. Petroleum Eng. J.* **1** (1961) 153-169.
65. D. Stauffer and A. Aharony, *Introduction to Percolation Theory*, 2nd Ed. (Taylor and Francis, Washington, D.C., 1992).
66. D. W. Peaceman, *Fundamentals of Numerical Reservoir Simulation* (Elsevier, New York, 1977).
67. B. Derrida, J. G. Zabolitzky, J. Vannimenus, and D. Stauffer, *J. Stat. Phys.* **36** (1984) 31.
68. P. R. King, *Transport in Porous Media* **4** (1988) 37-58.
69. A. Aharony, E. L. Hinrichsen, A. Hansen, J. Feder, T. Jossang, and H. H. Hardy, *Physica A* **177** (1991) 260-266.
70. J. Bernasconi, *Phys. Rev. B* **18** (1978) 2185-2191.

71. M. Alonso, and E. J. Finn, *Fundamental University Physics, Volume II: Fields and Waves* (Addison-Wesley, Reading, Mass., 1967), p. 617.
72. E. L. Hinrichsen, A. Aharony, J. Feder, A. Hansen, T. Jossang, and H. H. Hardy, *Transport in Porous Media* **12** (1993) 55-72.
73. J. Carruthers, Master's of Science Thesis, Imperial College, London (1990).
74. R. G. Larson, L. E. Scriven, and H. T. Davis, *Chemical Eng. Science* **36** (1981) 57-73.
75. V. K. S. Shante and S. Kirkpatrick, *Adv. Phys.* **20** (1971) 325.
76. C. Wilson, Masters of Science Thesis, Imperial College, London (1991).
77. R. J. Norris, and J. J. M., Lewis, SPE Paper 22692, presented at SPE Ann. Tech. Conf., Dallas, TX, (1991).
78. K. Aziz and A. Settari, *Petroleum Reservoir Simulation* (Elsevier, New York, 1979).
79. B. C. Craft and M. F. Hawkins, *Applied Petroleum Reservoir Engineering* (Prentice-Hall, Englewood Cliffs, New Jersey, 1959).
80. T. A. Hewett and R. A. Behrens, in *Reservoir Characterization II*, eds. L. W. Lake, H. B. Carroll, Jr., and T. C. Wesson (Academic Press, San Diego, 1991) pp. 402-441.
81. J. L. LeBlanc and B. H. Caudle, *Soc. of Petroleum Eng. J.* **11** (1971) 7-12.
82. J. C. Martin and R. E. Wegner, *Soc. of Petroleum Eng. J.* **19** (1979) 313-323.
83. R. W. Tang, R. A. Behrens, and A. S. Emanuel, *SPE Reservoir Engineering* **6** (1991) 253-258.
84. R. A. Beier, *SPE Advanced Technology Series* **2** (No. 2, 1994) 132-139.
85. T. A. Hewett and R. A. Behrens, *SPE Formation Evaluation* **8** (1993) 258-266.
86. D. E. Harris and R. L. Perkins, SPE Paper 22760 presented at the SPE Ann. Tech. Conf., Dallas, Texas (1991).
87. J. R. Kyte and D. W. Berry, *Soc. of Petroleum Eng. J.* **15** (1975) 269-276.
88. C. A. Kossack, *SPE Formation Evaluation* **5** (1990) 226-232.
89. P. R. Ballin and A. G. Journel, *J. Canadian Petroleum Tech.* **31** (1992) 52-62.
90. D. R. Guerillot and I. F. Morelon, SPE Paper 24891 presented at SPE Ann. Tech. Conf., Washington, DC (1992).
91. R. Giordano, S. Salter, and K. Mohanty, SPE Paper 14365, presented at SPE Ann. Tech. Conf., Las Vegas (1985).
92. *ECLIPSE-100 Reference Manual*, Exploration Consultants Ltd., Oxfordshire, England, 1992.

93. A. J. Katz, and A. H. Thompson, *Physical Review Letters* **54** (1985) 1325-1328.
94. A. J. Katz and A. H. Thompson, *J. Geophysical Research* **92** (1987) 599-607.
95. C. E. Krohn and A. H. Thompson, *Physical Review B* **33** (1986) 6366-6374.
96. C. C. Barton, C. H. Scholz, T. A. Schutter, P. R. Herring, and W. J. Thomas, AAPG Ann. Convention, Dallas, Texas, 1991.
97. W. J. Lee, *Well Testing* (Soc. of Petroleum Eng., Richardson, Texas, 1982).
98. A. C. Gringarten, H. J. Ramey, Jr., and R. Raghavan, *Soc. Petroleum Eng. J.* **14** (1974) 347-60.
99. D. Bourdet, J. A. Ayoub, and Y. M. Pirard, *SPE Formation Evaluation* **4** (1989) 293-302.
100. H. E. Stanley, in *On Growth and Form, Fractal and Non-Fractal Patterns in Physics*, eds. H. E. Stanley and N. Ostrowsky (Martinus Nijhoff Publishers, Boston, 1986) pp. 21-53.
101. J. Chang and Y. C. Yortsos, *SPE Formation Evaluation* **5** (1990) 31-38.
102. R. A. Beier, SPE Paper 21553, presented at CIM/SPE International Tech. Meeting, Calgary, Alberta (1990).
103. J. A. Barker, *Water Resources Research* **24** (1988) 1796-1804.
104. R. A. Beier, *SPE Formation Evaluation* **9** (1994) 122-128.
105. J. A. Acuna, I. Ershaghi, and Y. C. Yortsos, SPE Paper 24705 presented at SPE Ann. Tech. Conf., Washington, DC (1992).
106. J. T. Trocchio, *J. Petroleum Tech.* **42** (1990) 1038-1045.
107. J. Walsh, J. Watterson, and G. Yielding, *Nature* **351** (1991) 391-393.
108. J. Walsh and J. Watterson, *J. Structural Geology* **10** (1988) 239-247.
109. K. J. Heffer and T. G. Bevan, SPE Paper 20981, presented at SPE European Petroleum Conf., The Hague, Netherlands (1990).
110. B. D. M. Gauthier and S. D. Lake, *AAPG Bulletin* **77** (1993) 761-777.
111. H. J. Herrmann and S. Roux, eds., *Statistical Models for the Fracture of Disordered Media* (North-Holland, New York, 1990).
112. S. R. Brown and C. H. Scholz, *J. Geophysical Research* **B90** (1985) 12575-12582.
113. R. F. Voss and J. Clarke, *J. Acoust. Soc. Am.* **63** (1978) 1.
114. *Reference Guide, Basic Version 3* (Microsoft Corporation, Redmond, WA, 1987).
115. Y. Gefen, A. Aharony, and S. Alexander, *Physical Review Letters* **50** (1983) 77-80.
116. R. Orbach, *Science* **231** (1986) 814-819.
117. S. Alexander and R. Orbach, *Journal de Physique-Letters* **43** (1982) 625-631.

118. R. V. Churchhill, *Operational Mathematics* (McGraw-Hill, New York, 1958).
119. H. Stehfest, *Communication of the ACM* **13** (1970) 47-49.
120. H. J. Ramey, *J. Petroleum Technology* **17** (1965) 223-233.

This page is intensionally left blank

NOMENCLATURE

a	= constant
A	= Area
$A_{j,\ell}$	= array for layers in cross section
$A_{j,k,\ell}$	= array for layers in 3D distributions
A_n	= Area of nth step in Koch Island
A_{SD}	= constant in spectral density equation
A_Δ	= area of triangle bounded by two vectors
b	= constant
B	= failure strength coefficient
$B_{j,\ell}$	= array in 2D combining noise and layers
$B_{j,k,\ell}$	= array in 3D by combining noise and layers
B_o	= formation volume factor for oil
B_{SD}	= constant in spectral density equation
$\bar{B}$	= average value of cross-sectional array
$\bar{B}_A$	= average of entire B array
$\bar{B}_H$	= average of well values in B array
$\bar{B}_L$	= average of left edge values in B array
$\bar{B}_q$	= average of values in B array at well q
$\bar{B}_R$	= average of right edge values in B array
$\bar{B}_W$	= average of well values
c	= constant
c_t	= compressibility
$\mathbf{C}$	= covariance matrix
$C_d(\xi_j)$	= cumulative probability distribution function
$C_g(\xi)$	= cum Gaussian distribution function

$C_{j,\ell}$ = array for final cross section
$C_{j,k,\ell}$ = array for final 3D distribution
C_s = wellbore storage coefficient
$Cov(f_j,g_k)$ = covariance
d = diameter
d_f = fractal dimension
d_s = spectral or fracton dimension
D = ensemble average of differences D_k
D = vector from covariance equation
D(k) = difference of expected and true value for realization k
D_Δ = area of triangle defined by 3 wells
E_y = size ratio in x and y directions squared
E_z = size ratio in x and z directions squared
$\hat{E}_z$ = ratio of number of horizontal and vertical grids
f(x) = function of x
$f(\rho)$ = transformed variable related to pressure
$f_g(\xi)$ = Gaussian distribution
f_j = data sequence in 1D
$\bar{f}_j(x,t)$ = vertically averaged fractional flow of phase j
$f_{j,k}$ = array in 2D
$f_\ell(\xi)$ = log normal distribution
$f_p(\xi)$ = p-normal distribution
f = frequency
f' = constant value of frequency
f_x = frequency in x direction
f_y = frequency in y direction
f_z = frequency in z direction
F_c = force coefficient
$F_{j,\ell}$ = 2D noise array after linear scaling
$F_{j,k,\ell}$ = 3D noise array after linear scaling
F_m = discrete Fourier transform of f_j
$F_{m,p}$ = discrete Fourier transform of $f_{j,\ell}$
$\mathscr{F}$ = continuous Fourier operator in 2D
g = gravitational constant
g(x) = function in 1D

$g(x,y)$ = function in 2D
$\bar{g}$ = average value in g array
$\hat{g}$ = conductance
g_j = data sequence in 1D
$\breve{g}_j$ = modified data sequence in 1D
$g_{j,\ell}$ = array for 2D distributions
$g_{j,k,\ell}$ = array for 3D distribution
$g_j(k)$ = data value at point j, realization (or trace) k
g_{min} = minimum value in g array
g_{max} = maximum value in g array
g_0 = average true value over many ensembles
g_0' = average expected value over many ensembles
$g_0'(k)$ = expected value of realization k at x
$<g>_j$ = ensemble average at jth time step
$\{g\}_k$ = time average of kth trace
$G(\omega)$ = Fourier transform of g(x)
G_m = discrete Fourier transform of g_j
$\breve{G}_m$ = rearranged discrete Fourier transform of g_j
$G_{m,p}$ = discrete Fourier transform of $g_{j,\ell}$
$G_{m,n,p}$ = discrete Fourier transform in $g_{j,k,\ell}$
h = net effective pay of reservoir
$h(x_s)$ = integral of spectral density
$h(x-\chi)$ = function for smoothing
$h_{j,\ell}$ = pixel value in core forgery
$h_{j',\ell'}$ = function assigned at grid locations
h_{j-p} = discrete function for smoothing
H = Hurst exponent
H(x) = local height of cross-section
$H(\omega)$ = Fourier transform of h(x)
$H_{j,\ell}$ = linear interpolation based on fBm noise array edges
i = $(-1)^{1/2}$
$I_v(\rho)$ = modified Bessel function of the first kind
k = permeability
$\bar{k}$ = vertically averaged permeability
k_{eff} = effective permeability

$\tilde{k}_{rj}$ = pseudorelative permeability of phase j
k_{ro} = relative permeability to oil
k_{rw} = relative permeability to water
$K(s)$ = correlation function
$K_\nu(\rho)$ = modified bessel function of the second kind
ℓ = unit of measurement
L = length
L_r = reference length scale
L_x = size of grid in x direction
L_y = size of grid in y direction
L_z = size of grid in z direction
L_1 = lower cutoff length scale
L_2 = upper cutoff length scale
m = mass of object
M = mobility ratio
$M_T(x,t)$ = total mobility
$\bar{M}_T(x,t)$ = vertically averaged total mobility
N = note number
N = number of points or measuring units
N_e = number of traces or realizations
N_p = number of pairs
N_{pt} = number of points in traces
N_W = number of wells
N_x = number of points or grids in x direction
N_{xs} = number of grids in x direction for large scale array
N_y = number of points or grids in y direction
N_z = number of points or grids in z direction
N_{zs} = number of grids in z direction for large scale array
N_l = number of lags
p = pressure
p_j = pressure of phase j
$\tilde{p}_D$ = Laplace transform of p_D
Δp = pressure change
$PV(x)$ = cumulative pore volume at streamwise position x
q = volume flow rate

q_o = volume flow rate of oil
q_w = volume flow rate of water
$Q(t)$ = cumulative injection in pore volumes
r = radial coordinate
r_{inv} = radius of investigation
$R(t)$ = total resistance along streamtube
R_g = range for kriging
$R_j(\tau)$ = range for R/S analysis
R_m = discrete amplitude in 1D
$R_{m,n}$ = discrete amplitude in 2D
$R_{m,n,p}$ = discrete amplitude in 3D
$R_T(t)$ = sum of all resistances over streamtubes
s = separation distance or lag
s_n = sum of geometric series
s_{noise} = scaling factor for noise array
s_x = constant for anisotropic noise
s_y = constant for anisotropic noise
S = spectral density
$S_{j,k,\ell}$ = discrete form of spectal density
S_{or} = residual oil saturation
S_w = water saturation
S_{wi} = initial water saturation
t = time
Δt = shut-in time
T = period
$\mathbf{v}_j$ = jth vector for trilinear interpolation
v_x = velocity component in x direction
v_y = velocity component in y direction
V = variance for well pairs
V_A = variance of entire B array
V_H = variance of B array at well locations
V_{DP} = Dystra-Parson's coefficient
V_n = variance of the noise array
V_W = variance of well values
Var = variance

$\mathbf{w}$ = vector of weights
w_j = weight of the jth point
$W(x)$ = width of streamtube
$W_{L,\ell}$ = left edge well log data
$W_{q,k}$ = data at well q and depth ℓ
$W_{R,\ell}$ = right edge well log data
$\bar{W}_L$ = average of left edge well data
$\bar{W}_R$ = average of right edge well data
$\bar{W}_q$ = vertical average of data in well q
x = horizontal spatial coordinate
x_j = x coordinate for jth well
x_L = endpoint of interval
x_s = continuous values of x from discrete values
x_0 = constant value
x_0 = beginning endpoint of interval
Δx = spacing of points in x direction
y = horizontal spatial coordinate
y_j = horizontal coordinate for jth well
Δy = spacing of points in y direction
z = vertical coordinate
α = constant
β = constant
γ = exponent in coordinate transformation
$\gamma(f_j,g_k)$ = variogram
$\gamma_e(j,s)$ = ensemble variogram at jth point and lag s
$\gamma_t(k,s)$ = time avg variogram of kth trace and lag s
$\Gamma(\alpha)$ = gamma function
$\delta(x)$ = delta function
Δ = unit measuring area
ϵ = constant
ζ = streamwise coordinate
ζ_j = normal score transformed variable of ξ_j
Z = maximum value of streamwise coordinate
η = hydraulic diffusivity
θ_m = discrete phase in 1D

$\theta_{m,p}$ = discrete phase in 2D
$\theta_{m,n,p}$ = discrete phase in 3D
Θ = exponent related to topology of network
κ_1 = constant
κ_2 = constant
λ = running variable in R/S analysis
λ = Lagrange multiplier
λ_{SD} = constant prefactor in SD equation
μ = viscosity
μ_H = mean of original distribution
μ_j = viscosity of phase j
μ_W = desired mean
ξ = continuous data variable
ξ_m = discrete data variable
$\bar{\xi}$ = mean of ξ
Ξ = Fourier transform of ξ
$\Xi_{m,p}$ = discrete Fourier transform of $\xi_{j,\ell}$
ρ = transformed coordinate
ρ_j = density of phase j
$\rho(f_j,g_k)$ = correlation coefficient
σ = standard deviation
σ_D = standard deviation of differences
τ = R/S analysis length
τ = Laplace transform coordinate for time
ϕ = porosity
$\phi_{q,\ell}$ = porosity in well q at depth ℓ
$\phi_{L,\ell}$ = porosity on left edge at depth ℓ
$\phi_{R,\ell}$ = porosity on right edge at depth ℓ
Φ = flow potential
Φ_{av} = average flow potential in reservoir
Φ_j = flow potential of phase j
$\bar{\Phi}$ = vertically averaged flow potential
$\Delta\bar{\Phi}(t)$ = flow potential drop along streamtube
χ = dummy variable for x
$\psi_{j,\ell}$ = residual noise for fBm cross section

ω = angular frequency
ω_m = discrete angular frequency
ω_r = angular frequency in radial direction
ω_x = angular frequency in x direction
ω_y = angular frequency in y direction
ω_z = angular frequency in z direction
$\Delta\omega$ = discrete angular frequency interval
Ω_M = finite frequency range in the x direction
Ω_N = finite frequency range in the y direction
Ω_P = finite frequency range in the z direction

Subscripts

D = dimensionless
m = match point
o = oil
r = reference
w = water
we = effective wellbore

Superscripts

* = complex conjugate

INDEX

Amplitude
  of 1D sequence, 37, 39, 57
  of 2D array, 118
  of 3D array, 185
  finger in 2D, 118
  of sine and cosine curves, 37-39
  of smoothed sequence, 157-159
  relationship to spectral density, 58

BASIC, 315, 319
Bed resolution, 84-85, 88

Capillary pressure, 215, 239, 288
Compressibility, 326, 329
Core
  analyses, 3-4
  filling in data from, 163
  geological units from, 146
  permeability, 197-204, 275
  relative permeability, 245, 257, 263
Core photo
  forgery of, 14, 102-107
  fGn character of, 94-96
  permeability distributions, 206-208
  spectral density of, 118-121
Correlation
  among wells, 139, 141
  between porosity-permeability, 198-202, 206
  between two points, 13
  coefficient, 52
  function, 52
  with respect to lag, 13, 36
Correlogram, 52
Covariance
  definition of, 36, 51-52
  examples of, 32-35, 80
  from spectral density in 1D, 67-77
  from spectral density in 2D, 133-135
  matrix, 169-170
  of fBm, 42
  of fGn, 42
Cross sections
  in flow simulations, 241-269
  pressure drawdown on, 323
  of fBm porosity, 151-152, 178-180
  of fGn porosity, 141-149, 171-177
Cumulative distribution, 280

Darcy's law, 197
Diffusion equation, 326-328
Diffusion limited aggregation, 11-12
Dystra-Parson's coefficient, 29, 50

Effective permeability
  correlation with porosity, 199-201
  differences in 2D and 3D, 219
  effect of grid size on, 222-225
  from pressure transient test, 298, 337
  from FRACTAM, 204-211
  function of distance, 225-229
  of fGn cross sections, 258
  of each phase, 238
  of heterogeneous media, 197-198
  of layered media, 203-204
  of random media, 203-204
Ensemble
  definition of, 24
  ergodic, 26, 40, 49
  of earth strata, 24, 139
  of realizations, 144
Ensemble average
  definition of, 25-26, 49-50
  examples of, 25-29
  variogram, 66-67

Ergodicity, 25-26, 49
Euclidian dimension, 7

Failure strength, 309, 314
Faults
  in cross sections, 139
  models of, 324
  properties of, 306
  scaling relationships, 307-308
Fractional Brownian motion
  clouds from, 13, 103
  ergodicity of, 65-67
  generated cross sections, 151-152, 178-180
  generators of, 58-60
  in flow simulations, 253-257
  mountains from, 13, 103
  stationarity of, 65-67
  statistical measures of, 12, 41-48
Fractional Gaussian noise
  covariance of, 67-77
  ergodicity of, 65-67
  generators of, 58-60, 77
  generated cross sections, 141-149, 171-177
  generated 3D distributions, 152-156, 180-193
  in cores, 94
  in core photos, 95,102-111
  in flow simulations, 251-257, 269-275
  in music, 316
  in outcrop photos, 94-96
  in outcrops, 96-102
  in well logs, 87-94
  stationarity of, 65-67
  statistical measures of, 13, 41-47
  variogram of, 67-77
Filter
  effects on phase, 157-159
  effects on spectral density, 124-129
  effect in well logs, 84
  smoothing, 123
  used to combine data sets, 157-159
  used in generating pictures, 103
Finite difference
  flow simulation, 241, 259, 268
  techniques, 237
Flux-weighted method, 267, 295-296
Formation volume factor, 329, 338
Fourier transform
  definition of, 37, 54-55
  discrete, 56
  in 1D, 54-57
  in 2D, 102, 118-123
  in 3D, 190-191
  inverse in 1D, 55
  inverse in 2D, 131-133
  inverse in 3D, 190-191
  some useful properties of, 60-65
Fractal
  definition of, 4
  exact, 11
  property of an object, 16-17
  statistical, 11-15
FRACTAM
  accuracy of, 208-209
  algorithm in 2D for, 234
  algorithm in 3D for, 235-236
  description of, 204-211
  equations for, 232-233
  in 2D, 205-207
  in 3D, 210-211
fractional flow
  curve, 248-249
  equation for, 289
  in pseudorelative permeability, 293
  in streamtube, 251, 289
fractures
  properties of, 306
  propagation model of, 309-314
  scaling relationships, 307-308
frequency
  angular, 42, 54-55
  in spectral density model, 39, 102, 118
  of sound, 315
  of sine or cosine curves, 37-38
  mapped to musical notes, 319-322

Gaussian distribution
  histogram, 29
  equation of, 51
  transformation to, 85-86, 128-130
Geostatistics
  kriging, *see* Kriging
  simulation, 163-171
  variogram, 32
Grayburg formation
  deposition of, 143-144
  waterflood in, 241

Histogram
  definition of, 29
  of fBm, 42
  of fGn, 42

Horizontal well log, 80-82
Hurst exponent, 40
Hydraulic diffusivity, 326

Intermittency exponent, 40
Inverted-nine spot, 241, 260-261

Kriging
  indicator methods, 156
  smooth map from, 152
  to fill-in data trace, 163
Kyte and Berry method, 259, 293-294

Lag
  in covariance, 36, 51-52
  in R/S analysis, 40, 53-54
  in variogram, 36-37, 52
Laplace transform, 330-332
Layer cake model
  in flow simulations, 252, 264
  permeability, 258
  porosity, 140, 264
Linear scaling
  for sequences, 65
  in adding noise, 177-179, 191-192
Log-normal distribution
  equation of, 51
  permeability, 202
  resistivity logs, 88
Logging tools
  definition and examples of, 3
  bed resolution of, 84
  wells, *see* Well logs

Mean
  definition of, 29, 50
  ergodicity of, 65, 67
  in histogram, 85
  linear scaling of, 65, 142-143
  of generated sequences, 64, 121
  stationarity of, 65, 67
Minipermeameter, 96-102
Mobility
  in flux-weighted method, 295
  in Kyte and Berry method, 293
  in streamtube, 248-249, 289-291
  total, 288
Mobility ratio
  definition of, 266
  effects on streamtubes, 249
  of miscible displacement, 259
  of waterflood, 263, 266
Multivariant measures,
  equations for, 51-54
  of example data sets, 32-41
  of fBm, 42-47
  of fGn, 42-45
Music, 315-323

Net effective pay, 298, 329-330
Noise array
  in 3D fGn distributions, 190
  in core forgeries, 131-133
  in fBm cross sections, 152, 178-179
  in fGn cross sections, 142, 173-174
Normal score transform
  effects on spectral density, 86
  example of, 96, 99
  method, 128-130
Numerical dispersion
  corrections for, 267-269
  in finite differences, 259, 261

Outcrop
  permeability measurements, 96-102
  photos, 94-96

P-normal distribution, 29, 51
Percolation theory, 204-211, 215
Percolation thresholds, 217-218, 223
Period, 37, 55
Permeability
  correlation to porosity, 199-201, 215
  cutoffs, 215-218
  definition of, 197
  dependence on scale, 220-223
  differences in 2D and 3D, 219-220
  effective, *see* Effective permeability
  from percolation theory, 215-218
  in outcrop, 96-102
  relative, *see* Relative permeability
  scaling up, 202-214
Permeability distribution
  generation of, 198-202
  in flow simulations, 241
Phase
  of 1D sequence, 37, 57
  of 2D array, 118, 127, 132
  of 3D array, 187-189
  of sine and cosine curves, 37-39
  of smoothed trace, 157-159

Porosity
correlation with permeability, 199-201, 215, 220-223
scaling up, 202-203
well logs, 4, 80, 86-93
Porosity distribution
fBm cross sections, 151-152, 178-180
fGn cross sections, 141-149, 171-177
in flow simulations, 241
in 3D, 152-156, 180-193
Pressure transient
buildup, 298
conventional model for, 300-303
drawdown, 299
falloff, 300
field data, 301
Pseudorelative permeability
flux-weighted method, 259, 295-296
for cross sections, 258-259
in miscible displacement, 259-262
in waterflood, 262-267
Kyte and Berry method, 259, 293-294
numerical dispersion correction, 267-269

Ranking realizations
method for, 279-283
example of, 283-286
R/S analysis
definition of, 53-54
example of, 40-41
for fBm, 47
for fGn, 42
Relative permeability
definition, 238-240
for grid blocks, 245, 277
from cores, 245-246
Renormalization, 204-211
Reservoir simulation
history of, 237
history matching, 240
scaling up input for, 240-241
Rhythm, 321
Roll-off of spectral density
due to bed resolution, 84
relationship to filter size, 124-128

San Andres formation
deposition of, 143-144
pressure buildup in, 301-303
thin section from, 304-305
waterflood in, 241
Saturation
in cross section, 245-247
in streamtube
starting values in flow simulations, 251
Scaling up
limitations of, 276-278
multiple phases, 241
permeability, 202-214
porosity, 202-203
Seismic data
integration with well data, 156-162, 194-196
related to porosity, 156
showing faults, 306-308, 324
Sierpinski carpet, 5-9
Spectral density
covariance from, 133-135
definition of, 37, 53
examples of, 39
model for core photos, 102, 118-122
of fBm, 42-46
of fGn, 42-46
related to amplitude, 58
roll-off of, *see* Roll-off of spectral density
Splotches
in core photos, 111
in cross sections, 142, 144-146
in 3D distributions, 153
noise, *see* Noise array
Standard deviation
definition of, 29, 50
dependence on grid size, 141
linear scaling of, 65
Stationarity, 26, 80
Stehfest algorithm, 331-332
Streamlines
effect of mobility ratio on, 249
for well pattern, 243
generation of, 291-292
Streamtube method
comparison with field data, 251-255
equations of, 289-293
limitations of, 245
major steps in, 241-244

Time average
definition of, 25-26, 49
examples of, 25-28
variogram, 66-67
Tracer simulations
in cross sections, 259
for ranking realizations, 280-286

Trend
areal mapping of, 24
effects on time average, 25-28
effects on ensemble average, 25-28
in well logs, 27, 80
removing, 80, 140
Trilinear interpolation, 152, 181-183
Tube network, 204-205, 230
Type curves
for conventional model, 301
for fractal model, 332-334

Univariant measures,
equations of, 50-51
seen in histogram, 28-30
of fBm, 42
of fGn, 42

Variance, 36-37, 50
Variogram
definition of, 52
ergodicity of, 65-67
examples of, 32-37
of fBm, 42
of fGn, 42, 67-77
stationarity of, 65-67

Well logs
definition of, 3
fGn character of, 80-94
input for cross sections, 138-149
input for 3D distributions, 152-154
music from, 315-323
tools, *see* Logging tools
Wellbore storage, 298, 301, 328

FRACTALS IN RESERVOIR ENGINEERING

H. H. Hardy
Richard A. Beier
Conoco Inc.

During the past decade considerable evidence has been accumulated to support the application of fractals to describe sedimentary rocks. Such descriptions have direct applications to the production of oil and gas from hydrocarbon-bearing rocks. This book reviews the supporting data and develops models for reservoir heterogeneity that are based on these findings. Specific applications treat porosity and permeability distributions that are needed for subsequent reservoir flow simulations.

Although the authors adopt the viewpoint of the reservoir engineer, the book is general enough to be of interest to geologists, geophysicists, or anyone trying to characterize disordered media and its transport properties. Engineers in other fields will find this book helpful in applying fractals to their respective applications. The main body of each chapter is a verbal description of the concepts. Mathematical developments are given in appendices which are not required for the understanding of the verbal descriptions. This organization makes the concepts assessible without requiring the readers to have any specific mathematical background.

ISBN-13: 978-981-02-2069-3
ISBN-10: 981-02-2069-3

2574 hc